

The Origin of Risk^{*}

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Abstract

We propose a tractable model in which risk, at both the micro and macro levels, is endogenous and driven by incentives. In the model, each firm chooses the mean and the variance of its productivity process, as well as how it covaries with the productivity of other firms. Aggregate risk arises when firms select productivity processes that are correlated with one another. The theory predicts that larger firms and those with lower markups are less volatile and less correlated with aggregate productivity. We find support for these predictions in the data. Through their impact on risk-taking decisions, distortions such as taxes and markups can make GDP more volatile in equilibrium. In a calibrated version of the model, removing distortions significantly reduces GDP volatility.

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1 Introduction

Risk affects a wide range of economic activities. At the micro level, it influences how households save and consume and how firms invest and innovate. When aggregated, these decisions shape the business cycle and long-run growth. Understanding the determinants of risk is therefore essential. Yet most macroeconomic models treat risk as an exogenous *primitive*, assuming that key objects like productivity follow predefined stochastic processes. In contrast, this paper explores the idea that risk is an endogenous *outcome*, and that its quantity and properties are driven by incentives. Under this view, risk depends on economic conditions, and policymakers can exert some control over it.

That economic agents can influence the risk they face seems natural. Firms, for instance, affect how much risk they are exposed to by deciding whom to hire, how to organize production, what projects to undertake, and where to locate plants. A US firm that relocates part of its production to China might reduce costs, but it also becomes more exposed to geopolitical risk and trade wars. Similarly, a farmer who grows crops near a river might benefit from reliable irrigation, but also becomes more vulnerable to flooding. Taken together, these individual exposure decisions shape the risk profile of the entire economy: if many farms locate near the river, crop yields become correlated through shared flood exposure, turning a localized flood into an aggregate shock.

Some stylized features of the data support the idea that risk is, at least in part, endogenous. At the aggregate level, the volatility of a country’s GDP is related to its income level, its political system, and the quality of its institutions. At the firm level, the volatility of productivity and its correlation with GDP vary systematically with firm size and markup levels. These patterns are hard to reconcile with risk that is purely exogenous.¹

To explore the origin of risk and its aggregate implications, we develop a parsimonious general equilibrium model in which risk is endogenous. We focus on the decisions made by firms and, given its importance, on the productivity risk they face. Rather than modeling every single decision affecting productivity risk, we abstract from underlying micro-mechanisms and let firms choose their productivity process directly. Specifically, we let them choose the mean and the variance of their productivity, as well as how it correlates with that of other firms. We make this choice operational in the model by assuming that there are underlying sources of risk, and that firms can adjust their exposure to these fundamental risk factors.

In the model, managing a firm’s risk exposure can reduce its productivity. For example, a firm might adopt rules and procedures to guard against accidents, cyberattacks, or data loss, but the

¹See Ramey and Ramey (1995), Alesina et al. (1996), Acemoglu and Zilibotti (1997), and Koren and Tenreyro (2007; 2013) for work on the link between GDP volatility and country characteristics. In Appendix A.4, we show cross-country evidence that aggregate TFP volatility decreases with per capita income, while it increases with taxes and markups. In Section 8, we provide firm-level evidence that the productivity of larger firms and those with lower markups is less volatile and less correlated with aggregate GDP.

constraints these measures impose on normal workplace operations can lower productivity. Risk management might also require resources: avoiding droughts by planting crops near the shore may involve renting an expensive piece of land, and navigating a rapidly changing market may require hiring skilled managers. In the model, these resources are supplied by a representative risk-averse household at a cost in terms of utility. When making risk-exposure decisions, firms balance these productivity and resource costs against the benefits of a safer risk profile. Consequently, while eliminating aggregate risk entirely is possible in this economy, the high cost that this would entail generally pushes agents to tolerate some amount of risk.

We assume that markets are complete, and firms therefore use the household’s stochastic discount factor when comparing cash flows in different states of the world. As a result, firms are concerned with how their risk exposure correlates with the aggregate economy. During downturns, the household’s marginal utility of consumption is high, making cash flows in those states particularly valuable. Firms therefore aim to lower their exposure to pro-cyclical risk factors. Since consumption risk is influenced by the aggregation of firms’ exposure decisions, firms diversify by seeking exposure to risk factors that are less correlated with those affecting other firms.

One of our objectives is to evaluate the impact of firm-level risk-exposure decisions on aggregate volatility. To properly capture how micro shocks translate into macro fluctuations, we embed the firms in an otherwise standard production network economy. The input-output structure implies that risk decisions by one firm affect its neighbors through supply chain linkages. The model also features exogenous wedges, potentially from taxes or markups, that create a gap between the price at which goods are sold and their production costs. These wedges are the only source of inefficiency in our economy. By varying them, we can examine how taxes and markups affect risk-taking decisions and, through that channel, aggregate risk.

We show that there exists a unique equilibrium in this economy, and that this equilibrium can be characterized as the solution to a distorted planning problem. This problem implies that equilibrium risk-exposure decisions balance the goals of increasing expected GDP, reducing the variance of GDP, and lowering the costs of managing risk. Because of the wedges, risk-exposure decisions are generally inefficient, and the economy tends to be overexposed to harmful risk factors. We also establish conditions under which this overexposure decreases expected GDP and increases aggregate volatility.

Because it admits a closed-form solution, our model delivers sharp predictions for how risk-taking behavior varies with firm characteristics. Since productivity multiplies the input bundle in a firm’s production function, the marginal benefit of managing productivity risk increases naturally with the size of the firm. In contrast, the cost of risk management, which includes the productivity and resource components, might not scale as fast with firm size. In such a case, larger firms have stronger incentives to manage risk, and they tend to be less exposed to harmful risks in equilibrium. As wedges reduce firm sizes, more distorted firms also tend to be more exposed to harmful risks.

We use detailed firm-level data that cover the near-universe of Spanish firms to evaluate these predictions. Consistent with the theory, we find that the variance of a firm’s TFP growth and its covariance with GDP growth decrease with its size and increase with its markup. Our model thus implies that a single force, firms’ incentive to manage risk, can account for these four key firm-level regularities.

Well-established facts about stock returns are also consistent with our model. As documented by Fama and French (1992), larger firms tend to comove less with the aggregate stock market. The mechanisms of the model offer an explanation for this stylized fact, provided that increases in productivity benefit a firm’s stock price. Additionally, we find that firms with higher markups exhibit greater return covariance with the stock market, in line with our theoretical predictions.

To evaluate the quantitative implications of endogenous risk for the macroeconomy, we calibrate the model to the Spanish economy. Our data cover most revenue-generating firms in Spain, and we replicate each of them in the calibrated model. We choose key parameters to precisely match data features relevant to the model’s mechanisms, including each firm’s markup, the volatility of its productivity, and the covariance of its productivity with GDP.

One key prediction of our model is that wedges tend to make GDP more volatile through their impact on risk-taking decisions. To evaluate the importance of this mechanism, we conduct an experiment in which we remove all wedges from the calibrated model. Without wedges, firms find it valuable to manage risk more aggressively, and the standard deviation of GDP falls substantially. This suggests that the quantitative impact of distortions such as taxes and markups on aggregate volatility can be substantial.

We also conduct a second experiment that mimics an uncertainty shock by raising the volatility of the economy’s underlying risk factor. When firms cannot adjust their risk-taking decisions, a 1% rise in this fundamental volatility raises GDP volatility by 0.40%. When firms are free to manage their risk, they reduce their exposure in response to the higher fundamental risk, and the same 1% rise raises GDP volatility by only 0.11%. This finding suggests that frictions and policies that impede firms’ ability to manage risk might have sizable adverse effects on aggregate volatility.

We test the robustness of these results to alternative wedge measures and to the inclusion of disaster risk in an augmented model (Rietz, 1988; Barro, 2006). The latter model is able to match key asset-pricing moments, such as the equity premium. We find that removing wedges significantly reduces the equity premium, as firms find it more valuable to manage risk more aggressively.

Literature review

The main contribution of this paper is to propose a parsimonious theory of aggregate fluctuations driven by endogenous risk exposure. In contrast, standard representative agent models of the business cycle such as Kydland and Prescott (1982) assume that aggregate TFP follows an exogenous

process. In models with individual firms, firm risk is generally exogenous but aggregate TFP risk can be endogenously driven by firms' decisions (Khan and Thomas, 2008, Clementi and Palazzo, 2016, Bloom et al., 2018, among many others). In contrast, our paper builds on the idea that firms can choose their own productivity processes.

The idea that firms have some control over their productivity has a long tradition in the growth literature. In endogenous (Romer, 1990; Grossman and Helpman, 1991; Aghion and Howitt, 1992) and semi-endogenous (Jones, 1995) growth models, firms invest in R&D to increase their productivity. A similar process is at work in models with firm dynamics in the spirit of Klette and Kortum (2004). In our model, each firm can also select the mean of its productivity process but, in contrast to those works, it can also pick its variance and its covariance structure. Aggregate fluctuations arise as a consequence.

Closer to our work are papers in the development and growth literature including Greenwood and Jovanovic (1990), Acemoglu and Zilibotti (1997), and Cole, Greenwood, and Sanchez (2016). In those models, agents can choose between low-risk, low-return projects and high-risk, high-return projects. When risk is undiversifiable, agents favor safer projects. As markets become more complete, however, riskier projects are preferred, leading to faster economic growth. In contrast, markets are always complete in our model, and we focus on business-cycle risk rather than long-run growth. We also investigate how distortions such as markups and taxes influence aggregate risk-taking behavior.

In the corporate finance literature, managers often select between projects with different risk levels, but the focus of these papers is usually on the agency problem between firm owners and managers (Jensen and Meckling, 1976; Ross, 1977). One distinguishing feature of our work is that firms also select the correlation of their productivity with that of other firms.

Our model shares features with network economies such as Long and Plosser (1983) and Acemoglu et al. (2012). Some of the propagation mechanisms at work in these models are also active in our paper. We also relate to the literature on wedges in production network economies, which includes Jones (2011), Baqaee and Farhi (2019), Liu (2019), and Bigio and La'O (2020). Unlike in standard Cobb-Douglas network economies, aggregate volatility depends on wedges in our setup through their impact on exposure decisions. Finally, our paper is related to Pellet and Tahbaz-Salehi (2023) and Kopytov et al. (2024) who study (exogenous) uncertainty in network economies.

The rest of the paper is organized as follows. We introduce our model of endogenous risk in Section 2 and derive some of its aggregate properties in Section 3. We characterize firms' equilibrium risk-taking decisions in Section 4 and discuss the existence, uniqueness, and efficiency of the equilibrium in Section 5. We then explore how primitives of the model shape aggregate risk (Section 6) and GDP (Section 7). In Section 8, we provide firm-level evidence that supports the predictions of the model. In Section 9, we calibrate the model and evaluate its quantitative implications. Section 10 concludes.

2 A model of endogenous risk

We study the origin of risk in an otherwise standard production network economy. The economy is populated by a set of firms, each producing a differentiated good that can be used as an intermediate input or for final consumption. Firms can choose the distribution of their productivity, and economic conditions influence those choices. Collectively, these micro-level decisions shape the aggregate risk profile of the economy.

2.1 Firms and production technologies

There are N firms, indexed by $i \in \{1, \dots, N\}$, each producing a differentiated good. Firm i has access to the constant-returns-to-scale Cobb-Douglas production function

$$F_i(\delta_i, L_i, X_i) = e^{a_i(\varepsilon, \delta_i)} \zeta_i L_i^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N X_{ij}^{\alpha_{ij}}, \quad (1)$$

where L_i is labor, $X_i = (X_{i1}, \dots, X_{iN})$ is a vector of intermediate inputs, and ζ_i is a normalization term.² We denote by α the matrix of intermediate input shares, $0 \leq \alpha_{ij} < 1$, and impose that $\sum_{j=1}^N \alpha_{ij} < 1$ for all i . That matrix describes the production network in this economy.³

Firms can choose the probability distribution of their total factor productivity $a_i(\varepsilon, \delta_i)$. To make this idea operational, we assume that $a_i(\varepsilon, \delta_i)$ depends on several sources of fundamental risk, collected in the $M \times 1$ vector ε , and on firm i 's exposure to those risks, captured by the $M \times 1$ vector δ_i . Specifically, we let

$$a_i(\varepsilon, \delta_i) = \delta_i^\top \varepsilon - b_i(\delta_i), \quad (2)$$

such that δ_{im} determines i 's exposure to risk factor ε_m . We do not restrict δ_i to be positive, so that firms can be negatively exposed to a factor. We assume that $\varepsilon \sim \mathcal{N}(\mu, \Sigma)$ follows a normal distribution, where the vector μ is the expected level of the risk factors, and the positive definite matrix Σ determines both the uncertainty about individual elements of ε and their correlations. In addition to the random component $\delta_i^\top \varepsilon$, firm i 's productivity depends on a productivity cost of risk exposure $b_i(\delta_i)$, which we discuss in more detail below.

By choosing δ_i , firm i implicitly selects the mean, the variance, and the covariance of its productivity,

$$\mathbb{E}[a_i] = \delta_i^\top \mu - b_i(\delta_i), \quad \mathbb{V}[a_i] = \delta_i^\top \Sigma \delta_i, \quad \text{and} \quad \text{Cov}[a_i, a_j] = \delta_i^\top \Sigma \delta_j. \quad (3)$$

²We set $\zeta_i^{-1} = \left(1 - \sum_{j=1}^N \alpha_{ij}\right)^{1 - \sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N \alpha_{ij}^{\alpha_{ij}}$ to simplify the unit cost expression given by (10) below. We adopt the convention that vectors $x = (x_1, \dots, x_N)$ are columns.

³It is natural to think that the risk-taking decisions of a firm affect its (perhaps remote) suppliers and customers and, through that channel, aggregate quantities. We include input-output linkages in the model to study that interaction, but they are not essential for some of the mechanisms of the model.

It follows that exposure to more volatile or more correlated factors leads to higher TFP volatility. The last term in (3) shows that the choice of δ_i also affects how i 's productivity covaries with that of other firms. As we will see, these correlations create aggregate risk, so that firm decisions matter for the risk profile of the overall economy. For example, if many firms choose to operate in the same geographic region, a localized earthquake could lead to a large drop in GDP.

Our specification of the fundamental sources of risk ε is purposefully abstract, and we do not take a stance on what exactly a specific risk factor ε_m is. In reality, firms make a large number of decisions that affect their risk profile: where to locate a plant, whom to hire, what project to develop, how to get financing, among many others. Each of these decisions involves particular tradeoffs, and including them all in the model would make it intractable. We instead adopt a holistic approach and focus on *how much* risk firms take on, and on how *correlated* that risk is across firms. We introduce the risk factor structure as a way to tractably model these choices.^{4,5}

2.2 Costs of risk exposure

The term $b_i(\delta_i)$ in the productivity process (2) captures the fact that managing risk might lead to a loss in productivity. For instance, a firm might reduce its risk by putting in place rules to avoid mistakes and accidents, but those rules might also constrain workers, which would lower their productivity. We assume that $b_i(\delta_i)$ takes the quadratic form

$$b_i(\delta_i) = \frac{1}{2} (\delta_i - \delta_i^\circ)^\top B_i (\delta_i - \delta_i^\circ), \quad (4)$$

where B_i is an $M \times M$ positive-definite matrix. As firm i can achieve risk exposure $\delta_i = \delta_i^\circ$ at no cost, we refer to δ_i° as the *natural risk exposure* of firm i .

In addition to this productivity cost, managing risk might also require the use of resources. For instance, renting a piece of land away from the shore might be needed to reduce flood risk or vulnerability to climate change, and hiring an experienced lobbyist might alleviate political risk. In the model, these resources are provided by the representative household at a cost in terms of utility, and they are available to firms at a price W_R . To achieve risk exposure δ_i , firm i requires $g_i(\delta_i)$ units of risk management resources. We further let $g_i(\delta_i)$ take the form

$$g_i(\delta_i) = \frac{1}{2} (\delta_i - \delta_i^\circ)^\top G_i (\delta_i - \delta_i^\circ), \quad (5)$$

⁴Ideally, we would let firms pick their productivity distributions in an arbitrary way, but modeling correlations across firms creates some issues. For instance, should firm i or firm j decide how their two productivity processes correlate? We use the underlying risk factors as a tractable modeling device to sidestep these problems while allowing firms to influence their correlation structure.

⁵While we keep the risk factors abstract for most of the paper, we provide an empirical exercise in Appendix A.6 in which ε corresponds to climate change risk. We show there that the response of Spanish firms to temperature shocks is in line with our theoretical predictions.

where G_i is an $M \times M$ positive-definite matrix. As with the productivity cost b_i , setting $\delta_i = \delta_i^\circ$ requires no resources.⁶

The cost structure implied by b_i and g_i is rich enough to allow for aggregate and firm-specific sources of risk. Indeed, there can be a risk factor ε_m for which deviating from $\delta_{jm}^\circ = 0$ is extremely costly for any firm $j \neq i$. In this case, ε_m is effectively specific to firm i . Likewise, the model can accommodate risks that cannot be mitigated by making b_i or g_i sufficiently steep. A constant firm-specific TFP shifter can also be introduced by assuming that there is a degenerate risk factor m with $\Sigma_{mm} = 0$ and that deviating from a fixed exposure to m is arbitrarily costly. Finally, firms can choose $\delta_i = 0$ and eliminate risk completely, although this might be costly. As such, the presence of risk in this economy is a choice made by economic agents responding to incentives.

The cost functions b and g imply that risk-taking incentives vary with firm size. Since productivity a_i multiplies the input bundle, the benefit of risk management increases with firm size. On the cost side, the two functions b_i and g_i affect firms of different sizes differently. Since b_i is a productivity cost, its influence also scales up with the size of the input bundle and, as we will see, its impact on risk decisions is size-invariant. In contrast, g_i does not scale with the size of the firm, and it therefore provides larger firms with a risk management advantage. When combined, b_i and g_i allow us to flexibly capture the link between firm size and risk-taking decisions.

Events in this economy unfold in two stages. Before ε is realized, firms choose how much risk management resources to purchase and pick δ . The household also decides how many risk management resources to supply at that time. After ε is drawn, consumption, labor, and intermediate inputs are chosen and their respective markets clear. This timing reflects the fact that risk management decisions (where to locate a plant, which technology to adopt, what workplace rules to impose, etc.) are long-lived and cannot be changed rapidly once shocks are realized.

2.3 Household preferences

A risk-averse representative household owns the firms and supplies one unit of labor inelastically. The household derives utility from a composite consumption good, Y , and incurs disutility from supplying R units of risk management resources. The consumption good is a Cobb-Douglas aggregate of the differentiated goods: $Y = \prod_{i=1}^N (\beta_i^{-1} C_i)^{\beta_i}$. We impose $\sum_{i=1}^N \beta_i = 1$, so that $\beta_i \geq 0$ is good i 's expenditure share in equilibrium. As this aggregate Y also corresponds to the economy's value added, we refer to it as real GDP.

⁶We do not take a stance on the underlying adjustment margins that give rise to the cost functions b_i and g_i . In Appendix F.3, we propose different possible microfoundations that involve 1) researching a new technology, 2) a location problem, 3) workplace rules, and 4) hiring managers with different skills. In reality, many such microfoundations probably interact to give rise to b_i and g_i . Assuming quadratic costs allows us to explore how endogenous risk interacts with the macroeconomy in a tractable manner.

The household has King, Plosser, and Rebelo (1988) preferences that are given by⁷

$$\mathcal{U}(Y) \mathcal{V}(R), \quad (6)$$

where $\mathcal{U}(Y) = Y^{1-\rho}/(1-\rho)$ is CRRA with relative risk aversion parameter $\rho > 1$.⁸ The disutility from supplying risk management resources is given by $\mathcal{V}(R) = V(R)^{1-\rho}$, where we assume the functional form $V(R) = \exp(-\eta R)$. The parameter $\eta > 0$ controls how costly it is for the household to supply risk management resources.

The household's decisions unfold in two stages. Before uncertainty is realized, it chooses the quantity of risk management resources R to supply. After the shock ε is drawn, it makes its consumption decisions. In each state of the world, the household faces the budget constraint

$$\sum_{i=1}^N P_i C_i \leq W_L + W_R R + \Pi, \quad (7)$$

where P_i is the price of good i , Π is total firm profits, W_R is the price of risk management resources, and W_L is the wage, which we normalize to one as the numeraire (see Appendix F.2 for details). We further define $\bar{P} = \prod_{i=1}^N P_i^{\beta_i}$ as the consumption price index.

Solving the problem of the household provides a relationship between the supply of risk management resources R and their price W_R :

$$-\bar{P} Y \frac{V'(R)}{V(R)} = W_R. \quad (8)$$

Because of the exponential form for $V(R)$, this expression simplifies to $\eta \bar{P} Y = W_R$. This implies that the price of risk management resources is the product of η and nominal GDP. A higher η therefore makes risk management more expensive.

The problem of the household also allows us to derive another important quantity. We assume that markets are complete in this economy, meaning that there is a complete set of state- ε contingent claims. Since the household can trade these claims, their prices reflect the marginal utility of consumption in each state of the world. As usual, this implies that firms use the household's stochastic discount factor,

$$\Lambda = \frac{1}{\bar{P}} \frac{d}{dY} [\mathcal{U}(Y) \mathcal{V}(R)], \quad (9)$$

to discount profits (see Appendix F.1 for a derivation).

⁷The balanced-growth utility of King, Plosser, and Rebelo (1988) implies that the substitution and the income effects associated with the price of risk management resources cancel each other and allows us to derive closed-form expressions for aggregate quantities. Appendix G.3 shows how to extend the analysis to more general preferences.

⁸When $\log Y$ is normally distributed, maximizing $E[(1-\rho)^{-1} Y^{1-\rho}]$ amounts to maximizing $E[\log Y] - \frac{1}{2}(\rho-1)V[\log Y]$ such that $\rho \leq 1$ indicates whether the household prefers or is averse to uncertainty in log consumption. This is a consequence of the usual increase in the mean of a log-normal variable from an increase in the variance of the underlying normal variable.

2.4 Firm problem

As in any competitive model with constant returns to scale, firms take prices $P = (P_1, \dots, P_N)$ and the demand for their good Q_i as given. As we will see, these two variables are random in equilibrium.⁹

We solve the problem of firm i in two stages: 1) before ε is realized, the firm chooses its risk exposure δ_i , and 2) after ε is realized, it decides on how much labor to hire and on how many intermediate inputs to purchase, given its δ_i . In that second stage, ε is known and δ_i is fixed, and firm i 's cost minimization problem implies a standard Cobb-Douglas unit cost of production

$$K_i(\delta_i, P) = \frac{1}{e^{a_i(\varepsilon, \delta_i)}} \prod_{j=1}^N P_j^{\alpha_{ij}}. \quad (10)$$

The cost of producing one unit of good i is equal to the geometric average of the individual input prices, weighted by their respective shares, and adjusted for total factor productivity. Total factor productivity, in turn, depends on the firm's risk-exposure decision δ_i .

Given (10), the first stage of the firm's problem involves choosing risk exposure δ_i to maximize expected discounted profits

$$\delta_i^* \in \arg \max_{\delta_i} \mathbb{E} [\Lambda [P_i Q_i - K_i(\delta_i, P) Q_i - g_i(\delta_i) W_R]]. \quad (11)$$

The terms in the inner square bracket are sales, cost of goods sold, and risk management expenditure, respectively. They are multiplied by the stochastic discount factor Λ , which reflects the value of cash flows in different states of the world from the perspective of the representative household. As a result, firms are influenced by the representative household's attitude toward risk.

2.5 Equilibrium conditions

Prices are set at an exogenous wedge $\tau_i \geq 0$ over unit costs, such that

$$P_i = (1 + \tau_i) K_i(\delta_i, P) \text{ for all } i. \quad (12)$$

These wedges can be interpreted as markups, taxes, or other distortions. When $\tau_i = 0$ for all i , prices are fully competitive.¹⁰

An equilibrium is defined by the optimality conditions of the household and the firms holding

⁹Firms behave atomistically, with no control over prices or the demand they face. Kopytov et al. (2024) micro-founds this competitive structure in a similar economy by assuming that each index i is a sector with a continuum of measure-zero firms, with buyers and sellers meeting through a search process. An information friction prevents buyers from targeting specific firms within a sector, so individual firms cannot influence the demand they face.

¹⁰Due to constant returns to scale, risk management expenses lead to negative profits for τ near zero. To keep firms in operation, lump-sum transfers can be introduced without affecting the forces of the model.

simultaneously, and by all markets clearing.

Definition 1. An *equilibrium* is a choice of risk-exposure decisions $\delta^* = (\delta_1^*, \dots, \delta_N^*)$ and a stochastic tuple $(P^*, W_R^*, C^*, L^*, R^*, X^*, Q^*)$ such that

1. (Optimal risk-exposure choice) For each i , the risk-exposure decision δ_i^* solves (11) given prices P^* , the price of risk management resources W_R^* , demand Q_i^* , and the stochastic discount factor Λ^* given by (9).
2. (Optimal input choice) For each i , factor demands per unit of output L_i^*/Q_i^* and X_i^*/Q_i^* are a solution to the cost minimization problem, $K_i(\delta_i, P) = \min_{L_i, X_i} \left(L_i + \sum_{j=1}^N P_j X_{ij} \right)$ subject to $F_i(\delta_i, L_i, X_i) \geq 1$, given prices P^* and W_R^* , and the chosen risk exposure δ_i^* .
3. (Consumer maximization) The consumption vector C^* and the supply of risk management resources R^* maximize expected utility (6) subject to (7) given prices P^* and W_R^* .
4. (Unit cost pricing) For each i , P_i^* solves (12) where $K_i(\delta_i^*, P^*)$ is given by (10).
5. (Market clearing) For each i ,

$$C_i^* + \sum_{j=1}^N X_{ji}^* = Q_i^* = F_i(\delta_i^*, L_i^*, X_i^*), \quad \sum_{i=1}^N L_i^* = 1, \quad \text{and} \quad \sum_{i=1}^N g_i(\delta_i^*) = R^*. \quad (13)$$

Conditions 2 to 5 are standard and imply that firms and the household optimize and that all markets clear at equilibrium prices. Condition 1 emphasizes that risk-taking decisions are equilibrium objects that depend on the primitives of the economy. Together, these decisions shape the aggregate risk structure of the economy.

We have kept the model parsimonious to make the exposition transparent, but it is straightforward to extend it in several ways. For instance, many of our results still hold when we relax the assumption that b_i and g_i are quadratic. We can similarly relax the functional form assumptions on the utility function $\mathcal{U}$ and the disutility of supplying risk management resources $\mathcal{V}$. Furthermore, the model can be written with many different risk-management resources, each with its own supply elasticity. We work out these extensions in Appendix G. Other assumptions, however, are required for tractability. Without the Cobb-Douglas structure, the price vector P has no closed-form distribution given δ , which rules out an analytical solution. Departing from marginal-cost pricing also introduces complications. Even so, we explore a CES generalization in Appendix E in which firms understand the impact of their decisions on the demand they face. The main mechanisms carry over, but that model cannot be solved in closed form.

Some features of our setup resemble the classical CCAPM with normal shocks (Lucas Jr, 1978). While highly tractable, this model is well known to struggle to match standard asset pricing moments, such as the equity premium, implying that agents in our economy may not exhibit a realistic

attitude toward risk (Mehra and Prescott, 1985). To address this limitation, Appendix D extends the model to incorporate disaster risk (Rietz, 1988; Barro, 2006). We also explore the implications of disasters when we calibrate the model in Section 9.

3 Domar weights, equilibrium prices, and GDP

When making risk decisions, firms take into account the equilibrium distributions of prices, demands, and real GDP, the latter being relevant for the stochastic discount factor. In this section, we characterize these objects.

3.1 Domar weights and prices

Following Baqaee and Farhi (2019), we define the revenue-based input-output matrix Ω and the cost-based input-output matrix $\tilde{\Omega}$ as

$$\Omega_{ij} := \frac{P_j X_{ij}}{P_i Q_i} \quad \text{and} \quad \tilde{\Omega}_{ij} := \frac{P_j X_{ij}}{W_L L_i + \sum_{k=1}^N P_k X_{ik}}.$$

The elements Ω_{ij} and $\tilde{\Omega}_{ij}$ represent firm i 's expenditure on good j as a share of its total sales and of its input cost excluding risk management expenses, respectively. Because firms have Cobb-Douglas production functions, these shares are constant, with $\tilde{\Omega}_{ij} = \alpha_{ij}$ and $\Omega_{ij} = \alpha_{ij}/(1 + \tau_i)$. The latter equation follows since τ_i captures the gap between a firm's revenue and its production cost.

From Ω and $\tilde{\Omega}$, we also define the revenue-based and cost-based Leontief inverse matrices as

$$\mathcal{L} := (I - \Omega)^{-1} = I + \Omega + \Omega^2 + \dots \quad \text{and} \quad \tilde{\mathcal{L}} := (I - \tilde{\Omega})^{-1} = I + \tilde{\Omega} + \tilde{\Omega}^2 + \dots \quad (14)$$

While Ω_{ij} and $\tilde{\Omega}_{ij}$ capture the direct exposure of firm i to firm j as a share of revenues and costs, $\mathcal{L}_{ij}$ and $\tilde{\mathcal{L}}_{ij}$ capture both direct and indirect exposures through network linkages.

We also introduce two measures of firm importance. Firm i 's revenue-based Domar weight $\omega_i := P_i Q_i / (\bar{P} Y)$ is the ratio of its sales to nominal GDP. Market clearing (13) and household optimality imply that $\omega^\top = (\omega_1, \dots, \omega_N)^\top = \beta^\top \mathcal{L}$. Its cost-based counterpart is $\tilde{\omega}^\top := \beta^\top \tilde{\mathcal{L}}$. Both ω_i and $\tilde{\omega}_i$ capture firm i 's importance as a supplier, in terms of the revenue or the cost of its customers, and are large when the household favors i 's good (β_i large) or when another firm the household favors relies on i (large α_{ji} and β_j). Higher-order linkages also matter, as the expansions in (14) show. Without input-output linkages ($\alpha = 0$), both vectors equal β . The two measures differ only through wedges: because the revenue-based input share $\Omega_{ij} = \alpha_{ij}/(1 + \tau_i)$ falls with τ_i while the cost-based share $\tilde{\Omega}_{ij} = \alpha_{ij}$ does not, revenue-based weights are weakly decreasing in τ and cost-based weights are independent of it. In the absence of wedges, the two measures coincide. Neither measure depends on firms' risk-exposure decisions.

Combining these definitions with (10) and (12), we can write the vector of log prices as

$$p = -\tilde{\mathcal{L}}(\delta\varepsilon - b(\delta) - \log(1 + \tau)), \quad (15)$$

where $p = (\log P_1, \dots, \log P_N)$, $\log(1 + \tau) = (\log(1 + \tau_1), \dots, \log(1 + \tau_N))$, $b(\delta) = (b_1(\delta_1), \dots, b_N(\delta_N))$, and δ is the $N \times M$ matrix with typical element δ_{im} . It follows that the price p_i of good i is high if firm i 's productivity $a_i(\varepsilon, \delta_i) = \delta_i^\top \varepsilon - b_i(\delta_i)$ is low, or if the productivity of one of its direct or indirect suppliers is low. Equation (15) also makes clear that wedges affect prices in the same way as a decline in productivity. Finally, since ε is normally distributed, so is p .

3.2 Moments of GDP

While δ describes firms' *individual* risk exposure, it will be convenient to work with an *aggregate risk-exposure* measure, which we define as $\Delta := \delta^\top \tilde{\omega}$. An element $\Delta_m = \sum_{i=1}^N \tilde{\omega}_i \delta_{im}$ of that vector is the (cost-based) Domar-weighted sum of each firm's exposure to factor m .

With that definition in hand, we can derive an expression for real GDP.

Lemma 1. *Log (real) GDP $y = \log Y$ is given by*

$$y = \Delta^\top \varepsilon - \tilde{\omega}^\top b(\delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L, \quad (16)$$

where $\Gamma_L := W_L L / (\bar{P}Y) = 1 - \tau^\top (\text{diag}(1 + \tau))^{-1} \omega$ is the labor share of income.

The first term in (16) shows that the contribution of factor ε_m to GDP is proportional to the aggregate exposure Δ_m to this factor. A factor is therefore a strong driver of GDP when many firms, or firms with large Domar weights, are heavily exposed to it. The remaining terms in (16) are deterministic: the second reflects productivity losses from risk management, while the last two capture distortions from wedges. When $\tau = 0$, cost-based and revenue-based Domar weights coincide, such that (16) collapses to $y = \omega^\top a(\varepsilon, \delta)$, and Hulten's (1978) theorem applies.

Because ε is normally distributed, so is log GDP. Its moments are

$$\mathbb{E}[y] = \Delta^\top \mu - \tilde{\omega}^\top b(\delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L \quad \text{and} \quad \mathbb{V}[y] = \Delta^\top \Sigma \Delta. \quad (17)$$

The first moment shows that increasing aggregate exposure Δ_m raises expected log GDP by the factor's expected productivity μ_m . Thus, higher exposure to high-mean factors is generally beneficial for $\mathbb{E}[y]$, though this may be counteracted by the TFP cost $b(\delta)$. That equation also implies that an increase in μ_m has a beneficial impact on $\mathbb{E}[y]$ if the economy is positively exposed to ε_m ($\Delta_m > 0$), but reduces expected log GDP otherwise.

The second moment in (17) describes the determinants of aggregate risk and is central to our analysis. It shows that an increase in the variance Σ_{mm} of a factor always leads to an increase in

the volatility of GDP. The magnitude of that increase is proportional to Δ_m^2 , such that its impact is stronger when Δ_m is very positive or very negative. In both cases, the economy is particularly vulnerable to ε_m . Correlations also matter. The impact of Σ_{mn} on $V[y]$ depends on the sign of $\Delta_m \Delta_n$. If the economy is positively or negatively exposed to *both* risk factors, a higher covariance increases aggregate risk. In contrast, with positive exposure to one factor and negative exposure to the other, an increase in covariance stabilizes the economy and $V[y]$ declines with Σ_{mn} .

Beyond the effect of Σ , changes in exposure decisions also affect the variance of GDP. One can show that

$$\frac{dV[y]}{d\Delta_m} = 2 \text{Cov}[y, \varepsilon_m] = 2 \sum_{n=1}^M \Delta_n \text{Cov}[\varepsilon_n, \varepsilon_m]. \quad (18)$$

The first equality implies that if ε_m is positively correlated with GDP, an increase in Δ_m adds risk to the economy and raises $V[y]$. The second equality follows since the stochastic part of GDP is $\Delta^\top \varepsilon$. It implies that the response of $V[y]$ to Δ_m depends on how correlated ε_m is with other risk factors, and on the economy's exposure to those factors. If the economy is positively exposed to a factor ε_n ($\Delta_n > 0$), and if ε_m and ε_n are positively correlated, an increase in Δ_m adds to the risk generated by ε_n , which contributes to a higher $V[y]$. In contrast, if ε_m and ε_n are negatively correlated, increasing Δ_m offsets some of the fluctuations generated by ε_n .

Finally, (17) also shows that wedges have no *direct* impact on $V[y]$. Any influence on aggregate risk that they might have must therefore operate *indirectly* through their impact on firms' endogenous risk-taking decisions.

4 Firm risk-exposure decisions

Since firms take P and Q as given, problem (11) reduces to minimizing a risk-adjusted version of production costs. We can rewrite it as

$$\delta_i^* \in \arg \min_{\delta_i} E[K_i(\delta_i, P) Q_i] + \text{Cov}(K_i(\delta_i, P) Q_i, \Lambda) / E[\Lambda] + g_i(\delta_i) W_R. \quad (19)$$

When firm i chooses its risk exposure δ_i , it naturally aims to minimize the expected cost of goods sold $E[K_i Q_i]$ and risk management expenses $g_i(\delta_i) W_R$. Risk also plays a role: the second term in (19) indicates that the firm seeks to minimize the covariance of its production cost $K_i Q_i$ with the stochastic discount factor Λ . The firm prefers to have low costs when the marginal utility of consumption is high or, equivalently, when GDP is low. Extra production is particularly valuable to the household in those states. If the firm is pro-cyclical, which is the case for the typical firm, minimizing $\text{Cov}(K_i Q_i, \Lambda)$ involves lowering the variance of the cost of goods sold $V[K_i(\delta_i, P) Q_i]$. This can be achieved by relying less on volatile risk factors, or by diversifying through exposure to

risk factors that offset each other.¹¹

4.1 The marginal value of risk exposure

The first-order condition from (19) provides more insight into the determinants of risk exposure.

Lemma 2. *In equilibrium, the risk exposure decision of firm i solves¹²*

$$\underbrace{\mathcal{E} K_i Q_i}_{\text{Marginal benefit of risk exposure}} = \underbrace{\nabla b_i(\delta_i) K_i Q_i + \nabla g_i(\delta_i) W_R}_{\text{Marginal cost of risk exposure}}, \quad (20)$$

where the marginal value of risk exposure $\mathcal{E}$ is defined as

$$\mathcal{E} := \mathbb{E}[\varepsilon] + \text{Cov}[\lambda, \varepsilon], \quad (21)$$

and where $\lambda := \log \Lambda$ is the log of the stochastic discount factor.

Firm optimization implies that the marginal cost of risk exposure, given by the right-hand side of (20), must be equal to its marginal benefit, $\mathcal{E} K_i Q_i$. This benefit, in turn, depends on the vector $\mathcal{E}$ that captures the marginal benefit of exposure per unit of firm size, and on the size of the firm itself, measured by its cost of goods sold $K_i Q_i$. Intuitively, since productivity multiplies the input bundle, changing its distribution is more consequential for large firms.

The marginal value of risk exposure $\mathcal{E}$ plays an important role in our analysis. Equation (21) shows that, unsurprisingly, risk factors with higher expected productivity $\mathbb{E}[\varepsilon] = \mu$ provide a higher marginal benefit of exposure $\mathcal{E}$. This benefit is also higher for risk factors that deliver high productivity in states of the world with high λ . These factors perform well when the household is poor, and they thus provide insurance in high marginal utility states. We will later see that $\mathcal{E}$ also corresponds to the marginal *welfare* benefit of aggregate exposure to ε . As a result, we say that a risk factor ε_m is *good* if $\mathcal{E}_m > 0$, and that it is *bad* if $\mathcal{E}_m < 0$.

The right-hand side of (20) shows that the marginal cost of exposure depends on the curvature of the productivity and resource cost functions b_i and g_i . Importantly, these two costs interact with firm size differently. Since b_i is a productivity cost, its importance rises as the firm grows, and so its impact in (20) scales with the size of the firm $K_i Q_i$. In contrast, the resource cost g_i does not, so it matters less for larger firms. Larger firms in the model therefore enjoy a risk-management advantage and manage risk more intensively. We provide direct empirical support for this prediction in Section 8.

¹¹ Although *equilibrium* $K_i Q_i$ is a constant fraction of nominal GDP, as we show in (25) below, our competition assumptions imply that the firm does not internalize this identity when solving (19). From its perspective, K_i varies with its choice δ_i , so both the expectation and the covariance in (19) depend on it.

¹² As we discuss below, the ratio $K_i Q_i / W_R$, and therefore the whole expression (20), is deterministic *in equilibrium*.

4.2 Individual risk-taking decisions

The right-hand side of (20) shows the separate influence of b_i and g_i on risk-exposure decisions, but it will often be convenient to combine their impact into an *effective* cost of exposure function¹³

$$h_i(\delta_i) := \frac{1}{2} (\delta_i - \delta_i^\circ)^\top H_i (\delta_i - \delta_i^\circ), \text{ where } H_i := B_i + \frac{W_R}{K_i Q_i} G_i. \quad (22)$$

Using this notation, the firm's first-order condition (20) simplifies to equating the marginal benefit of risk exposure to its marginal cost:

$$\mathcal{E} = \nabla h_i(\delta_i). \quad (23)$$

Solving this equation for δ_i yields the firm's optimal risk-taking decision.

Lemma 3. *The equilibrium risk-taking decision of the firm is*

$$\delta_i = \delta_i^\circ + H_i^{-1} \mathcal{E}. \quad (24)$$

Proof. The result follows immediately from solving (23) for δ_i . □

The firm's optimal risk exposure δ_i deviates from its natural level δ_i° whenever the risk factors are not neutral ($\mathcal{E} \neq 0$). The magnitude of this deviation depends on H_i^{-1} , which captures the curvature of the productivity and resource cost functions b_i and g_i . Flatter cost functions (a smaller H_i) make the firm more responsive to risk-taking incentives.

Since H_i is positive definite, its inverse has a positive diagonal. Consequently, an increase in $\mathcal{E}_m$ always leads to an increase in δ_{im} . Firms are therefore more exposed to attractive risk factors. In contrast, the response of δ_{in} for $n \neq m$ can be positive or negative depending on the sign of $[H_i^{-1}]_{mn}$. If $[H_i^{-1}]_{mn} > 0$, we say that risk factors m and n are *local complements*. In this case, an increase in $\mathcal{E}_m$ also leads to an increase in δ_{in} . If, instead, $[H_i^{-1}]_{mn} < 0$, m and n are *local substitutes*, and a beneficial change to $\mathcal{E}_m$ makes the firm move away from risk factor n .

These substitution patterns can be used to describe physical constraints a firm might face when managing different risks. For instance, suppose that technological constraints push a firm to locate its plants in a single location: either by the shore, where floods can happen, or in a plain, where droughts sometimes occur. The Hessian H_i can be parametrized to capture that situation by imposing that exposure to either flood or drought risk is affordable, but that exposure to both risks is prohibitively costly. In this case, flood and drought risks would be local substitutes.

¹³Expression (25) below, together with the household optimality condition $\eta \bar{P}Y = W_R$, implies that $W_R / (K_i Q_i) = \eta(1 + \tau_i) / \omega_i$, so that in equilibrium H_i is only a function of parameters.

4.3 Sales, wedges, and risk management

Since a firm's cost of goods sold $K_i Q_i$ matters for its risk-exposure decisions, it is important to understand its determinants. In equilibrium, we can write

$$K_i Q_i = \frac{P_i Q_i}{1 + \tau_i} = \frac{\omega_i}{1 + \tau_i} \bar{P} Y, \quad (25)$$

where the second equality follows from the definition of revenue-based Domar weights. Equation (25) makes clear that higher sales ω_i lead to higher production costs $K_i Q_i$. As a result, higher demand from the household (β) or the firms (α) increases $K_i Q_i$ through their impact on ω_i .

Wedges also play a crucial role, with an increase in τ_i leading to a decline in $K_i Q_i$. The intuition is straightforward. Since a firm's unit cost K_i is driven by the price of its inputs, it does not depend on τ_i in partial equilibrium. An increase in τ_i therefore leads directly to an increase in the price $P_i = (1 + \tau_i) K_i$, and that higher price pushes for a decline in demand Q_i . Since fewer units now have to be produced, the total cost of goods sold $K_i Q_i$ also declines.

Through their impact on firm size, Domar weights ω and wedges τ directly affect the incentives to manage risk. Recall from (20) that the benefit of managing risk grows one-for-one with $K_i Q_i$, while its cost grows at a slower pace. As a result, firms with higher ω_i or lower τ_i are willing to spend more on risk management, and they achieve better risk outcomes in return.

Corollary 1. *Firms with higher revenue-based Domar weights ω_i and lower wedges τ_i manage risk more aggressively, in the sense that¹⁴*

$$\frac{\partial [\mathcal{E}^\top \delta_i]}{\partial \omega_i} \geq 0 \quad \text{and} \quad \frac{\partial [\mathcal{E}^\top \delta_i]}{\partial \tau_i} \leq 0.$$

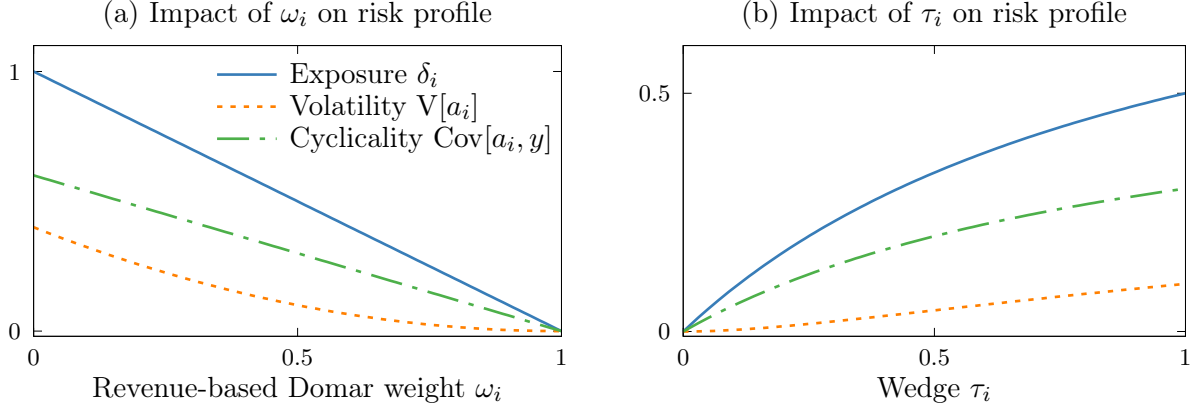
This result implies that firms with larger sales ω_i and smaller wedges τ_i tend to be more exposed to good risk factors ($\mathcal{E}_m > 0$) and less exposed to bad ($\mathcal{E}_m < 0$) ones. Since bad risk factors are more correlated with GDP, firms with less exposure to those factors tend to be less procyclical.

Example. Figure 1 illustrates these forces in a simple economy with a single risk factor ε . To fix ideas, we assume that ε is bad ($\mathcal{E} < 0$), and that the aggregate economy is positively exposed to it ($\Delta > 0$). A risk factor that drives the business cycle could fit that description. Panels (a) and (b) show the impact of increasing ω_i and τ_i on firm i 's risk profile. As ω_i increases or τ_i decreases, the production cost $K_i Q_i$ of the firm rises per (25) and so does its incentive to manage risk. As a result, firm i reduces its exposure δ_i to the bad risk factor ε , in line with Corollary 1. Figure 1 also shows the impact of this change on the variance of the firm's productivity a_i , given by (3), and its covariance with GDP, given by $\text{Cov}[a_i, y] = \delta_i^\top \Sigma \Delta$. Since both quantities increase with δ_i , high- ω_i and low- τ_i firms are less volatile and covary less with GDP. We will see in Section 8 that

¹⁴The partial derivatives imply a partial equilibrium analysis in which $\mathcal{E}$ is kept constant.

these patterns are also visible in firm-level Spanish data.

Figure 1: Impact of the sales share ω_i and wedge τ_i on firm i 's risk profile



Notes: Parameters are $\mathcal{E} = -1$, $G_i = W_R = \bar{P}Y = \delta_i^\circ = 1$, $\Sigma = 0.4$, $B_i = 0$ and $\Delta = 1.5$. Panel (a): Responses of δ_i , $V[a_i]$ and $\text{Cov}[a_i, y]$ to a change in ω_i from 0 to 1. Panel (b): Responses of δ_i , $V[a_i]$ and $\text{Cov}[a_i, y]$ to a change in τ_i from 0 to 1.

5 Equilibrium existence, uniqueness, and efficiency

In the previous section, we characterized firm risk-exposure decisions δ as a function of the marginal value of exposure $\mathcal{E}$. In equilibrium, however, $\mathcal{E}$ itself depends on exposure decisions. In this section, we consider the full equilibrium mapping in which $\mathcal{E}$ is endogenous. We begin with the social planner's problem and the efficient allocation. We then characterize the equilibrium of the model as the outcome of a *distorted* planner's optimization problem. We also show that there exists a unique equilibrium and that, because of the wedges, it is in general inefficient.

5.1 A planning problem

We consider the problem of a social planner that maximizes the expected utility of the household subject to the resource constraints (13). To describe that problem, it is convenient to introduce the *aggregate cost function* $\bar{h}_{SP}$, defined as the value function

$$\bar{h}_{SP}(\Delta) := \min_{\delta} \tilde{\omega}^\top b(\delta) - \log V \left(\sum_{i=1}^N g_i(\delta_i) \right), \quad \text{subject to } \Delta = \delta^\top \tilde{\omega}. \quad (26)$$

For a given aggregate risk-exposure vector Δ , $\bar{h}_{SP}(\Delta)$ corresponds to the smallest possible (log) utility cost required to achieve Δ . That cost is the sum of a TFP loss (first term) and a utility cost of supplying risk-management resources (second term). The minimization problem (26) also implicitly defines a function $\delta_{SP}(\Delta)$ that provides the firm-level risk-exposure matrix δ that minimizes the

welfare cost of achieving Δ . We can further decompose $\bar{h}_{SP}$ into its components

$$\bar{h}_{SP}(\Delta) = \underbrace{\tilde{\omega}^\top b(\delta_{SP}(\Delta))}_{\bar{b}_{SP}(\Delta)} - \underbrace{\log V\left(\sum_{i=1}^N g_i(\delta_{SP,i}(\Delta))\right)}_{\bar{g}_{SP}(\Delta)}, \quad (27)$$

where $\bar{b}_{SP}(\Delta)$ and $\bar{g}_{SP}(\Delta)$ are the optimal aggregate productivity and resource costs of achieving aggregate risk exposure Δ .

With those definitions in hand, the problem of the social planner can be written as

$$\mathcal{W}_{SP} := \max_{\Delta} \underbrace{\Delta^\top \mu - \bar{b}_{SP}(\Delta)}_{E[y_{SP}]} - \frac{1}{2}(\rho - 1) \underbrace{\Delta^\top \Sigma \Delta}_{V[y_{SP}]} - \bar{g}_{SP}(\Delta), \quad (28)$$

where $\mathcal{W}_{SP}$ provides a measure of welfare in the efficient allocation. This problem shows that the efficient aggregate risk exposure seeks to maximize the expected value of log GDP $E[y_{SP}]$, while minimizing its volatility $V[y_{SP}] = \Delta^\top \Sigma \Delta$ and the risk management utility cost $\bar{g}_{SP}$. Unsurprisingly, the importance of the variance term increases with the risk aversion ρ of the household.

5.2 A distorted planning problem

In the presence of wedges, the equilibrium allocation generally differs from the efficient one. We can, however, formulate a *distorted* version of the planning problem (28) whose solution coincides with the equilibrium. This alternative problem relies on a distorted version of $\bar{h}_{SP}$, defined as

$$\bar{h}(\Delta) := \min_{\delta} \tilde{\omega}^\top b(\delta) - \log V\left(\sum_{i=1}^N \kappa_i g_i(\delta_i)\right), \quad \text{subject to } \Delta = \delta^\top \tilde{\omega}. \quad (29)$$

We also define $\delta^*(\Delta)$ as the solution to (29), and the functions $\bar{b}(\Delta)$ and $\bar{g}(\Delta)$ as the analogues of their efficient counterparts from (27). The cost functions $\bar{h}$ and $\bar{h}_{SP}$ differ solely through the *efficiency gaps* $\kappa_i := (1 + \tau_i) \tilde{\omega}_i / \omega_i \geq 1$, which multiply g_i in (29). It follows that a higher κ_i raises the cost of allocating risk-management resources to firm i . Without wedges, the efficiency gaps are equal to one and $\bar{h} = \bar{h}_{SP}$. Higher wedges raise κ_i and amplify the distortion in $\bar{h}$.¹⁵

The distortion has a clear interpretation in terms of social versus private benefits of risk management. From (16), firm i 's contribution to GDP is proportional to its cost-based Domar weight $\tilde{\omega}_i$, and therefore so is the social value of its risk management. The firm's private incentive, by contrast, scales with $K_i Q_i \propto \omega_i / (1 + \tau_i)$ via the FOC (20). The two coincide when $\tau = 0$ but diverge as τ rises: $\tilde{\omega}_i$ stays fixed while $K_i Q_i$ shrinks. The ratio κ_i captures exactly this wedge between social

¹⁵As we have seen, increasing τ_j lowers the revenue-based Domar weights of all firms. It follows that $d\kappa_i/d\tau_j \geq 0$ (see (80) in the appendix for the exact formula). Without input-output linkages ($\alpha = 0$), the efficiency gaps take a particularly simple form, $\kappa_i = 1 + \tau_i$, and a change in wedge τ_i only affects the efficiency gap of firm i .

and private incentives.

We will use $\bar{h}$ to characterize the equilibrium, so it is useful to first derive some of its properties.

Lemma 4. *The aggregate cost function $\bar{h}$ is given by*

$$\bar{h}(\Delta) = \frac{1}{2} (\Delta - \Delta^\circ)^\top \bar{H} (\Delta - \Delta^\circ), \quad (30)$$

where $\Delta^\circ := (\delta^\circ)^\top \tilde{\omega}$. Furthermore, the Hessian matrix of $\bar{h}$ is given by

$$\bar{H} = \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \right]^{-1}, \quad (31)$$

where H_i is given by (22).

This result shows that the distorted aggregate cost $\bar{h}(\Delta)$ of reaching aggregate exposure Δ is a quadratic function of Δ . When all firms adopt their natural risk-exposure levels, $\delta = \delta^\circ$, the aggregate exposure is also at its natural level $\Delta^\circ = (\delta^\circ)^\top \tilde{\omega}$, and the cost of reaching that exposure is $\bar{h}(\Delta^\circ) = 0$. Whenever Δ departs from Δ° , the utility cost increases by an amount that depends on the curvature $\bar{H}$, given by the weighted harmonic aggregate of the firm-level cost matrices H_i . Intuitively, if all firms must pay a large cost—either in terms of productivity or in terms of resources—to gain exposure to some risk factor m (large $[H_i]_{mm}$ for all i), then the aggregate cost of increasing Δ_m is also large.

We can finally use the definition of $\bar{h}$ to characterize the equilibrium and its basic properties.

Proposition 1. *There exists a unique equilibrium, and its aggregate risk exposure Δ^* solves*

$$\mathcal{W}_{dist} := \max_{\Delta} \underbrace{\Delta^\top \mu - \bar{b}(\Delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L}_{\mathbb{E}[y]} - \frac{1}{2} (\rho - 1) \underbrace{\Delta^\top \Sigma \Delta}_{\mathbb{V}[y]} - \bar{g}(\Delta), \quad (32)$$

where $\bar{b}(\Delta) + \bar{g}(\Delta) = \bar{h}(\Delta)$ is given by (30). Without wedges ($\tau = 0$), the equilibrium is efficient.

Proposition 1 shows that the equilibrium allocation is the solution to a *fictitious planning problem* distorted by wedges. Wedges enter (32) by distorting expected log GDP $\mathbb{E}[y]$ and by increasing the cost of risk exposure embedded in $\bar{h} = \bar{b} + \bar{g}$. The last part of the proposition shows that τ is the only source of distortions in this environment.

6 Forces shaping risk-exposure decisions

Primitives of the economy, such as the moments of the risk factors and the wedges, affect aggregate risk exposure Δ through their influence on firms' risk-management incentives. In this

section, we solve the model and provide closed-form expressions for Δ in terms of model primitives. Our analysis relies on the optimality condition of the fictitious planner's problem (32).

Lemma 5. *The equilibrium aggregate risk exposure Δ solves*

$$\mathcal{E}(\Delta) = \nabla \bar{h}(\Delta), \quad (33)$$

where the marginal value of aggregate risk exposure $\mathcal{E}$ can be written as

$$\mathcal{E} = \underbrace{\mu}_{E[\varepsilon]} - \underbrace{(\rho - 1) \Sigma \Delta}_{\text{Cov}[\lambda, \varepsilon]}. \quad (34)$$

The first part of the lemma is the aggregate analogue of the firm's optimality condition (23). It states that in equilibrium the marginal cost $\nabla \bar{h}$ of aggregate exposure is equal to its marginal benefit to the household's expected utility, which is given by the vector $\mathcal{E}$. Recall that $\mathcal{E}$ also captures the benefit of marginal exposure from the firm's perspective.

The second part of the lemma provides an expression for $\mathcal{E}$ and its two components, $E[\varepsilon]$ and $\text{Cov}[\lambda, \varepsilon]$. To better understand that expression, consider first an economy with a single risk factor. In this case, we can represent the relationship between $\mathcal{E}$ and Δ on a simple graph, as in the left panel of Figure 2. That figure also shows the quadratic relationship between Δ and $V[y]$ implied by (17). It is clear from this figure that if the economy is positively exposed to the risk factor ($\Delta > 0$), any further increase in Δ would add to aggregate volatility. In contrast, under negative exposure ($\Delta < 0$), any marginal increase in Δ lowers aggregate volatility. Since the household dislikes uncertainty, the risk adjustment $\text{Cov}[\lambda, \varepsilon]$ is therefore negative under positive exposure and positive under negative exposure. In fact, $\mathcal{E}$ is always linearly decreasing in Δ , as (34) shows. When there is more than one risk factor, correlations matter as well. If the economy is heavily exposed to factors that positively covary with ε_m , any increase in Δ_m would raise aggregate volatility. In that case, the marginal value of exposure to ε_m would be low.

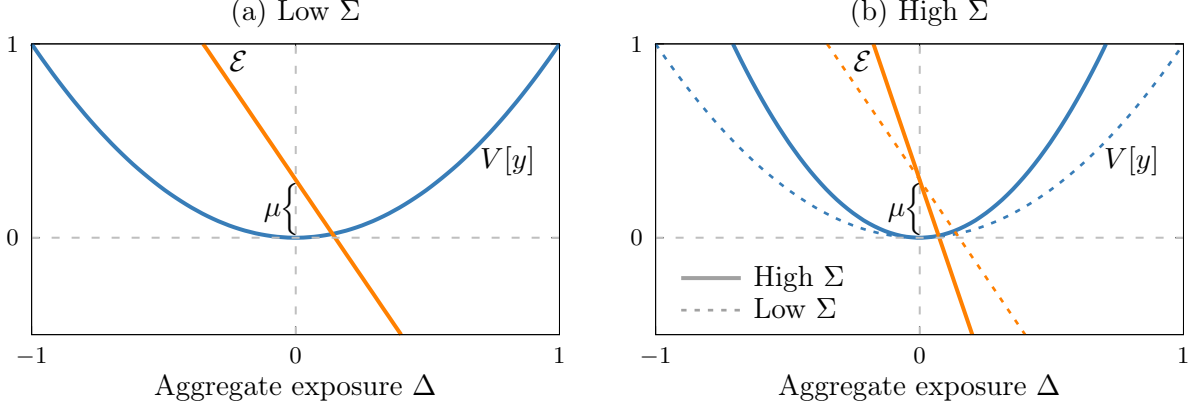
The moments of the risk factors also affect $\mathcal{E}$. Let $\mathbf{1}_m$ be the m -th standard basis vector. Then¹⁶

$$\frac{\partial \mathcal{E}}{\partial \mu_m} = \mathbf{1}_m \quad \text{and} \quad \frac{\partial \mathcal{E}}{\partial \Sigma_{mn}} = -\frac{1}{2}(\rho - 1)(\Delta_m \mathbf{1}_n + \Delta_n \mathbf{1}_m). \quad (35)$$

An increase in μ_m makes factor m more attractive but leaves the marginal value of exposure to all other factors unchanged. To understand the intuition behind the second equation in (35), consider the right panel of Figure 2, which shows what happens when uncertainty Σ increases. In this case, Δ has a stronger impact on the variance of GDP, so that marginal increases in Δ are now more harmful for $\Delta > 0$ and more beneficial for $\Delta < 0$. As a result, the relationship between $\mathcal{E}$ and Δ

¹⁶Whenever we take derivatives with respect to off-diagonal elements of Σ , we simultaneously change Σ_{mn} and Σ_{nm} to keep Σ symmetric, and divide the result by two.

Figure 2: Impact of Δ on $V[y]$ and $\mathcal{E}$ in an economy with a single risk factor



becomes steeper, as Figure 2 and equation (35) show. When there is more than one factor, similar reasoning applies to changes in covariances.¹⁷

6.1 Aggregate risk exposure

Using (33), we can solve for the equilibrium exposure decisions.

Proposition 2. *The equilibrium aggregate risk-exposure decisions are given by*

$$\Delta = \Delta^\circ + \mathcal{H}^{-1} \mathcal{E}^\circ, \quad (36)$$

where $\mathcal{E}^\circ := \mu - (\rho - 1) \Sigma \Delta^\circ$ and where the $M \times M$ positive definite matrix $\mathcal{H}^{-1}$ is

$$\mathcal{H}^{-1} := (\bar{H} + (\rho - 1) \Sigma)^{-1}. \quad (37)$$

Equation (36) provides a *closed-form* expression for equilibrium aggregate risk exposure Δ , and is the aggregate analogue of (24), which describes firm-level risk decisions. With Δ in hand, the entire equilibrium can also be solved in closed form.¹⁸ Equation (36) expresses Δ as the sum of its natural level Δ° and the term $\mathcal{H}^{-1} \mathcal{E}^\circ$, where $\mathcal{E}^\circ$ is the marginal benefit of exposure $\mathcal{E}$ evaluated at Δ° and the matrix $\mathcal{H}^{-1}$ translates these benefits into exposures. Since $\mathcal{H}^{-1}$ is positive definite, its diagonal is positive, so a naturally attractive factor ($\mathcal{E}_m^\circ > 0$) raises the economy's exposure to that same factor, all else equal. Its effect on exposure to a *different* factor $n \neq m$ depends instead on the off-diagonal entries of $\mathcal{H}^{-1}$, which encode general-equilibrium substitution across factors. If $[\mathcal{H}^{-1}]_{mn} > 0$, factors m and n are *global complements* and a higher $\mathcal{E}_m^\circ$ raises Δ_n . If instead $[\mathcal{H}^{-1}]_{mn} < 0$, they are *global substitutes* and a higher $\mathcal{E}_m^\circ$ lowers Δ_n .

¹⁷Specifically, an increase in Σ_{mn} decreases the marginal value of exposure to both factors m and n if $\Delta_m > 0$ and $\Delta_n > 0$. In this case, the larger covariance implies that a marginal increase in Δ_m or Δ_n would translate into a larger increase in aggregate risk. The opposite happens when $\Delta_m < 0$ and $\Delta_n < 0$.

¹⁸From (36), we can compute $\mathcal{E}$ using (34) and δ using (24). Other objects can be computed from these expressions.

As (37) shows, these substitution patterns have two sources. The first is the Hessian $\bar{H}$ of the aggregate cost function $\bar{h}$, whose element $\bar{H}_{mn}$ measures how greater exposure to factor m changes the *marginal* cost of exposure to factor n . Recall from (31) that $\bar{H}$ is a Domar-weighted harmonic sum of the firm-level cost Hessians $(H_1, \dots, H_N)$, so it aggregates the *local* substitution patterns of individual firms into the global ones. The second source is the covariance matrix Σ . A positive correlation between factors m and n tends to make them global substitutes, since raising Δ_m leads the fictitious planner to lower Δ_n to contain aggregate risk. Unsurprisingly, the strength of this second channel grows with the household's risk aversion ρ .

6.2 Beliefs and aggregate risk exposure

The moments (μ, Σ) affect how attractive the risk factors are, which in turn influences risk-exposure decisions.

Proposition 3. *Let γ denote either the mean μ_m or an element Σ_{mn} of the covariance matrix. The response of the equilibrium aggregate risk exposure Δ to a change in γ is given by*

$$\frac{d\Delta}{d\gamma} = \mathcal{H}^{-1} \frac{\partial \mathcal{E}}{\partial \gamma}, \quad (38)$$

where $\partial \mathcal{E} / \partial \gamma$ is given by (35).

Proposition 3 shows that the impact on Δ of a change in a moment γ operates through its *direct* influence on the marginal value of exposure $\mathcal{E}$, as captured by the partial derivative $\partial \mathcal{E} / \partial \gamma$. That is, a change in γ makes some risk factors more or less attractive than before, and this triggers an adjustment of risk-taking decisions through the matrix $\mathcal{H}^{-1}$. Since $\mathcal{H}^{-1}$ is positive definite, it is straightforward to characterize the outcome of that adjustment process.

Corollary 2. *An increase in the expected value μ_m of risk factor m leads to an increase in aggregate risk exposure Δ_m . An increase in the variance Σ_{mm} of risk factor m leads to a decrease in Δ_m if $\Delta_m > 0$ and to an increase in Δ_m if $\Delta_m < 0$.*

Increasing μ_m makes exposure to ε_m more attractive without affecting the cost $\bar{h}$ of achieving that exposure. Consequently, the fictitious planner chooses a higher Δ_m . The impact of Σ is more subtle. Recall from (17) that the sensitivity of $V[y]$ to Σ_{mm} depends on the absolute value of Δ_m . Indeed, an increase in Σ_{mm} makes GDP more volatile even if $\Delta_m < 0$. It follows that a higher Σ_{mm} makes the fictitious planner reduce $|\Delta_m|$, which explains the second part of the corollary.

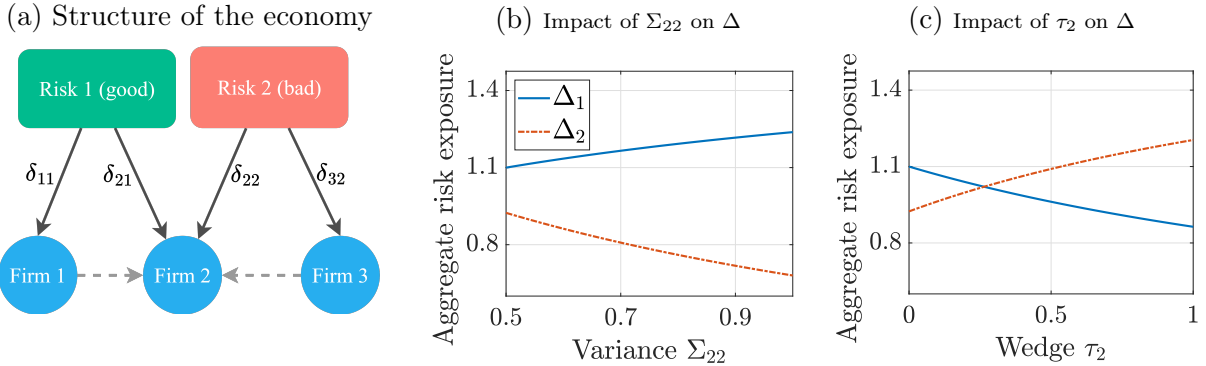
Example. As in Proposition 2, the global substitution matrix $\mathcal{H}^{-1}$ controls how changes in the attractiveness of a risk factor ε_m affect exposure to another factor, say ε_n . To better understand how these substitution patterns work, consider a simple economy in which three firms can be exposed to two sources of risk, as in panel (a) of Figure 3. Risk 1 is good with $\mathcal{E}_1 > 0$, while risk 2 is bad

with $\mathcal{E}_2 < 0$. The cost functions h are parametrized so that firm 1 is only exposed to risk 1 and cannot adjust its exposure. Similarly, firm 3 is only exposed to risk 2. Firm 2, in contrast, is free to change its exposure to the two risk factors, and its cost function h_2 is such that δ_{21} and δ_{22} are substitutes. All firms use labor to produce but, in addition, firm 2 uses goods from firms 1 and 3.

One can think of many real-world examples that would give rise to such substitution patterns. For instance, firm 2 might have only a limited amount of resources (management attention or specialized scientists) that can be allocated to researching two different technologies. Another example may involve two regions, where region 1 is more productive than region 2 (for instance, due to a better climate). Firm 2 must decide where to locate its plants. In this context, δ_{21} and δ_{22} are substitutes if it is challenging for firm 2 to manage geographically dispersed plants.

Panel (b) in Figure 3 shows what happens to the equilibrium aggregate risk exposure when the bad factor becomes riskier. When Σ_{22} increases, $\mathcal{E}_2$ declines and risk factor 2 becomes less attractive. As a result, firm 2 reduces its exposure to it, and since the other firms' exposures are fixed, Δ_2 declines. As the two risk factors are substitutes, increasing δ_{21} then becomes cost-effective, and the overall economy's exposure to the good factor increases.

Figure 3: Impact of parameters in a simple economy



Notes: Panel (a): The structure of the economy; there is an arrow from firm j to firm i if $\alpha_{ij} > 0$, and from risk factor m to firm i if $\delta_{im} \neq 0$. Panels (b) and (c): effects of changes in parameters. Initial parametrization is as follows. Household: $\rho = 2$ and $\beta_2 = 0.8$, $\beta_1 = \beta_3 = 0.1$. Network: $\alpha_{21} = \alpha_{23} = 0.25$, all other entries of α are zero. Beliefs: $\mu = (0.75, 0)$, Σ is diagonal with $\text{diag}(\Sigma) = (0.5, 0.5)$. Risk exposures: $\delta_{11}^\circ = \delta_{32}^\circ = 1$, $\delta_{22}^\circ = 1.9$, $\delta_{12}^\circ = \delta_{21}^\circ = \delta_{31}^\circ = 0$, $G_1 = G_3$ is diagonal with very large entries on the main diagonals; $G_{2,11} = G_{2,22} = 1$, $G_{2,12} = G_{2,21} = 0.75$. We set $B_i = 0$ for all i . Wedges: $\tau = 0$. In panel (b), Σ_{22} changes from 0.5 to 1. In panel (c), τ_2 changes from 0 to 1.

6.3 Wedges and aggregate risk exposure

The next result describes how wedges affect aggregate risk exposure.

Proposition 4. *The response of the equilibrium aggregate risk exposure Δ to a change in wedge τ_i is given by*

$$\frac{d\Delta}{d\tau_i} = \frac{d\mathcal{H}^{-1}}{d\tau_i} \mathcal{E}^\circ, \quad (39)$$

where

$$\frac{d\mathcal{H}^{-1}}{d\tau_i} = -\mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) \mathcal{H}^{-1},$$

and where the response $d\kappa_j/d\tau_i$ of the efficiency gap is nonnegative, and $\partial \bar{H}/\partial \kappa_j$ is a positive definite matrix.¹⁹

Wedges affect the equilibrium risk exposure Δ through the curvature of the aggregate cost function $\bar{h}$. Specifically, an increase in τ_i leads to a higher efficiency gap κ_j for all j . This implies that firms find risk management less appealing, and the curvature of the aggregate cost function increases, in the sense that $\partial \bar{H}/\partial \kappa_j$ is a positive definite matrix. These changes in curvature then affect $\mathcal{H}^{-1}$ and, hence, Δ .

Proposition 4 provides a general characterization of the impact of τ , but we can get sharper results under a particular parametrization of the model.

Definition 2. An economy is *diagonal* if the risk factors are uncorrelated (diagonal Σ), and the individual cost functions $(b_1, \dots, b_N)$ and $(g_1, \dots, g_N)$ feature neither local complementarity nor local substitutability (diagonal B_i and G_i for all i).

Correlations between risk factors and complex substitution patterns can give rise to interesting mechanisms, but they often obscure some simpler forces that are at work in the economy. To highlight those forces, we will sometimes simplify the analysis by focusing on diagonal economies. The following result describes the impact of τ in such a setting.

Corollary 3. In a diagonal economy, a higher wedge τ_i increases Δ_m for all m such that $\mathcal{E}_m < 0$ (bad risks) and decreases Δ_m for all m such that $\mathcal{E}_m > 0$ (good risks).

This result shows that wedges push firms to increase their exposure to bad risks and to decrease their exposure to good risks. Intuitively, an increase in τ_i makes firms shrink in terms of their cost of goods sold. This implies that risk management becomes less cost-effective: since the firm is smaller, managing its TFP is now less rewarding. As a result, firms spend less to increase exposure to good risk factors and to reduce exposure to bad risk factors. As we will see in the next section, this has important implications for welfare.

Example. The last panel of Figure 3 shows what happens to the example economy of panel (a) when firm 2's wedge increases. As a result of its higher sales price, firm 2 shrinks, which makes risk management relatively more costly. It follows that aggregate exposure to the good risk factor declines and exposure to the bad risk factor increases.

Proposition 4 implies that distortions matter for risk-taking decisions. As a result, the equilibrium Δ departs from its efficient level whenever $\tau > 0$.

¹⁹We give expressions for $d\kappa_j/d\tau_i$ and $\partial \bar{H}/\partial \kappa_j$ in the proof of this proposition.

Lemma 6. *Suppose that $\tau_j > 0$ for at least one firm j and that $\mathcal{E}^\circ \neq 0$. Then $(\Delta - \Delta^{SP})^\top \mathcal{E}^\circ < 0$, where Δ and Δ^{SP} are the aggregate risk exposure vectors in the equilibrium and the efficient allocation, respectively. Furthermore, in a diagonal economy the sign of $\Delta_m - \Delta_m^{SP}$ is the opposite of that of $\mathcal{E}_m^\circ$.*

When there is a single risk factor, the first part of Lemma 6 implies that the equilibrium is overexposed to bad natural risks ($\mathcal{E}^\circ < 0$) and underexposed to good natural risks ($\mathcal{E}^\circ > 0$), compared to the efficient allocation. Intuitively, wedges distort firms' incentives to manage their risk exposures and, when aggregated, those decisions lead to a departure from Δ^{SP} . When there are multiple risk factors, these forces operate on average. For instance, if there are two risk factors, one good and one bad, the equilibrium can be overexposed to the good natural risk only if it is severely overexposed to the bad natural risk as well. The second part of the lemma shows that this over/underexposure result applies factor by factor in a diagonal economy.²⁰

7 GDP and welfare

Since changes in the environment affect the economy's risk exposure, they also affect GDP and welfare. In this section, we study the impact of endogenous risk on these quantities.

7.1 Moments of GDP

An increase in a parameter χ can influence the moments of GDP through two channels. First, it might trigger an adjustment in Δ , and that adjustment would, in turn, affect GDP. This is the main channel we focus on in this paper. Second, an increase in χ may also have a direct impact on GDP that operates independently of χ 's influence on Δ . For instance, an increase in wedges τ , on its own, can reduce GDP. This second channel is not specific to our model and, in many cases, has already been studied in the literature. In what follows, we therefore focus on the role played by changes in risk-exposure decisions, and filter out the impact of the second channel.

The following result describes how a change in a parameter affects GDP through Δ .²¹

Proposition 5. *Let χ denote one of μ_m , Σ_{mn} , or τ_i . Then the impact of a change in χ on the moments of log GDP is given by*

$$\frac{dE[y]}{d\chi} - \frac{\partial E[y]}{\partial \chi} = (\mu - \nabla \bar{b})^\top \frac{d\Delta}{d\chi} \quad \text{and} \quad \frac{dV[y]}{d\chi} - \frac{\partial V[y]}{\partial \chi} = 2\Delta^\top \Sigma \frac{d\Delta}{d\chi}, \quad (40)$$

²⁰In addition to μ , Σ and τ , it is straightforward to describe how other parameters affect Δ . For instance, in Appendix F.4 we explore the role of the production network α . We find that an increase in α_{ij} typically leads to higher revenue-based Domar weights, which implies that firms manage risk more aggressively in response.

²¹Proposition 5 also applies when χ denotes a network link α_{ij} , as we show in Appendix F.4.

where the partial derivatives indicate that Δ is kept fixed, and where $d\Delta/d\chi$ is given by (38) for μ_m and Σ_{mn} , and by (39) for τ_i .

Proof. The result follows directly from (17). $\square$

Because the partial derivatives in (40) hold Δ fixed, the left-hand sides isolate the *indirect* effect of χ , which operates through the induced change in aggregate exposure $d\Delta/d\chi$. For expected log GDP, this indirect effect multiplies the exposure response $d\Delta/d\chi$ by the marginal effect of exposure on expected output, $\mu - \nabla \bar{b}$. The latter balances two forces: more exposure to a factor m contributes its mean μ_m to expected output, while also affecting the aggregate productivity cost $\bar{b}$. For instance, recall that by Corollary 2 an increase in μ_m raises Δ_m ; when $\mu_m > 0$, the extra exposure lifts $E[y]$, but the accompanying rise in $\bar{b}$ can work against it and, if large enough, can make $E[y]$ fall. Because χ typically shifts exposure to many factors at once, the first equation in (40) sums these contributions across factors.

The variance of log GDP responds in a similar way, except that correlations between factors now matter. Rewriting the second expression in (40) as $2 \text{Cov} \left[y, (d\Delta/d\chi)^\top \varepsilon \right]$ shows that the sign of the response depends on whether the factors whose exposure changes covary positively or negatively with log GDP. An increase in μ_m , for example, raises Δ_m and hence $V[y]$ whenever ε_m is positively correlated with y .

We can further characterize the impact of (μ, Σ) on GDP in a diagonal economy. To keep our discussion concise, we focus on the case with resource costs only ($B_i = 0$ for all i). Empirical evidence presented in the next section suggests that resource costs are important, which motivates this choice. We tackle the general case with $B_i \neq 0$ in the proofs of Corollaries 4 and 5.

Corollary 4. *Let $B_i = 0$ for all i . In a diagonal economy, the following statements hold.*

1. *The impact of an increase in μ_m on GDP satisfies*

$$\text{sign} \left(\frac{dE[y]}{d\mu_m} - \frac{\partial E[y]}{\partial \mu_m} \right) = \text{sign}(\mu_m) \quad \text{and} \quad \text{sign} \left(\frac{dV[y]}{d\mu_m} - \frac{\partial V[y]}{\partial \mu_m} \right) = \text{sign}(\Delta_m). \quad (41)$$

2. *The impact of an increase in Σ_{mm} on GDP satisfies*

$$\text{sign} \left(\frac{dE[y]}{d\Sigma_{mm}} - \frac{\partial E[y]}{\partial \Sigma_{mm}} \right) = -\text{sign}(\mu_m \Delta_m) \quad \text{and} \quad \frac{dV[y]}{d\Sigma_{mm}} - \frac{\partial V[y]}{\partial \Sigma_{mm}} \leq 0. \quad (42)$$

The first result describes how an increase in μ_m affects GDP through Δ . Without productivity costs, $E[y]$ rises in response if $\mu_m > 0$ and decreases otherwise, as explained above. The response of $V[y]$, in contrast, depends on whether the economy is positively or negatively exposed to m . If $\Delta_m > 0$, the increase in μ_m makes Δ_m even more positive which exposes the economy to more risk,

and $V[y]$ rises as a result. If $\Delta_m < 0$ instead, the increase in μ_m makes Δ_m less negative, and so the economy is less vulnerable to shock m .

The second part of Corollary 4 offers similar expressions for the impact of an increase in variance. From Corollary 2, we know that an increase in Σ_{mm} leads to a decline in $|\Delta_m|$. This raises $E[y]$ if $\mu_m \Delta_m < 0$, but lowers it otherwise. In contrast, the impact of Σ_{mm} on the variance of GDP via Δ is unambiguous. Since $|\Delta_m|$ shrinks, the economy becomes less sensitive to ε_m , and $V[y]$ declines as well.

We can similarly characterize how wedges affect GDP. When there is a single risk factor, this characterization is straightforward.

Corollary 5. *Let $B_i = 0$ for all i , and suppose that there is a single risk factor. Then*

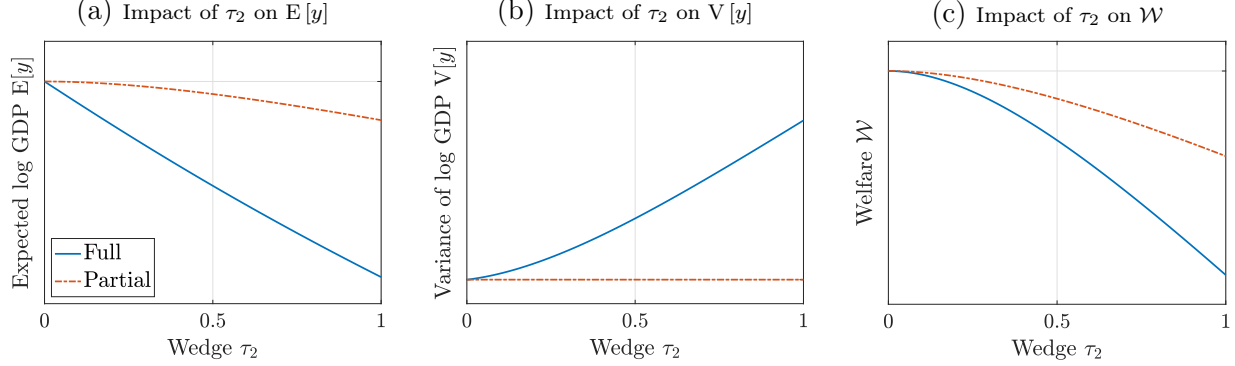
$$\text{sign}\left(\frac{dE[y]}{d\tau_i} - \frac{\partial E[y]}{\partial \tau_i}\right) = -\text{sign}(\mu\mathcal{E}) \quad \text{and} \quad \text{sign}\left(\frac{dV[y]}{d\tau_i} - \frac{\partial V[y]}{\partial \tau_i}\right) = -\text{sign}(\Delta\mathcal{E}). \quad (43)$$

To get some intuition for this result, suppose there is a single risk factor that is bad ($\mathcal{E} < 0$), and that firms are on average positively exposed to it ($\Delta > 0$). That risk factor could be, for example, an underlying productivity shock that drives the business cycle. Since wedges raise exposure to bad risk factors (Corollary 3), a higher τ_i increases Δ . Whether this has a positive impact on $E[y]$ depends on the sign of μ . If $\mu > 0$, then $E[y]$ also increases, while it declines if $\mu < 0$. Wedges also affect GDP volatility. In this case, since $\Delta > 0$, increasing τ_i makes the economy even more exposed to the risk factor, and $V[y]$ increases.

Example. When there is more than one risk factor, the response of the economy to a change in wedges depends on how correlated the risk factors are and on the substitution patterns embedded in the firms' cost functions. To get a better sense of the forces at work, we can go back to the example economy of Figure 3. The first panel of Figure 4 shows the impact of raising τ_2 on $E[y]$. Recall from the third panel of Figure 3 that increasing τ_2 makes the economy more exposed to risk 2 (the bad risk) and less exposed to risk 1 (the good risk). Since under our parametrization $\mu_1 > 0$ and $\mu_2 = 0$, this triggers a decline in $E[y]$. Notice that this decline is more pronounced than if Δ remained fixed (red line). In that case, the increase in τ_2 would lower GDP only through its usual distortionary effect. We see that the endogenous risk decisions of the firms make the response to an increase in wedges more severe.

The second panel of Figure 4 shows the response of $V[y]$ to the same increase in τ_2 . Both risk factors have the same variance, but given our parametrization, the increase in Δ_2 is stronger than the decline in Δ_1 . It follows that the increase in τ leads to a rise in the volatility of log GDP. Notably, since $V[y]$ does not directly depend on τ (see (17)), $V[y]$ would be unaffected by wedges if Δ remained constant. This is illustrated by the red line in the panel. This example highlights that endogenous uncertainty is essential for wedges to affect aggregate volatility in our model.

Figure 4: The moments of GDP and welfare react to changes in the environment



Notes: The structure of the economy is given in panel (a) of Figure 3. Solid blue curve: full model. Red dashed curve: model with fixed Δ . Initial parametrization is as follows. Household: $\rho = 2$ and $\beta_2 = 0.8$, $\beta_1 = \beta_3 = 0.1$. Network: $\alpha_{21} = \alpha_{23} = 0.25$, all other entries of α are zero. Risk-factor moments: $\mu = (0.75, 0)$, Σ is diagonal with $\text{diag}(\Sigma) = (0.5, 0.5)$. Risk exposures: $\delta_{11}^\circ = \delta_{32}^\circ = 1$, $\delta_{22}^\circ = 1.9$, $\delta_{12}^\circ = \delta_{21}^\circ = \delta_{31}^\circ = 0$, $G_1 = G_3$ are diagonal with very large entries on the main diagonals; $G_{2,11} = G_{2,22} = 1$, $G_{2,12} = G_{2,21} = 0.75$. We set $B_i = 0$ for all i . Initial wedges: $\tau = 0$. In all panels, τ_2 changes from 0 to 1.

7.2 Welfare

Model primitives affect GDP through their impact on risk-exposure decisions, but what the household ultimately cares about is welfare, which combines the moments of GDP with the disutility of managing risk. In this section, we evaluate the consequences of endogenous risk for welfare. We first characterize how welfare behaves in an environment without distortions.

Proposition 6. *Without wedges ($\tau = 0$), the impact of the moments (μ, Σ) on welfare is given by*

$$\frac{d\mathcal{W}}{d\mu_m} = \frac{\partial \mathcal{W}}{\partial \mu_m} = \Delta_m \quad \text{and} \quad \frac{d\mathcal{W}}{d\Sigma_{mn}} = \frac{\partial \mathcal{W}}{\partial \Sigma_{mn}} = -\frac{1}{2}(\rho - 1)\Delta_m \Delta_n.$$

Furthermore, the impact of the wedge τ_i on welfare is given by

$$\frac{d\mathcal{W}}{d\tau_i} = \frac{\partial \mathcal{W}}{\partial \tau_i} = 0.$$

Proof. When $\tau = 0$, the objective function $\mathcal{W}_{dist}$ of the fictitious planner coincides with welfare. The result then follows from the envelope theorem. $\square$

This result shows that making ε_m more productive on average (higher μ_m) benefits welfare if the economy is positively exposed to ε_m . In contrast, if $\Delta_m < 0$, increasing μ_m leads to a welfare loss. The proposition also shows that making ε_m more risky (higher Σ_{mm}) is always detrimental to welfare. Finally, an increase in the covariance Σ_{mn} hurts welfare if the economy is positively or negatively exposed to both risk factors. If, instead, Δ_m and Δ_n have opposite signs, the increase in Σ_{mn} is beneficial since the shocks are more likely to offset each other in this case.

Proposition 6 also shows that welfare responds to a marginal change in a parameter *as if* the

risk-exposure decisions were kept fixed (partial derivatives). This is a consequence of the envelope theorem. In the absence of wedges, the equilibrium is efficient and so Δ maximizes welfare. Any marginal change in Δ around its optimum therefore has no first-order welfare effect.²² The situation is, however, different in the presence of wedges, in which case the response of risk exposure can have a first-order effect.

The next result characterizes how welfare responds to changes in (μ, Σ) in a distorted equilibrium. To keep the analysis concise, we focus on diagonal economies here and derive the general results in Appendix C.15.

Lemma 7. *Suppose that the economy is diagonal and that $\tau_j > 0$ for at least one firm j . Then,*

$$\text{sign}\left(\frac{d\mathcal{W}}{d\mu_m} - \frac{\partial\mathcal{W}}{\partial\mu_m}\right) = \text{sign}(\mathcal{E}_m) \quad \text{and} \quad \text{sign}\left(\frac{d\mathcal{W}}{d\Sigma_{mm}} - \frac{\partial\mathcal{W}}{\partial\Sigma_{mm}}\right) = -\text{sign}(\Delta_m \mathcal{E}_m).$$

The first part of the lemma shows that increasing the expected value μ_m of a good risk factor ($\mathcal{E}_m > 0$) leads to a bigger welfare gain when risk is endogenous. To understand why, it helps to illustrate how the marginal benefit and marginal cost of exposure interact, as in the left panel of Figure 5. The solid blue curve shows the function $\bar{h}$, which reflects the cost of exposure as it is *perceived* by the fictitious planner, while the slope of the orange line is equal to $\mathcal{E}$. In equilibrium, we know from (33) that $\mathcal{E} = \nabla \bar{h}$ (black point in the figure). But recall from (29) that because of the distortions, the fictitious planner perceives the cost of exposure to be higher than it really is. The dashed blue curve in the figure represents the true utility cost of exposure, which we denote by $\bar{h}_V(\Delta)$.²³ It follows that at the equilibrium Δ^* , the true marginal cost $\nabla \bar{h}_V$ is smaller than the marginal benefit $\mathcal{E}$, and an increase in Δ would be welfare-increasing. This is precisely what happens when μ_m increases: Δ_m goes up, which leads to an additional increase in welfare, as Lemma 7 shows. The right panel of Figure 5 shows the same quantities when the risk factor is bad ($\mathcal{E}_m < 0$). In this case, the perceived marginal cost $\nabla \bar{h}$ is *more negative* than the true marginal cost $\nabla \bar{h}_V$. As a result, increasing Δ would lead to a decline in welfare.

The second part of Lemma 7 shows that a similar result holds for an increase in Σ_{mm} . If ε_m is good, a higher exposure Δ_m would improve welfare. After an increase in Σ_{mm} , the fictitious planner reduces $|\Delta_m|$ to limit the economy's vulnerability to ε_m . It follows that if $\Delta_m > 0$, there is a decline in exposure, which lowers welfare. If instead $\Delta_m < 0$, the increase in risk Σ_{mm} leads to a less negative exposure Δ_m , which improves welfare.

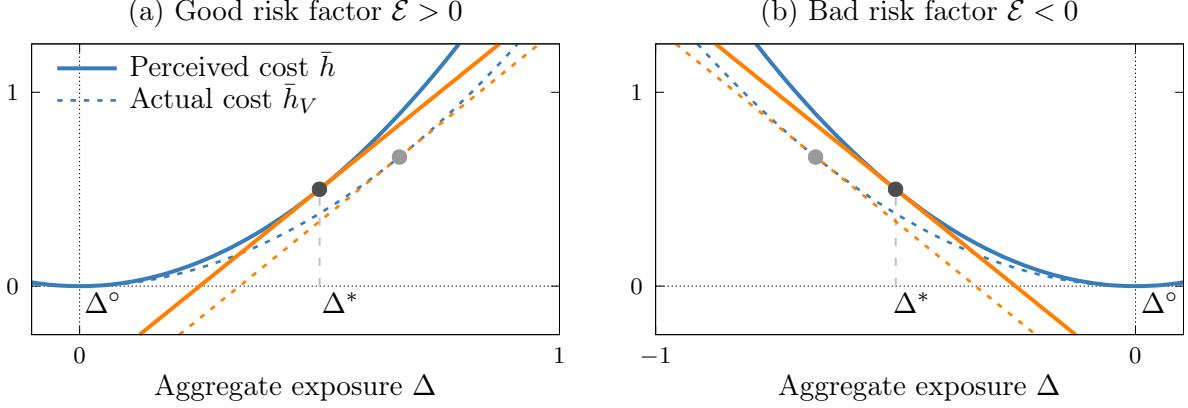
A similar result holds for the impact of wedges.

Lemma 8. *In a diagonal economy, an increase in a wedge is more detrimental to welfare when risk-exposure decisions can adjust, that is, $d\mathcal{W}/d\tau_i \leq \partial\mathcal{W}/\partial\tau_i$.*

²²Similarly, since $\tau = 0$ maximizes $\mathcal{W}$, a marginal increase in wedges from that point has no impact on welfare, as Proposition 6 shows (Baqaee and Farhi, 2019; Bigio and La'O, 2020).

²³We define $\bar{h}_V$ formally in (83) in Appendix C.15.

Figure 5: Actual and perceived costs of adjusting risk exposure



This result shows that endogenous risk increases the welfare cost of wedges. By Corollary 3, a higher τ_i increases exposure to bad risks and reduces exposure to good risks. The lemma follows because additional exposure to bad risks and reduced exposure to good risks are both welfare-decreasing.

The right panel of Figure 4 illustrates how these forces operate in the model economy of Figure 3. If Δ is kept fixed, an increase in the wedge τ_2 of the central firm amplifies the distortions, which is detrimental to welfare (red line). But if the risk exposure Δ of the economy is free to adjust, the adverse impact of τ_2 on welfare is exacerbated (blue line) due to the higher exposure to the bad risk factor 2 and to the lower exposure to the good risk factor 1 (see third panel of Figure 3).

These results describe a new channel through which distortions like taxes and markups might reduce welfare. Beyond their conventional distortionary effects, these wedges also interfere with firm risk-taking incentives, altering the economy's aggregate risk profile in potentially harmful ways. This suggests that incorporating endogenous risk responses into the evaluation of tax policy and other interventions might be necessary to assess their welfare consequences accurately.

8 Reduced-form evidence

In the model, the resource cost of risk management does not scale with firm size while its benefit does. Firms with high production costs---those with high sales and low markups---therefore find it worthwhile to spend more on risk management, as Corollary 1 shows. In a business cycle context, this implies that those firms are less volatile and covary less with GDP. In this section, we verify that these predictions hold in the data.

As we focus on business cycle risk, we embed our model into a dynamic setting. To keep the analysis simple, we assume there is a single risk factor $\varepsilon_t \sim \text{iid } \mathcal{N}(0, \Sigma)$, and let firm i 's TFP be

$$\log TFP_{it} = \delta_{it}\varepsilon_t - b_i(\delta_{it}) + \gamma_i t + v_{it}, \quad (44)$$

where γ_i is a deterministic trend, and $v_{it} \sim \text{iid } \mathcal{N}(\mu_i^v, \Sigma_i^v)$ is a firm-specific shock to capture variation that is orthogonal to the aggregate factor ε_t .²⁴

In this setup, risk-taking decisions are time invariant ($\delta_{it} = \delta_i$) and Lemma 3 implies that

$$\delta_i = \delta_i^\circ + \left(B_i + \eta \frac{1 + \tau_i}{\omega_i} G_i \right)^{-1} \mathcal{E}. \quad (45)$$

It is natural to think that the aggregate economy is positively exposed to business cycle risk, and so we assume that $\Delta > 0$.²⁵ In this case, since $\mu = 0$, aggregate risk is bad ($\mathcal{E} < 0$), and (45) implies that firms with larger revenue-based Domar weights ω_i or smaller wedges τ_i choose a lower risk exposure δ_i .

Combining (16) and (44), we can compute the moments

$$V[\Delta \log TFP_{it}] = 2\delta_i^2 \Sigma + 2\Sigma_i^v \quad \text{and} \quad \text{Cov}[\Delta \log TFP_{it}, \Delta y_t] = 2\Delta \Sigma \delta_i + 2\tilde{\omega}_i \Sigma_i^v, \quad (46)$$

where $\Delta \log TFP_{it} = \log TFP_{it} - \log TFP_{it-1}$ and $\Delta y_t = y_t - y_{t-1}$. Combining these expressions with (45) implies that if $\delta_i > 0$, the TFP growth of firms with larger Domar weights and smaller wedges is less volatile and covaries less with GDP growth.²⁶

To test these predictions, we need a measure of firm TFP. Because we do not observe firm prices, we generalize the baseline model by introducing sectors and a CES demand structure within each sector. This extension, detailed in Appendix E, makes variation in firm sales informative about firm TFP and additionally lets the model capture substitution patterns across firms that are absent in the baseline. Crucially, although the CES structure is less tractable, the same comparative statics carry over: firms with larger Domar weights and smaller wedges are less volatile and covary less with GDP.

We use detailed firm-level data from Orbis that cover the near-universe of Spanish firms between 1995 and 2019. Our sample contains 7,683,409 firm-year observations. We use those data to construct firm-level measures of TFP growth, markups (our measure of wedges) and Domar weights. We briefly describe how we construct our sample below, and include more detail in Appendix A.1.

Two ingredients enter our measure of firm TFP. On the demand side, the mapping from sales

²⁴For simplicity, we assume that firms cannot adjust their exposure to v_{it} and γ_i . The analysis is similar if they can, but requires additional assumptions on the cost functions b_i and g_i . We can also allow for autocorrelation in ε_t and v_{it} , and for correlations between the v_{it} of different firms at the cost of extra complications.

²⁵Since $E[\varepsilon_t] = 0$, the choice of $\Delta > 0$ is essentially a normalization that implies that firms have, on average, positive exposure $\delta_{it} > 0$. Flipping the signs of δ° , δ , Δ and $\mathcal{E}$ yields an economy that is indistinguishable from the original in terms of the variance of firm-level TFP and its covariance with GDP, as (46) shows below. Indeed, our results in Section 7 on the effect of Σ on the moments of GDP and welfare depend on the sign of the product of Δ and $\mathcal{E}$, not on their individual signs.

²⁶With $\delta_i < 0$, a higher ω_i or a lower τ_i implies more volatile TFP. In our calibration, $\delta_i > 0$ for most firms. The covariance also includes a granular term $2\tilde{\omega}_i \Sigma_i^v$ that grows linearly with firm size $\tilde{\omega}_i$. In our data, however, individual Domar weights are typically tiny (the median is on the order of 10^{-7}), so this term is negligible relative to the endogenous component $2\delta_i \Delta \Sigma$.

to TFP depends on the within-sector elasticity of substitution σ_s , which we set to $\sigma_s = 3$ in our baseline, a value that lies within the range of existing estimates. We show in the appendix that our findings are robust to alternative values.²⁷ On the production side, consistent with the constant returns-to-scale Cobb-Douglas production function in the model, we estimate sector-specific output elasticities of capital, labor, and materials with a cost-share approach, following Foster, Haltiwanger, and Syverson (2008) and De Loecker, Eeckhout, and Unger (2020).²⁸ We combine these ingredients to compute each firm’s model-implied TFP growth. Throughout all of our reduced-form exercises, we rely only on within-sector-year variation in TFP growth.

We compute each firm’s revenue-based Domar weight as nominal sales divided by Spain’s nominal GDP, and estimate markups using the production approach of De Loecker and Warzynski (2012). Specifically, $1 + \tau_{it} = \hat{\alpha}_{s(i)}^L / (\text{Wage Bill}_{it} / \text{Sales}_{it})$, where $\text{Wage Bill}_{it} / \text{Sales}_{it}$ is the share of labor expenditure in sales and $\hat{\alpha}_{s(i)}^L$ is the estimated output elasticity of labor. Across all firm-year observations, the average markup is 1.33 and the median Domar weight is 4.81×10^{-7} .

8.1 Volatility of firm-level TFP

With these data, we first explore how firm characteristics are related to the volatility of a firm’s TFP growth. To do so, we compute the standard deviation of TFP growth for each firm, σ_i^{TFP} , and the time-series averages of its markups and Domar weights. We then construct within-sector quintiles based on these averages to create a set of dummy variables, D_{ji}^{Domar} and D_{ji}^{Markup} , such that $D_{ji}^{Domar} = 1$ if firm i ’s Domar weight is in quintile j , and analogously for markups.²⁹ We then run the cross-sectional regression

$$\sigma_i^{TFP} = \sum_{j=1}^5 \beta_j^{Domar} D_{ji}^{Domar} + \sum_{j=1}^5 \beta_j^{Markup} D_{ji}^{Markup} + \theta_s + \varepsilon_i, \quad (47)$$

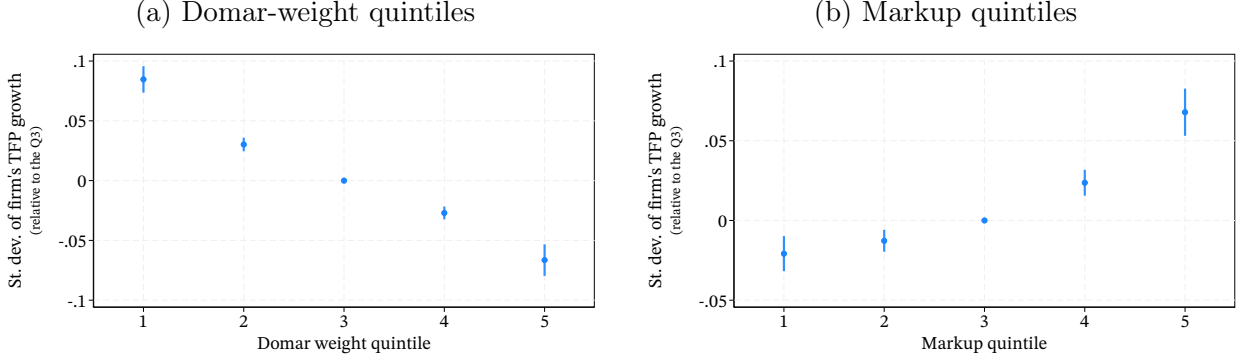
where θ_s are sector-specific fixed effects. We plot the estimated coefficients β_j^{Domar} in panel (a) and β_j^{Markup} in panel (b) of Figure 6. Consistent with the theory, firms with lower Domar weights and

²⁷Estimates of this elasticity vary across literatures: Carvalho et al. (2021) report values around 1.2 in a production-network setting; Broda and Weinstein (2006) and Hottman, Redding, and Weinstein (2016) estimate values mostly between 3 and 6 in the trade and firm-heterogeneity literatures; estimates on Spanish firm data are around 2 in Doraszelski and Jaumandreu (2013) and 3 in Almunia et al. (2021). In the knife-edge case of $\sigma_s = 1$, the revenues of the firm are independent of its TFP, which therefore cannot be estimated without data on prices and quantities. In practice, the estimates become noisy when σ_s is close to one.

²⁸As noted by Raval (2023a), this cost-share approach avoids biases from estimating output elasticities based on revenue production functions without price data (Bond et al., 2021).

²⁹We consider variation within 2-digit sectors in our reduced-form analysis for two reasons. First, firms in the same sector are more likely to have similar natural risk exposure δ_i° and risk management cost functions g_i and b_i . Second, measured TFP growth and markups may be affected by sector-specific deflators and production elasticities. Consistent with this concern, De Ridder, Grassi, and Morzenti (2024) show that within-sector markup dispersion is more accurately measured than cross-sector dispersion.

Figure 6: TFP volatility, Domar weights, and markups



Notes: Estimation results of (47) using a sample of Spanish firms from Orbis with the normalization $\beta_3^{Domar} = \beta_3^{Markup} = 0$ to avoid perfect multicollinearity. Sample construction is described in Appendix A.1. The sample excludes firms with average markups below 1 and is winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight, and markup within each sector in the panel before computing the firm-level moments. Panel (a): β_j^{Domar} by Domar weight quintile j ($j = 1$ is lowest Domar weight, $j = 5$ is highest Domar weight). Panel (b): β_j^{Markup} by markup quintile ($j = 1$ is lowest markup, $j = 5$ is highest markup). 90% confidence intervals are constructed using standard errors that are clustered at the industry level.

higher markups tend to have more volatile TFP growth.³⁰ These relationships are statistically and economically significant. A firm in the top quintile of the Domar weight distribution is about 15 p.p. less volatile than a firm in the bottom quintile. In contrast, firms in the top quintile of the markup distribution are about 9 p.p. more volatile than firms in the bottom quintile.

In Appendix A.2, we run a simple cross-sectional regression of the standard deviation of firm-level TFP growth on Domar weights and markups. We also conduct an analogous analysis with US firms from Compustat. In both cases, we find similar results.

8.2 Cyclicity of firm-level TFP

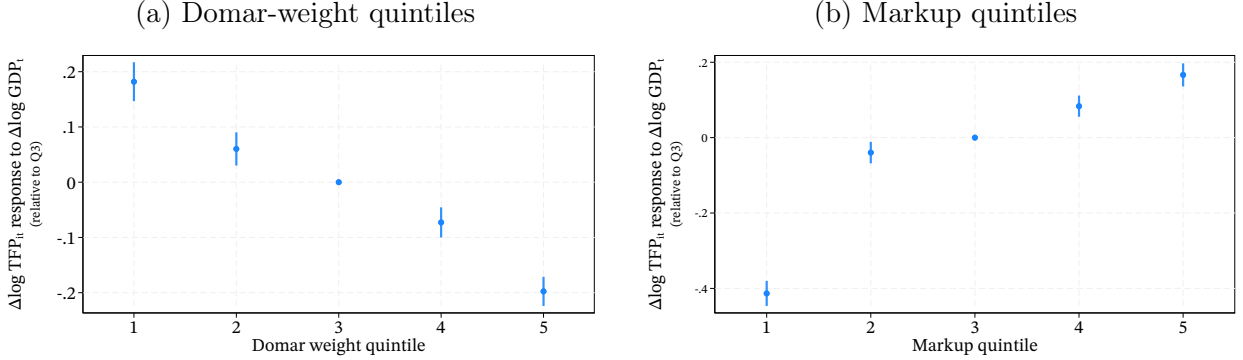
Next, we turn to our model's prediction that the TFP growth of firms with higher Domar weights and lower markups covaries less with GDP growth. Again, we construct a set of dummy variables, $D_{ji,t}^{Domar}$ and $D_{ji,t}^{Markup}$, such that $D_{ji,t}^{Domar} = 1$ if firm i 's Domar weight is in within-sector quintile j in year t , and analogously for markups. We then run the following panel regression, which interacts lagged Domar weight and markup quintiles with log GDP growth:

$$\begin{aligned} \Delta \log TFP_{it} = & \sum_{j=1}^5 \beta_j^{Domar} (D_{ji,t-1}^{Domar} \times \Delta \log GDP_t) + \sum_{j=1}^5 \beta_j^{Markup} (D_{ji,t-1}^{Markup} \times \Delta \log GDP_t) \\ & + \sum_{j=1}^5 \alpha_j^{Domar} D_{ji,t-1}^{Domar} + \sum_{j=1}^5 \alpha_j^{Markup} D_{ji,t-1}^{Markup} + \theta_{st} + \varepsilon_{it}, \end{aligned} \quad (48)$$

³⁰Calvino et al. (2018) find that larger firms have lower sales growth volatility in a cross-country sample. Yeh (2023) shows that large firms are less volatile in US Census data.

where θ_{st} are sector-year fixed effects. The coefficients of interest, β_j^{Domar} and β_j^{Markup} , are reported in Figure 7. Consistent with the model, firms with higher Domar weights and lower markups covary less with GDP.³¹ In Appendix A.2.2, we confirm these results through panel regressions with various sets of fixed effects.

Figure 7: Sensitivity of firm-level TFP to GDP



Notes: Estimation results of (48) using a sample of Spanish firms from Orbis with the normalization $\beta_3^{Domar} = \beta_3^{Markup} = 0$ to avoid perfect multicollinearity. Sample construction is described in Appendix A.1. The estimation sample is winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight and markup in each sector. Panel (a): β_j^{Domar} by Domar weight quintile j ($j = 1$ is lowest Domar weight, $j = 5$ is highest Domar weight). Panel (b): β_j^{Markup} by markup quintile ($j = 1$ is lowest markup, $j = 5$ is highest markup). Quintiles of lagged Domar weight and markup are computed only using firms with non-missing observations in the current period. 90% confidence intervals are constructed using standard errors that are clustered at the firm level.

8.3 Robustness, additional exercises, and discussion

In Appendix A, we provide several robustness exercises. First, we estimate markups using materials instead of labor as the flexible input (Raval, 2023b). Second, to avoid relying only on markups inferred from the production approach, we compute an alternative markup measure that compares total costs directly to sales (Baqee and Farhi, 2019; Burstein, Carvalho, and Grassi, 2024). Third, we replace TFP growth with sales growth in both the panel and cross-sectional exercises. Fourth, we provide estimates that control for firm age. In all cases, we find that firms with higher Domar weights and lower markups are less volatile and covary less with GDP.

Through the lens of the model, these findings imply that larger firms manage risk more intensively. An empirical literature on corporate risk management points in the same direction: larger firms are more likely to have Chief Risk Officers (Liebenberg and Hoyt, 2003), implement holistic enterprise-wide risk management (ERM) systems (Beasley, Clune, and Hermanson, 2005; Hoyt and Liebenberg, 2011), and hedge risks using derivatives (Géczy, Minton, and Schrand, 1997; Bodnar, Hayt, and Marston, 1998).

³¹Sales of large firms also covary less with GDP in US data (Crouzet and Mehrotra, 2020).

We complement this literature with new evidence on the resources firms devote to risk management. While there is no comprehensive accounting data on these expenditures, many of them are recorded as general operating expenses. In Appendix A.5, we show that operating expenses shrink with firm size as a proportion of sales in Orbis data from Spain. We also use Brazil’s administrative matched employer–employee data (RAIS) to measure risk spending more directly, through the employment and wage-bill shares of risk-management occupations; these shares also decline with firm size. Combined with our results, these findings indicate that large firms achieve better risk outcomes at a lower share of resources, consistent with the scale advantage in risk management provided by g_i in the model.

Our model also rationalizes a well-known fact about stock returns: larger firms tend to comove less with the aggregate stock market (Fama and French, 1992). The mechanisms of the model provide an explanation for this stylized fact, as long as productivity increases generally raise a firm’s stock price. Additionally, we show in Appendix A.3 that firms with higher markups exhibit greater covariance with the aggregate stock market, consistent with our theoretical predictions.

There are also indications that the model mechanisms are visible in aggregate data. In Appendix A.4, we show that richer countries have lower TFP volatility. If those countries have access to better risk-management technologies (smaller H_i or η), they should be better at reducing their risk exposure, consistent with this pattern. We also find that countries with higher markups feature more volatile TFP, in line with the forces of the model.

Overall, the findings of this section suggest that key predictions of our theory are in line with firm-level productivity data, stock market return data, and stylized country-level evidence. Of course, other mechanisms could also be at work, and we view our mechanism as complementing these alternative explanations. Importantly, our theory jointly accounts for the four main regularities we document in this section: firm-level TFP volatility and cyclicalities both fall with sales and rise with markups. To our knowledge, no other single mechanism explains all four.³²

9 Calibration to the Spanish economy

To evaluate the quantitative importance of endogenous risk-taking decisions for the macroeconomy, we calibrate the model to the Spanish economy. We use the firm-level data introduced in the previous section together with Spanish input-output data to identify the parameters of the model. We work with the CES generalization of the baseline model, which was introduced in Section 8. With the calibrated model in hand, we conduct two exercises: we evaluate how the economy

³²Several mechanisms could account for individual patterns. Larger firms may be less volatile because they diversify idiosyncratic shocks across more products, as in models of multi-product firms (Klette and Kortum, 2004), or across more trading partners in a production network (Herskovic et al., 2020). Smaller firms may comove more with the aggregate because they are more financially constrained (Gertler and Gilchrist, 1994). Finally, the granular channel through which large firms mechanically drive aggregate fluctuations (Gabaix, 2011) implies they should comove *more*, not less, with the business cycle.

responds to an increase in fundamental uncertainty, and how removing wedges affects aggregate volatility. We also explore the implications of endogenous uncertainty in a calibrated economy with rare disasters. We summarize our calibration strategy below and provide more detail in Appendix B. Appendix E.6 describes an approximation procedure that we use to solve the CES model.

9.1 Sectoral CES model

To better map model quantities to the data, we modify the model of Section 2 in two ways. First, we introduce a sectoral structure: in each sector, an aggregator combines the output of individual firms into a sector-specific composite good, and individual firms purchase intermediate inputs from these aggregators rather than directly from each other. This lets us use Spanish sectoral input-output data to discipline the matrix of network connections. Second, we specify CES sectoral aggregators, which generalize the Cobb-Douglas aggregation of the baseline model. The CES specification is consistent with how we estimate firms' TFP in Section 8. We analyze the resulting model in detail in Appendix E and provide an overview below.

There are S sectors. In each sector s , an aggregator converts the output of the N_s firms in that sector into a sector-specific good according to

$$Q_s = \left(\sum_{i=1}^{N_s} \theta_{si} \left(\frac{Q_{si}}{\theta_{si}} \right)^{\frac{\sigma_s-1}{\sigma_s}} \right)^{\frac{\sigma_s}{\sigma_s-1}},$$

where $\sum_{i=1}^{N_s} \theta_{si} = 1$, and where Q_{si} is the output of firm i in sector s . Firms operate the production function

$$Q_{si} = \exp(\delta_{sit}\varepsilon_t - b_{si}(\delta_{sit}) + \gamma_s t + v_{sit}) \zeta_{si} L_{si}^{1-\sum_{s'=1}^S \hat{\alpha}_{ss'}} \prod_{s'=1}^S X_{si,s'}^{\hat{\alpha}_{ss'}}, \quad (49)$$

where $X_{si,s'}$ denotes the use of the composite good of sector s' by firm i in sector s . Notice that the input elasticities are sector-specific, in line with the available data. As before, prices are set at a constant markup τ_{si} above marginal cost.³³ Finally, the household only consumes goods produced by the sectoral aggregators, so that GDP is given by $Y = \prod_{s=1}^S (\beta_s^{-1} C_s)^{\beta_s}$.

As in Section 8, the TFP of firm i is the sum of a trend $\gamma_s t$, an idiosyncratic shock $v_{sit} \sim \text{iid } \mathcal{N}(\mu_{si}^v, \Sigma_{si}^v)$, a risk-exposure cost $b_i(\delta_{sit})$, and the firm's exposure δ_{sit} to an aggregate shock $\varepsilon_t \sim \text{iid } \mathcal{N}(0, \Sigma)$. The stationary structure of the model implies that firms make the same risk-exposure decisions every period, and so we drop the subscript t and write δ_{si} .

Below, we look at how changes in the environment affect the variance of GDP growth. In the CES model, GDP is approximately log-linear in the shocks (see Lemma 12 in Appendix E), with the approximation being accurate when Σ is small and there are many firms per sector. Under that

³³In Appendix E.5, we allow firms to have market power and to internalize the demand curve they face. We show that for our counterfactual exercises the resulting model is equivalent to the one with exogenous markups.

approximation, the variance of GDP growth admits the decomposition

$$V[y_t - y_{t-1}] \approx \underbrace{2\Sigma\Delta^2}_{\text{endogenous risk}} + 2 \underbrace{\sum_{s=1}^S \sum_{i=1}^{N_s} \Sigma_{si}^v \varsigma_{si}^2}_{\text{exogenous risk}}. \quad (50)$$

The first term in (50) is where the endogenous risk mechanism is at work. It depends on aggregate risk exposure $\Delta = \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} \delta_{si}$, which is the weighted sum of individual exposures δ_{si} . The weights ς_{si} are given by (142) in Appendix E. The second term reflects the granular contribution of firm-level idiosyncratic shocks to aggregate volatility (Gabaix, 2011).

Two features of the CES specification differ from the Cobb-Douglas baseline and shape the analysis below. First, the weights ς_{si} in (50) differ from the cost-based Domar weights $\tilde{\omega}_{si}$. Under CES, shocks reallocate sales across firms within a sector, which moves Domar weights and the labor share. Under Cobb-Douglas, Domar weights are independent of shocks. The labor share, given in Lemma 1, is therefore also fixed, and $\varsigma_{si} = \tilde{\omega}_{si}$. Second, under CES, firms whose risk exposure exceeds the sectoral average gain within-sector sales share when the aggregate shock is positive and lose it when it is negative. This positive comovement gives such firms an additional incentive to take on risk. This effect is absent under Cobb-Douglas, where sales shares do not respond to shocks.

9.2 Calibration strategy

Our calibration strategy aims to replicate the firm-level Orbis dataset in our model economy. Our calibrated economy therefore features 62 sectors and 531,944 individual firms. The largest sectors are “real estate (including imputed rents)” with a consumption share of 13% and “accommodation and food services” with a consumption share of 12%.

Several parameters map directly to aggregate data. We set $(\beta_1, \dots, \beta_S)$ to match the sectoral consumption shares in the 2010 Spanish National Accounts. The same data deliver the matrix of sectoral input shares $\hat{\alpha}$. Consistent with the optimization problem of the sectoral aggregators, we set θ_{si} to match each firm’s sales share within its sector.

Since our calibrated model is consistent with the reduced-form framework of Section 8, the firm-level markups τ_{si} are set to the markup estimates from that section. To pin down the firm’s risk exposure δ_{si} and the variance Σ_{si}^v of its idiosyncratic shock, we use two moments per firm: the variance of its TFP growth and its covariance with GDP growth. Together with (46), which holds approximately in the CES model (with ς_{si} instead of $\tilde{\omega}_{si}$), these two moments pin down the two unknowns. We provide more detail on this procedure in Appendix B.2.

Three parameters are set outside the model. As in Section 8, we use $\sigma_s = 3$ for all sectors. The coefficient of risk aversion is set to $\rho = 5$, as is standard in the literature. Finally, Σ is a free normalization. Changes in Σ are absorbed by rescaling other model objects, and we set $\Sigma = 1$.

The remaining parameters are those of the firm-level risk-management cost functions. To identify them, we rely on Lemma 10 in Appendix E, which shows that, when aggregate risk is small, the firm's risk exposure is approximately

$$\delta_{si} \approx \delta_{si}^\circ + \left(B_s + \eta \frac{1 + \tau_{si}}{\bar{\omega}_{si}} G_s \right)^{-1} \mathcal{E}_{si}^\circ, \quad (51)$$

where we impose that B_s and G_s are sector-specific. Here $\bar{\omega}_{si}$ is the firm's Domar weight at the steady state, and $\mathcal{E}_{si}^\circ$ is the steady-state marginal benefit of risk exposure (both derived in Appendix E). In the CES economy, $\mathcal{E}_{si}^\circ$ is firm-specific: firms with above-average risk exposure gain within-sector sales share when the aggregate shock is positive and lose it when it is negative. This positive covariance between their sales share and the shock creates an additional incentive for such firms to increase their risk exposure. This channel is absent under Cobb-Douglas, where firm sales shares are fixed, and so $\mathcal{E}^\circ$ is common to all firms.

We perform a curve-fitting exercise and find the values of B_s and ηG_s that best fit (51) sector by sector. Firm-level variation in sales and markups identifies the respective roles of B_s and ηG_s in shaping risk-taking decisions: B_s enters as a sector-wide constant in the curvature, while ηG_s enters scaled by $(1 + \tau_{si}) / \bar{\omega}_{si}$, which varies across firms within a sector.³⁴

Overall, our calibration procedure exactly matches six key data features: 1) the sectoral consumption shares, 2) the firm-level sales shares within sectors, 3) the sectoral input-output cost shares, 4) the firm-level markups, 5) the variance of each firm's TFP growth, and 6) the covariance of each firm's TFP growth with GDP growth.³⁵ Appendix B.4 provides more details about the calibrated economy.

9.3 Increases in aggregate risk

When risk-exposure decisions are endogenous, the economy has more flexibility to absorb an increase in fundamental uncertainty. To gauge the importance of this mechanism, we increase the variance Σ of the aggregate risk factor and compare the response of GDP volatility under two regimes: one in which firms keep δ at its calibrated value, and one in which they re-optimize. When δ is held fixed, (50) implies that the endogenous component of the variance of GDP growth scales linearly with Σ , so its standard deviation scales linearly with $\sqrt{\Sigma}$. As Table 1 shows, the elasticity of the standard deviation of GDP growth to $\sqrt{\Sigma}$ is 0.40 in this case, implying that a 1% rise in the

³⁴Our calibration procedure cannot distinguish between η and G_s , and only provides an estimate for ηG_s . But since these quantities enter the model equations only through their product, there is no need for us to determine η and G_s separately. We therefore remain agnostic about the supply elasticity of risk-management resources η .

³⁵The model overshoots the standard deviation of GDP growth: it is 2.35% in the data, while it is 3.29% in our calibrated model. In the model, a version of Baqaee and Farhi's (2019) theorem is at work: the variance of GDP growth is determined by the variance of firm-level TFP, weighted by markup-distorted Domar weights. Once these objects are fitted, GDP volatility is pinned down. The fact that this delivers higher volatility than in the data is a consequence of the network structure common to much of the literature.

standard deviation of the aggregate risk factor raises that of GDP growth by 0.4%. Once firms can re-optimize, the higher Σ makes exposure to the aggregate risk factor more costly, so firms reduce δ and the elasticity falls to 0.11. The endogenous response of risk-taking decisions thus absorbs most of the rise in fundamental uncertainty.

Table 1: Standard deviation of GDP growth with and without disaster risk

	Benchmark			Model with disaster risk					
	Calib.	Fix. δ	Flex. δ	Cond. on no disasters			Unconditional		
				Calib.	Fix. δ	Flex. δ	Calib.	Fix. δ	Flex. δ
Increase in Σ (elast. st. dev. Δy)	–	0.40	0.11	–	0.40	0.11	–	0.14	0.01
Removing wedges (st. dev. Δy)	3.29%	3.51%	1.83%	3.29%	3.51%	2.09%	7.23%	7.34%	5.77%
Double disaster prob. (st. dev. Δy)	–	–	–	3.29%	3.29%	2.81%	7.23%	9.62%	9.10%

Notes: “Fixed δ ” refers to an economy in which δ is fixed at its calibrated value. “Cond. on no disasters” refers to the moments in a sample without disasters. “Unconditional” refers to the moments in a sample with disasters. “elast. st. dev. (Δy)” is computed as the percentage difference between the counterfactual and calibrated standard deviation of GDP growth, divided by the percentage change in $\sqrt{\Sigma}$.

9.4 Wedges and inefficient risk exposure

The presence of wedges pushes firms to make inefficient risk-exposure decisions. To evaluate the quantitative importance of these inefficiencies, we compute the equilibrium in a version of the calibrated economy without wedges. When wedges disappear, firms become larger and their revenue-based Domar weights converge to their cost-based counterparts. Keeping δ fixed, this would have no impact on GDP volatility in the Cobb-Douglas economy. Indeed, recall from (50) that GDP-growth volatility depends only on cost-based Domar weights (independent of τ), the risk exposure δ (held fixed by assumption), and the properties of the random variables (also independent of τ). In the CES economy, by contrast, removing wedges triggers within-sector reallocation, which shifts the GDP-shock weights ς_{si} in (50). As Table 1 shows, this raises the standard deviation of GDP growth slightly from 3.29% to 3.51%.

Once firms can re-optimize, the picture changes substantially. Since the removal of wedges makes firms larger, they manage risk more aggressively. As a result, the standard deviation of GDP growth drops to 1.83%. Wedges therefore have a substantial impact on aggregate volatility when risk is endogenous.

Taken together, our findings suggest that ignoring the endogenous nature of risk-taking can lead to an overestimation of the impact of fundamental risk on the economy and an underestimation of the adverse effect of taxes and markups on aggregate volatility. We interpret these findings with caution: our model is stylized, most notably in functional forms and the nature of competition, and

we rely on an approximation to solve the CES model, so the exercises should be read as an estimate of the magnitudes involved rather than as precise quantitative answers.

In Appendix B, we report several additional results for the calibrated economy. First, we repeat the counterfactual experiments under alternative measures of wedges. When wedges are large, the equilibrium is far from the efficient allocation and is less able to dampen an increase in Σ . Naturally, removing wedges also has a stronger impact on volatility when τ is large. Second, we examine the cross-sectional implications of changes in Σ and of wedge removal. Third, we show that raising the elasticity of substitution from $\sigma_s = 3$ to $\sigma_s = 4$ leaves our results largely unchanged. Fourth, we provide a calibration of the Cobb-Douglas model, which we can solve exactly. Endogenous risk has the same qualitative impact as in our CES benchmark, though its quantitative effect is smaller.

9.5 Disaster risk

Our benchmark model has the advantage of being simple and tractable, but as is well known, it cannot replicate traditional measures of the importance of aggregate risk, such as the equity premium (Mehra and Prescott, 1985). In this section, we consider an alternative model with disaster risk (Rietz, 1988; Barro, 2006). Namely, we assume that the single risk factor ε consists of two independent components, $\varepsilon = \varepsilon^n + \varepsilon^d$, where $\varepsilon^n \sim \mathcal{N}(0, \Sigma)$ and the disaster component ε^d takes two values: $\varepsilon^d = -d$ with probability π_d and $\varepsilon^d = 0$ otherwise.³⁶ We calibrate this economy through the same procedure used above and set π_d and d in line with international evidence (Barro, 2006).³⁷

Results from this calibration are reported in the “Model with disaster risk” panels of Table 1. The “Conditional on no disasters” columns refer to moments computed in samples without disaster events, while the “Unconditional” columns include them. Conditional on no disasters, an increase in Σ has an impact similar to that in the benchmark economy; unconditionally, the response is much smaller because ε^n accounts for only a fraction of the overall variability of ε . Removing wedges, in contrast, has a large impact in the economy with disasters. Given the destructive nature of ε^d , the social planner is willing to spend heavily to reduce its impact, but wedges impede that process. When they are removed, spending on risk mitigation increases, and aggregate volatility, conditional and unconditional, falls substantially.

We also consider the impact of doubling the likelihood of a disaster, setting $\pi_d = 3.4\%$. When firms are not able to change their risk exposure, this leads to a large increase of almost 2.5 percentage points in unconditional volatility. In contrast, when risk is endogenous, the increase in π_d pushes firms to spend more on risk mitigation, and the aggregate risk exposure declines. Unconditional volatility increases by only 1.9 percentage points in that case. Since aggregate exposure Δ governs

³⁶Appendix D considers a Cobb-Douglas model with disasters and extends our theoretical results to that setting. Appendix E.8 introduces disasters in the CES setting. In line with our baseline calibration, we adopt the CES specification for the current section.

³⁷We set $\pi_d = 0.017$ and describe the calibration of d and the rest of the model in Appendix B.8.

the economy’s exposure to the entire business-cycle risk factor ε , this change also reduces volatility associated with normal business cycles. Through this mechanism, a higher likelihood of disasters can tame run-of-the-mill business cycle volatility.

Because of the rare disasters, the model generates a sizable equity premium of 8.20%.³⁸ Since risk is endogenous, changes in the environment can influence this premium: when all wedges are removed, for instance, the equity premium falls to 5.95%. We interpret this number cautiously, as the model is stylized and our estimation of the cost of managing disaster risk is indirect. Nonetheless, it suggests that taxes and other distortions may have a sizable impact on the risk premium through their influence on risk-taking decisions.

10 Conclusion

This paper proposes a parsimonious theory of endogenous risk in which firms are free to choose the properties of their productivity processes. Despite its simplicity, the model can explain key features of the data related to how firms’ characteristics influence their risk profiles. In a calibration of the model, we find that changes in the environment can have a large impact on aggregate volatility through their influence on risk-taking decisions.

The model is stylized, and many extensions merit exploration. For instance, we have assumed that markets are complete so that firms value cash flows using the household’s stochastic discount factor. However, in a model with entrepreneurs who cannot diversify risks related to their businesses, risk-management decisions would likely become more consequential, potentially amplifying the model’s mechanisms (Greenwood and Jovanovic, 1990). More broadly, since our framework is tractable, we hope it can serve as a useful starting point for studying endogenous risk in other settings, such as firms’ exposure to supply-chain, geopolitical, and climate risks.

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³⁸See Appendix E.8 for details. We set the time discount factor to about 0.98 to match a risk-free rate of 2% and assume no trend growth in GDP, roughly in line with the Spanish economy since 2008.

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Online Appendix

A Appendix for Section 8

This appendix contains details about the reduced-form exercises of Section 8.

A.1 Data sources and variable construction

A.1.1 Orbis data

Our data on Spanish firms come from the Orbis Historical Disk Product. Orbis is the largest cross-country firm-level database that covers public and private firms’ financial and real activities. We choose Spain for our analysis due to its near-universal coverage of firms (covering more than 95% of total industry gross output after 2010) as detailed in Gopinath et al. (2017) and Kalemli-Özcan et al. (2024). We use the sample period 1995-2019 for our analysis.³⁹

Sample cleaning Following the procedure of Kalemli-Özcan et al. (2024), we link each firm’s descriptive information with its financial information via the unique BVD firm identifier (BVDID). We restrict our analysis to Spanish firms, defined as firms that satisfy two criteria: 1) their latest address is in Spain, and 2) their BVDID starts with the ISO-2 code ES. Within the Orbis Spanish sample, we apply the following standard cleaning procedure:

1. We harmonize the calendar year of each firm-year observation using the variable `closing_date`: if the closing date is after or on July 1, the current year is assigned as the calendar year. Otherwise, the previous year is assigned.⁴⁰
2. In a given year, a firm in the Orbis database might have multiple financial statements from different sources (local registry, annual report, or others), for consolidated or unconsolidated accounts. When several source-consolidation combinations exist for a firm, we deduplicate by selecting the account that, in order of priority, 1) shows the most consistent reporting frequency (closest to regular annual reporting), 2) offers the longest non-missing time series for key financial variables (fixed assets and/or sales), and 3) is consolidated, if the first two criteria are tied.
3. We only keep firms with non-missing and positive sales (`operating_revenue_turnover`), fixed assets (`fixed_assets`), wage bills (`costs_of_employees`), and material costs

³⁹Orbis offers good coverage of the Spanish economy starting from 1995. Moreover, the most recent observations in the version of Orbis Historical Disk Product that we use are from 2021. We therefore use 2019 as the last year of the sample since there is usually a two-year reporting lag for some variables (see Kalemli-Özcan et al. (2024) for details).

⁴⁰This adjustment matters little for the Spanish sample, as 99% of firms close their books on December 31.

(`material_costs`). We also harmonize the units of all monetary values to be in current euros.

4. To prevent outliers from affecting the production function and markup estimation, we exclude any firm-year observation whose average revenue product for any input (fixed assets, wage bills, or material costs) lies above the 99th percentile or below the 1st percentile of that year’s distribution. Furthermore, over the full sample period, we sequentially remove the top 0.1% of observations in sales, fixed assets, wage bills, and material costs (in that order).

Variable definitions and estimation of firm-level markup and TFP Our empirical construction of firm-level TFP, markups, and Domar weights relies on the sectoral CES extension of the baseline model, introduced in Section 9 and developed in full in Appendix E. We adopt the sector structure, CES demand system, and notation of those sections throughout.

We calculate a firm i ’s revenue-based Domar weight as $\omega_{it} = \frac{\text{sales}_{it}}{\text{GDP}_{\text{nom},t}}$, where $\text{GDP}_{\text{nom},t}$ is the Spanish nominal GDP (in euros) obtained from the Annual Spanish National Accounts produced by the National Statistics Institute (INE).

We now describe how we estimate firms’ production functions and use them to construct firm-level markups and productivity growth. Consistent with the constant returns-to-scale Cobb-Douglas production function in the model, we assume firm output takes the following form:⁴¹

$$\log Q_{it} = \alpha_{s(i)}^L \log L_{it} + \alpha_{s(i)}^M \log M_{it} + \alpha_{s(i)}^K \log K_{it} + \log TFP_{it}, \quad (52)$$

where the output elasticities are specific to firm i ’s NACE 2-digit sector $s(i)$ and sum to one. The inputs L_{it} , M_{it} , and K_{it} are, respectively, labor, intermediate input and capital.

We estimate sector-specific output elasticities using a cost-share approach following Foster, Haltiwanger, and Syverson (2008) and De Loecker, Eeckhout, and Unger (2020). As noted by Raval (2023), this estimator requires two assumptions, both consistent with our model, (1) constant returns to scale, and (2) first-order cost minimization conditions hold on average for all inputs. This approach also avoids the bias that Bond et al. (2021) identify when output elasticities are estimated from revenue production functions without firm-level price data. Denoting by w_{it} the wage, by p_{it}^M the price of intermediate inputs, by r_{it} the rental rate of capital, and by κ_{it} the firm’s marginal cost, we can write

$$\alpha_{s(i)}^L = \frac{\mathbb{E}[w_{it}L_{it}|s]}{\mathbb{E}[\kappa_{it}Q_{it}|s]}, \quad \alpha_{s(i)}^M = \frac{\mathbb{E}[p_{it}^M M_{it}|s]}{\mathbb{E}[\kappa_{it}Q_{it}|s]}, \quad \alpha_{s(i)}^K = \frac{\mathbb{E}[r_{it}K_{it}|s]}{\mathbb{E}[\kappa_{it}Q_{it}|s]} \quad \forall s. \quad (53)$$

For each firm-year, we define total cost $\kappa_{it}Q_{it}$ as the sum of wage bill, material costs, and capital

⁴¹This empirical specification is closely related to the production function (49) in the quantitative model, where the model’s “labor” input represents the composite value-added bundle (corresponding to L_{it} and K_{it} in the empirical specification), and the model’s composite intermediate input bundle corresponds to “materials” (M_{it}).

costs. Wage bill, $w_{it}L_{it}$, and material costs, $p_{it}^M M_{it}$, are taken directly from the data. We construct capital costs as $r_{it}K_{it} = \frac{r_{it}}{p_{it}^K} \times p_{it}^K K_{it}$, where $p_{it}^K K_{it}$ is the reported book value of the fixed assets. Following a similar approach to De Loecker, Eeckhout, and Unger (2020), we set the user cost of capital r_{it}/p_{it}^K to 20%.⁴² Within each NACE 2-digit sector, the elasticities $\hat{\alpha}_s^L$, $\hat{\alpha}_s^M$, and $\hat{\alpha}_s^K$ are then computed via equation (53) using all firm-year observations. The elasticities therefore sum to one by construction.

Following De Loecker and Warzynski (2012), we compute the markup as $1 + \tau_{it} = \hat{\alpha}_{s(i)}^L / \left(\frac{\text{Wage Bill}_{it}}{\text{Sales}_{it}} \right)$, where $\hat{\alpha}_{s(i)}^L$ is the estimated labor elasticity in production and $\frac{\text{Wage Bill}_{it}}{\text{Sales}_{it}}$ is the share of labor expenditure in firms' sales.⁴³ Our baseline markup measure regards labor as a variable input in production.⁴⁴ We report robustness to materials-based and total-cost markup measures in Appendix A.2.3.

To construct firm-level TFP growth, we rely on the CES demand structure within each sector, as described in Section 9 and Appendix E. Maintaining the assumptions that all firms in a sector use the same Cobb-Douglas production function of the form (52) and that taste shifters θ_{si} are time invariant, we can identify the change in a firm's productivity as:⁴⁵

$$\begin{aligned} \Delta \log TFP_{it} = & \left(\frac{\sigma_s}{\sigma_s - 1} \Delta \log (P_{it} Q_{it}) - \hat{\alpha}_{s(i)}^L \Delta \log L_{it} - \hat{\alpha}_{s(i)}^K \Delta \log K_{it} - \hat{\alpha}_{s(i)}^M \Delta \log M_{it} \right) \\ & - \frac{\sigma_s}{\sigma_s - 1} \Delta \log P_{st} - \frac{1}{\sigma_s - 1} \Delta \log Q_{st}, \end{aligned} \quad (54)$$

where $P_{it}Q_{it}$ denotes firm nominal revenue, P_{st} is the sectoral CES price index, and Q_{st} is sectoral output. The term in parentheses is firm-year specific. We set $\sigma_s = 3$ for all sectors in the baseline analysis and explore robustness to this value in Appendix A.2.5. Sales $P_{it}Q_{it}$ are observed directly, and the elasticities $\hat{\alpha}_s^L$, $\hat{\alpha}_s^M$, and $\hat{\alpha}_s^K$ come from our estimates in (53). The input quantities L_{it} , K_{it} , and M_{it} can be obtained by deflating the corresponding nominal cost series with sectoral price

⁴²Following De Loecker, Eeckhout, and Unger (2020), we compute the user cost of capital as $r_{it}/p_{it}^K = I_t + \text{Risk Premium}_t - \Pi_t + \Delta$, where I_t is the risk-free nominal rate, Risk Premium_t is the risk premium, Π_t is inflation, and Δ is the depreciation rate. We calibrate each component using its sample average for Spain over 1995–2019. We use sample-average values because the user cost enters only the construction of capital costs and sector-level output elasticities; holding it fixed avoids introducing high-frequency variation in asset returns into production-function and markup estimates. Using the Jordà-Schularick-Taylor Macrohistory Database (Jordà et al., 2019), we set $I_t + \text{Risk Premium}_t$ equal to the average nominal equity total return, 12%, and Π_t equal to average CPI inflation, 2%. We set the depreciation rate to $\Delta = 10\%$. These values imply a user cost of capital of 20%. This value is also consistent with common calibrations in the literature (see, for example, Hsieh and Klenow, 2009 and Peter and Ruane, 2025).

⁴³Under perfect competition, this estimator recovers the wedge $1 + \tau_{it}$. Under monopolistic competition with CES preferences, it instead recovers the total markup $\frac{\sigma_s}{\sigma_s - 1} (1 + \tau_{it})$, which combines the Dixit-Stiglitz markup with the exogenous wedge. Appendices E and E.5 show that the two formulations are observationally equivalent. Since our reduced-form exercises exploit only within-sector-year variation in markups, the distinction does not affect the empirical results.

⁴⁴Since $\alpha_{s(i)}^L$ is common within sector, our within-sector-year comparisons are insensitive to its level; only cross-firm differences in labor shares affect the labor-based markup variation used in the reduced-form regressions.

⁴⁵Equation (54) follows from combining the CES demand system (108) with the Cobb-Douglas production function (52), under time-invariant θ_{si} . Appendix E.7 provides the explicit derivation.

indices, as is standard in the literature. Our reduced-form exercises, however, exploit only within-sector-year variation in TFP growth, so the choice of deflators is immaterial.⁴⁶ The same logic applies to the last two terms in (54): they are common to all firms within a sector-year and are absorbed by the demeaning or fixed effects.

For the calibration exercise of Section 9, we instead need the *level* of $\Delta \log TFP_{it}$, not only its within-sector-year variation. We then construct L_{it} , K_{it} , and M_{it} under the assumption that all firms within a sector face the same input prices. We deflate material costs and fixed assets using EU-KLEMS industry-specific value-added and capital price indices, respectively, at the most detailed NACE 2-digit level when available, and we deflate the wage bill using the GDP deflator.⁴⁷ The sectoral price index P_{st} and sectoral output Q_{st} are constructed from sectoral data published by Spain’s National Statistics Institute (INE). Appendix B provides further detail.

For all reduced-form exercises that involve panel regressions, we restrict our sample to firms with at least two non-missing observations of measured TFP growth. Additionally, for all cross-sectional regressions, we only keep firms with at least 5 non-missing observations of TFP growth (such that TFP volatility can be precisely estimated) and an average markup above one, meaning that they do not constantly operate at a loss.

A.1.2 CRSP/Compustat Merged and US stock market data

This subsection describes the firm-level financial and stock market variables used in Appendix A.3 below. We use two main sources of data: 1) the CRSP/Compustat Merged Fundamentals Annual, which provides firm-level Domar weights and markups; and 2) the WRDS Beta Suite database, which provides stocks’ loadings on the aggregate market return (stock-market betas). Following De Loecker, Eeckhout, and Unger (2020), the analysis spans 1981–2016.

CRSP/Compustat Merged Our sample selection and cleaning procedure follows standard practices in the literature, particularly those carried out in De Loecker, Eeckhout, and Unger (2020).

1. We apply the following standard filters when processing the data: 1) consolidation level is C; 2) industry format is INDL; 3) data format is STD; 4) population source is D; 5) currency is USD; 6) we include both active and inactive companies; 7) we exclude financial, utilities and public sector firms (i.e. those with SIC code in the following range: 4900 – 4999, 6000 – 6999 and 9000 – 9999). We use the data from 1980 to 2016.

⁴⁶Specifically, we use sector-year-demeaned TFP growth in the cross-sectional volatility exercises and include sector-year fixed effects in the panel regressions.

⁴⁷As noted by Comin et al. (2025), Spain does not have consistent sectoral coverage for price indices for gross output and intermediate inputs. We therefore follow Comin et al. (2025) and use the price index for value added as the material price deflator.

2. We drop firm-year observations for which firms are not incorporated (**fic**) or headquartered (**loc**) in the USA.
3. We use PERMNO as firm identifier and drop the firm-year observations with duplicates.
4. We only keep firms with non-missing and positive sales (**sale**), gross book value of fixed assets (**ppegt**), employees (**emp**), and cost of goods sold (**cogs**).
5. We drop firm-year observations with COGS-to-sales ratio in the top and bottom 1% of the corresponding year’s distributions.

WRDS Beta Suite We estimate each stock’s equity beta in each year using the Beta Suite by WRDS. We use PERMNO as our firm identifier. For each firm-month pair, we estimate the CAPM market model using a 5-year backward-looking rolling window of monthly firm and aggregate stock market returns. Specifically, the equity beta β_{it} for each firm-month pair is obtained from the time series regression

$$r_{it} - rf_t = \alpha_i + \beta_{it}mktrf_t + \varepsilon_{it},$$

where r_{it} is firm i ’s stock return, $mktrf_t$ is the Fama-French Excess Return on the Market, and rf_t is the risk-free rate during month t . The estimation runs inside the Beta Suite with the following options: 1) frequency is **monthly**; 2) both the estimation window and the minimum window are 60 months; 3) risk model is **Market Model**; 4) return type is **Regular Return**. We retain only the December estimate as the firm’s equity beta for that calendar year.

Variable definitions and estimation of firm-level markup and TFP The variable construction for Compustat follows the Orbis procedure closely, but adapts it to the data in three respects: (i) cost of goods sold serves as a composite variable input, since labor and material costs are not separately reported; (ii) output elasticities are estimated sector-by-year rather than sector-only, following De Loecker, Eeckhout, and Unger (2020); and (iii) all variables are deflated by the GDP deflator, as granular sectoral deflators are not available for this sample.

We define firm i ’s revenue-based Domar weight as $\omega_{it} = \frac{\text{sales}_{it}}{\text{GDP}_{\text{nom},t}}$, where $\text{GDP}_{\text{nom},t}$ is US nominal GDP (the **GDPA** series from the Bureau of Economic Analysis). Our cleaning and variable definitions follow De Loecker, Eeckhout, and Unger (2020), except that we estimate output elasticities using a sector-year cost-share estimator rather than their production-function approach. We consider the Cobb-Douglas production function

$$\log Q_{it} = \alpha_{V,s(i)t} \log V_{it} + \alpha_{K,s(i)t} \log K_{it} + \varepsilon_{it}, \quad (55)$$

where Q_{it} , V_{it} , and K_{it} are deflated values of sales, cost of goods sold, and fixed assets, respectively. As in De Loecker, Eeckhout, and Unger (2020), we use costs of goods sold as a composite variable

input because material costs and wage bills are not separately reported in Compustat. All variables are deflated using the GDP deflator.

For each NAICS 2-digit sector s and year t , we compute the variable-input and capital elasticities as the corresponding sector-year cost shares:

$$\hat{\alpha}_{V,st} = \frac{\mathbb{E}[\text{COGS}_{it}|s, t]}{\mathbb{E}[\text{COGS}_{it} + r_t K_{it}|s, t]}, \quad \hat{\alpha}_{K,st} = \frac{\mathbb{E}[r_t K_{it}|s, t]}{\mathbb{E}[\text{COGS}_{it} + r_t K_{it}|s, t]},$$

where $r_t K_{it}$ is real capital expenditure, computed as the user cost of capital times deflated fixed assets. This estimator imposes constant returns to scale within each sector-year cell and recovers output elasticities from observed input expenditures via cost minimization.

Treating cost of goods sold as the flexible input, we follow De Loecker, Eeckhout, and Unger (2020) to compute markups as $1 + \tau_{it} = \hat{\alpha}_{V,s(i)t} / \left(\frac{\text{Costs of Goods Sold}_{it}}{\text{Sales}_{it}} \right)$. Under CES demand and the production function (55), firm-level TFP growth is

$$\begin{aligned} \Delta \log TFP_{it} = & \frac{\sigma_s}{\sigma_s - 1} \Delta \log \text{Sales}_{it} - \frac{1}{2} (\hat{\alpha}_{V,s(i)t} + \hat{\alpha}_{V,s(i),t-1}) \Delta \log V_{it} \\ & - \frac{1}{2} (\hat{\alpha}_{K,s(i)t} + \hat{\alpha}_{K,s(i),t-1}) \Delta \log K_{it} + \lambda_{st}, \end{aligned}$$

with $\sigma_s = 3$ in the baseline. As explained in Appendix E, the term λ_{st} collects the sector-year components implied by the CES demand system, including sectoral price and quantity growth. Since these components are common to all firms within a sector-year, they can be absorbed by sector-year fixed effects. For the cross-sectional volatility exercises, we first residualize TFP growth by NAICS 2-digit-sector-by-year fixed effects and then compute each firm's time-series standard deviation of the residual.

For the panel regression involving equity-beta of Appendix A.3, we restrict the sample to firms with at least two observations of measured TFP growth. For cross-sectional regressions, we require at least five observations and an average markup above one, mirroring the Orbis sample.

Constructing NYSE breakpoints for the deciles of Domar weights We now discuss how we construct the NYSE breakpoints for the deciles of Domar weights. Our approach resembles the procedure used in Fama and French (1992) to construct the NYSE breakpoints for market equity (ME), but instead is applied to Domar weights in our annual CRSP/Compustat Merged sample. We make the following restrictions to the sample: 1) we exclude all firms from the financial, utilities, and public sectors along with those with invalid annual sales data, 2) we drop all observations that do not have a matched 5-year CAPM beta at the PERMNO-year level, and (3) we only consider firms with a stock exchange code (`exchg`) of 11 such that we only keep the firms that are traded on the NYSE.

After restricting the sample, we compute the 10th, 20th, 30th, 40th, 50th, 60th, 70th, 80th, and

90th percentiles of Domar weights for each year. These annual percentile values are then used to form the Domar weight deciles in the main sample.

A.2 Additional results

This section contains additional results related to the reduced-form exercises of Section 8.

A.2.1 Volatility of TFP growth

To complement the results reported in Figure 6, we run a cross-sectional regression of the firm-level standard deviation of TFP growth on log average Domar weight and log average markup while controlling for sector fixed effects,

$$\sigma_i^{TFP} = \alpha + \beta^{Domar} \log(\overline{\text{Domar Weight}}_i) + \beta^{Markup} \log(\overline{\text{Markup}}_i) + \theta_s + \varepsilon_i. \quad (56)$$

The results are given in Table 2. Consistent with our theory, we find $\beta^{Domar} < 0$ and $\beta^{Markup} > 0$ for both the Orbis Spain sample and the Compustat sample.

Table 2: Volatility of TFP growth, Domar weights and markups

	Dependent variable: Volatility of firm-level TFP growth	
	(1): Orbis Spain	(2): Compustat
$\log(\overline{\text{Domar Weight}}_i)$	−0.043*** (0.003)	−0.035*** (0.001)
$\log(\overline{\text{Markup}}_i)$	0.077*** (0.013)	0.021*** (0.006)
Sector FE	Yes	Yes
Observations	304,244	6,442
R^2	0.303	0.376

Notes: The table presents estimation results of (56) using a sample of Spanish firms from Orbis in column (1) and US public firms from Compustat in column (2). See text for a detailed description of variable construction and sample selection. The Orbis Spain sample is winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight and markup in each sector before computing firm-level moments. The Compustat sample is winsorized at top and bottom 1%. Standard errors (in parentheses) are clustered at the NACE 2-digit industry level for the Orbis Spain sample in column (1), and are heteroskedasticity-robust for the Compustat sample in column (2). *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.2.2 Sensitivity of firm-level TFP to GDP

To complement the results reported in Figure 7, we run a panel regression to examine whether firms with different Domar weights and markups display different comovement patterns between

firm-level TFP growth and aggregate GDP growth:

$$\begin{aligned} \Delta \log TFP_{it} = & \beta_1 \log (\text{Domar Weight}_{it-1}) \times \Delta \log GDP_t + \beta_2 \log (\text{Markup}_{it-1}) \times \Delta \log GDP_t \\ & + \alpha_1 \log (\text{Domar Weight}_{it-1}) + \alpha_2 \log (\text{Markup}_{it-1}) + \text{Fixed Effects} + \varepsilon_{it}. \end{aligned} \quad (57)$$

The coefficients of interest are β_1 and β_2 , and we expect to find $\beta_1 < 0$ and $\beta_2 > 0$. Column (1) of Table 3 shows the results of estimating (57) with NACE 2-digit sector-by-year fixed effects, our benchmark specification. In column (2), we add firm fixed effects to this specification; in column (3), we instead use NACE 4-digit sector-by-year fixed effects; and in column (4), we add firm fixed effects to the NACE 4-digit sector-by-year specification. In all specifications, we find statistically significant coefficients whose signs are consistent with our theory. Together, these specifications show that the implications of our model are not sensitive to the sectoral definition and hold both within and across firms.

Table 3: Sensitivity of firm-level TFP to GDP

	Dependent variable: Firm-level TFP growth			
	(1)	(2)	(3)	(4)
$\log (\text{Domar Weight}_{it-1}) \times \Delta \log GDP_t$	-0.111*** (0.005)	-0.108*** (0.005)	-0.106*** (0.005)	-0.100*** (0.006)
$\log (\text{Markup}_{it-1}) \times \Delta \log GDP_t$	0.313*** (0.011)	0.415*** (0.013)	0.380*** (0.012)	0.466*** (0.014)
Firm FE	No	Yes	No	Yes
NACE-2-Sector-Year FE	Yes	Yes	No	No
NACE-4-Sector-Year FE	No	No	Yes	Yes
Observations	7,488,988	7,488,988	7,488,589	7,488,553
R^2	0.040	0.216	0.046	0.220

Notes: Table presents estimation results of (57) using a sample of Spanish firms from Orbis. See text for a detailed description of variable construction and sample selection. The estimation sample is winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight and markup in each sector. Coefficients on levels of GDP growth, (log) Domar weight and (log) markup are not reported for brevity. Standard errors (in parentheses) are clustered at the firm level. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.2.3 Robustness with alternative markup measures

Our baseline markups are constructed from cost-share-based production function estimates, treating labor as the flexible input. A large body of work has emphasized potential pitfalls in estimating markups with standard production data. Bond et al. (2021) argue that markups estimated using production functions with revenue data can be biased. However, this concern does not directly apply in our case. Under our maintained assumptions of a constant returns-to-scale Cobb-Douglas production function and cost minimization, the cost-share estimator identifies the relevant output

elasticities from factor-cost data. Our benchmark markup measure therefore does not rely on the same revenue-based production-function variation that motivates the bias concern.

Nevertheless, we investigate robustness of our findings to alternative markup measures. First, Raval (2023) shows that estimated markups can depend on whether labor or materials are treated as the flexible input. We therefore repeat our analysis using materials as the flexible factor. We compute this materials-based markup as $1 + \tau_{it}^M = \hat{\alpha}_{s(i)}^M / \left(\frac{\text{Material Costs}_{it}}{\text{Sales}_{it}} \right)$, where $\hat{\alpha}_{s(i)}^M$ is the estimated output elasticity of materials obtained from the same cost-share-based production function estimation. Results from estimating (56) and (57) with this alternative markup measure are reported in columns (2) and (5) of Table 4, alongside our benchmark estimates in columns (1) and (4).

Second, we construct a markup measure based directly on total costs. Under constant returns to scale, frictionless and competitively supplied inputs, and no fixed costs, the inverse markup can be recovered as total cost divided by revenue. Following Baqaee and Farhi (2019) and Burstein, Carvalho, and Grassi (2024), we compute markups as the ratio of a firm’s sales to total cost, where total cost is the sum of capital, labor, and material expenditures: $1 + \tau_{it}^{\text{Total Cost}} = \frac{\text{Sales}_{it}}{\text{Total Cost}_{it}}$. Results from estimating (56) and (57) with these total-cost-based markups are reported in columns (3) and (6) of Table 4.

The estimates reported in Table 4 show that the results are qualitatively similar and statistically significant across all markup measures. As in the benchmark regressions, smaller firms and firms with higher markups exhibit greater TFP volatility and greater sensitivity of firm-level TFP to GDP.

A.2.4 Results with sales growth as dependent variable

Our baseline reduced-form measure of firm-level TFP growth is constructed from sales, input expenditures, and the CES elasticity σ_s through the inversion in (54). To show that our patterns are not an artifact of this construction, we re-run the reduced-form exercises using firm sales growth – a directly observed quantity – as the outcome.

To formalize the link between firm-level TFP growth and sales growth in the CES model of Appendix E, let $\log(P_{sit}Q_{sit})$ denote log sales, a_{sit} log TFP, and τ_{si} the firm’s output wedge. The CES demand system implies that firm i ’s within-sector sales share is

$$\chi_{sit} := \frac{P_{sit}Q_{sit}}{P_{st}Q_{st}} = \theta_{si} \frac{P_{sit}^{1-\sigma_s}}{P_{st}^{1-\sigma_s}}.$$

Substituting the pricing equation, $P_{sit} = \frac{1+\tau_{si}}{\exp(a_{sit})} \prod_{s'=1}^S P_{s't}^{\hat{\alpha}_{ss'}}$, gives

$$\log \chi_{sit} = \log \theta_{si} + (\sigma_s - 1) a_{sit} - (\sigma_s - 1) \log(1 + \tau_{si}) + B_{st},$$

Table 4: Results with alternative markup measures

Dependent variable	Firm-level TFP growth			Vol. of firm-level TFP growth		
	(1)	(2)	(3)	(4)	(5)	(6)
Markup estimate	Labor	Materials	Total Cost	Labor	Materials	Total Cost
$\Delta \log GDP_t$ $\times \log (\text{Domar Weight}_{it-1})$	-0.111*** (0.005)	-0.040*** (0.005)	-0.150*** (0.005)			
$\Delta \log GDP_t$ $\times \log (\text{Markup}_{it-1})$	0.313*** (0.011)	0.070*** (0.012)	0.349*** (0.035)			
$\log (\overline{\text{Domar Weight}}_i)$				-0.043*** (0.003)	-0.027*** (0.003)	-0.036*** (0.003)
$\log (\overline{\text{Markup}}_i)$				0.077*** (0.013)	0.127*** (0.013)	0.302*** (0.042)
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,488,988	7,488,988	7,488,988	304,244	514,513	522,179
R^2	0.040	0.074	0.076	0.303	0.404	0.283

Notes: Table presents estimation results of (57) (columns (1), (2) and (3)) and (56) (columns (4), (5) and (6)) using a sample of Spanish firms from Orbis. See text for a detailed description of variable construction and sample selection. The estimation samples for columns (1), (2) and (3) are winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight and markup in each sector. The estimation samples for columns (4), (5) and (6) are winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight and markup in the panel in each sector before computing the firm-level moments. In columns (1) and (4), firms' markups are computed using wage bill-based markups. In columns (2) and (5), firms' markups are computed using material-based markups. In columns (3) and (6), firms' markups are computed using total-cost-based markups. Coefficients on levels of (log) Domar weight and (log) markup are not reported for brevity in columns (1), (2) and (3). Standard errors (in parentheses) are clustered at the firm level for the panel regressions in columns (1), (2) and (3), and are clustered at the NACE 2-digit industry level for the cross-sectional regressions in columns (4), (5) and (6). *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

where $B_{st} = (\sigma_s - 1)(p_{st} - \sum_{s'} \hat{\alpha}_{ss'} p_{s't})$ is common to all firms in sector s at time t . Since $\log(P_{sit}Q_{sit}) = \log(P_{st}Q_{st}) + \log \chi_{sit}$, log sales satisfies

$$\log(P_{sit}Q_{sit}) = \log \theta_{si} + (\sigma_s - 1)a_{sit} - (\sigma_s - 1)\log(1 + \tau_{si}) + \log(P_{st}Q_{st}) + B_{st}.$$

Differencing under the maintained assumption that taste shifters θ_{si} and wedges τ_{si} are time-invariant yields

$$\Delta \log(P_{sit}Q_{sit}) = (\sigma_s - 1)\Delta a_{sit} + \underbrace{\Delta \log(P_{st}Q_{st}) + \Delta B_{st}}_{\text{Sector-year common terms}},$$

where the sector-year term is absorbed by our fixed effects. Thus, when heterogeneity in firms' exposure to aggregate shocks operates through TFP, sales growth should display the same qualitative comparative statics as TFP growth.

We re-run both reduced-form exercises using sales growth directly. In the panel specification, we replace firm-level TFP growth in (57) with $\Delta \log(P_{sit}Q_{sit})$ as the dependent variable. In the cross-sectional specification, we replace the volatility of firm-level TFP growth in (56) with the

volatility of sector-year-residualized sales growth.

The results are reported in Table 5. Columns (1) and (3) reproduce the baseline TFP-growth exercises, while columns (2) and (4) use sales growth instead. We find strong evidence of the same patterns in sales growth: smaller firms and firms with higher markups have more volatile sales growth and a stronger sales-growth response to aggregate GDP growth, mirroring the TFP results.

Table 5: Results with sales growth as dependent variable

Dependent variable	Firm-level growth		Vol. of firm-level growth	
	(1)	(2)	(3)	(4)
Outcome	TFP growth	Sales growth	TFP growth	Sales growth
$\Delta \log GDP_t$	−0.111***	−0.163***		
$\times \log(\text{Domar Weight}_{it-1})$	(0.005)	(0.005)		
$\Delta \log GDP_t$	0.313***	0.039***		
$\times \log(\text{Markup}_{it-1})$	(0.011)	(0.011)		
$\log(\overline{\text{Domar Weight}}_i)$			−0.043***	−0.026***
			(0.003)	(0.002)
$\log(\overline{\text{Markup}}_i)$			0.077***	0.067***
			(0.013)	(0.009)
Fixed effects	Yes	Yes	Yes	Yes
Observations	7,488,988	7,412,451	304,244	300,302
R^2	0.040	0.075	0.303	0.300

Notes: Table presents the panel and cross-sectional robustness exercises using sales growth directly. Columns (1) and (3) use firm-level TFP growth and its sector-year-residualized volatility, while columns (2) and (4) replace these outcomes with log sales growth and its sector-year-residualized volatility. The panel regressions correspond to (57); the cross-sectional regressions correspond to (56). See text for a detailed description of variable construction and sample selection. Standard errors (in parentheses) are clustered at the firm level for the panel regressions in columns (1) and (2) and are clustered at the NACE 2-digit industry level for the cross-sectional regressions in columns (3) and (4). *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.2.5 Robustness to the elasticity of substitution

To construct firm-level TFP growth, we use the CES demand system to map revenue growth into quantity growth. Our baseline sets the sector-level elasticity of substitution to $\sigma_s = 3$. We now vary this parameter and recompute the dependent variables used in both the panel and cross-sectional regressions. Our robustness grid starts from $\sigma_s = 1.2$, a low short-run value estimated in a production-network setting by Carvalho et al. (2021), and extends to values between $\sigma_s = 2$ and $\sigma_s = 8$, which span the range typically estimated or calibrated in the trade and firm-dynamics literatures (e.g. Broda and Weinstein, 2006 and Hottman, Redding, and Weinstein, 2016). Notably, the 5th and 95th percentiles of the firm-level elasticity-of-substitution estimates across the 98 product groups in Hottman, Redding, and Weinstein (2016) are 2.3 and 8.5, respectively, which closely corresponds

to the range we examine here.

Table 6 reports the estimation results of the cross-sectional volatility regressions, where the volatility of firm-level TFP growth is recomputed under each value of the elasticity of substitution. Table 7 reports the panel estimates, where GDP growth is interacted with lagged sales and labor-based markups, and NACE 2-digit sector-year fixed effects are absorbed. Across these alternatives, the signs and statistical significance are stable: smaller firms and firms with higher markups exhibit higher TFP volatility and greater sensitivity of firm-level TFP growth to aggregate GDP growth. The magnitudes of the coefficients decrease as σ_s increases. As goods become more substitutable (higher σ_s), firm sales respond more strongly to changes in TFP. Consequently, firm TFP growth identified from (54) becomes less volatile.

Table 6: Robustness across elasticities of substitution: volatility of firm-level TFP growth

	(1) $\sigma_s = 1.2$	(2) $\sigma_s = 2$	(3) $\sigma_s = 3$	(4) $\sigma_s = 4$	(5) $\sigma_s = 6$	(6) $\sigma_s = 8$
$\log(\overline{\text{Domar Weight}}_i)$	−0.145*** (0.013)	−0.050*** (0.003)	−0.043*** (0.003)	−0.042*** (0.003)	−0.041*** (0.003)	−0.041*** (0.003)
$\log(\overline{\text{Markup}}_i)$	0.449*** (0.058)	0.117*** (0.017)	0.077*** (0.013)	0.064*** (0.012)	0.054*** (0.012)	0.050*** (0.011)
NACE 2-digit FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	304,244	304,244	304,244	304,244	304,244	304,244
R^2	0.242	0.276	0.303	0.314	0.320	0.322

Notes: Dependent variable is the standard deviation of firm-level TFP growth under the value of the elasticity of substitution indicated by the parameter row. Each column absorbs NACE 2-digit fixed effects and clusters standard errors at the NACE 2-digit level. Markups are labor-based. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 7: Robustness across elasticities of substitution: sensitivity of firm-level TFP to GDP

	(1) $\sigma_s = 1.2$	(2) $\sigma_s = 2$	(3) $\sigma_s = 3$	(4) $\sigma_s = 4$	(5) $\sigma_s = 6$	(6) $\sigma_s = 8$
$\log(\overline{\text{Domar Weight}}_{it-1}) \times \Delta \log GDP_t$	−1.251*** (0.030)	−0.237*** (0.007)	−0.111*** (0.005)	−0.067*** (0.004)	−0.030*** (0.004)	−0.014*** (0.004)
$\log(\overline{\text{Markup}}_{it-1}) \times \Delta \log GDP_t$	0.992*** (0.064)	0.382*** (0.016)	0.313*** (0.011)	0.290*** (0.010)	0.270*** (0.009)	0.261*** (0.009)
Sector-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,488,988	7,488,988	7,488,988	7,488,988	7,488,988	7,488,988
R^2	0.077	0.059	0.040	0.030	0.021	0.018

Notes: Table presents estimates of (57) using alternative elasticities of substitution to construct firm-level TFP growth. All columns use labor-based markups; the parameter row gives the elasticity of substitution used in the TFP measure. GDP growth is omitted because it is collinear with absorbed NACE 2-digit sector-year fixed effects. Standard errors (in parentheses) are clustered at the firm level. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.2.6 Robustness with age controls and age interactions

Fort et al. (2013) emphasize that firm age can account for part of the relationship between a firm's size and its sensitivity to the business cycle. Since firm age is not explicitly modeled in our theory, we verify that our reduced-form panel results are robust to controlling for firm age and allowing the cyclical of TFP growth to vary with age.

We measure firm age as the current year minus the incorporation year reported in Orbis. We group firms into five age bins: 0–5, 6–10, 11–15, 16–25, and more than 25 years. We then estimate the following age-augmented version of the binned panel specification:

$$\begin{aligned}
\Delta \log TFP_{it} = & \sum_{j=1}^5 \beta_j^{Domar} \left(D_{ji,t-1}^{Domar} \times \Delta \log GDP_t \right) + \sum_{j=1}^5 \beta_j^{Markup} \left(D_{ji,t-1}^{Markup} \times \Delta \log GDP_t \right) \\
& + \sum_a \beta_a^{Age} \left(D_{ai,t}^{Age} \times \Delta \log GDP_t \right) + \sum_{j=1}^5 \alpha_j^{Domar} D_{ji,t-1}^{Domar} + \sum_{j=1}^5 \alpha_j^{Markup} D_{ji,t-1}^{Markup} \\
& + \sum_a \alpha_a^{Age} D_{ai,t}^{Age} + \theta_{st} + \varepsilon_{it}.
\end{aligned} \tag{58}$$

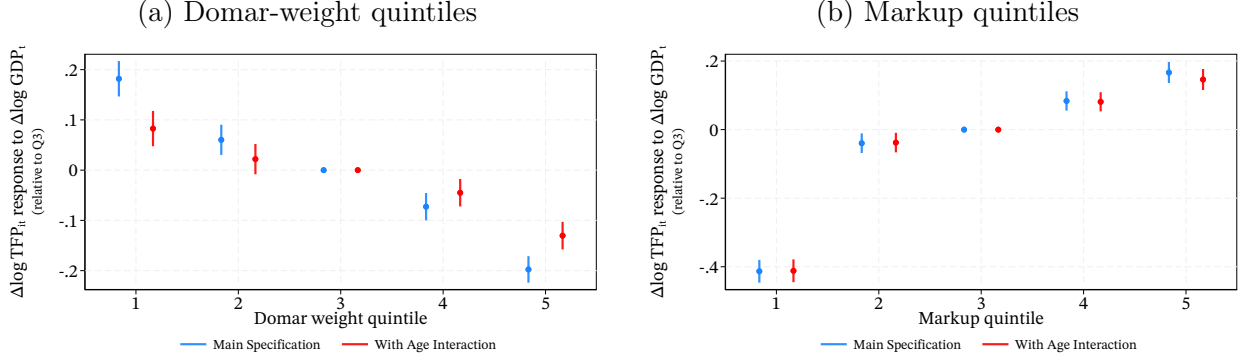
The Domar-weight and markup indicators are quintiles defined within sector-year cells, and the age indicators are the age bins described above; the 11–15 year age bin is the omitted category. Sector-year fixed effects absorb the main effect of aggregate GDP growth and common sector-year shocks, and standard errors are clustered at the firm level.

Figure 8 compares the Domar-weight and markup coefficient profiles from the age-augmented specification with the baseline binned specification. The profiles of Domar weights and markups are largely unchanged, while the effect of firm size is somewhat dampened, potentially because of its correlation with firm age. Since our mechanism affects aggregate volatility through firms' Domar-weighted contributions to aggregate output, we also repeat this exercise weighting observations by current Domar weights. In this weighted specification, the Domar-weight and markup profiles are statistically indistinguishable from their corresponding weighted counterparts without age controls (see Figure 9). This indicates that, when it comes to the importance of our mechanism for aggregate fluctuations, the role of age is likely limited.

A.2.7 Robustness on the scaling of TFP volatility

We now show that our cross-sectional findings – that smaller firms and firms with higher markups exhibit greater TFP volatility – are robust to an alternative scaling of the dependent variable. Specif-

Figure 8: Robustness with age controls: Domar-weight and markup quintiles



Notes: The figure plots the estimated coefficients on lagged Domar-weight quintiles and lagged markup quintiles interacted with aggregate GDP growth. The age-augmented specification controls for firm age-bin indicators and their interactions with aggregate GDP growth. The baseline specification is shown for comparison. Standard errors are clustered at the firm level.

ically, we re-estimate (56) using $\log(\sigma_i^{TFP})$ in place of σ_i^{TFP} as the dependent variable. Column (2) of Table 8 presents the estimates under this alternative scaling and column (1) reproduces our baseline results with σ_i^{TFP} as the dependent variable. Results under this alternative scaling remain statistically significant and qualitatively similar to the baseline.

Table 8: Cross-sectional results with alternative scaling of TFP volatility

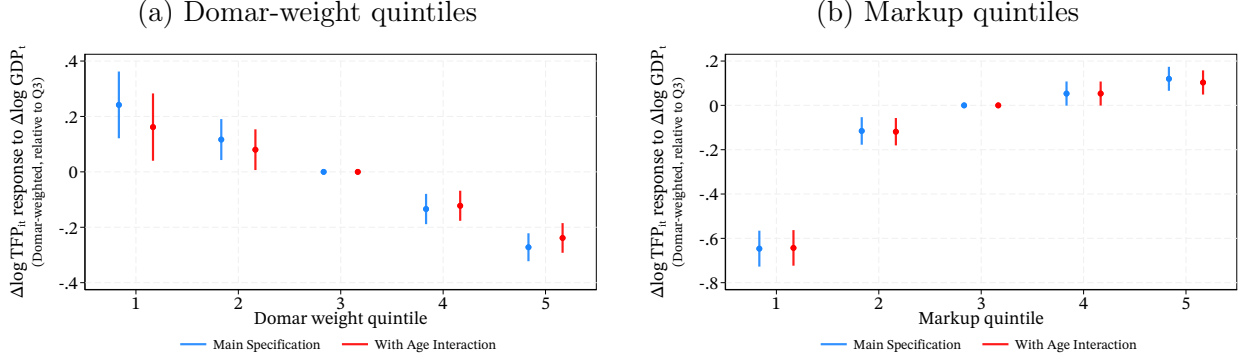
	(1): σ_i^{TFP}	(2): $\log(\sigma_i^{TFP})$
$\log(\overline{\text{Domar Weight}_i})$	-0.043*** (0.003)	-0.146*** (0.007)
$\log(\overline{\text{Markup}_i})$	0.077*** (0.013)	0.248*** (0.025)
Sector FE	Yes	Yes
Observations	304,244	304,244
R^2	0.303	0.323

Notes: Table presents estimation results of (56) with different scaling of TFP volatility using a sample of Spanish firms from Orbis. See text for a detailed description of variable construction and sample selection. The estimation sample is winsorized at the top and bottom 0.5% of observations of TFP growth, Domar weight and markup in the panel in each sector before computing the firm-level moments. Standard errors (in parentheses) are clustered at the NACE 2-digit industry level. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.3 US stock market evidence

Our theory predicts that, all else equal, larger firms (as measured by Domar weights) and those with lower markups tend to exhibit weaker comovement with aggregate shocks. One common way to investigate this covariance is to use stock market equity betas that can be estimated using high-frequency financial data. As is well known in the finance literature (e.g., Fama and French, 1992),

Figure 9: Domar-weighted regression with age controls: Domar-weight and markup quintiles



Notes: The figure plots the estimated coefficients on lagged Domar-weight quintiles and lagged markup quintiles interacted with aggregate GDP growth. The age-augmented specification controls for firm age-bin indicators and their interactions with aggregate GDP growth. The baseline specification is shown for comparison. Both specifications use Domar weights as observation weights. Standard errors are clustered at the firm level.

firms' equity betas are strongly negatively correlated with their stock market capitalizations. Since firms with high market capitalizations tend to have high Domar weights, we expect to find a similar relationship between betas and Domar weights, as our theory predicts. We also explore how equity betas vary with firms' wedges.

We use a sample of US public firms from the CRSP/Compustat Merged database (see Appendix A.1.2 for a detailed description of the data). For each firm, we estimate its 5-year rolling-window CAPM beta. As in the previous section, we use markups to proxy for wedges, where markups are constructed as in De Loecker, Eeckhout, and Unger (2020). To explore how firms' equity betas correlate with their Domar weights and markups, we use the same approach as in the previous section. Specifically, for each year we construct firms' lagged within-sector markup deciles and lagged NYSE-based Domar-weight deciles.⁴⁸ We then construct a set of dummy variables, D_{jit}^{Domar} and D_{jit}^{Markup} , such that $D_{jit}^{Domar} = 1$ if firm i 's Domar weight is in decile j in year t , and analogously for markups. We then run the following regression with sector and year fixed effects,

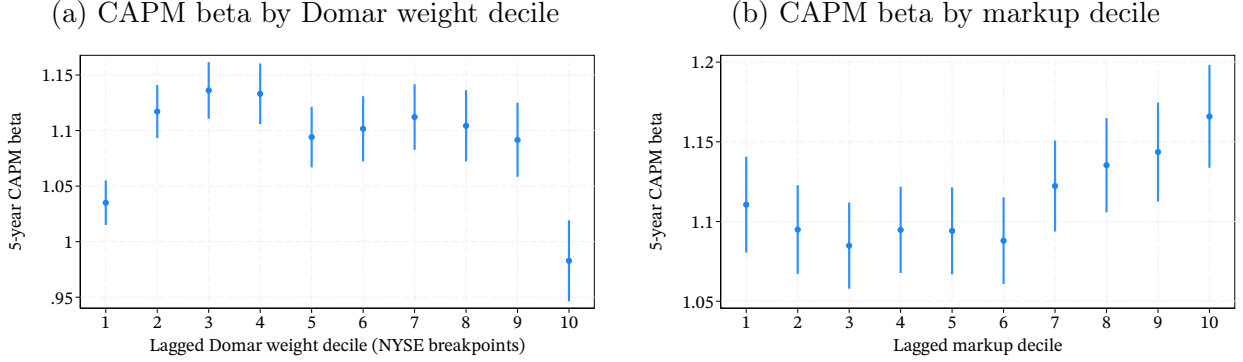
$$\beta_{it}^{CAPM} = b_0 + \sum_{j=1}^{10} b_j^{Domar} D_{jit-1}^{Domar} + \sum_{j=1}^{10} b_j^{Markup} D_{jit-1}^{Markup} + \theta_s + \theta_t + \varepsilon_{it}, \quad (59)$$

and plot the estimated regression coefficients b_j^{Domar} and b_j^{Markup} in Figure 10. Panel (a) shows that larger firms tend to have lower CAPM betas. That relationship holds almost monotonically except for firms in the first decile. For such small firms, the estimated betas can be noisy or even biased downwards due to liquidity issues (Ibbotson, Kaplan, and Peterson, 1997). Panel (b) shows

⁴⁸Following Fama and French (1992), we use NYSE breakpoints when defining the Domar weight deciles. The size distribution of firms in the other large exchange, Nasdaq, has changed substantially over time (historically, Nasdaq was dominated by small tech companies). In contrast, the size distribution of the NYSE firms is more stable.

that firms with higher markups tend to have higher CAPM betas. Both these results are in line with our theory as long as increases in productivity growth and stock market returns are positively correlated.

Figure 10: CAPM betas, Domar weights and markups



Notes. Estimation results of (59) using a sample of US firms from Compustat with the normalization $b_5^{Domar} = b_5^{Markup} = 0$ to avoid perfect multicollinearity. See text for a detailed description of variable construction and sample selection. The estimation sample is trimmed at the top and bottom 3% of observations of equity beta, lagged Domar weight, and lagged markup. Panel (a): $b_j^{Domar} + b_0$ by Domar weight decile j ($j = 1$ is lowest Domar weight, $j = 10$ is highest Domar weight). Panel (b): $b_j^{Markup} + b_0$ by markup decile ($j = 1$ is lowest markup, $j = 10$ is highest markup). 90% confidence intervals are constructed using standard errors that are clustered at the firm level.

A.4 Cross-country evidence

In this appendix, we show that countries with higher GDP per capita and smaller distortions tend to have lower TFP growth volatility. We use two measures of country-level distortions: 1) the share of government consumption in GDP (as a proxy for tax burden) and 2) country-level aggregate markups. We combine data from the Penn World Table (version 10.01) with the country-level markup estimates from De Loecker and Eeckhout (2018). For each country, we compute the standard deviation of TFP growth σ_{it}^{TFP} using a rolling window from year $t - 10$ through year t , requiring at least ten non-missing annual observations. We use TFP at constant national prices. The cross-country median of σ_{it}^{TFP} in the Penn World Table is 2.69%. Following Acemoglu and Zilibotti (1997), we then run the following regression:

$$\log \sigma_{it}^{TFP} = \alpha + \beta^{GDP} \log(\text{GDP}_{pc,it}) + \beta^\tau \log \tau_{it} + FE_i + FE_t + \varepsilon_{it}, \quad (60)$$

where $\log(\text{GDP}_{pc,it})$ is log of GDP per capita of country i in year t (we use expenditure-side real GDP divided by total population), and $\tau_{it} \in \left\{ \frac{\text{Gov Cons}}{\text{GDP}}_{it}, \text{Markup}_{it} \right\}$ represents the two measures of distortions. The variables FE_i and FE_t are country and year fixed effects. The estimated coefficients are given in Table 9, and the associated binscatters are shown in Figure 11.

We find that countries with larger government consumption shares (heavier tax burdens) and

higher aggregate markups tend to have higher TFP volatility, while richer economies tend to experience lower TFP volatility. Our model can rationalize these findings. Insofar as some of those distortions affect the wedge between the price at which goods are sold and their production cost, our theory would suggest that firms in these countries are less aggressive with their risk management, and aggregate volatility should be higher as a result. Similarly, richer countries might have better risk management capabilities (lower B_i , G_i , or η), which would explain the lower aggregate volatility.

Table 9: Cross-country TFP volatility, GDP per capita and distortions

	Dependent variable: Country-specific TFP volatility				
	(1)	(2)	(3)	(4)	(5)
$\log(\text{GDP}_{pc,it})$	-0.170** (0.075)			-0.178** (0.072)	-0.164 (0.200)
$\log(\frac{\text{Gov Cons}}{\text{GDP}}_{it})$		0.114* (0.062)		0.126** (0.057)	
$\log(\text{Markup}_{it})$			0.287* (0.151)		0.253* (0.143)
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,227	5,227	1,382	5,227	1,382
R^2	0.674	0.671	0.733	0.677	0.736

Notes: Results of estimating (60) using a cross-country sample from Penn World Tables. See text for a detailed description of variable construction and sample selection. Standard errors (in parentheses) are clustered at the country level. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

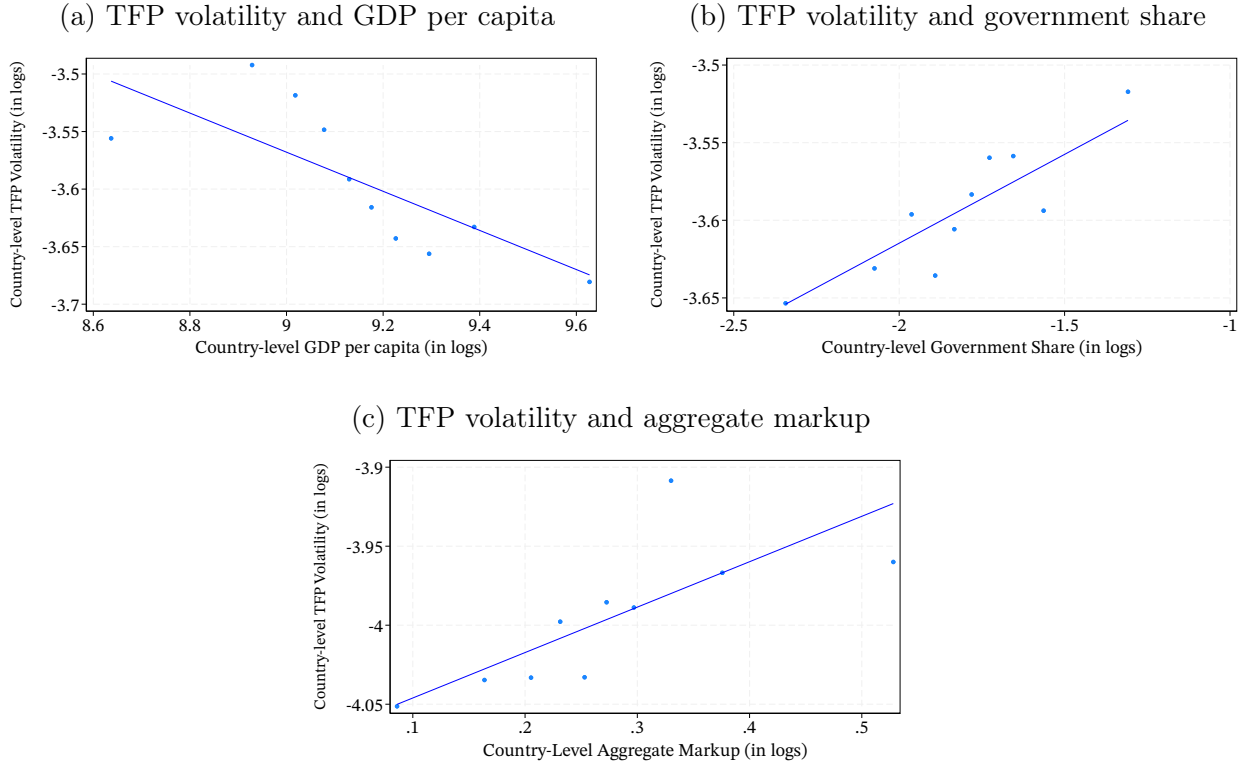
A.5 Evidence on the returns-to-scale of risk management inputs

As discussed in Section 4, the resource cost g_i gives larger firms a scale advantage in risk management. This implies that the share of sales a firm spends on risk management, $W_R g_i(\delta_i) / (P_i Q_i)$, falls with firm size for large firms. This prediction hinges on the cost matrix G_i being size-independent. If instead the risk-management technology scaled with size (say, $G_i = \hat{G}_i \omega_i$, where ω_i is the firm's Domar weight), the resource cost would grow in proportion to sales, eliminating the scale advantage, and the spending share would be *independent* of firm size.⁴⁹

In this section, we present two empirical exercises showing that the risk-spending share does vary with firm size, consistent with the baseline model and at odds with this size-scaling alternative. First,

⁴⁹Substituting (24) and $H_i = B_i + \eta \frac{1+\tau_i}{\omega_i} G_i$ into the share, and using $W_R = \eta \bar{P} Y$ and $P_i Q_i = \omega_i \bar{P} Y$, gives $W_R g_i(\delta_i) / (P_i Q_i) = \frac{\eta}{2\omega_i} \mathcal{E}^\top H_i^{-1} G_i H_i^{-1} \mathcal{E}$. With G_i fixed, $H_i \rightarrow B_i$ as ω_i grows, so the share decreases in ω_i for large firms. With $G_i = \hat{G}_i \omega_i$, $H_i = B_i + \eta(1 + \tau_i) \hat{G}_i$ no longer depends on ω_i , so the share is independent of firm size.

Figure 11: Binscatter plots of cross-country TFP volatility, GDP per capita and government share



Notes: Binned scatter plots of the log of cross-country TFP volatility against the log of GDP per capita (panel a), the log of government share (panel b), the log of country-level aggregate markup (panel c), using a cross-country sample from the Penn World Tables. See text for a detailed description of variable construction and sample selection. Country and year fixed effects are controlled for in all binscatter plots.

we show that management expenditures, which include risk-management costs, fall as a fraction of sales as firms grow. Second, we show that larger firms employ relatively fewer workers in occupations related to risk management.

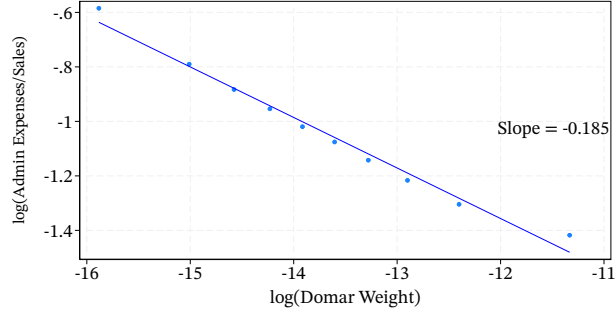
A.5.1 Management expenditures

From an accounting perspective, risk-management expenditures are not available in standard firm datasets, as their disclosure is not a GAAP requirement. Some risk-management expenditure would show up, however, in a firm’s “general administrative costs”.⁵⁰ That category includes expenses related to running a risk management department, the salaries of managers responsible for mitigating risk, etc. Figure 12 reports how those costs vary with firm size in the Spanish Orbis data. We clearly see from the figure that as a firm’s sales (revenue-based Domar weight) increase, the ratio of its administrative expenses to sales is not constant, contrary to what a model with no scale

⁵⁰What we refer to as “general administrative costs” is “other operating expenses” in Orbis. They use that terminology to harmonize different country-level datasets. In Compustat, the equivalent variable is “Selling, General and Administrative expenses (SG&A)”.

advantage in risk management would imply. Instead, the ratio declines as sales increase. While that measure is imperfect, it suggests some form of economies of scale in administrative expenses and in the risk-management expenditures that they include. Those economies of scale are provided by the resource cost function g_i in the model.⁵¹

Figure 12: Administrative expenditure share and firm size



Notes: Binned scatter plots of the log of administrative expenditure share against the log of Domar weight using a sample of Spanish firms from Orbis. The administrative expenditure share is defined as the ratio between “other operating expenses” and sales. The estimation sample is winsorized at the top and bottom 0.5% of the distributions of both variables and includes NACE 2-digit sector fixed effects. See text for a detailed description of variable construction and sample selection.

A.5.2 Risk-related occupations

The types of workers that firms hire also provide evidence for a size advantage in risk management. To show this, we rely on the RAIS dataset from Brazil. RAIS is a comprehensive administrative matched employer–employee dataset from the Brazilian Ministry of Labor that provides a high-quality census of the Brazilian formal labor market. It includes on average about 40 million workers per year, close to all formal employment in Brazil. Of particular interest to us, it features a highly granular occupational classification that allows us to identify workers who perform tasks directly related to risk-management activities. We can then examine how the employment and wage shares of these workers vary systematically with firm size.

We use RAIS data from 2002–2008 and harmonize occupations to the ISCO-88 (International Standard Classification of Occupations-1988) four-digit level using the CBO-ISCO crosswalk created by Muendler et al. (2004).⁵² ISCO-88 defines 390 four-digit occupations, each with a detailed description of the tasks associated with that occupation. We then use a large language model

⁵¹In the data, the slope coefficient is -0.19 and strongly statistically significant. We find a similar coefficient in the analogous regression for large firms in the calibrated model of Section 9.

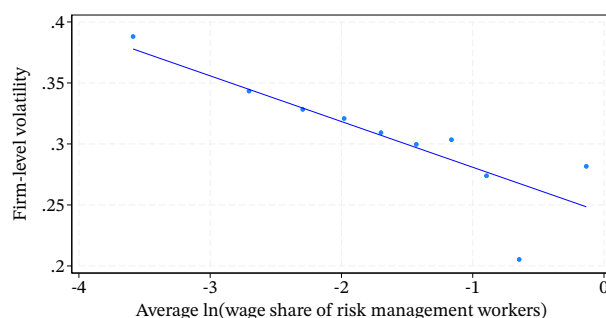
⁵²We have access to the RAIS data for 2002–2018, but we only use 2002–2008 to ensure consistent occupational mapping via the CBO-94-to-ISCO-88 crosswalk in Muendler et al. (2004). We follow the cleaning procedure of Dahis (2024) for the RAIS data.

(OpenAI o3) to identify occupations directly involved in managing risks in firms’ operations or production processes.⁵³ This procedure identifies 64 risk-related occupations. Examples include *Safety, Health & Quality Inspectors* and *Building and Fire Inspectors*, which are classified based on task descriptions like “oversee work-safety procedures,” “identify fire hazards,” and “ensure compliance with specifications and regulations.”

For each establishment, we compute the employment and wage-bill shares of these risk-management occupations. We restrict the analysis to establishments reporting at least one risk-management worker. This is to address the fact that many small firms do not employ dedicated risk-management staff and instead assign such tasks to employees in other roles, often combining multiple responsibilities.⁵⁴ With this restriction, our sample covers more than 75% of the total wage bill of formal employees for an average year.

To validate our classification of risk-related occupations, we show in Figure 13 that, even conditional on size, firms that hire a higher share of risk-management workers experience lower volatility of their total wage bills (a proxy for revenues, which is not in our dataset). Although this is only correlational evidence, it supports the idea that our measure captures the intensity of a firm’s risk-management efforts.

Figure 13: Firm-level volatility and risk-management wage share



Notes: Binned scatter plots of firm-level volatility of log total wage bill against log wage-bill share of risk-management occupations, controlling for firm size (proxied by the log total wage bill). The sample includes all firms in the 2002–2008 Brazilian matched employer-employee data (RAIS) with at least five employees on average and at least one risk-management worker. See text for a detailed description of variable construction and sample selection.

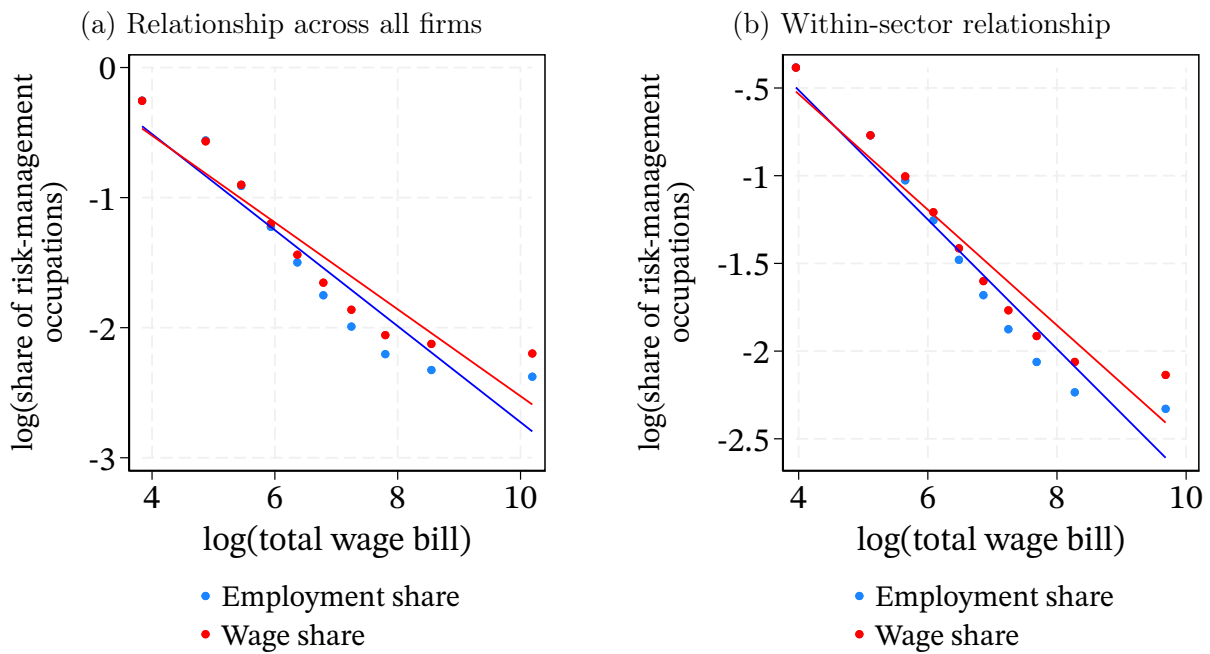
Figure 14 presents our core results. It shows binned scatter plots of the employment and wage-

⁵³We use the following prompt in OpenAI o3 after importing the official ISCO-88 structure and definitions: – *Given the following occupation definitions in the International Classification of Occupations in this Excel table, which occupations are directly involved in managing risks in a firm’s operations or production processes? Please provide the list of “Occupation Codes” and “Occupation Titles”, along with your rationale for this classification.* We tried alternative, semantically similar prompts; the classification results remained largely unchanged. We also experimented with an *intensive* measure of how risk related an occupation is, with similar results.

⁵⁴We find that the likelihood of employing a worker in a risk-management occupation increases with firm size, consistent with the presence of fixed costs and scale advantages in risk management among larger firms.

bill shares of risk-management occupations against firm size (measured as total wage bill). Panel (a) shows the raw relationship across all firms; Panel (b) includes 2-digit sector fixed effects, exploiting within-sector variation only. In both panels, we find strong evidence that the wage and employment shares of risk-management workers vary with firm size. This finding, combined with evidence from Figures 6 and 7, suggests that large firms achieve better risk outcomes while dedicating a smaller fraction of their workforce to risk reduction. This is consistent with the scale-advantage in risk-management provided by the resource cost g_i in the model.

Figure 14: Binscatter plots of employment and wage share of risk-management occupations and firm size



Notes: Binned scatter plots of the log employment share and log wage-bill share of risk-management occupations against the log total wage bill, using a sample of the 2002–2008 Brazilian matched employer-employee data (RAIS). Panel (a) uses the raw data, while Panel (b) uses the data residualized with CNAE 2-digit sector fixed effects. See text for details on variable construction and sample selection.

A.6 Empirical evidence with climate risk

In this appendix, we show that our model has broader quantitative applications to risk factors beyond business-cycle fluctuations. Specifically, we demonstrate that the model's predictions continue to hold when applied to climate risk.

We consider the implications of our endogenous risk model for climate change. It is well known that climate shocks (mostly heat shocks) can cause significant productivity damage at the firm level

(Zhang et al., 2018) and at the macro level (Dell, Jones, and Olken, 2012). In the context of our model, we therefore think of climate risk as a bad risk ($\mathcal{E} < 0$) and consider different adaptation actions that firms can take to lower their climate risk exposure (e.g. install air-conditioning units, or take measures against floods and storms). Corollary 1 implies that smaller firms and firms with higher markups should be more exposed to climate change since those firms find it less cost-effective to aggressively manage their climate risk exposure.

We test whether the model’s key predictions are supported by the data. To do so, we merge our firm panel with detailed weather data for Spain and firm-location data, compiled by Liu and Xu (2024) (see that paper for more details about the data). The weather data are derived from ERA5 hourly data and provide daily temperatures for each of Spain’s 50 NUTS-3 regions (province equivalents) from 1996 to 2018. The firm-location data were constructed from the postcode information in the Orbis dataset, using Eurostat’s postcode–NUTS-3 crosswalk. Using the merged dataset, we can therefore explore how the link between firm-level productivity and temperature varies with firm characteristics.

We focus on a simple exercise to test the model prediction that smaller firms and firms with larger markups are more exposed to climate-related temperature shocks. The productivity of these firms should decline more in years with particularly warm weather. To test that prediction, we first define the temperature bin variables, $B = \{\text{Tbin}_{r,t}^{<15^\circ\text{C}}, \text{Tbin}_{r,t}^{15-25^\circ\text{C}}, \text{Tbin}_{r,t}^{>25^\circ\text{C}}\}$, where $\text{Tbin}_{r,t}^b$ is the number of days in geographical region r and year t with a mean daily temperature that falls in the range b . The cold and hot bins are chosen to be close to the official Eurostat thresholds used for heating and cooling degree days ($\leq 15^\circ\text{C}$ and $\geq 24^\circ\text{C}$, respectively). We then run the following regression:

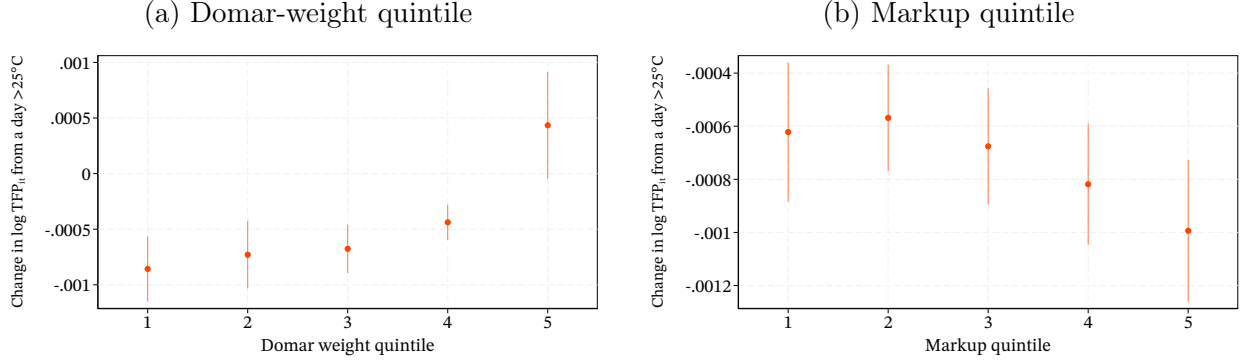
$$\begin{aligned} \log(TFP_{i,t}) = & \sum_{b \in B} \delta_b \times \text{Tbin}_{r(i),t}^b + \sum_{j=1}^5 \alpha_j^{Domar} \times D_{ji,t-1}^{Domar} + \sum_{j=1}^5 \alpha_j^{Markup} \times D_{ji,t-1}^{Markup} \\ & + \sum_{b \in B} \sum_{j=1}^5 \delta_{b,j}^{Domar} \times \text{Tbin}_{r(i),t}^b \times D_{ji,t-1}^{Domar} + \sum_{b \in B} \sum_{j=1}^5 \delta_{b,j}^{Markup} \times \text{Tbin}_{r(i),t}^b \times D_{ji,t-1}^{Markup} \\ & + \delta_i + \alpha_{s(i),t} + \varepsilon_{i,t}, \end{aligned} \tag{61}$$

where $r(i)$ is the NUTS-3 region in which firm i resides. As in Section 8, $D_{ji,t}^{Domar}$ is a dummy variable such that $D_{ji,t}^{Domar} = 1$ if firm i ’s Domar weight is in within-sector quintile j in year t , and analogously for markups. The regression uses their one-year lags.⁵⁵ Conditional on firm and sector-year fixed effects, $\text{Tbin}_{r(i),t}^{>25^\circ\text{C}}$ can be thought of as a measure of region-level heat shock. The coefficients of interest are thus $\delta_{>25^\circ\text{C},j}^{Domar}$ and $\delta_{>25^\circ\text{C},j}^{Markup}$. The first one captures the differential impact on productivity of an extra day of high temperature on firms in Domar weight quintile j , compared

⁵⁵We choose 15–25°C as the reference temperature bin.

to those in the third quintile. The coefficient $\delta_{>25^\circ\text{C},j}^{Markup}$ provides similar information but for markups.

Figure 15: Impact of high temperature on productivity by Domar weights and markups



Notes: Estimation results of (61) using a sample of Spanish firms from Orbis, with the normalization $\delta_{b,3}^{Domar} = \delta_{b,3}^{Markup} = 0$ for all $b \in B \setminus \{15-25^\circ\text{C}\}$, to avoid perfect multicollinearity. The sample contains firms that have valid postcodes residing in Spain (excluding the Canary Islands) from 1996 to 2018. See text for a detailed description of variable construction. The estimation sample is winsorized at the top and bottom 0.5% of observations of TFP, Domar weight and markup in each sector. Panel (a): $\delta_{>25^\circ\text{C}} + \delta_{>25^\circ\text{C},j}^{Domar}$ by Domar weight quintile j ($j = 1$ is lowest Domar weight, $j = 5$ is highest Domar weight). Panel (b): $\delta_{>25^\circ\text{C}} + \delta_{>25^\circ\text{C},j}^{Markup}$ by markup quintile ($j = 1$ is lowest markup, $j = 5$ is highest markup). 90% confidence intervals are constructed using standard errors that are clustered at the NUTS-3 region level.

The estimates are plotted in Figure 15. As we can see, the productivity of firms with higher Domar weights and lower markups responds less to high temperatures. This is consistent with our model. Indeed, in the model firms find it more beneficial to take measures to counteract the adverse effects of climate change on their operations. As a result, hot temperatures are less detrimental to their productivity.

We can also gauge the quantitative importance of our mechanisms using future climate-change projections. Under the SSP3-7.0 warming scenario, the average European NUTS-3 region is projected to experience about 8 additional days per year with temperatures above 25°C by 2100 relative to 1951-1980. Our estimates indicate that, due to climate change, firms in the bottom Domar-weight quintile will suffer 1.03 percentage points higher productivity loss than those in the top quintile, while firms in the top markup quintile will face 0.30 percentage points more damage than those in the bottom quintile. These results suggest that our mechanism could play a quantitatively important role when assessing the heterogeneous costs of climate change in the cross-section.

B Appendix for Section 9

This appendix contains details about the calibration of Section 9.

B.1 Construction of the sample

This appendix describes the datasets used in the calibration and how the associated aggregate and firm-level moments are computed.

1. We measure real GDP using the chain-linked volume index from the Annual Spanish National Accounts (Gross domestic product at market prices and its components, Table 4). To be consistent with the Orbis data, we use the mean and variance of GDP growth, $y_t - y_{t-1}$, during the 1995–2019 period.
2. We calibrate the sectoral parameters using the 2010 input-output table from the Annual Spanish National Accounts. This table partitions the Spanish economy into 62 sectors, most of which correspond to 2-digit NACE industries, while some combine closely related 2-digit industries.⁵⁶ Conforming to the accounting conventions in the data, we calibrate the input elasticities of good s' in the production of sector s as

$$\hat{\alpha}_{ss'} = \frac{\text{Input from } s' \text{ at basic prices}_s}{\text{Total input at basic prices}_s} \times \frac{\text{Intermediate consumption at purchaser's prices}_s}{\text{Output at basic prices}_s},$$

such that the residual labor share, $1 - \sum_{s'} \hat{\alpha}_{ss'}$, corresponds to the value-added share of output at basic price in the data.⁵⁷ We calibrate the consumption share β_s to be the share of final consumption expenditure of good s in the sum of consumption expenditure spent on the 62 sectors.

3. We compute firm-level objects from the Orbis sample. After steps 1-4 in Section A.1.1, we perform a few additional steps.
 - (a) We estimate the production-function output elasticities using the cost-share approach at the NACE 2-digit level and then compute firm-level markups, as described in Appendix A.1.1 for the reduced-form exercises. For the aggregate price and quantity terms below, we map firms separately to the input-output sectors used in the Spanish aggregate data.
 - (b) We winsorize nominal sales growth and markup at the top and bottom 0.5% of the pooled firm-year distribution. We then compute model-implied firm-level TFP growth,

$$\begin{aligned} \Delta \log TFP_{it} = & \left(\frac{\sigma_s}{\sigma_s - 1} \Delta \log (P_{it} Q_{it}) - \hat{\alpha}_{s(i)}^L \Delta \log L_{it} - \hat{\alpha}_{s(i)}^K \Delta \log K_{it} - \hat{\alpha}_{s(i)}^M \Delta \log M_{it} \right) \\ & - \frac{\sigma_s}{\sigma_s - 1} \Delta \log P_{st} - \frac{1}{\sigma_s - 1} \Delta \log Q_{st}, \end{aligned} \quad (62)$$

⁵⁶Sector 63 (household-related production activities) and sector 64 (services by extraterritorial organizations and bodies) are also present in the 2010 input-output table, but their input-output data are missing.

⁵⁷Because sector-to-sector data at purchasers' prices (i.e., adjusted for taxes less subsidies on products) are unavailable, we calibrate the intermediate-input expenditure share of inputs from sector s' used by sector s with the share computed in basic prices, $\frac{\text{Input from } s' \text{ at basic prices}_s}{\text{Total input at basic prices}_s}$.

where $P_{it}Q_{it}$ is observed nominal sales. The variables L_{it} , K_{it} , and M_{it} are constructed by deflating the wage bill, fixed assets, and material costs under the assumption that firms within a sector face common input prices. Specifically, we use the GDP deflator for the wage bill and EU-KLEMS sectoral price indices for fixed assets and materials, at the most detailed NACE 2-digit level available. We construct the sectoral price and quantity terms from Spanish INE sectoral accounts. We map firms to the 62 input-output sector classification in the INE tables. For each sector s and year t , we compute value-added price growth as the difference between nominal value-added growth and real value-added growth: $\Delta \log P_{st} = \Delta \log (VA_{st}^{nom}/VA_{st}^{real})$. Nominal gross-output growth is $\Delta \log GO_{st}^{nom}$, and gross-output quantity growth is measured as $\Delta \log Q_{st} = \Delta \log GO_{st}^{nom} - \Delta \log P_{st}$.⁵⁸ We set $\sigma_s = 3$ for all sectors, and provide robustness checks for this value in Appendices B.7 and B.9. We also compute each firm's revenue-based Domar weight as the sample average of its annual nominal sales divided by Spain's aggregate nominal GDP.

- (c) We keep only firms with more than five non-missing TFP-growth observations. We then compute the covariance between firm-level TFP growth and aggregate GDP growth, as well as the firm-level variance of TFP growth.

B.2 Identifying δ_{si} and Σ_{si}^v

Combining firm i 's TFP from (49) with log real GDP from (130), we can write

$$\begin{aligned} A_{si} &:= \mathbb{V} [\log TFP_{si,t} - \log TFP_{si,t-1}] = \mathbb{V} [\delta_{si}\varepsilon_t + v_{sit} + \gamma_s t - \delta_{si}\varepsilon_{t-1} - v_{sit-1} - \gamma_s (t-1)] \\ &= \mathbb{V} [\delta_{si}(\varepsilon_t - \varepsilon_{t-1}) + v_{sit} - v_{sit-1}] \\ &= 2\delta_{si}^2 \Sigma + 2\Sigma_{si}^v. \end{aligned} \tag{63}$$

Similarly, for the covariance, we have

$$\begin{aligned} D_{si} &:= \text{Cov} [y_t - y_{t-1}, \log TFP_{si,t} - \log TFP_{si,t-1}] = \text{Cov} \left[\Delta_t \varepsilon_t - \Delta_{t-1} \varepsilon_{t-1} + \varsigma^\top (v_t - v_{t-1}), \right. \\ &\quad \left. \delta_{si}\varepsilon_t - \delta_{si}\varepsilon_{t-1} + v_{sit} - v_{sit-1} \right] \\ &= 2\Delta \Sigma \delta_{si} + 2\varsigma_{si} \Sigma_{si}^v. \end{aligned} \tag{64}$$

⁵⁸As noted by Comin et al. (2025), Spain does not have consistent sectoral coverage for price indices for gross output and intermediate inputs. We therefore follow Comin et al. (2025) and use the value-added price index as gross-output deflator.

In these equations, we define A_{si} and D_{si} to simplify the notation in what follows. Notice also that (63) and (64) are the same as in Section 8, and that we can therefore measure both A_{si} and D_{si} directly from the Spanish data, as explained in that section.

From the observed A_{si} and D_{si} , we can then identify key objects of the calibrated economy. From (63), we can write

$$\Sigma_{si}^v = \frac{A_{si}}{2} - \delta_{si}^2 \Sigma. \quad (65)$$

Combining with (64), we find a quadratic equation in δ_{si} , whose solutions are given by

$$\delta_{si} = \frac{\Delta}{2\varsigma_{si}} \left(1 \pm \sqrt{1 - 2\varsigma_{si} \frac{D_{si} - \varsigma_{si} A_{si}}{\Delta^2 \Sigma}} \right). \quad (66)$$

Since $\Delta = \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} \delta_{si}$, we find

$$2\Delta = \sum_{s=1}^S \sum_{i=1}^{N_s} \left(\Delta \pm \sqrt{\Delta^2 - 2\varsigma_{si} \frac{D_{si} - \varsigma_{si} A_{si}}{\Sigma}} \right),$$

or

$$1 = \frac{1}{N-2} \sum_{s=1}^S \sum_{i=1}^{N_s} \sqrt{1 - 2\varsigma_{si} \frac{D_{si} - \varsigma_{si} A_{si}}{\Sigma \Delta^2}}.$$

Given a normalization for Σ , we can solve this equation numerically for Δ^2 . Since aggregate exposure is positive, this gives us Δ . Combining with (66), we find δ_{si} (we pick the negative root because it corresponds to a positive Δ , in line with our normalization). Finally, combining with (65) gives us Σ_{si}^v .

We use this procedure to pin down the risk exposure δ_{si} and the idiosyncratic volatility Σ_{si}^v of about 99% of the firms in the sample, but there are some firms that are too extreme to fit our setup. Those are firms with covariance D_{si} that are extremely large or extremely small compared to their variance A_{si} .⁵⁹ Those firms are generally small and we suspect that measurement errors might contribute to their extreme properties. To include these firms in the model, we winsorize their D_{si} . We have experimented with other ways to include these firms and have found that they only matter minimally for the counterfactual exercises.

B.3 Identifying B_s and ηG_s

To estimate the parameter of the exposure cost functions, we rely on Lemma 10, which implies

$$\delta_{si} \approx \delta_s^\circ + u_{si} + \left(B_s + \eta \frac{1 + \tau_{si}}{\bar{\omega}_{si}} G_s \right)^{-1} \mathcal{E}_{si}^\circ, \quad (67)$$

⁵⁹This happens when a firm covaries strongly with GDP while having a small variance. From (64), the strong covariance implies that δ_{si} must be large, but then a negative Σ_{si}^v might be required to match the variance through (63).

where we have imposed that B_s and G_s are sector-specific, and that $\delta_{si}^\circ = \delta_s^\circ + u_{si}$ can be written as the sum of a sector-specific term δ_s° and a residual u_{si} . At this stage in the calibration, we have already identified δ_{si} , τ_{si} , $\bar{\omega}_{si}$, and $\mathcal{E}_{si}^\circ$ ($\bar{\omega}_{si}$ and $\mathcal{E}_{si}^\circ$ are given in the proof of Lemma 10), and so we can estimate B_s and G_s as the values that best fit (67). The inverse function in (67) introduces strong nonlinearities when B_s and G_s are small, which makes this curve-fitting exercise challenging. We therefore work instead with an approximated version of (67). Since Domar weights $\bar{\omega}_{si}$ are small, we define $x_{si} := \bar{\omega}_{si} / (1 + \tau_{si})$ and approximate (67) with a quadratic function around $x_{si} = 0$ as

$$\delta_{si} \approx \delta_s^\circ + u_{si} + (\eta G_s)^{-1} \mathcal{E}_{si}^\circ x_{si} - (\eta G_s)^{-2} B_s \mathcal{E}_{si}^\circ x_{si}^2.$$

We then estimate ηG_s and B_s as those terms that minimize the sum of the squares of the residuals in sector s . Finally, with those values in hand, we set $\delta_s^\circ + u_{si} = \delta_{si} - \left(B_s + \eta \frac{1 + \tau_{si}}{\bar{\omega}_{si}} G_s \right)^{-1} \mathcal{E}_{si}^\circ$ so that (67) is satisfied.

B.4 Calibrated economy

This appendix provides more detail about the calibrated economy.

B.4.1 Distributions

The calibrated economy matches several features of the data perfectly. Figure 16 shows the distributions of four such features: 1) the sales share of each firm within its sector, given by (136) in the model (panel a), 2) the firm-level markups (panel b), 3) the standard deviation of firm-level TFP growth (panel c) and 4) the covariance between firm-level TFP growth and GDP growth (panel d).

In contrast, our calibration strategy implies that some other quantities are not fitted perfectly. This is the case for revenue-based Domar weights. We show the distribution of those weights in the data and in the model in Figure 17. The calibration matches well the right tail of the distribution, where the biggest firms are located. This is reassuring as those firms are important drivers of GDP fluctuations. In contrast, the calibration features too many small firms and too few mid-size firms. The right panel of Figure 17 shows that the calibrated distribution of cost-based Domar weights, which we do not observe in the data, follows a similar shape.

Figure 16: Data distributions that the calibration matches exactly

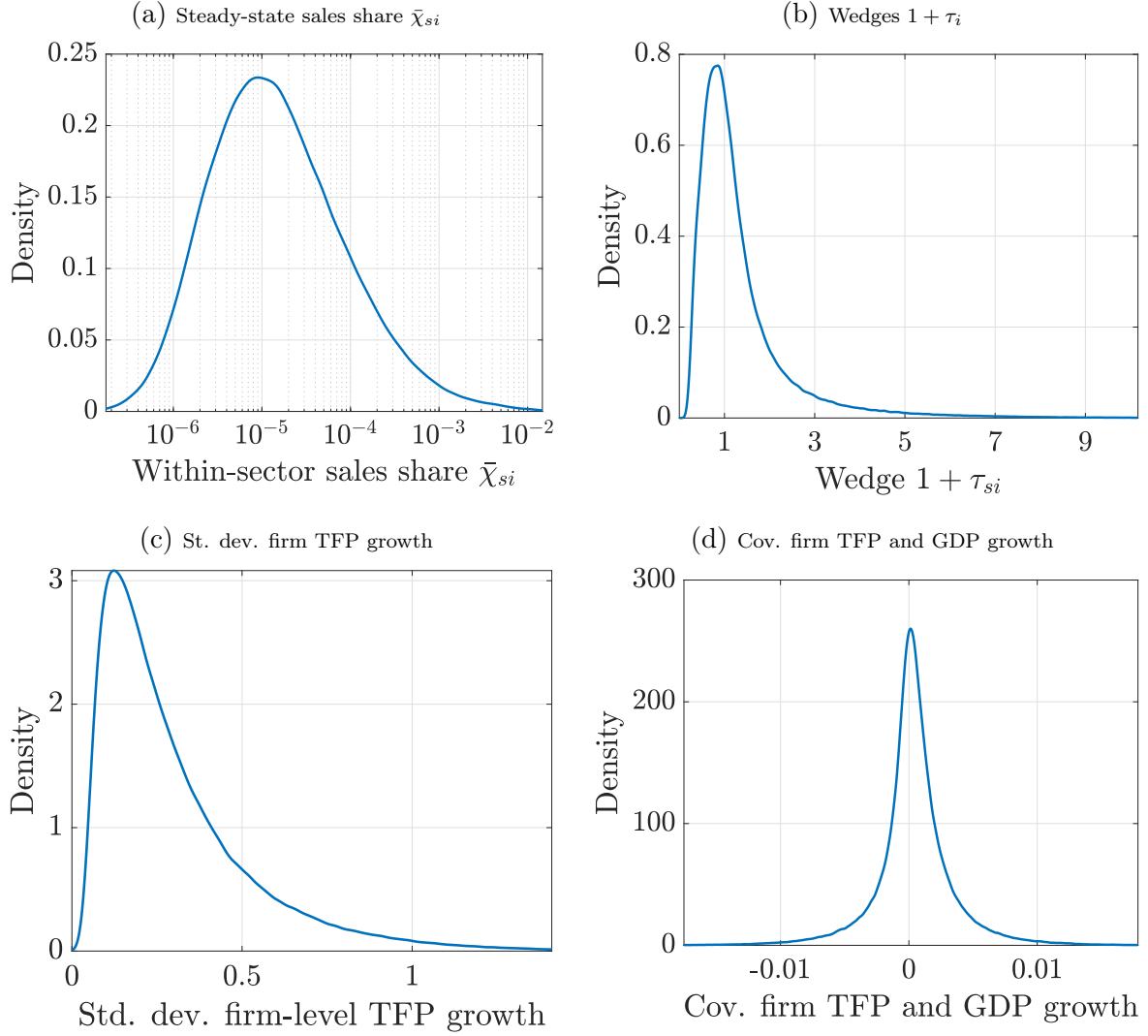
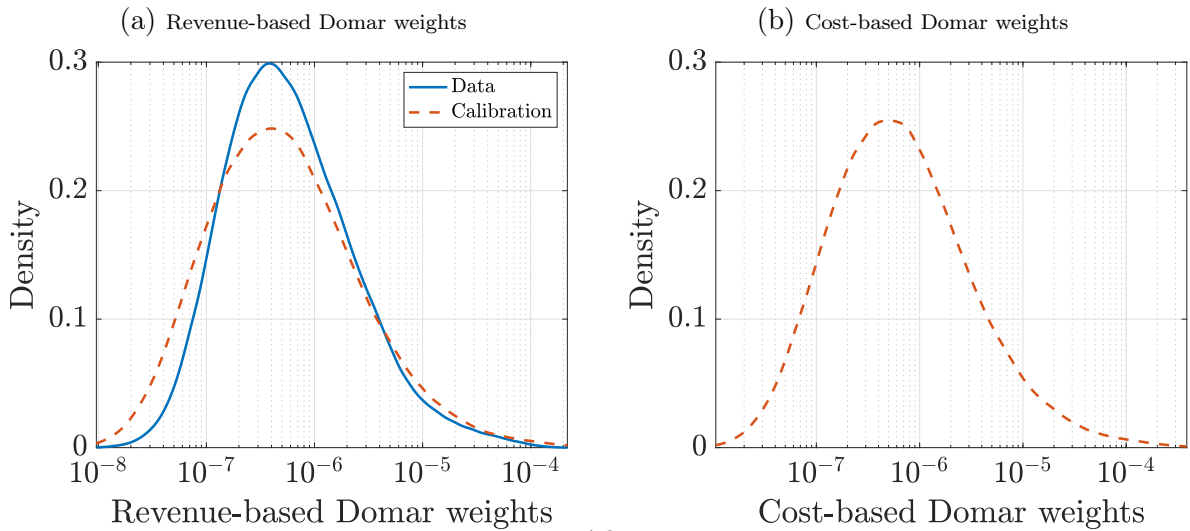


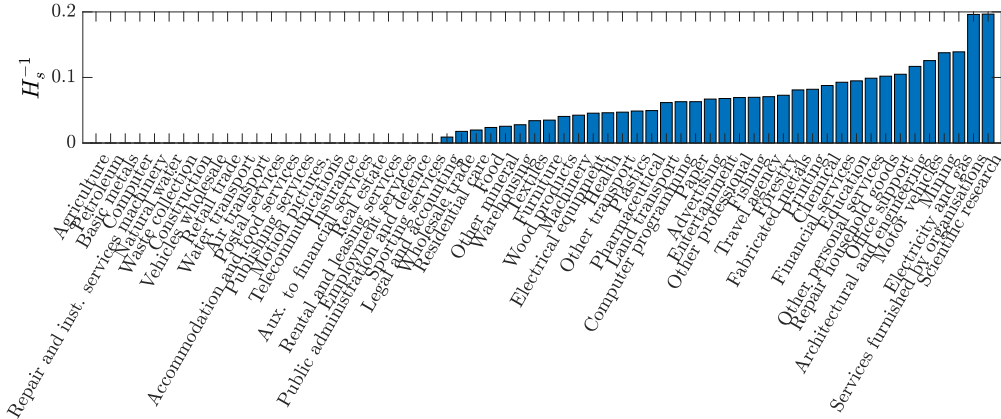
Figure 17: Domar weights of the firms in the data and in the model



B.4.2 Cost of risk exposure

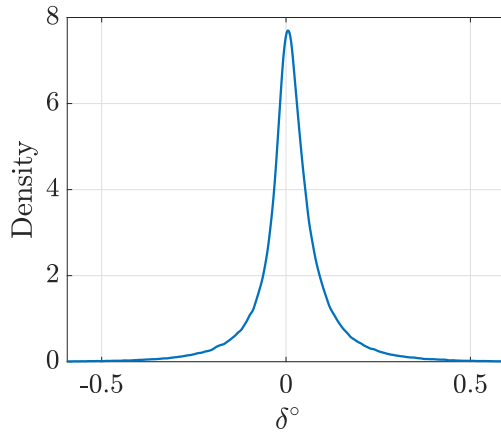
Figure 18 shows the estimated value of H_s^{-1} for each sector, where $H_{si} = B_s + \eta \frac{1+\tau_{si}}{\bar{\omega}_{si}} G_s$, and H_s^{-1} is the Domar-weighted average of H_{si}^{-1} . Several sectors have an estimated value of H_s^{-1} near zero, implying that their risk-taking behavior is rigid. Figure 19 reports the distribution of the estimated firm-level natural risk exposure δ_{si}° .

Figure 18: Estimated value of H_s^{-1} for each sector



Notes: The scale of H_s^{-1} depends on our choice of Σ . We set $\Sigma = 1$ for this figure.

Figure 19: Distribution of the estimated firm-level natural risk exposure δ_i° .



Notes: The scale of δ_i° depends on our choice of Σ . We set $\Sigma = 1$ for this figure.

B.4.3 Link between firm size, wedges and risk

To verify that our calibrated economy is able to replicate the empirical relationship between firm-level volatility, firm size and markups, we run the regressions in Appendix A.2.7 on the simulated

data. The results are presented in Table 10. Reassuringly, the estimated coefficients are close to those of Table 8 which performs the same analysis on the data.

Table 10: Cross-sectional results with alternative scaling of TFP volatility using calibrated data

	(1): $\sigma_i(\Delta \log TFP_{it})$	(2): $\log(\sigma_i(\Delta \log TFP_{it}))$
$\overline{\log(\text{Domar Weight})}_i$	−0.0421*** (0.0002)	−0.1462*** (0.0007)
$\overline{\log(\text{Markup})}_i$	0.0171*** (0.0005)	0.0629*** (0.0015)
Sector FE	Yes	Yes
Observations	531,944	531,944
R^2	0.200	0.261

Notes: The table presents estimation results of (56) with different scaling of TFP volatility using simulated data from the calibrated economy. Standard errors are in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

B.5 Cross-sectional effects of wedges and uncertainty

In this appendix, we investigate how increasing Σ by 10% and removing wedges affect the cross-section of firms. It is useful to rewrite (50) as

$$V[y_t - y_{t-1}] \approx \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} \text{Cov}[y_t - y_{t-1}, \log TFP_{si,t} - \log TFP_{si,t-1}]. \quad (68)$$

It follows that the variance of GDP growth is driven by the Domar-weighted average of the covariance of firm TFP growth with GDP growth. We will rely on that relation below to explore the impact of changes in the environment.

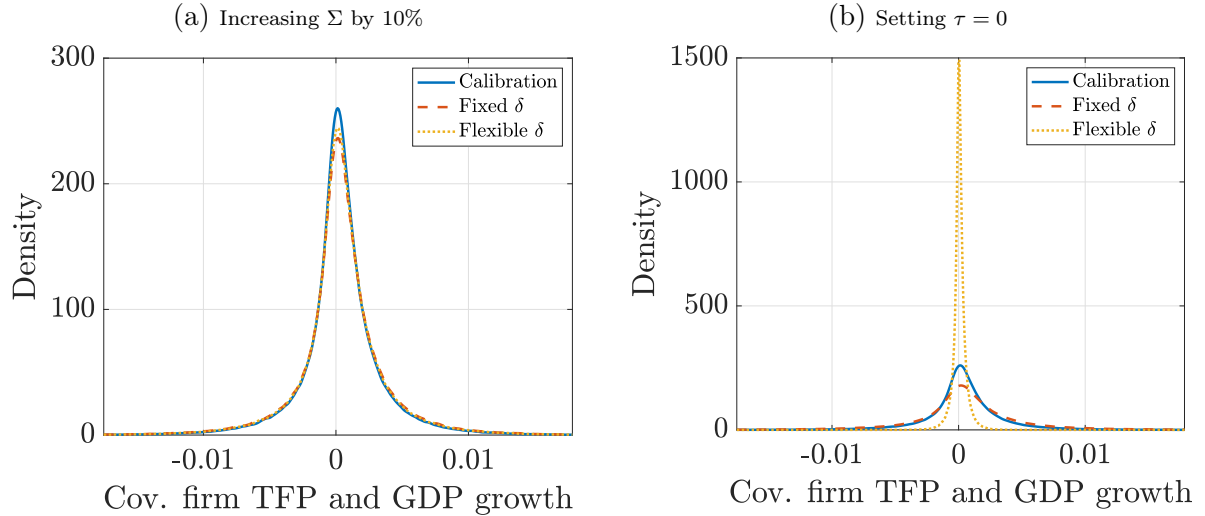
Consider the effect of an increase in Σ first. The left panel in Figure 20 shows that, keeping δ fixed, the distribution of the covariance between firm-level TFP growth and GDP growth widens slightly. Intuitively, as Σ increases, ε_t becomes a more important driver of firm-level TFP. As a result, firms that are positively exposed to ε_t now covary more with GDP while the opposite happens for negatively exposed firms.⁶⁰ Since on average firms are positively exposed to ε_t , this widening leads to an increase in the average covariance and, by (68), to an increase in aggregate volatility. When δ is flexible, firms respond to this large increase in aggregate volatility by managing their risk more aggressively, which undoes some of the widening, and mitigates the overall increase in aggregate risk.

Next, we look at the impact of removing the wedges. With δ fixed, the distribution of the

⁶⁰This can be seen more readily in (46). When Σ increases, the term $2\Delta\delta_i\Sigma$ grows for firms with $\delta_i > 0$ but shrinks for firms with $\delta_i < 0$, which explains the widening of the covariance distribution. In equilibrium, δ responds to mitigate that effect without offsetting it fully.

covariance of firm-level TFP growth with GDP growth widens. Since firm TFP growth rates, on average, comove positively with GDP growth, this translates into higher GDP volatility, in line with the result in Table 1. At the same time, when firms are free to adjust δ , they respond to the removal of wedges by managing risk substantially more actively, which makes their TFPs comove much less with GDP, and GDP growth volatility declines as a result. Consistent with this, the distribution becomes more peaked, as the figure shows.

Figure 20: Changes in the environment and the correlation distribution



Notes: Both panels show the cross-sectional distribution of the covariance of firm-level TFP growth with GDP growth. Panel (a) shows how that distribution changes when Σ increases by 10%. Panel (b) shows how that distribution changes under $\tau = 0$.

B.6 Calibration using different wedge measures

In this appendix, we recalibrate the benchmark economy of Section 9.5 under alternative wedge measures: 1) no wedges (efficient allocation), 2) markups estimated using a total-cost approach, and 3) wedges set to replicate the Spanish value-added tax. In each case, we recalibrate the model and repeat the experiments of Section 9.5. We report the results in Table 11.

Efficient allocation

For this exercise, we recalibrate the model imposing $\tau = 0$, so that there are no wedges. Calibrated GDP-growth volatility falls from 3.29% to 2.87%. The reason is that, in the CES economy, cross-firm dispersion in wedges generates dispersion in the GDP-shock weights ς_{si} in (50), and the granular term in that equation is quadratic in ς_{si} , so this dispersion adds to aggregate volatility. Removing wedges flattens the distribution of ς_{si} and shrinks the granular contribution to GDP volatility. This shift in the variance decomposition also shapes how the efficient economy responds

Table 11: Standard deviation of GDP growth under different wedge measures

	Benchmark			Efficient allocation		
	Calib.	Fixed δ	Flex. δ	Calib.	Fixed δ	Flex. δ
	Increase in Σ					
Elast. of St. dev. (Δy)	–	0.40	0.11	–	0.75	0.04
	Removing wedges					
St. dev. (Δy)	3.29%	3.51%	1.83%	2.87%	–	–
	Total-cost markups			Value-added tax		
	Calib.	Fixed δ	Flex. δ	Calib.	Fixed δ	Flex. δ
	Increase in Σ					
Elast. of St. dev. (Δy)	–	0.79	0.13	–	0.75	0.04
	Removing wedges					
St. dev. (Δy)	2.90%	2.62%	2.25%	2.87%	2.87%	2.78%

Notes: “Fixed δ ” refers to an economy in which δ is fixed at its calibrated value. “Elast. of St. dev. (Δy)” is computed as the percentage difference between the counterfactual and calibrated standard deviation of GDP growth, divided by the percentage change in $\sqrt{\Sigma}$.

to an increase in Σ . With δ fixed, a higher Σ scales up only the endogenous-risk term in (50), which now accounts for a larger share of total variance than in the baseline; the same shock therefore raises GDP-growth volatility by more in the efficient economy. Once δ adjusts, however, firms in the efficient economy manage risk more aggressively, and the response of GDP volatility is much smaller. The efficient allocation thus absorbs increases in fundamental uncertainty more easily than the baseline.

Alternative markup measures

One worry about our baseline markups is that our estimates might be biased or imprecise. To alleviate these concerns, we recalibrate the model using markups based on the total-cost approach, as described in Appendix A.2.3. These markups are smaller and less dispersed than our baseline estimates. By the same logic as in the efficient calibration, smaller wedge dispersion shrinks the granular contribution to GDP volatility: calibrated GDP-growth volatility is 2.90%, somewhat below the baseline 3.29%. For the same reason, an increase in Σ raises GDP volatility by more under fixed δ than in the baseline calibration, while the response under flexible δ remains modest. Since total-cost markups are small, removing wedges leads to a smaller decline in GDP volatility than in the baseline calibration.

Value-added tax

Spain has a value-added tax (VAT) that can be as high as 21% for certain sectors. We calibrate a version of our baseline model in which wedges are set to match the VAT instead of the estimated wedges. Since the VAT varies across sectors, we rely on data from the European Commission to set a sector-level wedge τ_s that is in line with statutory information.⁶¹ Since wedges in the model correspond to a sales tax instead of a value-added one, we adjust the statutory VAT rate with the value-added cost share of each firm. That is, we set

$$\tau_s = VAT_s \times \left(1 - \sum_{s'=1}^S \hat{\alpha}_{ss'} \right).$$

These wedges are fairly small, with a mean of 8%. It follows that the economy handles increases in Σ well, and that removing wedges does not lead to a large decline in aggregate volatility.⁶²

B.7 Calibration using $\sigma_s = 4$

In our baseline calibration, we set the elasticity of substitution to $\sigma_s = 3$. This appendix recalibrates the model with $\sigma_s = 4$, following the same steps as in the baseline. In particular, we re-estimate firm-level TFP using the procedure of Section 8. Firm-level TFP growth is typically less volatile under $\sigma_s = 4$, which translates into lower GDP-growth volatility: 3.00% versus 3.29% under $\sigma_s = 3$. We then repeat the counterfactual exercises of Section 9 (increase in Σ and removing wedges). The results, reported in Table 12, are qualitatively and quantitatively similar to those obtained under $\sigma_s = 3$.

Table 12: Standard deviation of GDP growth under $\sigma_s = 4$

	Calib.	Fixed δ	Flex. δ
Increase in Σ			
Elast. of St. dev. (Δy)	–	0.29	0.11
Removing wedges			
St. dev. (Δy)	3.00%	3.25%	1.62%

Notes: “Fixed δ ” refers to an economy in which δ is fixed at its calibrated value. “Elast. of St. dev. (Δy)” is computed as the percentage difference between the counterfactual and calibrated standard deviation of GDP growth, divided by the percentage change in $\sqrt{\Sigma}$.

⁶¹Those data are available at the website https://taxation-customs.ec.europa.eu/system/files/2021-06/vat_rates_en.pdf. We downloaded them on February 26th 2025.

⁶²Note that the fixed- δ economy features the same GDP growth volatility with and without wedges. That is because there is no within-sector dispersion in wedges.

B.8 Calibration in the disaster-risk exercise

In this appendix, we describe the calibration procedure used in Section 9.5. We first identify δ_{si} and Σ_{si}^v following the same steps as in Appendix B.2. We assume that the observed variances and covariance of firm TFP and GDP growth do not include disaster episodes, consistent with Barro (2006)’s definition that requires at least a 15% GDP decline. Annual real GDP growth never reached -15% in our 1996–2019 sample.

We then calibrate the cost function parameters following the same steps as in Appendix B.3. The only difference is that, in the presence of disaster risk, firm risk exposure is given by $\delta_{si} \approx \delta_{si}^\circ + H_{si}^{-1}(\mathcal{E}_{si}^\circ + \mathcal{F}_{si}^\circ)$, where $\mathcal{F}_{si}^\circ$ is derived in Lemma 13.

Recall that in the disaster exercise firm TFP is given by (see Appendix E.8)

$$a_{si}(\delta_{si}) = \delta_{si} \left(\varepsilon + \varepsilon^d \right) + \bar{\delta} \varepsilon^d - b_i(\delta_{si}) + \gamma_s t + v_{si},$$

where $\bar{\delta} > 0$ is a fixed disaster exposure common to all firms. We introduce $\bar{\delta}$ for the following reason. In the data, some firms have negative covariance between their TFP growth and GDP growth, which the model rationalizes with $\delta_{si} < 0$. Such firms benefit from negative shocks and, in particular, from disasters. In the CES model, this would lead to reallocation toward such firms during disasters and could even make disasters beneficial for the aggregate economy. Since this scenario is economically implausible, we assume that, in addition to their endogenous risk exposure δ_{si} , all firms share a positive fixed disaster exposure $\bar{\delta} > 0$.

We calibrate $\bar{\delta}$ and d to match two moments. First, following Barro (2006), we target a 29% real GDP drop during disasters. Second, we compute the sectoral cross-sectional standard deviation of firm TFP growth in disaster episodes and in normal times. The former standard deviation does not depend on $\bar{\delta}$ because $\bar{\delta}$ is common to all firms. It does, however, depend on the disaster size d , since firms have heterogeneous endogenous risk exposure δ_{si} . We extend our sample to include the COVID shocks, and find that the average cross-sectional standard deviation of firm TFP growth was higher during that episode than its pre-COVID analogue by 0.069. We set d to match this value. This calibration procedure implies $d = 2.23$ and $\bar{\delta} = 0.092$.

B.9 Solving the Cobb-Douglas model precisely

The CES quantitative exercises in Section 9 rely on a small-risk approximation because the exact CES model admits no closed-form solution (Appendix E). The Cobb-Douglas model ($\sigma_s = 1$ for all s) can be solved exactly. In this appendix, we redo the counterfactual exercises under the exact Cobb-Douglas solution. This confirms that the small-risk approximation is not driving our quantitative findings, and isolates the role of the sales-share channel that is present in CES but absent under Cobb-Douglas.

We use the firm-level TFP estimates obtained under $\sigma_s = 3$, as in Sections 8 and 9, to calibrate the model. As Appendix A.1 explains, our TFP estimation requires $\sigma_s \neq 1$: if $\sigma_s = 1$, any change in firm-level productivity is fully offset by a corresponding change in prices, leaving firm Domar weights constant. Identifying firm TFP would then require firm-level price data that we do not observe. In practice, any σ_s close to one makes the estimation procedure imprecise. We find, however, that the choice of σ_s for the TFP estimation only has a limited impact on our quantitative findings. This is because the estimated correlation of firm-level TFP growth with GDP growth and the estimated firm-level TFP growth volatility, which together discipline firm-level risk exposure in the calibration, do not change much with σ_s in the data for $\sigma_s > 2$ (Appendix A.2.5).⁶³

We repeat the two counterfactual exercises discussed in Section 9: an increase in the variance of the underlying risk factor and removing wedges. Table 13 shows the results. The qualitative messages of Section 9 carry over: higher Σ raises aggregate volatility and removing wedges reduces it. The magnitudes differ, in ways that we explain below.

GDP volatility is lower under Cobb-Douglas (2.87%) than under CES (3.29%). Under CES, firms whose risk exposure exceeds the sectoral average gain within-sector sales share when the aggregate shock is positive and lose it when it is negative. This positive co-movement gives such firms an additional incentive to take on risk, which contributes to aggregate volatility. This effect is absent under Cobb-Douglas, where sales shares do not respond to shocks.

The Cobb-Douglas economy is more sensitive to an increase in Σ when risk exposures are held fixed (elasticity 0.75, against 0.40 in CES). Under Cobb-Douglas, cost-based Domar weights are independent of wedges, so the wedge-induced dispersion in the GDP-shock weights ς_{si} that inflates the granular term of (50) in the CES benchmark is absent. A larger share of GDP variance therefore scales with Σ . The 0.75 elasticity coincides with that of the CES efficient allocation in Table 11, where wedges are absent and therefore the same logic applies. Once risk exposures are allowed to adjust, the elasticity collapses in both models (0.18 in Cobb-Douglas, 0.11 in CES).

The Cobb-Douglas economy responds less to removing wedges. Under Cobb-Douglas, firms' contributions to the aggregate risk exposure are shaped by their cost-based Domar weights, which are independent of shocks and wedges. Removing wedges therefore does not change aggregate volatility if risk exposures are held fixed (Table 13, 2.87% in both columns). When risk exposures adjust, aggregate volatility falls to 2.50%. This amounts to a 0.37 percentage point decline, against a 1.46 percentage point decline in the CES model. In both models, removing wedges makes firms larger and pushes them to manage risk more aggressively. Under CES, the sales-share co-movement channel amplifies this adjustment: a larger sales share sharpens the firm's private incentive to lower its risk exposure. Under Cobb-Douglas, where sales shares do not respond to shocks, this amplifier

⁶³Indeed, if we reproduce the Cobb-Douglas exercise of this section with TFP data estimated using $\sigma_s = 4$ instead of $\sigma_s = 3$, we find similar results. For instance, the elasticities of the standard deviation of GDP to changes in Σ are 0.77 and 0.19 under fixed and flexible δ 's, respectively. The corresponding numbers for $\sigma_s = 3$ are 0.75 and 0.18 (Table 13).

is absent, leaving the direction of the effect preserved but its magnitude muted.

Table 13: Standard deviation of GDP growth: exact Cobb-Douglas vs. CES

	CES, $\sigma_s = 3$ (baseline)			CD model, $\sigma_s = 1$		
	Calib.	Fixed δ	Flex. δ	Calib.	Fixed δ	Flex. δ
Elast. of St. dev. (Δy)	–	0.40	0.11	–	0.75	0.18
St. dev. (Δy)	3.29%	3.51%	1.83%	2.87%	2.87%	2.50%

Notes: “Fixed δ ” refers to an economy in which δ is fixed at its calibrated value. “Elast. of St. dev. (Δy)” is computed as the percentage difference between the counterfactual and calibrated standard deviation of GDP growth, divided by the percentage change in $\sqrt{\Sigma}$. The CES model is solved under the small-risk approximation; the Cobb-Douglas model is solved exactly, using TFP moments estimated under $\sigma_s = 3$.

C Proofs

C.1 Proof of Lemma 1

Lemma 1. *Log (real) GDP $y = \log Y$ is given by*

$$y = \Delta^\top \varepsilon - \tilde{\omega}^\top b(\delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L, \quad (16)$$

where $\Gamma_L := W_L L / (\bar{P}Y) = 1 - \tau^\top (\text{diag}(1 + \tau))^{-1} \omega$ is the labor share of income.

Proof. Total profit in this economy is

$$\Pi = \sum_{i=1}^N \Pi_i = \sum_{i=1}^N \tau_i K_i Q_i - W_R \sum_{i=1}^N g_i(\delta_i) = \sum_{i=1}^N \frac{\tau_i}{1 + \tau_i} \omega_i \bar{P}Y - W_R R,$$

where the last equality follows from the definition of revenue-based Domar weights, the unit-pricing condition (12), and market clearing for risk-management resources (13). This equation implies that the share of profit, gross of risk management expenditure, in nominal GDP is

$$\Gamma_\Pi := \frac{\Pi + W_R R}{\bar{P}Y} = \sum_{i=1}^N \frac{\tau_i}{1 + \tau_i} \omega_i = \tau^\top (\text{diag}(1 + \tau))^{-1} \omega.$$

The household’s budget constraint (7) gives $\bar{P}Y = W_L L + W_R R + \Pi$, so $\Gamma_L + \Gamma_\Pi = 1$ and the expression for Γ_L follows. Next, from the definition of the labor share, we can write $\bar{P}Y = \Gamma_L^{-1} W_L L$. Taking the log and combining (15) with the consumption price-index definition $\bar{P} = \prod_{i=1}^N P_i^{\beta_i}$, we find

$$y = \log(\bar{P}Y) - \log \bar{P} = \log(\Gamma_L^{-1} W_L L) + \tilde{\omega}^\top (\delta \varepsilon - b(\delta) - \log(1 + \tau)).$$

Since $W_L L = 1$ given our choice of numeraire, we find (16). $\square$

C.2 Proof of Lemma 2

Lemma 2. *In equilibrium, the risk exposure decision of firm i solves*

$$\underbrace{\mathcal{E} K_i Q_i}_{\text{Marginal benefit of risk exposure}} = \underbrace{\nabla b_i(\delta_i) K_i Q_i + \nabla g_i(\delta_i) W_R}_{\text{Marginal cost of risk exposure}}, \quad (20)$$

where the marginal value of risk exposure $\mathcal{E}$ is defined as

$$\mathcal{E} := \mathbb{E}[\varepsilon] + \text{Cov}[\lambda, \varepsilon], \quad (21)$$

and where $\lambda := \log \Lambda$ is the log of the stochastic discount factor.

Proof. One can show that the firm's problem (19) is strictly convex, such that its first-order conditions are necessary and sufficient. They are given by

$$\mathbb{E}[(\varepsilon - \nabla b_i(\delta_i)) K_i Q_i] + \text{Cov}\left((\varepsilon - \nabla b_i(\delta_i)) K_i Q_i, \frac{\Lambda}{\mathbb{E}[\Lambda]}\right) = \nabla g_i(\delta_i) W_R,$$

where we used the fact that $\partial K_i / \partial \delta_i = -(\varepsilon - \nabla b_i(\delta_i)) K_i$. Next, the definition of revenue-based Domar weights implies that $\omega_i = \frac{P_i Q_i}{P_Y} \Leftrightarrow Q_i = \frac{\omega_i \Gamma_L^{-1}}{P_i}$. Together with (12), we can therefore write $K_i Q_i = \omega_i \Gamma_L^{-1} / (1 + \tau_i)$, such that $K_i Q_i$ is deterministic in equilibrium. Since $\nabla b_i(\delta_i)$ is also deterministic, the first-order conditions simplify to

$$\mathbb{E}[\varepsilon] - \nabla b_i(\delta_i) + \text{Cov}(\varepsilon, \lambda) = \nabla g_i(\delta_i) \frac{W_R}{\omega_i \Gamma_L^{-1} / (1 + \tau_i)}, \quad (69)$$

where we have used Stein's lemma, which gives $\text{Cov}[\varepsilon, \Lambda] / \mathbb{E}[\Lambda] = \text{Cov}[\varepsilon, \lambda]$ since λ is jointly Gaussian with ε (cf. (78)). Multiplying both sides by $K_i Q_i$ and rearranging using the definition of $\mathcal{E}$ yields (20). $\square$

C.3 Proof of Corollary 1

Corollary 1. *Firms with higher revenue-based Domar weights ω_i and lower wedges τ_i manage risk more aggressively, in the sense that⁶⁴*

$$\frac{\partial [\mathcal{E}^\top \delta_i]}{\partial \omega_i} \geq 0 \quad \text{and} \quad \frac{\partial [\mathcal{E}^\top \delta_i]}{\partial \tau_i} \leq 0.$$

⁶⁴The partial derivatives imply a partial equilibrium analysis in which $\mathcal{E}$ is kept constant.

Proof. From (24), we can write

$$\mathcal{E}^\top \delta_i = \mathcal{E}^\top \delta_i^\circ + \mathcal{E}^\top H_i^{-1} \mathcal{E},$$

where in equilibrium $H_i = B_i + \eta \frac{1+\tau_i}{\omega_i} G_i$ by (8) and (25). Since $\mathcal{E}$ is held fixed in partial equilibrium and δ_i° does not depend on ω_i or τ_i , only the second term varies. Differentiating with respect to ω_i , we get

$$\frac{\partial [\mathcal{E}^\top \delta_i]}{\partial \omega_i} = -\mathcal{E}^\top H_i^{-1} \frac{\partial H_i}{\partial \omega_i} H_i^{-1} \mathcal{E} = \eta \frac{1+\tau_i}{\omega_i^2} \mathcal{E}^\top H_i^{-1} G_i H_i^{-1} \mathcal{E} \geq 0,$$

where the inequality follows since G_i is positive definite. Similarly, $\partial H_i / \partial \tau_i = (\eta / \omega_i) G_i$, which yields

$$\frac{\partial [\mathcal{E}^\top \delta_i]}{\partial \tau_i} = -\frac{\eta}{\omega_i} \mathcal{E}^\top H_i^{-1} G_i H_i^{-1} \mathcal{E} \leq 0.$$

□

C.4 Proof of Lemma 4

Lemma 4. *The aggregate cost function $\bar{h}$ is given by*

$$\bar{h}(\Delta) = \frac{1}{2} (\Delta - \Delta^\circ)^\top \bar{H} (\Delta - \Delta^\circ), \quad (30)$$

where $\Delta^\circ := (\delta^\circ)^\top \tilde{\omega}$. Furthermore, the Hessian matrix of $\bar{h}$ is given by

$$\bar{H} = \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \right]^{-1}, \quad (31)$$

where H_i is given by (22).

Proof. Since the objective in (29) is strictly convex in δ and the constraint is linear, the first-order conditions are necessary and sufficient. The Lagrangian of (29) is

$$\mathcal{L} = \sum_{i=1}^N \tilde{\omega}_i b_i(\delta_i) + \eta \sum_{i=1}^N \kappa_i g_i(\delta_i) - \sum_{m=1}^M \nu_m \left(\Delta_m - \mathbf{1}_m^\top \delta^\top \tilde{\omega} \right), \quad (70)$$

where ν_m is the Lagrange multiplier on the m th row of the constraint $\Delta = \delta^\top \tilde{\omega}$. The first-order condition with respect to δ_i is

$$\tilde{\omega}_i \nabla b_i + \eta \kappa_i \nabla g_i = -\nu \tilde{\omega}_i, \quad (71)$$

which implies that for all i, k ,

$$\nabla b_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} \nabla g_i = \nabla b_k + \eta \frac{\kappa_k}{\tilde{\omega}_k} \nabla g_k = \nabla \bar{h}, \quad (72)$$

where the last equality comes from the envelope theorem, which gives $\nabla \bar{h} = \partial \mathcal{L} / \partial \Delta = -\nu$. Together with (5) and (4), this implies that

$$\delta_i = \delta_i^\circ + \left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i \right)^{-1} \left(B_k + \eta \frac{\kappa_k}{\tilde{\omega}_k} G_k \right) (\delta_k - \delta_k^\circ).$$

Writing $H_i = B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i$, which coincides with the matrix in (22) since $W_R / (K_i Q_i) = \eta \kappa_i / \tilde{\omega}_i$ in equilibrium by (8) and (25), the constraint $\Delta = \delta^\top \tilde{\omega}$ can be rewritten as

$$\begin{aligned} \Delta &= \sum_{i=1}^N \tilde{\omega}_i \delta_i = \Delta^\circ + \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \right] H_k (\delta_k - \delta_k^\circ) \Leftrightarrow \\ \delta_k - \delta_k^\circ &= H_k^{-1} \bar{H} (\Delta - \Delta^\circ), \end{aligned} \quad (73)$$

where $\bar{H} = \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \right]^{-1}$ is given by (31). Plugging this expression into (29), we obtain

$$\bar{h}(\Delta) = \frac{1}{2} (\Delta - \Delta^\circ)^\top \bar{H} \left[\sum_{i=1}^N H_i^{-1} (\tilde{\omega}_i B_i + \eta \kappa_i G_i) H_i^{-1} \right] \bar{H} (\Delta - \Delta^\circ).$$

Using the identity $\tilde{\omega}_i B_i + \eta \kappa_i G_i = \tilde{\omega}_i H_i$, the bracketed sum simplifies to $\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} = \bar{H}^{-1}$, and the result follows. $\square$

C.5 Proof of Proposition 1

Proposition 1. *There exists a unique equilibrium, and its aggregate risk exposure Δ^* solves*

$$\mathcal{W}_{dist} := \max_{\Delta} \underbrace{\Delta^\top \mu - \bar{b}(\Delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L}_{E[y]} - \frac{1}{2} (\rho - 1) \underbrace{\Delta^\top \Sigma \Delta}_{V[y]} - \bar{g}(\Delta), \quad (32)$$

where $\bar{b}(\Delta) + \bar{g}(\Delta) = \bar{h}(\Delta)$ is given by (30). Without wedges ($\tau = 0$), the equilibrium is efficient.

Proof. We first show that the set of equilibrium allocations coincides with the set of solutions to a maximization problem. Consider the maximization problem

$$\max_{\delta} \tilde{\omega}^\top \delta \mu - \tilde{\omega}^\top b(\delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L - \frac{1}{2} (\rho - 1) \tilde{\omega}^\top \delta \Sigma \delta^\top \tilde{\omega} + \log \left(V \left(\sum_{i=1}^N \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} g_i(\delta_i) \right) \right). \quad (74)$$

Notice that we can write this problem as

$$\begin{aligned} \max_{\Delta} \Delta^\top \mu - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L - \frac{1}{2} (\rho - 1) \Delta^\top \Sigma \Delta - \\ \min_{\delta \text{ s.t. } \Delta = \delta^\top \tilde{\omega}} \left\{ \tilde{\omega}^\top b(\delta) - \log \left(V \left(\sum_{i=1}^N \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} g_i(\delta_i) \right) \right) \right\}, \end{aligned}$$

which is the maximization problem (32). Note also that the objective function of this problem is strictly concave if $\bar{h}(\Delta) = \bar{b}(\Delta) + \bar{g}(\Delta)$ is strictly convex. But recall from Lemma 4 that $\bar{h}$'s Hessian satisfies

$$\bar{H} = \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \right]^{-1}. \quad (75)$$

Since H_i is positive definite, so is H_i^{-1} and so is the right-hand side of this equation. It follows that $\bar{H}$ is also positive definite, and $\bar{h}$ is therefore strictly convex. This implies that there is a unique solution to the maximization problems (74) and (32) and that the first-order conditions are sufficient to characterize it. To complete the proof, we will show that the equilibrium conditions coincide with these first-order conditions. This will imply that there is a unique equilibrium.

The first-order condition of the fictitious planner with respect to δ_{im} is

$$\tilde{\omega}_i \mu_m - \tilde{\omega}_i \frac{db_i(\delta_i)}{d\delta_{im}} - (\rho - 1) \tilde{\omega}_i \tilde{\omega}^\top \delta \Sigma \mathbf{1}_m + \frac{V' \left(\sum_{i=1}^N \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} g_i(\delta_i) \right)}{V \left(\sum_{i=1}^N \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} g_i(\delta_i) \right)} \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} \frac{dg_i(\delta_i)}{d\delta_{im}} = 0.$$

Because of V 's exponential form, this expression simplifies to

$$\tilde{\omega}_i \mu_m - \tilde{\omega}_i \frac{db_i(\delta_i)}{d\delta_{im}} - (\rho - 1) \tilde{\omega}_i \tilde{\omega}^\top \delta \Sigma \mathbf{1}_m - \eta \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} \frac{dg_i(\delta_i)}{d\delta_{im}} = 0. \quad (76)$$

The system of these $M \times N$ equations fully characterizes the unique solution to the fictitious planner's problem.

Now consider the equilibrium. Recall from (69) that the first-order condition of the firm implies that

$$\mathbb{E}[\varepsilon] - \nabla b_i(\delta_i) + \text{Cov}[\varepsilon, \lambda] = \nabla g_i(\delta_i) \frac{W_R}{\omega_i \Gamma_L^{-1} / (1 + \tau_i)}.$$

Then, using (9) together with $\mathcal{U}(Y) = Y^{1-\rho}/(1-\rho)$, $\mathcal{V}(R) = V(R)^{1-\rho}$, and $\bar{P}Y = \Gamma_L^{-1} W_L L$, the stochastic discount factor is

$$\Lambda = \mathcal{U}'(Y) (V(R))^{1-\rho} \bar{P}^{-1} = (\Gamma_L^{-1} W_L L)^{-\rho} (V(R))^{1-\rho} \bar{P}^{\rho-1}. \quad (77)$$

Taking the log, we get

$$\lambda = \log \left((\Gamma_L^{-1} W_L L)^{-\rho} (V(R))^{1-\rho} \right) - (\rho - 1) \beta^\top \tilde{\mathcal{L}} (\delta \varepsilon - b(\delta) - \log(1 + \tau)), \quad (78)$$

where we have used the fact that $\bar{p} = \beta^\top p$ together with (15). The only random term in λ is $-(\rho - 1) \tilde{\omega}^\top \delta \varepsilon$, and so $\text{Cov}[\varepsilon, \lambda] = -(\rho - 1) \Sigma \Delta$ (using $\beta^\top \tilde{\mathcal{L}} = \tilde{\omega}^\top$ and $\Delta = \delta^\top \tilde{\omega}$). The first-order condition therefore becomes

$$\mu - \nabla b_i(\delta_i) - (\rho - 1) \Sigma \Delta = \nabla g_i(\delta_i) \frac{W_R}{\omega_i \Gamma_L^{-1} / (1 + \tau_i)}.$$

Finally, from the definition of the labor share, we can write $\bar{P}Y = \Gamma_L^{-1} W_L L = \Gamma_L^{-1}$. It follows from (8) that $\eta \Gamma_L^{-1} = W_R$, and so the first-order conditions of the firms are equivalent to those of the fictitious planner (76).

When $\tau = 0$, revenue-based and cost-based Domar weights coincide, so $\kappa_i = 1$ for all i , $\log(1 + \tau) = 0$, and $\Gamma_L = 1$. The distorted cost function $\bar{h}$ in (29) reduces to the efficient $\bar{h}_{SP}$ in (26), and (32) reduces to (28). The equilibrium is therefore efficient. $\square$

C.6 Proof of Lemma 5

Lemma 5. *The equilibrium aggregate risk exposure Δ solves*

$$\mathcal{E}(\Delta) = \nabla \bar{h}(\Delta), \quad (33)$$

where the marginal value of aggregate risk exposure $\mathcal{E}$ can be written as

$$\mathcal{E} = \underbrace{\mu}_{\text{E}[\varepsilon]} - \underbrace{(\rho - 1) \Sigma \Delta}_{\text{Cov}[\lambda, \varepsilon]}. \quad (34)$$

Proof. Differentiating (32) with respect to Δ , the fictitious planner's first-order condition is

$$\mu - (\rho - 1) \Sigma \Delta = \nabla \bar{h}(\Delta).$$

Combining the definition of $\mathcal{E}$ given by (21) with (78), we can write

$$\mathcal{E} = \text{E}[\varepsilon] + \text{Cov} \left[-(\rho - 1) \tilde{\omega}^\top \delta \varepsilon, \varepsilon \right],$$

which with the definition of Δ yields the result. $\square$

C.7 Proof of Proposition 2

Proposition 2. *The equilibrium aggregate risk-exposure decisions are given by*

$$\Delta = \Delta^\circ + \mathcal{H}^{-1} \mathcal{E}^\circ, \quad (36)$$

where $\mathcal{E}^\circ := \mu - (\rho - 1) \Sigma \Delta^\circ$ and where the $M \times M$ positive definite matrix $\mathcal{H}^{-1}$ is

$$\mathcal{H}^{-1} := (\bar{H} + (\rho - 1) \Sigma)^{-1}. \quad (37)$$

Proof. Using $\nabla \bar{h}(\Delta) = \bar{H}(\Delta - \Delta^\circ)$ from (30) together with the planner's FOC (33), we can write

$$\Delta - \Delta^\circ = \bar{H}^{-1} \mathcal{E}(\Delta). \quad (79)$$

Substituting $\mathcal{E}(\Delta) = \mu - (\rho - 1) \Sigma \Delta$ from (34), it follows that

$$\begin{aligned} \mathcal{E} - \bar{H}(\Delta - \Delta^\circ) &= 0 \Leftrightarrow \Delta = \Delta^\circ + \bar{H}^{-1}(\mu - (\rho - 1) \Sigma \Delta) \Leftrightarrow \\ \mathcal{H} \Delta &= \bar{H} \Delta^\circ + \mu \Leftrightarrow \Delta = \mathcal{H}^{-1}(\bar{H} \Delta^\circ + \mu) = \Delta^\circ + \mathcal{H}^{-1} \mathcal{E}^\circ, \end{aligned}$$

where $\mathcal{E}^\circ = \mu - (\rho - 1) \Sigma \Delta^\circ$, and the last equality uses $\bar{H} = \mathcal{H} - (\rho - 1) \Sigma$ to rewrite $\bar{H} \Delta^\circ + \mu = \mathcal{H} \Delta^\circ + \mathcal{E}^\circ$. Finally, $\mathcal{H} = \bar{H} + (\rho - 1) \Sigma$ is positive definite since $\bar{H}$ is positive definite (Lemma 4), Σ is positive definite, and $\rho \geq 1$. Therefore, $\mathcal{H}^{-1}$ is also positive definite. $\square$

C.8 Proof of Proposition 3

Proposition 3. *Let γ denote either the mean μ_m or an element Σ_{mn} of the covariance matrix. The response of the equilibrium aggregate risk exposure Δ to a change in γ is given by*

$$\frac{d\Delta}{d\gamma} = \mathcal{H}^{-1} \frac{\partial \mathcal{E}}{\partial \gamma}, \quad (38)$$

where $\partial \mathcal{E} / \partial \gamma$ is given by (35).

Proof. Recall that Δ is given by (36). Since $\mathcal{H}$ does not depend on μ and $d\mathcal{E}^\circ / d\mu = \partial \mathcal{E} / \partial \mu = I$, we have

$$\frac{d\Delta}{d\mu} = \mathcal{H}^{-1} \frac{d\mathcal{E}^\circ}{d\mu} = \mathcal{H}^{-1} \frac{\partial \mathcal{E}}{\partial \mu}.$$

For Σ ,

$$\begin{aligned} \frac{d\Delta}{d\Sigma_{mn}} &= \frac{d\mathcal{H}^{-1}}{d\Sigma_{mn}} \mathcal{E}^\circ + \mathcal{H}^{-1} \frac{d\mathcal{E}^\circ}{d\Sigma_{mn}} = -(\rho - 1) \mathcal{H}^{-1} \frac{d\Sigma}{d\Sigma_{mn}} \mathcal{H}^{-1} \mathcal{E}^\circ - (\rho - 1) \mathcal{H}^{-1} \frac{d\Sigma}{d\Sigma_{mn}} \Delta^\circ \\ &= -(\rho - 1) \mathcal{H}^{-1} \frac{d\Sigma}{d\Sigma_{mn}} \Delta, \end{aligned}$$

where the second equality uses the matrix-inverse identity together with $\bar{H}$ being independent of Σ (so $d\mathcal{H}/d\Sigma_{mn} = (\rho - 1)d\Sigma/d\Sigma_{mn}$), and the third uses $\mathcal{H}^{-1}\mathcal{E}^\circ + \Delta^\circ = \Delta$ from (36). The result follows since $\partial\mathcal{E}/\partial\Sigma_{mn} = -(\rho - 1)(d\Sigma/d\Sigma_{mn})\Delta$ matches (35). $\square$

C.9 Proof of Corollary 2

Corollary 2. *An increase in the expected value μ_m of risk factor m leads to an increase in aggregate risk exposure Δ_m . An increase in the variance Σ_{mm} of risk factor m leads to a decrease in Δ_m if $\Delta_m > 0$ and to an increase in Δ_m if $\Delta_m < 0$.*

Proof. By Proposition 2, $\mathcal{H}$ is positive definite and, therefore, so is $\mathcal{H}^{-1}$. It follows that the diagonal elements of $\mathcal{H}^{-1}$ are positive. From (35) and Proposition 3, $\frac{d\Delta_m}{d\mu_m} = (\mathcal{H}^{-1})_{mm} > 0$ and $\frac{d\Delta_m}{d\Sigma_{mm}} = -(\rho - 1)\Delta_m(\mathcal{H}^{-1})_{mm}$, which has the opposite sign of Δ_m when $\rho > 1$. $\square$

C.10 Proof of Proposition 4

Proposition 4. *The response of the equilibrium aggregate risk exposure Δ to a change in wedge τ_i is given by*

$$\frac{d\Delta}{d\tau_i} = \frac{d\mathcal{H}^{-1}}{d\tau_i}\mathcal{E}^\circ, \quad (39)$$

where

$$\frac{d\mathcal{H}^{-1}}{d\tau_i} = -\mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) \mathcal{H}^{-1},$$

and where the response $d\kappa_j/d\tau_i$ of the efficiency gap is nonnegative, and $\partial \bar{H}/\partial \kappa_j$ is a positive definite matrix.

Proof. The first equation follows from (36), since Δ° and $\mathcal{E}^\circ$ do not depend on τ . Differentiating $\mathcal{H}^{-1}$, given by (37), with respect to τ_i , we find

$$\frac{d\mathcal{H}^{-1}}{d\tau_i} = -\mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) \mathcal{H}^{-1}.$$

Differentiating $\kappa_i = (1 + \tau_i) \frac{\tilde{\omega}_i}{\omega_i}$ with respect to τ_j yields

$$\frac{d\kappa_i}{d\tau_j} = \frac{\kappa_i}{\omega_i} \frac{\omega_j}{1 + \tau_j} \mathcal{L}_{ji} \geq 0, \quad (80)$$

where the inequality follows from $\mathcal{L}_{ji} \geq 0$. Finally, using (31), we find

$$\frac{\partial \bar{H}}{\partial \kappa_i} = \bar{H} (\eta H_i^{-1} G_i H_i^{-1}) \bar{H}, \quad (81)$$

which is positive definite since for any $x \neq 0$, $x^\top \bar{H} H_i^{-1} G_i H_i^{-1} \bar{H} x = (H_i^{-1} \bar{H} x)^\top G_i (H_i^{-1} \bar{H} x) > 0$ by positive definiteness of G_i and invertibility of $H_i^{-1} \bar{H}$. $\square$

C.11 Proof of Corollary 3

Corollary 3. *In a diagonal economy, a higher wedge τ_i increases Δ_m for all m such that $\mathcal{E}_m < 0$ (bad risks) and decreases Δ_m for all m such that $\mathcal{E}_m > 0$ (good risks).*

Proof. By Proposition 4 and (81),

$$\frac{d\Delta}{d\tau_i} = -\mathcal{H}^{-1} \left(\sum_{j=1}^N \bar{H} \left(\eta H_j^{-1} G_j H_j^{-1} \right) \bar{H} \frac{d\kappa_j}{d\tau_i} \right) \mathcal{H}^{-1} \mathcal{E}^\circ.$$

Recall that $\frac{d\kappa_j}{d\tau_i} \geq 0$ and $\frac{d\kappa_i}{d\tau_i} > 0$ by (80). In a diagonal economy (Σ , B_i , and G_i diagonal for all i), each H_j and $\bar{H}$ is also diagonal, so each term $\bar{H}(\eta H_j^{-1} G_j H_j^{-1}) \bar{H}$ is positive diagonal. Together with $d\kappa_i/d\tau_i > 0$, this implies that

$$\mathcal{H}^{-1} \left(\sum_{j=1}^N \bar{H} \left(\eta H_j^{-1} G_j H_j^{-1} \right) \bar{H} \frac{d\kappa_j}{d\tau_i} \right) \mathcal{H}^{-1}$$

is a positive diagonal matrix. Therefore, the sign of $\frac{d\Delta_m}{d\tau_i}$ is the opposite of the sign of $\mathcal{E}_m^\circ$. Finally, from the first-order condition of the fictitious planner we have

$$\mathcal{E} - \bar{H} (\Delta - \Delta^\circ) = 0 \Leftrightarrow \mathcal{E} = \bar{H} \mathcal{H}^{-1} \mathcal{E}^\circ. \quad (82)$$

In a diagonal economy, $\bar{H} \mathcal{H}^{-1}$ is diagonal with positive entries, so $\mathcal{E}_m$ and $\mathcal{E}_m^\circ$ have the same sign and the result follows. $\square$

C.12 Proof of Lemma 6

Lemma 6. *Suppose that $\tau_j > 0$ for at least one firm j and that $\mathcal{E}^\circ \neq 0$. Then $(\Delta - \Delta^{SP})^\top \mathcal{E}^\circ < 0$, where Δ and Δ^{SP} are the aggregate risk exposure vectors in the equilibrium and the efficient allocation, respectively. Furthermore, in a diagonal economy the sign of $\Delta_m - \Delta_m^{SP}$ is the opposite of that of $\mathcal{E}_m^\circ$.*

Proof. We first show that $\mathcal{H}_{SP}^{-1} - \mathcal{H}^{-1}$ is positive definite. By Loewner-order monotonicity of the matrix inverse, for any two positive definite matrices A and B , if $A - B$ is positive definite, so is $B^{-1} - A^{-1}$. It follows that $\mathcal{H}_{SP}^{-1} - \mathcal{H}^{-1}$ is positive definite if $\mathcal{H} - \mathcal{H}_{SP}$ is. From (37),

$$\mathcal{H} - \mathcal{H}_{SP} = \bar{H} - \bar{H}_{SP}.$$

It follows that $\mathcal{H}_{SP}^{-1} - \mathcal{H}^{-1}$ is positive definite if $\bar{H}_{SP}^{-1} - \bar{H}^{-1}$ is. From (31), we can write

$$\bar{H}_{SP}^{-1} - \bar{H}^{-1} = \sum_{i=1}^N \tilde{\omega}_i \left(\left(B_i + \eta \frac{1}{\tilde{\omega}_i} G_i \right)^{-1} - \left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i \right)^{-1} \right).$$

Recall that $\kappa_i \geq 1$, with the equality holding for all firms simultaneously only if $\tau = 0$. Then $\left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i \right) - \left(B_i + \eta \frac{1}{\tilde{\omega}_i} G_i \right) = \eta \frac{\kappa_i - 1}{\tilde{\omega}_i} G_i$ is a positive semidefinite matrix, and so is $\left(B_i + \eta \frac{1}{\tilde{\omega}_i} G_i \right)^{-1} - \left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i \right)^{-1}$. Then $\bar{H}_{SP}^{-1} - \bar{H}^{-1}$ is positive definite, and so is $\mathcal{H}_{SP}^{-1} - \mathcal{H}^{-1}$. Since Δ° and $\mathcal{E}^\circ$ depend only on primitives, they are the same in the equilibrium and the efficient allocation, so applying Proposition 2 to Δ and Δ^{SP} , we can write

$$(\Delta - \Delta^{SP})^\top \mathcal{E}^\circ = (\mathcal{H}^{-1} \mathcal{E}^\circ - \mathcal{H}_{SP}^{-1} \mathcal{E}^\circ)^\top \mathcal{E}^\circ = (\mathcal{E}^\circ)^\top (\mathcal{H}^{-1} - \mathcal{H}_{SP}^{-1}) \mathcal{E}^\circ.$$

This expression is negative provided $\mathcal{E}^\circ \neq 0$, which is the first part of this lemma.

Furthermore, if the risk factors are uncorrelated (diagonal Σ), and individual risk exposures are neither complements nor substitutes in the cost functions $(b_1, \dots, b_N)$ and $(g_1, \dots, g_N)$ (diagonal B_i and G_i for all i), then $\mathcal{H}$ and $\mathcal{H}_{SP}$ are diagonal matrices. Therefore, the sign of $\Delta_m - \Delta_m^{SP}$ is the opposite of the sign of $\mathcal{E}_m^\circ$, which is the result. $\square$

C.13 Proof of Corollary 4

Corollary 4. *Let $B_i = 0$ for all i . In a diagonal economy, the following statements hold.*

1. *The impact of an increase in μ_m on GDP satisfies*

$$\text{sign} \left(\frac{dE[y]}{d\mu_m} - \frac{\partial E[y]}{\partial \mu_m} \right) = \text{sign}(\mu_m) \quad \text{and} \quad \text{sign} \left(\frac{dV[y]}{d\mu_m} - \frac{\partial V[y]}{\partial \mu_m} \right) = \text{sign}(\Delta_m). \quad (41)$$

2. *The impact of an increase in Σ_{mm} on GDP satisfies*

$$\text{sign} \left(\frac{dE[y]}{d\Sigma_{mm}} - \frac{\partial E[y]}{\partial \Sigma_{mm}} \right) = -\text{sign}(\mu_m \Delta_m) \quad \text{and} \quad \frac{dV[y]}{d\Sigma_{mm}} - \frac{\partial V[y]}{\partial \Sigma_{mm}} \leq 0. \quad (42)$$

Proof. We derive general expressions for the four sign comparisons and then specialize to $B_i = 0$ (the corollary's hypothesis) and $G_i = 0$. The strict sign claims assume $\Delta_m \neq 0$. From the definition of $\bar{b}$ and using (73), we get

$$\bar{b}(\Delta) = \tilde{\omega}^\top b(\delta^*(\Delta)) = \frac{1}{2} (\Delta - \Delta^\circ)^\top \bar{H} \left(\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} B_i H_i^{-1} \right) \bar{H} (\Delta - \Delta^\circ).$$

Therefore,

$$\nabla \bar{b} = \bar{H} \left(\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} B_i H_i^{-1} \right) \mathcal{E},$$

where we have used (79). From (31), we see that given our assumptions, $\bar{H}$ is diagonal with positive entries. This implies that $\mathcal{H}^{-1}$ is also diagonal with positive entries. Combining the first equation in (40) with (38) and (35), we can write

$$\text{sign} \left(\frac{d \mathbb{E}[y]}{d \mu_m} - \frac{\partial \mathbb{E}[y]}{\partial \mu_m} \right) = \text{sign} \left(\left(\mu - \bar{H} \left(\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} B_i H_i^{-1} \right) \mathcal{E} \right)^\top \mathcal{H}^{-1} \mathbf{1}_m \right).$$

This expression is general, but if there are no productivity costs of adjusting risk exposure, $B_i = 0$ for all i , we get

$$\text{sign} \left(\frac{d \mathbb{E}[y]}{d \mu_m} - \frac{\partial \mathbb{E}[y]}{\partial \mu_m} \right) = \text{sign}(\mu_m).$$

If instead there are no resource costs of adjusting risk exposure, $G_i = 0$ for all i , then $H_i = B_i$ and $\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} B_i H_i^{-1} = \sum_{i=1}^N \tilde{\omega}_i B_i^{-1} = \bar{H}^{-1}$, so $\nabla \bar{b} = \mathcal{E}$. Then $\mu - \nabla \bar{b} = (\rho - 1) \Sigma \Delta$ and

$$\text{sign} \left(\frac{d \mathbb{E}[y]}{d \mu_m} - \frac{\partial \mathbb{E}[y]}{\partial \mu_m} \right) = \text{sign}(\Delta_m).$$

Combining the second expression in (40) with (38) and (35), we can write

$$\text{sign} \left(\frac{d \mathbb{V}[y]}{d \mu_m} - \frac{\partial \mathbb{V}[y]}{\partial \mu_m} \right) = \text{sign} \left(2 \Delta^\top \Sigma \mathcal{H}^{-1} \mathbf{1}_m \right) = \text{sign}(\Delta_m).$$

We can follow the same procedure for Σ_{mm} . From the first equation in (40) together with (38) and (35), we find

$$\text{sign} \left(\frac{d \mathbb{E}[y]}{d \Sigma_{mm}} - \frac{\partial \mathbb{E}[y]}{\partial \Sigma_{mm}} \right) = \text{sign} \left(- \left(\mu - \bar{H} \left(\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} B_i H_i^{-1} \right) \mathcal{E} \right)^\top \mathcal{H}^{-1} (\rho - 1) \Delta_m \mathbf{1}_m \right).$$

If $B_i = 0$ for all i , we have

$$\text{sign} \left(\frac{d \mathbb{E}[y]}{d \Sigma_{mm}} - \frac{\partial \mathbb{E}[y]}{\partial \Sigma_{mm}} \right) = \text{sign}(-\mu_m \Delta_m).$$

If $G_i = 0$ for all i , we have

$$\text{sign} \left(\frac{d \mathbb{E}[y]}{d \Sigma_{mm}} - \frac{\partial \mathbb{E}[y]}{\partial \Sigma_{mm}} \right) = \text{sign} \left(-\Delta^\top \Sigma \mathcal{H}^{-1} \Delta_m \mathbf{1}_m \right) < 0.$$

From the second expression in (40) together with (38) and (35), we can write

$$\frac{dV[y]}{d\Sigma_{mm}} - \frac{\partial V[y]}{\partial \Sigma_{mm}} = -2(\rho - 1) \Delta^\top \Sigma \mathcal{H}^{-1} \Delta_m \mathbf{1}_m = -2(\rho - 1) \Sigma_{mm} (\mathcal{H}^{-1})_{mm} \Delta_m^2,$$

which is strictly negative since $\rho > 1$ and $\Delta_m \neq 0$. $\square$

C.14 Proof of Corollary 5

Corollary 5. *Let $B_i = 0$ for all i , and suppose that there is a single risk factor. Then*

$$\text{sign} \left(\frac{dE[y]}{d\tau_i} - \frac{\partial E[y]}{\partial \tau_i} \right) = -\text{sign}(\mu \mathcal{E}) \quad \text{and} \quad \text{sign} \left(\frac{dV[y]}{d\tau_i} - \frac{\partial V[y]}{\partial \tau_i} \right) = -\text{sign}(\Delta \mathcal{E}). \quad (43)$$

Proof. Suppose that there is only one risk factor. Then all $M \times M$ positive definite matrices are simply positive scalars. From Proposition 5 together with (39), and using $\mathcal{H}^{-1} \mathcal{E}^\circ = \bar{H}^{-1} \mathcal{E}$ from (82), we can write

$$\begin{aligned} \text{sign} \left(\frac{dE[y]}{d\tau_i} - \frac{\partial E[y]}{\partial \tau_i} \right) &= \text{sign} \left(- \left(\mu - \bar{H} \left(\sum_{k=1}^N \tilde{\omega}_k H_k^{-1} B_k H_k^{-1} \right) \mathcal{E} \right) \mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) \bar{H}^{-1} \mathcal{E} \right) \\ &= \text{sign} \left(- \left(\mu - \bar{H} \left(\sum_{k=1}^N \tilde{\omega}_k H_k^{-1} B_k H_k^{-1} \right) \mathcal{E} \right) \mathcal{E} \right), \end{aligned}$$

where the second equality uses that $\mathcal{H}^{-1}$, $\bar{H}^{-1}$, and $\sum_{j=1}^N (\partial \bar{H} / \partial \kappa_j) (d\kappa_j / d\tau_i)$ are all positive scalars in the single-factor case. If there are no productivity costs of adjusting risk exposure, $B_k = 0$ for all k , we get

$$\text{sign} \left(\frac{dE[y]}{d\tau_i} - \frac{\partial E[y]}{\partial \tau_i} \right) = -\text{sign}(\mu \mathcal{E}).$$

Combining the second expression in (40) with (39), we can write

$$\text{sign} \left(\frac{dV[y]}{d\tau_i} - \frac{\partial V[y]}{\partial \tau_i} \right) = \text{sign} \left(-2\Delta \Sigma \mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) \bar{H}^{-1} \mathcal{E} \right) = -\text{sign}(\Delta \mathcal{E}),$$

where the last equality absorbs the positive scalars and uses $\Sigma > 0$. $\square$

C.15 Proof of Lemma 7

Before proving Lemma 7, we formulate a few auxiliary results. Define

$$\bar{h}_V(\Delta) := \tilde{\omega}^\top b(\delta(\Delta)) - \log V \left(\sum_{i=1}^N g_i(\delta_i(\Delta)) \right), \quad (83)$$

where $\delta_i(\Delta)$ is the distorted equilibrium exposure given by (24). Unlike $\bar{h}$, the weights of the different firms in $\bar{h}_V$ are not themselves distorted by the efficiency gaps $(\kappa_1, \dots, \kappa_N)$. As a result, $\bar{h}_V$ properly measures the disutility of equilibrium risk management decisions, and we can use it to compute welfare. The function $\bar{h}_V$ also differs from the planner's cost function $\bar{h}_{SP}$ since it uses the (distorted) equilibrium risk-exposure matrix δ as input instead of its efficient counterpart.

We can use $\bar{h}_V$ to describe how a change in the environment affects welfare. As before, we focus on the role played by the response of the exposure vector Δ by filtering out the fixed-exposure effect $\partial\mathcal{W}/\partial\chi$.

Proposition 7. *Let χ denote one of μ_m , Σ_{mn} , or τ_i . Then the impact of a change in χ on welfare is given by*

$$\frac{d\mathcal{W}}{d\chi} - \frac{\partial\mathcal{W}}{\partial\chi} = (\mathcal{E} - \nabla\bar{h}_V)^\top \frac{d\Delta}{d\chi} = (\nabla\bar{h} - \nabla\bar{h}_V)^\top \frac{d\Delta}{d\chi}, \quad (84)$$

where the partial derivatives indicate that Δ is kept fixed, and where $d\Delta/d\chi$ is given by (38) for $\chi = \mu_m$ or Σ_{mn} , and by (39) for $\chi = \tau_i$.

Proof. Combining the definition of $\bar{h}_V$ with (24) and (79), we can write

$$\bar{h}_V = \frac{1}{2} (\Delta - \Delta^\circ)^\top \bar{H} \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \left(B_i + \eta \frac{1}{\tilde{\omega}_i} G_i \right) H_i^{-1} \right] \bar{H} (\Delta - \Delta^\circ).$$

This equation implies that

$$\nabla\bar{h}_V = \bar{H}_V (\Delta - \Delta^\circ),$$

where

$$\bar{H}_V = \bar{H} \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \left(B_i + \eta \frac{1}{\tilde{\omega}_i} G_i \right) H_i^{-1} \right] \bar{H}.$$

Subtracting from the analogous bracket for $\bar{h}$ (which equals $\bar{H}^{-1}$ by the identity $\tilde{\omega}_i B_i + \eta \kappa_i G_i = \tilde{\omega}_i H_i$ from the proof of Lemma 4) yields a difference of $\sum_{i=1}^N \eta (\kappa_i - 1) H_i^{-1} G_i H_i^{-1}$, so

$$\nabla\bar{h} - \nabla\bar{h}_V = \bar{H} \left(\sum_{i=1}^N \eta (\kappa_i - 1) H_i^{-1} G_i H_i^{-1} \right) \bar{H} (\Delta - \Delta^\circ). \quad (85)$$

Log welfare with CRRA preferences and log-normal y equals $E[y] - \frac{1}{2}(\rho - 1)V[y] - \eta R$ with $R = \sum_{i=1}^N g_i(\delta_i)$ in equilibrium. Substituting $E[y]$ from (16) and using $\bar{h}_V$ from (83), log welfare equals $\Delta^\top \mu - \bar{h}_V(\Delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L - \frac{1}{2}(\rho - 1)\Delta^\top \Sigma \Delta$, and comparing with (32) gives

$$\mathcal{W} = \mathcal{W}_{dist} + \bar{h}(\Delta) - \bar{h}_V(\Delta).$$

Differentiating with respect to χ yields

$$\frac{d\mathcal{W}}{d\chi} - \frac{\partial\mathcal{W}}{\partial\chi} = (\nabla\bar{h} - \nabla\bar{h}_V)^\top \frac{d\Delta}{d\chi},$$

where $\frac{d\mathcal{W}_{dist}}{d\chi} - \frac{\partial\mathcal{W}_{dist}}{\partial\chi} = 0$ by the envelope theorem applied to (32). The first equality in (84) follows since $\mathcal{E} = \nabla\bar{h}$ in equilibrium by Lemma 5. $\square$

The first equality in (84) highlights the two channels through which a change in χ affects welfare via Δ . First, the response $d\Delta/d\chi$ of the risk-exposure vector triggers a change in the expected utility of consumption that is proportional to the marginal benefit of that exposure, as captured by $\mathcal{E}$. For instance, an increase in the expected value μ_m of a good risk factor triggers an increase in welfare through that channel. Second, the response of Δ also triggers a change in risk management costs that is proportional to $\nabla\bar{h}_V$. If the economy is already heavily exposed to ε_m ($\Delta_m > \Delta_m^\circ$), the same increase in μ_m leads to a large increase in costs, which reduces welfare.

The second equality in (84) follows since in equilibrium the benefit of exposure $\mathcal{E}$ must be equal to its marginal cost as it is *perceived* by the fictitious planner, $\nabla\bar{h}$. When there are distortions, this perceived cost deviates from the *true* marginal cost of exposure $\nabla\bar{h}_V$, and welfare can be affected by Δ as a result. The gap $\nabla\bar{h} - \nabla\bar{h}_V$ therefore provides a measure of how important distortions are for the impact of risk exposure on welfare.

To further explore the impact of χ on welfare, we therefore need to characterize the gap between $\nabla\bar{h}$ and $\nabla\bar{h}_V$. In general, one can compute these objects and look at their difference. We can however sign that gap in a simple way in a diagonal economy.

Lemma 9. *Suppose that the economy is diagonal and that $\tau_j > 0$ for at least one firm j . Then the sign of $[\nabla\bar{h}]_m - [\nabla\bar{h}_V]_m$ is the same as the sign of $\mathcal{E}_m$.*

Proof. Combining (85) with (79), we can write

$$\nabla\bar{h} - \nabla\bar{h}_V = \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1} \right) \mathcal{E}. \quad (86)$$

Under our assumptions, $\bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1} \right)$ is a positive definite diagonal matrix. Therefore, the sign of $[\nabla\bar{h}]_m - [\nabla\bar{h}_V]_m$ is the same as the sign of $\mathcal{E}_m$. $\square$

Since in equilibrium we must have $\mathcal{E} = \nabla\bar{h}$, the gradient $[\nabla\bar{h}]_m$ must be positive if ε_m is a good risk factor and negative otherwise. Because of the wedges, however, the fictitious planner perceives that adjusting Δ is costlier than it really is, which implies that $\bar{h}_V$ is flatter than $\bar{h}$. It follows that for a good risk factor, the true marginal cost is lower than the perceived one, and $[\nabla\bar{h}]_m - [\nabla\bar{h}_V]_m > 0$. The opposite is true for a bad risk factor.

Lemma 7. Suppose that the economy is diagonal and that $\tau_j > 0$ for at least one firm j . Then,

$$\text{sign}\left(\frac{d\mathcal{W}}{d\mu_m} - \frac{\partial\mathcal{W}}{\partial\mu_m}\right) = \text{sign}(\mathcal{E}_m) \quad \text{and} \quad \text{sign}\left(\frac{d\mathcal{W}}{d\Sigma_{mm}} - \frac{\partial\mathcal{W}}{\partial\Sigma_{mm}}\right) = -\text{sign}(\Delta_m \mathcal{E}_m).$$

Proof. Using (86), we can write (84) as

$$\frac{d\mathcal{W}}{d\chi} - \frac{\partial\mathcal{W}}{\partial\chi} = \left(\frac{d\Delta}{d\chi}\right)^\top \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right) \mathcal{E}.$$

From Proposition 3, we have

$$\frac{d\Delta}{d\gamma} = \mathcal{H}^{-1} \frac{\partial\mathcal{E}}{\partial\gamma}.$$

Setting $\chi = \gamma$ and combining the last two equations, we find

$$\frac{d\mathcal{W}}{d\gamma} - \frac{\partial\mathcal{W}}{\partial\gamma} = \left(\frac{\partial\mathcal{E}}{\partial\gamma}\right)^\top \mathcal{H}^{-1} \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right) \mathcal{E}.$$

If $\gamma = \mu_m$, this equation becomes

$$\frac{d\mathcal{W}}{d\mu_m} - \frac{\partial\mathcal{W}}{\partial\mu_m} = \mathbf{1}_m^\top \mathcal{H}^{-1} \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right) \mathcal{E}.$$

Given our assumptions, the matrix $\mathcal{H}^{-1} \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right)$ is diagonal with positive elements. It follows that the sign of $\frac{d\mathcal{W}}{d\mu_m} - \frac{\partial\mathcal{W}}{\partial\mu_m}$ is the same as the sign of $\mathcal{E}_m$. If instead $\gamma = \Sigma_{mm}$, the equation becomes

$$\frac{d\mathcal{W}}{d\Sigma_{mm}} - \frac{\partial\mathcal{W}}{\partial\Sigma_{mm}} = -(\rho - 1) \Delta_m \mathbf{1}_m^\top \mathcal{H}^{-1} \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right) \mathcal{E},$$

and so the sign of $\frac{d\mathcal{W}}{d\Sigma_{mm}} - \frac{\partial\mathcal{W}}{\partial\Sigma_{mm}}$ is the same as the sign of $-\Delta_m \mathcal{E}_m$. $\square$

C.16 Proof of Lemma 8

Lemma 8. In a diagonal economy, an increase in a wedge is more detrimental to welfare when risk-exposure decisions can adjust, that is, $d\mathcal{W}/d\tau_i \leq \partial\mathcal{W}/\partial\tau_i$.

Proof. Using (85) and (79), we can write (84) as

$$\frac{d\mathcal{W}}{d\chi} - \frac{\partial\mathcal{W}}{\partial\chi} = \left(\frac{d\Delta}{d\chi}\right)^\top \bar{H} \left(\sum_{k=1}^N \eta(\kappa_k - 1) H_k^{-1} G_k H_k^{-1}\right) \mathcal{E}.$$

From Proposition 4 together with (81) and (82) ($\bar{H}\mathcal{H}^{-1}\mathcal{E}^\circ = \mathcal{E}$), we have

$$\frac{d\Delta}{d\tau_i} = -\mathcal{H}^{-1}\bar{H} \left(\sum_{j=1}^N \eta H_j^{-1} G_j H_j^{-1} \frac{d\kappa_j}{d\tau_i} \right) \mathcal{E}.$$

Setting $\chi = \tau_i$ and combining the last two equations, we find

$$\frac{d\mathcal{W}}{d\tau_i} - \frac{\partial \mathcal{W}}{\partial \tau_i} = -\mathcal{E}^\top \left(\sum_{j=1}^N \eta H_j^{-1} G_j H_j^{-1} \frac{d\kappa_j}{d\tau_i} \right) \bar{H}\mathcal{H}^{-1}\bar{H} \left(\sum_{k=1}^N \eta (\kappa_k - 1) H_k^{-1} G_k H_k^{-1} \right) \mathcal{E}.$$

Under our assumptions, the matrix between $\mathcal{E}^\top$ and $\mathcal{E}$ is diagonal with positive elements (strictly positive when $\tau \neq 0$), and the result follows. $\square$

D Disaster risk

In this appendix, we explore the impact of disaster risk on the mechanisms of the model (Rietz, 1988; Barro, 2006; Barro and Ursúa, 2012).

D.1 Model

The model is the same as in the main text, except for a different distribution for the vector of fundamental risks ε . Specifically, we still assume that the elements $(\varepsilon_2, \dots, \varepsilon_M)$ of ε are normally distributed, but we allow $\varepsilon_1 = \varepsilon_1^n + \varepsilon_1^d$ to be the sum of a normal component ε_1^n and a disaster component ε_1^d . Disasters ε_1^d follow a Bernoulli distribution given by

$$\varepsilon_1^d = \begin{cases} 0 & \text{with probability } 1 - \pi_d, \\ -d & \text{with probability } \pi_d, \end{cases}$$

where π_d is the probability of a disaster and $d > 0$ is its magnitude. As in the main text, the normal components of ε are jointly normally distributed, $(\varepsilon_1^n, \varepsilon_2, \dots, \varepsilon_M) \sim \mathcal{N}(\mu, \Sigma)$. We further assume that ε_1^d is independent of all other shocks.

The presence of disaster risk alters the moments of the GDP process, such that

$$\mathbb{E}[y] = \Delta^\top \mu - \Delta_1 \pi_d d - \bar{b}(\Delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L, \quad (87)$$

$$\mathbb{V}[y] = \Delta^\top \Sigma \Delta + \Delta_1^2 (1 - \pi_d) \pi_d d^2. \quad (88)$$

The expression for the expected value of GDP is similar to that in the main text, with the exception of the disaster term $\Delta_1 \pi_d d$. Unsurprisingly, high exposure to disasters Δ_1 lowers $\mathbb{E}[y]$. Frequent (high π_d) and large (high d) disasters also lower the mean of log GDP. Disasters also increase

the variance of log GDP, and a higher Δ_1 or larger d further amplifies this increase. A disaster probability π_d close to one half also leads to higher volatility.

D.2 Equilibrium characterization

The equilibrium in the economy with disasters can still be characterized as the solution of a distorted planner problem. However, in this case, welfare can no longer be written as a function of the first two moments of log GDP. Specifically, if we define $\bar{h}(\Delta)$, $\bar{b}(\Delta)$ and $\bar{g}(\Delta)$ as in the baseline model, we have the following result.

Proposition 8. *There exists a unique equilibrium, and its aggregate risk exposure Δ^* solves*

$$\begin{aligned} \mathcal{W}_{dist} := & \max_{\Delta} \Delta^\top \mu - \bar{b}(\Delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L - \\ & \frac{1}{2} (\rho - 1) \Delta^\top \Sigma \Delta - \frac{1}{\rho - 1} \log(1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)) - \bar{g}(\Delta). \end{aligned} \quad (89)$$

Without wedges ($\tau = 0$), the equilibrium is efficient.

Compared to (32), we see that the presence of disaster risk lowers welfare. Unsurprisingly, that decline is larger when d and π_d are large, or when the economy is heavily exposed to the first risk factor (Δ_1 large).

The first-order condition associated with (89) is

$$\mathcal{E}(\Delta) = \nabla \bar{h}(\Delta), \quad (90)$$

where the marginal value of risk exposure $\mathcal{E}$ now takes the form

$$\mathcal{E}(\Delta) = \mu - (\rho - 1) \Sigma \Delta - \mathbf{1}_1 \frac{\pi_d d \exp((\rho - 1) \Delta_1 d)}{1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)}. \quad (91)$$

The last term in this expression captures the impact of disaster risk on $\mathcal{E}$. This term is always negative, so exposure to disaster risk is always detrimental to the household's utility. That term is also a decreasing function of Δ_1 , such that more disaster exposure further increases the disutility of risk exposure.

Since $\nabla \bar{h}(\Delta) = \bar{H}(\Delta - \Delta^\circ)$, we can rewrite (90) as

$$\Delta - \Delta^\circ = \bar{H}^{-1} \mathcal{E}(\Delta). \quad (92)$$

D.3 How the environment affects risk exposure

In the presence of disaster risk, $\mathcal{E}(\Delta)$ is no longer a linear function of Δ , and we cannot solve for Δ in closed form. Nevertheless, we can use (92) and the implicit function theorem to derive how

various changes in the environment affect equilibrium risk exposure.

Proposition 9. *The responses of the equilibrium aggregate risk exposure Δ to changes in μ_m , Σ_{mn} , π_d , τ_i are given by*

$$\frac{d\Delta_k}{d\mu_m} = (\mathcal{H}^{-1})_{km} - \frac{\varphi(\mathcal{H}^{-1})_{k1}(\mathcal{H}^{-1})_{1m}}{1 + \varphi(\mathcal{H}^{-1})_{11}}, \quad (93)$$

$$\frac{d\Delta_k}{d\Sigma_{mn}} = -(\rho - 1) \left(\mathbf{1}_k^\top \mathcal{H}^{-1} - \frac{\varphi(\mathcal{H}^{-1})_{k1} \mathbf{1}_1^\top \mathcal{H}^{-1}}{1 + \varphi(\mathcal{H}^{-1})_{11}} \right) \left(\frac{1}{2} \mathbf{1}_m \Delta_n + \frac{1}{2} \mathbf{1}_n \Delta_m \right), \quad (94)$$

$$\frac{d\Delta_k}{d\pi_d} = -\frac{(\mathcal{H}^{-1})_{k1}}{1 + \varphi(\mathcal{H}^{-1})_{11}} \frac{d \exp((\rho - 1) \Delta_1 d)}{[1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)]^2}, \quad (95)$$

$$\frac{d\Delta_k}{d\tau_i} = -\mathbf{1}_k^\top \left(I - \frac{\varphi(\mathcal{H}^{-1})_{\cdot 1} \mathbf{1}_1^\top}{1 + \varphi(\mathcal{H}^{-1})_{11}} \right) \mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \nabla^2 \bar{h}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) (\nabla^2 \bar{h})^{-1} \mathcal{E}, \quad (96)$$

where $\mathcal{H}$ is given by (37), $(\mathcal{H}^{-1})_{\cdot 1} = ((\mathcal{H}^{-1})_{11}, \dots, (\mathcal{H}^{-1})_{M1})$, and

$$\varphi = \frac{(1 - \pi_d) \pi_d d^2 (\rho - 1) \exp((\rho - 1) \Delta_1 d)}{(1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d))^2}. \quad (97)$$

This proposition provides expressions for how the parameters of the economy affect the equilibrium aggregate risk exposure vector Δ . In general, how Δ depends on those parameters is determined by the potentially complicated complementarity/substitutability patterns between risk factors embedded in matrices $\bar{H}$ and Σ . We can, however, make sharper analytical predictions when the economy is diagonal (see Definition 2).

Corollary 6. *In a diagonal economy, $\frac{d\Delta_k}{d\mu_k} > 0$, $\text{sign}\left(\frac{d\Delta_k}{d\Sigma_{kk}}\right) = -\text{sign}(\Delta_k)$, $\frac{d\Delta_1}{d\pi_d} < 0$, and $\text{sign}\left(\frac{d\Delta_k}{d\tau_i}\right) = -\text{sign}(\mathcal{E}_k)$.*

Unsurprisingly, an increase in the expected value μ_k of a risk factor leads to increased exposure Δ_k to that factor. If a factor k becomes more risky (higher Σ_{kk}) the equilibrium exposure to that factor Δ_k moves toward zero. Indeed, recall from (88) that the impact of ε_k on the variance of GDP depends on the square of Δ_k . Corollary 6 also shows that if disasters become more likely (higher π_d), then the economy will become less exposed to them. Finally, the impact of wedges is as in the main text (Corollary 3).

To further characterize how the presence of the disaster risk affects the economy, we analyze how an increase in π_d affects the sensitivity of Δ to changes in the environment. To keep the analysis concise, we focus on a diagonal efficient economy ($\tau_i = 0 \forall i$).

Corollary 7. *Consider a diagonal efficient economy with no disaster risk, $\pi_d = 0$. An increase in*

the disaster probability π_d has the following effects on the sensitivity of Δ to μ_1 and Σ_{11} :

$$\begin{aligned}\frac{d^2 \Delta_1}{d\mu_1 d\pi_d} &= - [(\mathcal{H}^{-1})_{11}]^2 d^2 (\rho - 1) \exp((\rho - 1) \Delta_1 d), \\ \frac{d^2 \Delta_1}{d\Sigma_{11} d\pi_d} &= (\rho - 1) [(\mathcal{H}^{-1})_{11}]^2 d (d (\rho - 1) \Delta_1 + 1) \exp((\rho - 1) \Delta_1 d).\end{aligned}$$

All other cross-derivatives are zero, i.e., $\frac{d^2 \Delta_k}{d\mu_m d\pi_d} = \frac{d^2 \Delta_k}{d\Sigma_{mn} d\pi_d} = 0$ unless $k = m = n = 1$.

Corollary 7 shows that disaster risk weakens the response of the economy to changes in the moments of the normal shocks. A higher π_d implies that the disaster component plays a larger role in $\mathcal{E}$, as evident from (91). Therefore, the equilibrium exposure is affected relatively less by changes in μ and Σ . Consider for instance an increase in μ_1 . In the absence of disasters, a higher μ_1 leads to a higher exposure to the first risk factor (Corollary 2). The presence of disasters attenuates this response. As π_d increases, a higher Δ_1 implies more disaster exposure. As a result, the same increase in μ_1 leads to a smaller increase in Δ_1 . Similarly, without disasters a higher Σ_{11} implies a lower exposure to the first risk factor if $\Delta_1 > 0$ (Corollary 2). When disaster risk is present, the same decline in Δ_1 has a more positive effect on $\mathcal{E}$ because of the reduced disaster component. Hence, $\frac{d^2 \Delta_1}{d\Sigma_{11} d\pi_d} > 0$ if $\Delta_1 > 0$.

D.4 Implications for GDP

Next, we analyze how the parameters affect the moments of GDP and welfare through the endogenous risk exposure channel.

Proposition 10. *Let χ denote either μ_m , Σ_{mn} , π_d , or τ_i . Then the impact of a change in χ on the moments of log GDP is given by*

$$\frac{dE[y]}{d\chi} - \frac{\partial E[y]}{\partial \chi} = (\mu - \nabla \bar{b})^\top \frac{d\Delta}{d\chi} - \pi_d d \frac{d\Delta_1}{d\chi}, \quad \frac{dV[y]}{d\chi} - \frac{\partial V[y]}{\partial \chi} = 2\Delta^\top \Sigma \frac{d\Delta}{d\chi} + 2\Delta_1 (1 - \pi_d) \pi_d d^2 \frac{d\Delta_1}{d\chi}, \quad (98)$$

where partial derivatives indicate that Δ is kept fixed, and $\frac{d\Delta}{d\chi}$ is given in Proposition 9.

Proof. The result follows directly from (87) and (88). $\square$

It is straightforward to combine these expressions with those of Proposition 9 to capture the full impact of χ on $E[y]$ and $V[y]$. Unsurprisingly, disaster risk introduces another channel for changes in the environment to affect the mean and the variance of log GDP. For instance, if a change in χ leads to an increase in Δ_1 , this change will result in more destructive disasters along with a decline in $E[y]$ and an increase in $V[y]$. However, in normal times when disasters do not occur, this additional channel is not operative. Therefore, it is also useful to evaluate the moments of GDP conditional on no disasters. These are given by (17). In the following corollary, we analyze how the presence of disaster risk affects these moments through the endogenous risk exposure channel.

Corollary 8. *Consider a diagonal efficient economy with no disaster risk, $\pi_d = 0$. Suppose also that the TFP cost of adjusting risk exposure is zero, $b_i = 0$ for all i . An increase in the disaster probability π_d has the following effects on the sensitivity of the moments of log GDP to $\chi \in \{\mu_1, \Sigma_{11}\}$:*

$$\begin{aligned} \frac{d}{d\pi_d} \left(\frac{dE[y|no disasters]}{d\chi} - \frac{\partial E[y|no disasters]}{\partial \chi} \right) &= \mu_1 \frac{d^2 \Delta_1}{d\pi_d d\chi}, \\ \frac{d}{d\pi_d} \left(\frac{dV[y|no disasters]}{d\chi} - \frac{\partial V[y|no disasters]}{\partial \chi} \right) &= 2\Sigma_{11} \left(\frac{d\Delta_1}{d\pi_d} \frac{d\Delta_1}{d\chi} + \frac{d^2 \Delta_1}{d\pi_d d\chi} \Delta_1 \right). \end{aligned}$$

To understand the intuition behind this corollary, suppose that $\mu_1 > 0$ and $\Delta_1 > 0$. From Corollary 7, we know that the presence of disaster risk attenuates how changes in μ and Σ affect equilibrium risk exposure. This attenuation effect is also present for the moments of GDP. In particular, an increase in π_d implies a weaker positive response of Δ_1 to μ_1 , $\frac{d^2 \Delta_1}{d\pi_d d\mu_1} < 0$, resulting in a smaller increase in the expected log GDP. At the same time, the variance of log GDP increases by less (we have $\frac{d\Delta_1}{d\pi_d} \frac{d\Delta_1}{d\mu_1} < 0$ and $\frac{d^2 \Delta_1}{d\pi_d d\mu_1} \Delta_1 < 0$). Next, $\frac{d^2 \Delta_1}{d\pi_d d\Sigma_{11}} > 0$, which means that an increase in Σ_{11} leads to a smaller decline in the expected log GDP and the variance of log GDP due to disasters.

Next, consider welfare implications.

Proposition 11. *Without wedges ($\tau = 0$), the impact of μ , Σ , and π_d on welfare is given by*

$$\begin{aligned} \frac{d\mathcal{W}}{d\mu_m} &= \frac{\partial \mathcal{W}}{\partial \mu_m} = \Delta_m, & \frac{d\mathcal{W}}{d\Sigma_{mn}} &= \frac{\partial \mathcal{W}}{\partial \Sigma_{mn}} = -\frac{1}{2} (\rho - 1) \Delta_m \Delta_n, \\ \frac{d\mathcal{W}}{d\pi_d} &= \frac{\partial \mathcal{W}}{\partial \pi_d} = -\frac{1}{\rho - 1} \frac{\exp((\rho - 1) \Delta_1 d) - 1}{1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)}. \end{aligned}$$

Furthermore, the impact of the wedge τ_i on welfare is given by

$$\frac{d\mathcal{W}}{d\tau_i} = \frac{\partial \mathcal{W}}{\partial \tau_i} = 0.$$

Proof. When $\tau = 0$, the objective function $\mathcal{W}_{dist}$ of the fictitious planner, given by (89), coincides with welfare. The result then follows from the envelope theorem. $\square$

Proposition 11 shows that the impact of μ , Σ and τ in an economy with disaster risk is as in an economy without disasters (Proposition 6). It also shows that an increase in the likelihood of disasters reduces welfare. It is also straightforward to see how an increase in the disaster probability affects the sensitivity of welfare to changes in μ and Σ .

$$\frac{d^2 \mathcal{W}}{d\mu_m d\pi_d} = \frac{d\Delta_m}{d\pi_d} \quad \text{and} \quad \frac{d^2 \mathcal{W}}{d\Sigma_{mm} d\pi_d} = -(\rho - 1) \Delta_m \frac{d\Delta_m}{d\pi_d}.$$

In a diagonal economy, $\frac{d\Delta_1}{d\pi_d} < 0$ and $\frac{d\Delta_m}{d\pi_d} = 0$ if $m \neq 1$ (Corollary 6). A higher μ_1 leads to higher

welfare if the economy is positively exposed to the first risk factor. The disaster risk implies a lower exposure Δ_1 , which leads to an attenuated response of $\mathcal{W}$ to μ_1 . A higher Σ_{11} , in contrast, harms welfare if the economy is strongly exposed to the first risk factor, i.e., $|\Delta_1|$ is high. A higher π_d lowers Δ_1 . Therefore, if $\Delta_1 > 0$, a higher π_d makes the impact of Σ_{11} on $\mathcal{W}$ less negative.

We can push this analysis further. Define $\bar{h}_V$ as in (83). Then Proposition 7 holds in the economy with disasters because, as in the main model, we have $\mathcal{W} = \mathcal{W}_{dist} + \bar{h}(\Delta) - \bar{h}_V(\Delta)$. Furthermore, Lemma 9 also holds, and its proof is unaffected by the presence of the disaster risk.⁶⁵

Finally, the analogues of Lemmas 7 and 8 hold in this economy as well.

Corollary 9. *Suppose that the economy is diagonal and that $\tau_j > 0$ for at least one firm j . Then the following holds.*

$$\begin{aligned} \text{sign}\left(\frac{d\mathcal{W}}{d\mu_m} - \frac{\partial\mathcal{W}}{\partial\mu_m}\right) &= \text{sign}(\mathcal{E}_m), & \text{sign}\left(\frac{d\mathcal{W}}{d\Sigma_{mm}} - \frac{\partial\mathcal{W}}{\partial\Sigma_{mm}}\right) &= -\text{sign}(\Delta_m\mathcal{E}_m), \\ \text{sign}\left(\frac{d\mathcal{W}}{d\pi_d} - \frac{\partial\mathcal{W}}{\partial\pi_d}\right) &= -\text{sign}(\mathcal{E}_1). \end{aligned}$$

Furthermore, an increase in wedges is more detrimental to welfare when risk-exposure decisions can adjust, that is, $\frac{d\mathcal{W}}{d\tau_i} \leq \frac{\partial\mathcal{W}}{\partial\tau_i}$.

In addition to the results described in the main text, this corollary describes how π_d affects welfare when the equilibrium is inefficient. When disaster risk is bad ($\mathcal{E}_1 < 0$), an increase in the likelihood of disaster has a smaller impact on welfare when risk exposure is endogenous. Intuitively, firms limit their exposure to ε_1 when disasters become more likely, which limits their adverse effects.

D.5 Asset pricing moments

As is well known in the literature (Rietz, 1988; Barro, 2006), the presence of disaster risk can resolve the equity premium and risk-free rate puzzles in a standard endowment economy without an implausibly high risk aversion.⁶⁶ The ability of the model to match these moments is important because it implies that it accurately captures the attitude of the representative household to changes in uncertainty. In this section, we show that our model augmented with disaster risk can generate a realistic equity premium and risk-free rate.

Specifically, suppose that there is just one risk factor and firm i 's TFP is given by (44). Then output growth is

$$y_t - y_{t-1} = \Delta(\varepsilon_t - \varepsilon_{t-1}) + \tilde{\omega}^\top(v_t - v_{t-1}) + \gamma,$$

⁶⁵In the proof of Lemma 9, we use the fact that $\delta_i - \delta_i^\circ = \tilde{\omega}_i(\eta\kappa_i G_i + \tilde{\omega}_i B_i)^{-1}(\nabla^2 \bar{h})(\Delta - \Delta^\circ)$. This is also the case in the model with the disaster risk, as follows from (101). Given that, the proof is unaffected by the presence of disaster risk. In particular, the expression for $\nabla \bar{h} - \nabla \bar{h}_V$ is still given by (86), but $\mathcal{E}$ is now given by (91).

⁶⁶Additional features, such as Epstein–Zin preferences, are needed to explain other asset pricing moments; see, e.g., Barro (2009) and Wachter (2013).

where $\varepsilon_t = \varepsilon_t^n + \varepsilon_t^d$, $\varepsilon_t^n \sim \mathcal{N}(\mu, \Sigma)$ and ε_t^d takes values 0 with probability $1 - \pi_d$ and $-d$ with probability π_d ; $v_{it} \sim \mathcal{N}(\mu_i^v, \Sigma_i^v)$. All shocks are iid. The Euler equation of the representative household is

$$1 = e^{-r} \mathbb{E}_t \left[\frac{D_{t+1} + S_{t+1}}{S_t} \frac{C_{t+1}^{-\rho} \exp(\eta(\rho-1)R_{t+1})}{C_t^{-\rho} \exp(\eta(\rho-1)R_t)} \right],$$

where e^{-r} is the time discount factor, D_t is the real dividend, S_t is the real asset price and R_t is the amount of risk management resources. Under our shock specification, risk exposures are time-invariant, and hence $R_{t+1} = R_t$. Then the Euler equation simplifies to

$$1 = e^{-r} \mathbb{E}_t \left[\frac{D_{t+1} + S_{t+1}}{S_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\rho} \right].$$

Consider first a risk-free asset that always pays out one, even if a disaster happens. Then the risk-free rate is

$$\log \mathbb{E} R_{f,t+1} = \log \mathbb{E} \left\{ e^{-r} \mathbb{E}_t [\exp(-\rho(y_{t+1} - y_t))] \right\}^{-1} = r + \rho\gamma.$$

Consider now a consumption claim, such that $D_t = C_t$.⁶⁷ Define the price-consumption ratio $\phi_t = \frac{S_t}{C_t}$. Then the Euler equation can be written as

$$1 = e^{-r} \mathbb{E}_t \left[\frac{1 + \phi_{t+1}}{\phi_t} \exp \left(-(\rho-1) \left(\Delta(\varepsilon_{t+1} - \varepsilon_t) + \tilde{\omega}^\top (v_{t+1} - v_t) + \gamma \right) \right) \right]. \quad (99)$$

Conjecture that

$$\phi_t = \phi_0 \exp \left(\phi_\varepsilon \varepsilon_t + \phi_v^\top v_t \right).$$

Plugging this into (99), we can immediately see that $\phi_\varepsilon = (\rho-1)\Delta$ and $\phi_v = (\rho-1)\tilde{\omega}$. Then

$$\begin{aligned} \phi_0 &= \frac{\exp(-r - (\rho-1)\gamma)}{1 - \exp(-r - (\rho-1)\gamma)} \mathbb{E}_t \left[\exp \left(-(\rho-1) \left(\Delta \varepsilon_{t+1} + \tilde{\omega}^\top v_{t+1} \right) \right) \right] = \\ &= \frac{\exp(-r - (\rho-1)\gamma)}{1 - \exp(-r - (\rho-1)\gamma)} \exp \left(-(\rho-1) \left(\Delta \mu + \tilde{\omega}^\top \mu^v \right) + \frac{1}{2} (\rho-1)^2 \left(\Delta^2 \Sigma + \tilde{\omega}^\top \Sigma^v \tilde{\omega} \right) \right) \times \\ &\quad [1 - \pi_d + \pi_d \exp((\rho-1)\Delta d)]. \end{aligned}$$

The expected return on the consumption claim is

$$\log \mathbb{E} R_{c,t+1} = \log \mathbb{E} \left(\frac{C_{t+1} + S_{t+1}}{S_t} \right) = \log \mathbb{E} \left(\frac{1 + \phi_{t+1}}{\phi_t} \frac{C_{t+1}}{C_t} \right).$$

⁶⁷Equivalently, we can consider a claim on total firm profits. The equity premium will be the same because real profits in our model are proportional to aggregate output.

Omitting tedious yet straightforward derivations, we get

$$\begin{aligned} \log \mathbb{E}R_{c,t+1} = & \gamma + \rho \left(\Delta^2 \Sigma + \tilde{\omega}^\top \Sigma^v \tilde{\omega} \right) + \log (1 - \pi_d + \pi_d \exp (\rho \Delta d)) \\ & + \log \left\{ \exp \left((\rho - 1) \rho \left(\Delta^2 \Sigma + \tilde{\omega}^\top \Sigma^v \tilde{\omega} \right) \right) (1 - \pi_d + \pi_d \exp (-\rho \Delta d)) + \right. \\ & \left. \frac{1 - \exp (-r - (\rho - 1) \gamma)}{\exp (-r - (\rho - 1) \gamma)} \frac{1 - \pi_d + \pi_d \exp (-\Delta d)}{1 - \pi_d + \pi_d \exp ((\rho - 1) \Delta d)} \right\}. \end{aligned}$$

D.6 Proofs

Proof of Proposition 8

Proposition 8. *There exists a unique equilibrium, and its aggregate risk exposure Δ^* solves*

$$\begin{aligned} \mathcal{W}_{dist} := & \max_{\Delta} \Delta^\top \mu - \bar{b}(\Delta) - \tilde{\omega}^\top \log (1 + \tau) - \log \Gamma_L \\ & - \frac{(\rho - 1)}{2} \Delta^\top \Sigma \Delta - \frac{1}{\rho - 1} \log (1 - \pi_d + \pi_d \exp ((\rho - 1) \Delta_1 d)) - \bar{g}(\Delta). \end{aligned} \quad (89)$$

Without wedges ($\tau = 0$), the equilibrium is efficient.

Proof. We first show that the set of equilibrium allocations coincides with the set of solutions to a maximization problem. Consider the maximization problem

$$\begin{aligned} \max_{\delta} & \tilde{\omega}^\top \delta \mu - \tilde{\omega}^\top b(\delta) - \tilde{\omega}^\top \log (1 + \tau) - \log \Gamma_L - \frac{1}{2} (\rho - 1) \tilde{\omega}^\top \delta \Sigma \delta^\top \tilde{\omega} - \\ & \frac{1}{\rho - 1} \log \left(1 - \pi_d + \pi_d \exp \left((\rho - 1) d \sum_{i=1}^N \delta_{i1} \tilde{\omega}_i \right) \right) + \log \left(V \left(\sum_{i=1}^N \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} g_i(\delta_i) \right) \right). \end{aligned} \quad (100)$$

Notice that we can write this problem as

$$\begin{aligned} \max_{\Delta} & \Delta^\top \mu - \tilde{\omega}^\top \log (1 + \tau) - \log \Gamma_L - \frac{1}{2} (\rho - 1) \Delta^\top \Sigma \Delta - \frac{1}{\rho - 1} \log (1 - \pi_d + \pi_d \exp ((\rho - 1) \Delta_1 d)) - \\ & \min_{\delta \text{ s.t. } \Delta = \delta^\top \tilde{\omega}} \left\{ \tilde{\omega}^\top b(\delta) - \log \left(V \left(\sum_{i=1}^N \frac{\tilde{\omega}_i (1 + \tau_i)}{\omega_i} g_i(\delta_i) \right) \right) \right\}, \end{aligned}$$

which is the maximization problem (89). Note also that the objective function of this problem is strictly concave because $\bar{h}(\Delta)$ is strictly convex. This implies that there is a unique solution to the maximization problems (100) and (89) and that the first-order conditions are sufficient to characterize it. To complete the proof, we will show that the equilibrium conditions coincide with these first-order conditions. This will imply that there is a unique equilibrium.

The first-order condition of the fictitious planner with respect to δ_{im} is

$$\delta_i - \delta_i^\circ = H_i^{-1} \left[\mu - (\rho - 1) \Sigma \Delta - \mathbf{1}_1 \frac{\pi_d d \exp((\rho - 1) \Delta_1 d)}{1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)} \right], \quad (101)$$

where H_i is given by (22) and $\mathbf{1}_1 = (1, 0, \dots, 0)$ is the $M \times 1$ basis vector. The system of these $M \times N$ equations fully characterizes the unique solution to the fictitious planner's problem.

Now consider the equilibrium. One can show that the firm's problem (19) is strictly convex, such that its first-order conditions are necessary and sufficient. As in the model of the main text, they are given by

$$\mathbb{E}[(\varepsilon - \nabla b_i(\delta_i)) K_i Q_i] + \text{Cov} \left((\varepsilon - \nabla b_i(\delta_i)) K_i Q_i, \frac{\Lambda}{\mathbb{E}[\Lambda]} \right) = \nabla g_i(\delta_i) W_R,$$

where we used the fact that $\partial K_i / \partial \delta_i = -(\varepsilon - \nabla b_i(\delta_i)) K_i$. Next, the definition of revenue-based Domar weights implies that $\omega_i = \frac{P_i Q_i}{PY} \Leftrightarrow Q_i = \frac{\omega_i \Gamma_L^{-1}}{P_i}$. Together with (12), we can therefore write $K_i Q_i = \omega_i \Gamma_L^{-1} / (1 + \tau_i)$, such that $K_i Q_i$ is deterministic. The first-order conditions become

$$\frac{dg_i}{d\delta_{im}} = \frac{1}{W_R} \frac{\mathbb{E} \left[\Lambda K_i Q_i \left(\varepsilon_m - \frac{db_i}{d\delta_{im}} \right) \right]}{\mathbb{E} \Lambda}.$$

Using (78) for the expression of Λ , we can simplify this equation to (101), which completes the proof. $\square$

Proof of Proposition 9

Proposition 9. *The responses of the equilibrium aggregate risk exposure Δ to changes in μ_m , Σ_{mn} , π_d , τ_i are given by*

$$\frac{d\Delta_k}{d\mu_m} = (\mathcal{H}^{-1})_{km} - \frac{\varphi(\mathcal{H}^{-1})_{k1}(\mathcal{H}^{-1})_{1m}}{1 + \varphi(\mathcal{H}^{-1})_{11}}, \quad (93)$$

$$\frac{d\Delta_k}{d\Sigma_{mn}} = -(\rho - 1) \left(\mathbf{1}_k^\top \mathcal{H}^{-1} - \frac{\varphi(\mathcal{H}^{-1})_{k1} \mathbf{1}_1^\top \mathcal{H}^{-1}}{1 + \varphi(\mathcal{H}^{-1})_{11}} \right) \left(\frac{1}{2} \mathbf{1}_m \Delta_n + \frac{1}{2} \mathbf{1}_n \Delta_m \right), \quad (94)$$

$$\frac{d\Delta_k}{d\pi_d} = -\frac{(\mathcal{H}^{-1})_{k1}}{1 + \varphi(\mathcal{H}^{-1})_{11}} \frac{d \exp((\rho - 1) \Delta_1 d)}{[1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)]^2}, \quad (95)$$

$$\frac{d\Delta_k}{d\tau_i} = -\mathbf{1}_k^\top \left(I - \frac{\varphi(\mathcal{H}^{-1})_{\cdot 1} \mathbf{1}_1^\top}{1 + \varphi(\mathcal{H}^{-1})_{11}} \right) \mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \nabla^2 \bar{h}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) (\nabla^2 \bar{h})^{-1} \mathcal{E}, \quad (96)$$

where $\mathcal{H}$ is given by (37), $(\mathcal{H}^{-1})_{\cdot 1} = ((\mathcal{H}^{-1})_{11}, \dots, (\mathcal{H}^{-1})_{M1})$, and

$$\varphi = \frac{(1 - \pi_d) \pi_d d^2 (\rho - 1) \exp((\rho - 1) \Delta_1 d)}{(1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d))^2}. \quad (97)$$

Proof. This proposition directly follows from the implicit function theorem applied to (92). Specifically, denote

$$\mathcal{F} = \Delta - \Delta^\circ - \bar{H}^{-1} \left(\mu - (\rho - 1) \Sigma \Delta - \mathbf{1}_1 \frac{\pi_d d \exp((\rho - 1) \Delta_1 d)}{1 - \pi_d + \pi_d \exp((\rho - 1) \Delta_1 d)} \right).$$

The Jacobian of $\mathcal{F}$ is

$$\begin{aligned} J_{\mathcal{F}} &= I + \bar{H}^{-1} (\rho - 1) \Sigma + \varphi (\bar{H}^{-1})_{\cdot 1} \mathbf{1}_1^\top = \\ &= \bar{H}^{-1} \left[\mathcal{H} + \varphi \mathbf{1}_1 \mathbf{1}_1^\top \right], \end{aligned}$$

where φ is given by (97). Using the Sherman-Morrison formula, we get

$$J_{\mathcal{F}}^{-1} = \left(I - \frac{\varphi (\mathcal{H}^{-1})_{\cdot 1} \mathbf{1}_1^\top}{1 + \varphi (\mathcal{H}^{-1})_{11}} \right) \mathcal{H}^{-1} \bar{H}.$$

Then, for a given parameter χ ,

$$\frac{d\Delta_k}{d\chi} = -\mathbf{1}_k^\top J_{\mathcal{F}}^{-1} \frac{\partial \mathcal{F}}{\partial \chi}.$$

In particular, if χ is μ_m , Σ_{mn} , or π_d , we have

$$\frac{d\Delta_k}{d\chi} = \mathbf{1}_k^\top \left(I - \frac{\varphi (\mathcal{H}^{-1})_{\cdot 1} \mathbf{1}_1^\top}{1 + \varphi (\mathcal{H}^{-1})_{11}} \right) \mathcal{H}^{-1} \frac{\partial \mathcal{E}}{\partial \chi},$$

where $\mathcal{E}$ is given by (91). Straightforward differentiation of $\mathcal{E}$ with respect to different parameters yields the result. If $\chi = \tau_i$, we have

$$\frac{d\Delta_k}{d\tau_i} = \mathbf{1}_k^\top \left(I - \frac{\varphi (\mathcal{H}^{-1})_{\cdot 1} \mathbf{1}_1^\top}{1 + \varphi (\mathcal{H}^{-1})_{11}} \right) \mathcal{H}^{-1} \bar{H} \frac{\partial \bar{H}^{-1}}{\partial \tau_i} \mathcal{E},$$

which immediately simplifies to (96). □

Proof of Corollary 6

Corollary 6. *In a diagonal economy, $\frac{d\Delta_k}{d\mu_k} > 0$, $\text{sign}\left(\frac{d\Delta_k}{d\Sigma_{kk}}\right) = -\text{sign}(\Delta_k)$, $\frac{d\Delta_1}{d\pi_d} < 0$, and $\text{sign}\left(\frac{d\Delta_k}{d\tau_i}\right) = -\text{sign}(\mathcal{E}_k)$.*

Proof. In a diagonal economy, $\mathcal{H}_{ij} = 0$ if $i \neq j$. Then

$$\frac{d\Delta_k}{d\mu_k} = \begin{cases} (\mathcal{H}^{-1})_{kk} & k \neq 1, \\ \frac{(\mathcal{H}^{-1})_{11}}{1+\varphi(\mathcal{H}^{-1})_{11}} & k = 1. \end{cases}$$

Since $\mathcal{H}^{-1}$ is a positive definite matrix, $\frac{d\Delta_k}{d\mu_k} > 0$.

Next,

$$\frac{d\Delta_k}{d\Sigma_{kk}} = \begin{cases} -(\rho-1)(\mathcal{H}^{-1})_{kk}\Delta_k & k \neq 1, \\ -(\rho-1)\frac{(\mathcal{H}^{-1})_{11}}{1+\varphi(\mathcal{H}^{-1})_{11}}\Delta_1 & k = 1, \end{cases}$$

and so $\text{sign}\left(\frac{d\Delta_k}{d\Sigma_{kk}}\right) = -\text{sign}(\Delta_k)$.

$\frac{d\Delta_1}{d\pi_d} < 0$ directly follows from (95). Finally,

$$\frac{d\Delta_k}{d\tau_i} = -\mathbf{1}_k^\top \begin{pmatrix} \frac{1}{1+\varphi(\mathcal{H}^{-1})_{11}} & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \mathcal{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i} \right) \bar{H}^{-1} \mathcal{E}. \quad (102)$$

In a diagonal economy, the matrix between $\mathbf{1}_k^\top$ and $\mathcal{E}$ is a diagonal matrix with a positive main diagonal. Hence, $\text{sign}\left(\frac{d\Delta_k}{d\tau_i}\right) = -\text{sign}(\mathcal{E}_k)$. $\square$

Proof of Corollary 7

Corollary 7. *Consider a diagonal efficient economy with no disaster risk, $\pi_d = 0$. An increase in the disaster probability π_d has the following effects on the sensitivity of Δ to μ_1 and Σ_{11} :*

$$\begin{aligned} \frac{d^2\Delta_1}{d\mu_1 d\pi_d} &= -[(\mathcal{H}^{-1})_{11}]^2 d^2(\rho-1) \exp((\rho-1)\Delta_1 d), \\ \frac{d^2\Delta_1}{d\Sigma_{11} d\pi_d} &= (\rho-1)[(\mathcal{H}^{-1})_{11}]^2 d(d(\rho-1)\Delta_1 + 1) \exp((\rho-1)\Delta_1 d). \end{aligned}$$

All other cross-derivatives are zero, i.e., $\frac{d^2\Delta_k}{d\mu_m d\pi_d} = \frac{d^2\Delta_k}{d\Sigma_{mn} d\pi_d} = 0$ unless $k = m = n = 1$.

Proof. First, notice that $\varphi(\pi_d)$ is given by (97), which implies

$$\frac{d\varphi}{d\pi_d} \stackrel{\pi_d=0}{=} d^2(\rho-1) \exp((\rho-1)\Delta_1 d).$$

Using this expression when differentiating (93) and (94) with respect to π_d , we get

$$\begin{aligned} \frac{d^2 \Delta_1}{d\mu_1 d\pi_d} \Big|_{\pi_d=0} &= [(\mathcal{H}^{-1})_{11}]^2 d^2 (\rho - 1) \exp((\rho - 1) \Delta_1 d) . \\ \frac{d^2 \Delta_1}{d\Sigma_{11} d\pi_d} \Big|_{\pi_d=0} &= (\rho - 1) \left[[(\mathcal{H}^{-1})_{11}]^2 d^2 (\rho - 1) \exp((\rho - 1) \Delta_1 d) \Delta_1 - (\mathcal{H}^{-1})_{11} \frac{d\Delta_1}{d\pi_d} \right] = \\ &= (\rho - 1) [(\mathcal{H}^{-1})_{11}]^2 d \exp((\rho - 1) \Delta_1 d) (d(\rho - 1) \Delta_1 + 1) , \end{aligned}$$

where $\frac{d\Delta_1}{d\pi_d}$ is given by (95). Clearly, in the diagonal economy all other cross-derivatives are zero because $\mathcal{H}$ is a diagonal matrix. $\square$

Proof of Corollary 8

Corollary 8. *Consider a diagonal efficient economy with no disaster risk, $\pi_d = 0$. Suppose also that the TFP cost of adjusting risk exposure is zero, $b_i = 0$ for all i . An increase in the disaster probability π_d has the following effects on the sensitivity of the moments of log GDP to $\chi \in \{\mu_1, \Sigma_{11}\}$:*

$$\begin{aligned} \frac{d}{d\pi_d} \left(\frac{dE[y|no disasters]}{d\chi} - \frac{\partial E[y|no disasters]}{\partial \chi} \right) &= \mu_1 \frac{d^2 \Delta_1}{d\pi_d d\chi}, \\ \frac{d}{d\pi_d} \left(\frac{dV[y|no disasters]}{d\chi} - \frac{\partial V[y|no disasters]}{\partial \chi} \right) &= 2\Sigma_{11} \left(\frac{d\Delta_1}{d\pi_d} \frac{d\Delta_1}{d\chi} + \frac{d^2 \Delta_1}{d\pi_d d\chi} \Delta_1 \right). \end{aligned}$$

Proof. From (17),

$$\begin{aligned} \frac{dE[y|no disasters]}{d\chi} - \frac{\partial E[y|no disasters]}{\partial \chi} &= \mu^\top \frac{d\Delta}{d\chi}, \\ \frac{dV[y|no disasters]}{d\chi} - \frac{\partial V[y|no disasters]}{\partial \chi} &= 2\Delta^\top \Sigma \frac{d\Delta}{d\chi}. \end{aligned}$$

In a diagonal economy, π_d has no effect on $\Delta_2, \dots, \Delta_M$. Furthermore, Σ is diagonal. Then the result follows from a straightforward differentiation of the above expressions with respect to π_d . $\square$

Proof of Corollary 9

Corollary 9. *Suppose that the economy is diagonal and that $\tau_j > 0$ for at least one firm j . Then the following holds.*

$$\begin{aligned} \text{sign} \left(\frac{d\mathcal{W}}{d\mu_m} - \frac{\partial \mathcal{W}}{\partial \mu_m} \right) &= \text{sign}(\mathcal{E}_m), \quad \text{sign} \left(\frac{d\mathcal{W}}{d\Sigma_{mm}} - \frac{\partial \mathcal{W}}{\partial \Sigma_{mm}} \right) = -\text{sign}(\Delta_m \mathcal{E}_m), \\ \text{sign} \left(\frac{d\mathcal{W}}{d\pi_d} - \frac{\partial \mathcal{W}}{\partial \pi_d} \right) &= -\text{sign}(\mathcal{E}_1). \end{aligned}$$

Furthermore, an increase in wedges is more detrimental to welfare when risk-exposure decisions can adjust, that is, $\frac{d\mathcal{W}}{d\tau_i} \leq \frac{\partial \mathcal{W}}{\partial \tau_i}$.

Proof. Using (86), we can write (84) as

$$\frac{d\mathcal{W}}{d\chi} - \frac{\partial\mathcal{W}}{\partial\chi} = \left(\frac{d\Delta}{d\chi}\right)^\top \bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right) \mathcal{E}. \quad (103)$$

From Proposition 9, in a diagonal economy we have $\frac{d\Delta_k}{d\mu_k} > 0$ and $\frac{d\Delta_k}{d\mu_m} = 0$ if $k \neq m$. Furthermore, $\frac{d\Delta_1}{d\pi_d} < 0$ and $\frac{d\Delta_k}{d\pi_d} = 0$ if $k \neq 1$. Finally, $\text{sign}\left(\frac{d\Delta_k}{d\Sigma_{kk}}\right) = -\text{sign}(\Delta_k)$ and $\frac{d\Delta_k}{d\Sigma_{mm}} = 0$ if $k \neq m$. The result for μ_m , Σ_{mm} , and π_d then follows from the fact that $\bar{H} \left(\sum_{i=1}^N \eta(\kappa_i - 1) H_i^{-1} G_i H_i^{-1}\right)$ is a diagonal matrix with a positive main diagonal.

Finally, from (102) we know that in a diagonal economy

$$\left(\frac{d\Delta}{d\tau_i}\right)^\top = -\mathcal{E}^\top \bar{H}^{-1} \left(\sum_{j=1}^N \frac{\partial \bar{H}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i}\right) \mathcal{H}^{-1} \begin{pmatrix} \frac{1}{1+\varphi(\mathcal{H}^{-1})_{11}} & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Combining this with (103), we get that $\frac{d\mathcal{W}}{d\tau_i} - \frac{\partial\mathcal{W}}{\partial\tau_i} \leq 0$ in a diagonal economy. $\square$

E Model with CES sectoral aggregator

In this appendix, we generalize the baseline model by replacing the Cobb-Douglas aggregation of firms within each sector with a CES demand system. The key new feature is that firms' sales shares within their sector are no longer constant: they respond endogenously to relative prices. When the elasticity of substitution σ_s exceeds one, firms that are more productive, and therefore charge lower prices, capture a larger share of their sector's revenue. This price sensitivity of market shares enriches the model and, as we show, preserves the overall structure of the baseline framework. We use this more flexible model when estimating firm-level TFP and markups in Section 8 and in the calibration exercise of Section 9.

We consider two related versions of the CES model. In the first, each firm behaves competitively: it does not internalize its impact on prices and so prices are equal to marginal costs plus an exogenous wedge. In the second version, the firm internalizes the CES demand curve it faces and, as a result, has market power and prices its good at an endogenous markup over its marginal cost. A central result of this appendix is that both versions of the model are governed by the same equations, with the only distinction being the source of the markup: exogenous in the competitive version or endogenous from monopolistic competition. We describe both models in parallel and establish their equivalence in Section E.5. All proofs for this appendix are in Section E.9.

E.1 Firms

There are S sectors. In each sector s , a competitive aggregator converts the output of the N_s firms in that sector into a sector-specific good according to the CES production function

$$Q_s = \left(\sum_{i=1}^{N_s} \theta_{si} \left(\frac{Q_{si}}{\theta_{si}} \right)^{\frac{\sigma_s-1}{\sigma_s}} \right)^{\frac{\sigma_s}{\sigma_s-1}}, \quad (104)$$

where $\sum_{i=1}^{N_s} \theta_{si} = 1$, $\sigma_s > 0$ is the elasticity of substitution between goods produced by different firms in sector s , and Q_{si} is the output of firm i in sector s . The parameters $\theta_{si} > 0$ govern the relative quality or taste weights of each firm's product. When $\sigma_s \rightarrow 1$, the aggregator reduces to the Cobb-Douglas specification of the baseline model; as σ_s increases, goods become more substitutable. Throughout, we assume that $\sigma_s \geq 1$, which is the empirically relevant case.

Firm i in sector s operates the Cobb-Douglas production function

$$Q_{si} = \exp(a_{si}(\delta_{si}, \varepsilon)) \left(\frac{L_{si}}{1 - \sum_{s'=1}^S \hat{\alpha}_{ss'}} \right)^{1 - \sum_{s'=1}^S \hat{\alpha}_{ss'}} \prod_{s'=1}^S \left(\frac{X_{si,s'}}{\hat{\alpha}_{ss'}} \right)^{\hat{\alpha}_{ss'}}, \quad (105)$$

where $X_{si,s'}$ denotes the use of the composite good of sector s' by firm i in sector s . The input elasticities $\hat{\alpha}$ are sector-specific to allow for a closer contact with the data.⁶⁸ The TFP of firm i in sector s is $a_{si}(\delta_{si}, \varepsilon)$, where $\varepsilon \sim \mathcal{N}(\mu, \Sigma)$. We write down an explicit equation for a_{si} below.

We think of the sectoral producers as an aggregation device and so we assume that they have no markups and that they make no risk-exposure decisions. Standard cost minimization for sectoral aggregator s yields the CES price index

$$P_s^{1-\sigma_s} = \sum_{i=1}^{N_s} \theta_{si} P_{si}^{1-\sigma_s}. \quad (106)$$

E.1.1 Firm pricing

The cost-minimization problem of firm i in sector s implies a standard unit cost. In the competitive version of the model, we assume that the firm prices its good at an exogenous markup τ_{si} over this cost, namely

$$P_{si} = (1 + \tau_{si}) K_{si} = \frac{1 + \tau_{si}}{\exp(a_{si}(\delta_{si}, \varepsilon))} \prod_{s'=1}^S P_{s'}^{\hat{\alpha}_{ss'}}. \quad (107)$$

This is the same pricing rule as in the baseline model. In Section E.5, we consider a model in which firms internalize the demand curve they face and show that this gives rise to an endogenous markup,

⁶⁸We do not have information about firm-to-firm linkages in Spain, so we rely on sectoral information to calibrate the model.

but that the resulting model is otherwise equivalent.

E.1.2 Sales shares

Under the CES sectoral aggregator, firms' sales shares within the sector are no longer constant but instead depend on relative prices:

$$\chi_{si} := \frac{P_{si}Q_{si}}{P_sQ_s} = \theta_{si} \frac{P_{si}^{1-\sigma_s}}{P_s^{1-\sigma_s}}. \quad (108)$$

If $\sigma_s > 1$, firm i 's sales share is higher when the price it charges is lower relative to its within-sector competitors. Intuitively, lower prices attract more demand, and since demand is elastic, the revenue gain from additional quantity more than offsets the revenue loss from the lower price. This is the key distinction from the baseline model: more productive firms not only have lower costs, but also capture a larger slice of their sector's revenue. Under Cobb-Douglas ($\sigma_s = 1$), the sales share is constant, $\chi_{si} = \theta_{si}$, regardless of the firm's relative productivity.

E.1.3 Sectoral prices

Combining (106) and (107), we can express sectoral prices in terms of firm-level productivities and markups. As in the baseline model, these prices depend on the full input-output network through the Leontief inverse:

$$p = -(I - \hat{\alpha})^{-1} \begin{pmatrix} \vdots \\ \frac{1}{\sigma_s - 1} \log \left(\sum_{i=1}^{N_s} \theta_{si} (\exp(a_{si}(\delta_{si}, \varepsilon) - \log(1 + \tau_{si})))^{\sigma_s - 1} \right) \\ \vdots \end{pmatrix}. \quad (109)$$

The expression inside the logarithm is a power mean of firm-level "adjusted productivity" $\exp(a_{si} - \log(1 + \tau_{si}))$, with the power governed by $\sigma_s - 1$. When $\sigma_s = 1$, this collapses to a weighted geometric average, recovering the Cobb-Douglas case. When $\sigma_s > 1$, more productive firms receive disproportionately higher weight in determining the sectoral price, reflecting the endogenous reallocation of market shares toward low-cost producers. The Leontief inverse $(I - \hat{\alpha})^{-1}$ then propagates these sectoral prices through the input-output network, capturing how a productivity improvement in one sector lowers costs for all its downstream users. With sectoral prices in hand, firm-level prices follow immediately from (107).

E.1.4 Risk-exposure choice of the competitive firm

The firm's problem unfolds in two stages. In the second stage, firm i chooses its inputs given its risk exposure and the shock realization. In the first stage, firm i in sector s chooses its risk exposure

δ_{si} to maximize expected discounted profits:

$$\delta_{si}^* \in \arg \max_{\delta_{si}} \mathbb{E} [\Lambda [P_{si} Q_{si} - K_{si}(\delta_{si}, P) Q_{si} - g_{si}(\delta_{si}) W_R]].$$

A key feature of the competitive version of the model is that the firm takes prices P_{si} and the demand for its good Q_{si} as given. It does not internalize how changes in its productivity (and therefore its unit cost) feed back into the price it charges and the demand it faces. From the firm's perspective, the sole benefit of higher TFP is a reduction in unit cost K_{si} .

Using the unit cost expression (107), we can write the first-order condition as

$$\frac{\partial g_{si}}{\partial \delta_{si,m}} W_R = \frac{1}{\mathbb{E}[\Lambda]} \mathbb{E} \left[\Lambda \frac{1}{1 + \tau_{si}} P_{si} Q_{si} \frac{\partial a_{si}}{\partial \delta_{si,m}} \right]. \quad (110)$$

The left-hand side is the marginal resource cost of deviating from the firm's natural risk exposure. The right-hand side is the expected benefit, discounted by the stochastic discount factor Λ : a marginal increase in exposure $\delta_{si,m}$ reduces the unit cost by $K_{si} \times \partial a_{si} / \partial \delta_{si,m}$, and this saving applies to Q_{si} units of output. Using $K_{si} = P_{si} / (1 + \tau_{si})$ from (107), the total cost saving is therefore $\frac{1}{1 + \tau_{si}} P_{si} Q_{si} \frac{\partial a_{si}}{\partial \delta_{si,m}}$. As we show in Section E.5, the monopolistically competitive firm, which does internalize the demand response, arrives at precisely the same first-order condition.

E.2 Household

The household's problem remains the same as in the baseline model. After uncertainty is realized, the household picks consumption to solve

$$\max_{\{C_s\}_{s=1}^S} \frac{1}{1 - \rho} \left(\prod_{s=1}^S \left(\frac{C_s}{\beta_s} \right)^{\beta_s} \right)^{1 - \rho} \mathcal{V}(R) \quad \text{s.t.} \quad \sum_{s=1}^S P_s C_s \leq 1 + W_R R + \Pi, \quad (111)$$

where $\Pi = \sum_{s=1}^S \sum_{i=1}^{N_s} ((P_{si} - K_{si}) Q_{si} - W_R R_{si})$ is the total profits earned by firms.

As before, we define nominal GDP as

$$\bar{P}Y = \sum_{s=1}^S P_s C_s = 1 + \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} P_{si} Q_{si}, \quad (112)$$

where the first term is the household's unit labor endowment and the second captures total markup revenue across all firms.

The first-order condition of (111) is

$$C^{-\rho} C \frac{\beta_s}{C_s} \mathcal{V}(R) = \Lambda P_s \implies C^{1-\rho} \mathcal{V}(R) = \Lambda \bar{P}Y, \quad (113)$$

where Λ is the Lagrange multiplier on the budget constraint. From this equation, we also get

$$C = \prod_{s=1}^S \left(\frac{C_s}{\beta_s} \right)^{\beta_s} = \bar{P}Y \prod_{s=1}^S P_s^{-\beta_s}, \quad (114)$$

which implies that the stochastic discount factor is

$$\Lambda = \mathcal{V}(R) (\bar{P}Y)^{-\rho} \left(\prod_{s=1}^S P_s^{\beta_s} \right)^{\rho-1}. \quad (115)$$

As in the baseline model, the SDF is high when the household is poor, which might correspond to a low nominal GDP $\bar{P}Y$ or a high price level

$$\bar{P} := \prod_{s=1}^S P_s^{\beta_s}.$$

Finally, R is chosen before uncertainty is realized, implying the following first-order condition:

$$\mathcal{V}'(R) = - \frac{\mathbb{E}[\Lambda]}{\mathbb{E}\left[\frac{C^{1-\rho}}{1-\rho}\right]} W_R.$$

Combining with (113), we get

$$\frac{\mathcal{V}'(R)}{\mathcal{V}(R)} = (\rho - 1) \frac{\mathbb{E}[\Lambda]}{\mathbb{E}[\Lambda \bar{P}Y]} W_R. \quad (116)$$

E.3 Market clearing

The market-clearing condition for good s is

$$Q_s = C_s + \sum_{s'=1}^S \sum_{i=1}^{N_{s'}} X_{s'i,s}.$$

Multiplying this expression by P_s and dividing by nominal GDP $\bar{P}Y$, we get

$$\omega^\top = \beta^\top (I - \tilde{\alpha})^{-1}, \quad (117)$$

where ω_s is the revenue-based Domar weight of sector s ,

$$\omega_s = \frac{P_s Q_s}{\bar{P}Y},$$

and

$$\tilde{\alpha}_{s's} = \hat{\alpha}_{s's} \sum_{i=1}^{N_{s'}} \frac{\chi_{s'i}}{1 + \tau_{s'i}}, \quad (118)$$

where χ_{si} is given by (108). The matrix $\tilde{\alpha}$ captures the *effective* input-output structure of the economy, adjusted for markups. A sector in which firms charge higher markups passes through a smaller fraction of its revenue to input purchases, reducing the effective intensity of intermediate-good linkages. Under the Cobb-Douglas specification ($\sigma_s \rightarrow 1$), $\chi_{si} = \theta_{si}$ is a constant, and so are $\tilde{\alpha}$ and ω_s . If $\sigma_s \neq 1$, sales shares χ_{si} respond to relative prices, making $\tilde{\alpha}$ and ω_s stochastic.

Using the definition of sectoral revenue-based Domar weights, we can write nominal GDP (112) as

$$\bar{P}Y = \frac{1}{1 - \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \chi_{si} \omega_s}. \quad (119)$$

E.4 Choice of risk exposure

As in the baseline model, we assume that $\mathcal{V}(R) = \exp((\rho - 1)\eta R)$. Furthermore, the TFP of firm i in sector s is

$$a_{si}(\delta_{si}) = \delta_{si}^\top \varepsilon - b_i(\delta_{si}) + \gamma_s t + v_{si}, \quad (120)$$

where $v_{si} \sim \text{iid } \mathcal{N}(0, \Sigma_{si}^v)$ is a firm-level idiosyncratic shock, and $\varepsilon \sim \mathcal{N}(\mu, \Sigma)$ collects the fundamental risk factors. The two cost functions are unchanged from the baseline:

$$b_{si}(\delta_{si}) = \frac{1}{2} (\delta_{si} - \delta_{si}^\circ)^\top B_{si} (\delta_{si} - \delta_{si}^\circ), \quad g_{si}(\delta_{si}) = \frac{1}{2} (\delta_{si} - \delta_{si}^\circ)^\top G_{si} (\delta_{si} - \delta_{si}^\circ). \quad (121)$$

For brevity, we define firm i 's revenue-based Domar weight (its nominal sales as a share of nominal GDP) as

$$\omega_{si} := \chi_{si} \omega_s = \frac{P_{si} Q_{si}}{\bar{P}Y}.$$

Unlike in the Cobb-Douglas baseline, where firm Domar weights are technological constants, ω_{si} is now *stochastic*: it depends on the realization of ε through equilibrium prices. Indeed, with $\sigma_s > 1$, a firm with high productivity charges a price below its sectoral average, captures a larger within-sector revenue share χ_{si} , and therefore grows in size ω_{si} . The expansion is greater the larger is the firm's exposure δ_{si} to ε : a firm heavily exposed to factor ε_m gains the most market share when ε_m is favorable, and loses the most when it is not. The strength of this co-movement is therefore firm-specific.

Substituting $P_{si} Q_{si} = \omega_{si} \bar{P}Y$ in the firm's first-order condition (110), exploiting the quadratic forms in (121), and using the household's optimality condition (116), the firm's exposure choice can

be written⁶⁹

$$\delta_{si} = \delta_{si}^{\circ} + H_{si}^{-1} \mathcal{E}_{si}, \quad (122)$$

where the firm-specific marginal value of risk exposure is

$$\mathcal{E}_{si} := \frac{\mathbb{E} [\Lambda \omega_{si} \bar{P} Y \varepsilon]}{\mathbb{E} [\Lambda \omega_{si} \bar{P} Y]}, \quad (123)$$

and the firm-specific curvature matrix of its effective cost of exposure is

$$H_{si} := B_{si} + (1 + \tau_{si}) \eta \frac{\mathbb{E} [\Lambda \bar{P} Y]}{\mathbb{E} [\Lambda \omega_{si} \bar{P} Y]} G_{si}. \quad (124)$$

Equation (122) has the same structure as its baseline counterpart (24): the firm departs from its natural exposure δ_{si}° in the direction of the risk-adjusted benefit $\mathcal{E}_{si}$, with the size of the deviation governed by the curvature H_{si}^{-1} of its effective cost function. What changes in the CES setting is the content of these two objects. In the baseline, $\mathcal{E}$ is an aggregate quantity, common to all firms, and the curvature H_{si}^{-1} only depends on parameters. Here, both objects are firm-specific and involve expectations over endogenous random quantities. These differences come from the fact that under CES, a firm's revenue $\omega_{si} \bar{P} Y$ co-moves with productivity shocks in a way that is unique to that firm.

Consider $\mathcal{E}_{si}$ first. The marginal benefit of higher productivity to firm i is the cost saving on production, $K_{si} Q_{si} \partial a_{si} / \partial \delta_{si}$, which scales with the firm's cost of goods sold. In the baseline, $K_{si} Q_{si}$ is deterministic, so it factors out of the firm's first-order condition. Under CES, $K_{si} Q_{si}$ is itself stochastic and co-moves with ε . The firm therefore cares about productivity in proportion to *how big it expects to be in each state*, and the size adjustment differs across firms because each firm's size tracks its own exposure δ_{si} . The marginal benefit of exposure, given by (123), formalizes this: $\mathcal{E}_{si}$ is the conditional expectation of ε under a probability measure tilted by the product of the household's marginal utility of consumption (Λ , as in the baseline) and the firm's own nominal revenue ($\omega_{si} \bar{P} Y$, the new ingredient). Similarly for H_{si} , the importance of the TFP cost of risk exposure $b_i(\delta_{si})$ depends on the size of the firm, which now varies across states of the world. The expression for H_{si} must therefore be adjusted so that the relative importance of the productivity (B_{si}) and resource (G_{si}) costs take this into account.

These forces generate a positive feedback in risk taking. A firm with high exposure to factor ε_m tends to be relatively large in states where ε_m realizes favorably. The tilted expectation in (123) therefore puts more weight on those states, which raises the firm's perceived attractiveness of ε_m , which in turn justifies the high exposure. The strength of this feedback grows with the elasticity

⁶⁹Note that δ_{si} is time-independent despite the presence of sectoral trends in (120). This is because all shocks are iid, and χ_{si} does not depend on sectoral trends. As a result, all other endogenous objects in (122) do not depend on sectoral trends.

σ_s and with the firm's existing exposure. It is muted under Cobb-Douglas, where exposure has no effect on relative size.

Because χ_{si} , ω_s , and $\bar{P}Y$ all depend on the realization of ε through equilibrium prices, and prices in turn depend on every firm's exposure decision, the expectations in (123)–(124) couple firm i 's decision to those of all other firms. Equation (122) therefore defines a fixed-point system over the entire vector of risk exposures. This precludes a closed-form solution. We instead solve this system numerically; Appendix E.6 develops a perturbation around the deterministic ($\Sigma = 0$) economy that yields a tractable approximation (51) that we use in our calibration.

Note that all elements of (122) have been derived above. Specifically, sectoral prices P_s are given by (109). Using these prices, we can determine firm-level prices P_{si} via (107), and firm sales shares χ_{si} via (108). Sectoral revenue-based Domar weights ω_s can be obtained by combining (117) and (118). Nominal GDP $\bar{P}Y$ is given by (119). Finally, the SDF Λ is given by (115). Therefore, (122) implicitly determines the vector of risk exposures for all firms.

E.5 A model in which firms have market power

In Section E.1, firms were price takers and the markup τ_{si} was treated as exogenous. We now consider an alternative formulation in which firms have market power: each firm internalizes the CES demand curve it faces and chooses its price (equivalently, its quantity) to maximize expected discounted profits. The main result of this subsection is that the two formulations are equivalent. The equilibrium objects, including risk-exposure choices, are governed by the same equations, provided τ_{si} is reinterpreted to absorb the endogenous markup that arises from monopolistic competition. This equivalence is what allows us to use empirical estimates of firm-level markups in our calibration without committing to any particular interpretation of where they come from.

E.5.1 Pricing

Cost minimization implies the same unit cost as in the competitive case,

$$K_{si} = \frac{1}{\exp(a_{si}(\delta_{si}))} \prod_{s'=1}^S P_{s'}^{\hat{\alpha}_{ss'}}.$$

With market power, firm i in sector s chooses its quantity Q_{si} to solve

$$\Pi_{si} = \max_{Q_{si}} \frac{1}{1 + \hat{\tau}_{si}} P_{si}(Q_{si}) Q_{si} - K_{si} Q_{si}, \quad (125)$$

where $\hat{\tau}_{si} \in (-1, \infty)$ is an exogenous wedge on revenue, e.g., due to taxes. Setting $\hat{\tau}_{si} = 0$ recovers the standard monopolistic-competition problem.

The inverse demand curve $P_{si}(Q_{si})$ is obtained by inverting the firm's CES demand $Q_{si} =$

$\theta_{si} (P_{si}/P_s)^{-\sigma_s} Q_s$ and is given by

$$P_{si} (Q_{si}) = P_s Q_s^{1/\sigma_s} \left(\frac{Q_{si}}{\theta_{si}} \right)^{-1/\sigma_s}.$$

This is the standard downward-sloping CES demand curve with constant elasticity σ_s . The associated marginal revenue is $\frac{1}{1+\hat{\tau}_{si}} \frac{\sigma_s-1}{\sigma_s} P_{si}$, and equating it to marginal cost K_{si} in (125) gives the familiar pricing rule

$$P_{si} = \frac{\sigma_s}{\sigma_s - 1} (1 + \hat{\tau}_{si}) K_{si}. \quad (126)$$

The price differs from the marginal cost by the product of two markups. The first, $\sigma_s/(\sigma_s - 1)$, is the standard CES markup that the firm chooses by restricting output along its demand curve. The second, $1 + \hat{\tau}_{si}$, is the exogenous wedge. Defining the combined markup

$$1 + \tau_{si} := \frac{\sigma_s}{\sigma_s - 1} (1 + \hat{\tau}_{si}),$$

the pricing equation (126) reduces to $P_{si} = (1 + \tau_{si}) K_{si}$, which is exactly the pricing rule of the competitive case in (107). The two models therefore generate identical pricing equations; only the interpretation of τ_{si} differs. In the competitive model, τ_{si} is fully exogenous. Here, it is the product of an endogenous component $\sigma_s/(\sigma_s - 1)$ and any additional exogenous wedge $1 + \hat{\tau}_{si}$.

E.5.2 Risk-exposure choice

The firm's stage-one problem is to choose δ_{si} to maximize expected discounted profits net of risk-management expenses,

$$\delta_{si}^* \in \arg \max_{\delta_{si}} \mathbb{E} [\Lambda (\Pi_{si} (\delta_{si}) - g_{si} (\delta_{si}) W_R)], \quad (127)$$

where $\Pi_{si} (\delta_{si})$ is the value of the inner problem (125) evaluated at the firm's optimal quantity. We now show that the resulting first-order condition for δ_{si} is identical to the competitive-case FOC (110), so the entire analysis of Section E.4 carries over.⁷⁰

The cleanest way to see this is via the envelope theorem. In problem (125), K_{si} enters only as a parameter that the firm takes as given when it chooses Q_{si} . The envelope theorem therefore implies

$$\frac{\partial \Pi_{si}}{\partial K_{si}} = -Q_{si},$$

i.e., a marginal increase in unit cost reduces profit by exactly the amount of output produced—the optimal Q_{si} adjustment has zero first-order effect at the optimum. Chaining through K_{si} and using

⁷⁰Concavity of (127) requires that the TFP risk management cost be sufficiently steep, i.e., $B_{si} > (\sigma_s - 1) \Sigma$. We therefore impose that condition in our calibration exercise.

$\partial K_{si}/\partial \delta_{si,m} = -K_{si} \partial a_{si}/\partial \delta_{si,m}$, we obtain

$$\frac{\partial \Pi_{si}}{\partial \delta_{si,m}} = \frac{\partial \Pi_{si}}{\partial K_{si}} \frac{\partial K_{si}}{\partial \delta_{si,m}} = K_{si} Q_{si} \frac{\partial a_{si}}{\partial \delta_{si,m}}. \quad (128)$$

Substituting (128) into the FOC of (127) and dividing by $E[\Lambda]$ reproduces the competitive FOC (110) exactly. The closed-form expression (122), with $\mathcal{E}_{si}$ and H_{si} as in (123)–(124), therefore applies unchanged.

The envelope step makes the deeper economic mechanism transparent. Both the competitive and the monopolistic firm benefit from a productivity gain through the same channel: a saving on the unit cost of producing a given output. In the competitive case, the firm takes Q_{si} as given, and the saving accrues directly. In the monopolistic case, the firm *also* adjusts its quantity in response to the lower marginal cost—it lowers its price and produces more—but because the firm was already at its profit-maximizing quantity, that adjustment has zero first-order value at the optimum. What remains is the same direct cost saving as in the competitive case, $K_{si} Q_{si} \partial a_{si}/\partial \delta_{si,m}$. The two firm types therefore have identical risk-management incentives.

This equivalence has a useful empirical implication. In our calibration, we estimate firm-level markups τ_{si} from the data (Section 8). The estimated τ_{si} absorbs both the CES component $\sigma_s/(\sigma_s - 1)$ and any exogenous wedge $\hat{\tau}_{si}$. Because the model’s predictions and comparative statics depend only on τ_{si} , we do not need to take a stance on how the empirical markup decomposes between these two sources.

E.6 Small-risk approximation

Equation (122) defines δ_{si} implicitly: the expectations on its right-hand side depend on the joint distribution of Λ , ω_{si} , and $\bar{P}Y$, which in turn depends on every firm’s risk-exposure decision through equilibrium prices. The exact CES model is therefore a fixed point in δ -space and admits no closed-form solution. To obtain tractable analytical results we linearize around the deterministic ($\Sigma = 0$) economy, in which ε is identically zero and all expectations collapse to deterministic values. Specifically, we assume $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small.⁷¹ The approximation is accurate when aggregate shocks are not too large. Throughout, we also assume that firms are small relative to their sectors, so that a law of large numbers applies.

In this regime, δ_{si} admits the expansion

$$\delta_{si} = \delta_{si}^\circ + \delta_{si}^{(1)} + o(\Sigma),$$

where the zero-order term equals the firm’s natural exposure δ_{si}° (with $\Sigma = 0$ and $\mu = 0$, the firm has no incentive to depart from δ_{si}°) and $\delta_{si}^{(1)}$ is linear in Σ and captures the firm’s first-order response

⁷¹We set the mean of ε to zero, in line with the assumption we make in the quantitative section of the paper.

once aggregate risk turns on.

Lemma 10. *Suppose that $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small. Then*

$$\delta_{si} \approx \delta_{si}^\circ + H_{si}^{-1} \mathcal{E}_{si}^\circ, \quad (129)$$

where $\mathcal{E}_{si}^\circ = \Sigma D_{si}(\delta^\circ)$ with $D_{si}(\delta^\circ)$ defined in the proof, and where

$$H_{si} = (1 + \tau_{si}) \frac{1}{\bar{\chi}_{si} \bar{\omega}_s} \eta G_{si} + B_{si},$$

where $\bar{\chi}_{si}$ and $\bar{\omega}_s$ are evaluated at $\varepsilon = 0$.

Equation (129) has the same shape as the exact result (122): the firm departs from δ_{si}° in the direction $H_{si}^{-1} \mathcal{E}_{si}^\circ$. What changes is that the two objects on the right-hand side are now evaluated at the deterministic equilibrium, which is what makes them tractable. The curvature H_{si} is the exact-CES expression (124) with the expectation ratio $\mathbb{E}[\Lambda \bar{P}Y] / \mathbb{E}[\Lambda \omega_{si} \bar{P}Y]$ collapsed to its deterministic value $1/\bar{\omega}_{si} = 1/(\bar{\chi}_{si} \bar{\omega}_s)$. The marginal value $\mathcal{E}_{si}^\circ$ is the first-order term in the Taylor expansion of (123) in Σ around the deterministic equilibrium: at $\Sigma = 0$, $\mathcal{E}_{si}$ vanishes (no aggregate risk, hence no risk-adjustment incentive), and at small Σ it scales linearly through a coefficient $D_{si}(\delta^\circ)$ that depends only on the natural exposures of all firms and the deterministic-equilibrium sectoral objects.⁷²

The dependence of D_{si} on the firm's own δ_{si}° is the small-risk version of the positive-feedback mechanism described in Section E.4. Under CES with $\sigma_s > 1$, a firm whose natural exposure exceeds the sectoral average has market share χ_{si} that co-moves positively with ε : when ε is favorable, its relative productivity rises, its relative price falls, and it captures more sectoral revenue. The firm's revenue therefore weights states with high ε more heavily, raising its perceived attractiveness of ε . Because the strength of this co-movement depends on δ_{si}° , $\mathcal{E}_{si}^\circ$ inherits the firm specificity. In the Cobb-Douglas limit $\sigma_s \rightarrow 1$, market shares no longer respond to relative prices, the channel disappears, and $\mathcal{E}_{si}^\circ$ collapses to the common vector

$$\mathcal{E}^\circ = -\Sigma(\rho - 1) \sum_{s=1}^S \tilde{\omega}_s \sum_{i=1}^{N_s} \chi_{si} \delta_{si}^\circ,$$

which coincides with $\mathcal{E}^\circ$ in the baseline model (Proposition 2).

A key prediction of the baseline model is that firms with higher Domar weights and lower wedges manage risk more aggressively (Corollary 1). Lemma 10 allows us to show that this prediction extends to the CES model.

⁷²The firm's response $\delta_{si}^{(1)} = H_{si}^{-1} \mathcal{E}_{si}^\circ$ is $O(\Sigma)$. Other firms' responses are also $O(\Sigma)$. The cross-firm interaction, i.e., firm i 's response depending on others' responses through equilibrium prices, is the product of two such terms and is therefore $O(\Sigma^2)$. Discarding it leaves a clean linear system: each firm responds to the deterministic-equilibrium $\mathcal{E}_{si}^\circ$ as if other firms remained at δ° .

Lemma 11. *Suppose that $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small. Firms with higher steady-state Domar weights $\bar{\omega}_{si} = \bar{\chi}_{si}\bar{\omega}_s$ and lower wedges τ_{si} manage risk more aggressively, in the sense that⁷³*

$$\frac{\partial \left[(\mathcal{E}_{si}^\circ)^\top \delta_{si} \right]}{\partial \bar{\omega}_{si}} > 0 \quad \text{and} \quad \frac{\partial \left[(\mathcal{E}_{si}^\circ)^\top \delta_{si} \right]}{\partial \tau_{si}} < 0.$$

In the lemma, we differentiate with respect to the steady-state Domar weight $\bar{\omega}_{si}$, which is itself an endogenous object. We can instead differentiate with respect to the exogenous parameter θ_{si} . From (108), $\bar{\omega}_{si} = \bar{\chi}_{si}\bar{\omega}_s$ is monotone in θ_{si} at the deterministic equilibrium, so the comparative static in θ_{si} translates directly to one in $\bar{\omega}_{si}$.

We now turn to output.

Lemma 12. *Suppose that $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small. Suppose also that firm-specific shocks are small, such that $v_{si} \sim \text{iid } \mathcal{N}(0, \Sigma_{si}^v)$ with Σ_{si}^v small. Then*

$$y \approx \bar{y} + \Delta^\top \varepsilon + \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} v_{si}, \quad (130)$$

where $\bar{y}$ does not depend on shocks, $\Delta = \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} \delta_{si}$, and ς_{si} is defined in the proof.

The log-linearized expression (130) is analogous to its baseline counterpart, given by (16). A notable difference is that the aggregate risk exposure Δ is now given by the sum of individual risk exposures weighted by ς_{si} , while in the baseline model those weights are simply firm cost-based Domar weights. The difference is due to the fact that, in the CES model, shocks affect firm Domar weights and, therefore, markup rebates to the household (second term in (112)). In the Cobb-Douglas model, Domar weights are fixed and do not depend on shocks, and so this channel is muted. In the absence of markups (i.e., if $\tau_{si} = 0$ for all firms), ς_{si} simplifies to cost-based Domar weight, as is evident from (142) in the proof of Lemma 12.

E.7 Specialization for the reduced-form analysis

The reduced-form analysis of Section 8 works in a specialization of the CES economy: a single aggregate risk factor, iid over time, and firm-specific idiosyncratic shocks. We apply Lemmas 10–12 to that setting, derive the moments of TFP growth used in the regressions, and verify that the cross-sectional predictions hold.

We assume a single aggregate risk factor $\varepsilon_t \sim \text{iid } \mathcal{N}(0, \Sigma)$ and a TFP process

$$\log TFP_{si,t} = \delta_{si} \varepsilon_t - b_i(\delta_{si}) + \gamma_s t + v_{si,t},$$

⁷³The partial derivatives imply a partial equilibrium analysis in which all sector- and economy-wide objects are taken as given.

where $v_{si,t} \sim \text{iid} \mathcal{N}(\mu_{si}^v, \Sigma_{si}^v)$ is a firm-specific idiosyncratic shock and γ_s is a deterministic trend. Risk-taking decisions are time-invariant in this setup: ε_t and $v_{si,t}$ are iid and the additive trends $\gamma_s t$ cancel in the firm's first-order condition (see the footnote following (122)). We therefore drop the time index and write δ_{si} .

Specializing Lemma 10 to one risk factor, all relevant matrices reduce to scalars. Letting $\bar{\omega}_{si} := \bar{\chi}_{si} \bar{\omega}_s$ denote firm i 's revenue-based Domar weight at $\varepsilon = 0$, the firm's exposure decision satisfies

$$\delta_{si} \approx \delta_{si}^\circ + \left(B_{si} + \eta \frac{1 + \tau_{si}}{\bar{\omega}_{si}} G_{si} \right)^{-1} \mathcal{E}_{si}^\circ, \quad (131)$$

where $\mathcal{E}_{si}^\circ = \Sigma D_{si}(\delta^\circ)$ is now a scalar, with D_{si} given by (140). As discussed in Section E.6, $\mathcal{E}_{si}^\circ$ is firm-specific whenever $\sigma_s > 1$: firms whose natural exposure differs from the sectoral average have market shares that co-move with ε , and they internalize this when choosing δ_{si} . In the Cobb-Douglas limit $\sigma_s \rightarrow 1$, $\mathcal{E}_{si}^\circ$ collapses to the common scalar $\mathcal{E}^\circ$ of Proposition 2, and (131) reduces to the simpler expression (45) used in the main text.

The qualitative predictions tested in Section 8, namely that firm-level TFP volatility and its covariance with GDP fall with size and rise with markups, do not require the Cobb-Douglas limit, as we show below.

Variance and covariance of TFP growth Differencing the TFP process and using the iid structure of ε_t and $v_{si,t}$, we can write

$$\begin{aligned} \text{V} [\log TFP_{si,t} - \log TFP_{si,t-1}] &= \text{V} [\delta_{si} (\varepsilon_t - \varepsilon_{t-1}) + v_{si,t} - v_{si,t-1}] \\ &= 2\delta_{si}^2 \Sigma + 2\Sigma_{si}^v. \end{aligned}$$

The deterministic trend $\gamma_s t$ and the productivity-cost term $b_i(\delta_{si})$ disappear when taking the variance.

For the covariance with GDP growth, Lemma 12 gives the approximated equation for GDP. Differencing and using mutual independence of the shocks gives

$$\text{Cov} [\log TFP_{si,t} - \log TFP_{si,t-1}, y_t - y_{t-1}] \approx 2\delta_{si} \Delta \Sigma + 2\varsigma_{si} \Sigma_{si}^v.$$

This delivers a version of (46), with ς_{si} playing the role of the cost-based Domar weight $\tilde{\omega}_{si}$ written in the main-text expression. The two coincide when $\tau_{si} = 0$; under markups, ς_{si} also captures how shocks reshuffle markup rebates to the household.

Comparative statics of the firm's exposure We can now derive the cross-sectional predictions tested in Section 8. The empirically relevant case for a single business-cycle risk factor has $\Delta > 0$, $\mathcal{E}_{si}^\circ < 0$, and $\delta_{si} > 0$: the household dislikes the procyclical variation in ε , and individual firms are

nonetheless on average positively exposed to it.⁷⁴ Lemma 11 specialized to one factor then implies $\partial\delta_{si}/\partial\bar{\omega}_{si} < 0$ and $\partial\delta_{si}/\partial\tau_{si} > 0$.⁷⁵ Larger firms (higher $\bar{\omega}_{si}$) take less risk; higher-markup firms take more risk. This is the within-CES analogue of the baseline Corollary 1.

Since $V[\Delta \log TFP_{si}] = 2\delta_{si}^2\Sigma + 2\Sigma_{si}^v$ is monotone in δ_{si}^2 and $\delta_{si} > 0$, the variance inherits the comparative statics of δ_{si} : it falls with firm size and rises with markups, which are the empirical pattern documented in Section 8.

The covariance decomposes into a systematic and an idiosyncratic component,

$$\text{Cov}[\Delta \log TFP_{si}, \Delta y_t] = \underbrace{2\delta_{si}\Delta\Sigma}_{\text{systematic}} + \underbrace{2\varsigma_{si}\Sigma_{si}^v}_{\text{idiosyncratic}}.$$

The systematic component inherits the comparative statics of δ_{si} : with $\Delta > 0$, it falls with $\bar{\omega}_{si}$ and rises with τ_{si} . The idiosyncratic component depends on ς_{si} , which is itself increasing in firm size, and therefore pulls the covariance *up* with $\bar{\omega}_{si}$. For the granular Spanish economy, however, this second contribution is quantitatively small: ς_{si} is at most on the order of a firm-level Domar weight (well below 1% even for the largest Spanish firms), while the systematic term is order one in the relevant parameters. The systematic component therefore dominates the cross-sectional variation in the covariance, and the predictions match those for the variance: covariance with GDP falls with size and rises with markups.

E.8 Disaster risk in the CES framework

The CES/Gaussian model studied so far is observationally close to the classical setup of Lucas Jr (1978) and, like it, struggles to match standard asset-pricing moments such as the equity premium (Mehra and Prescott, 1985). In this section we extend the model to incorporate *disaster risk* in the spirit of Rietz (1988) and Barro (2006): a low-probability, large-magnitude downward shock that gives the model enough non-Gaussian downside risk to match equity-premium moments. We use this extension to study the asset-pricing implications of endogenous risk in the calibration. Our analysis is similar to that of Appendix D which studies disasters in the Cobb-Douglas model.

We assume two independent risk factors. The first is the normal Gaussian risk $\varepsilon \sim \mathcal{N}(0, \Sigma)$ as before. The second is a binary disaster risk

$$\varepsilon^d = \begin{cases} 0 & \text{with probability } 1 - \pi_d, \\ -d & \text{with probability } \pi_d. \end{cases}$$

⁷⁴In our calibration, $\delta_{si} > 0$ and $\mathcal{E}_{si}^\circ < 0$ for most firms.

⁷⁵The PE assumption underlying the lemma holds $\mathcal{E}_{si}^\circ$ fixed in θ_{si} and τ_{si} : the focal firm is small relative to its sector, so changes in its parameters do not move the within-sector aggregates that enter $\mathcal{E}_{si}^\circ$.

with $d > 0$ the disaster size and $\pi_d \in (0, 1)$ its probability. Firm i in sector s has TFP

$$a_{si}(\delta_{si}) = \delta_{si} \left(\varepsilon + \varepsilon^d \right) + \bar{\delta} \varepsilon^d - b_i(\delta_{si}) + \gamma_s t + v_{si}.$$

The firm's risk-exposure choice δ_{si} shifts its loading on both normal and disaster risk in a tied way: it has exposure δ_{si} to the normal factor and $\delta_{si} + \bar{\delta}$ to the disaster factor. The exogenous baseline $\bar{\delta} \geq 0$ captures a disaster vulnerability common to all firms that the firm cannot adjust.⁷⁶

Risk-exposure decision. The analysis of Sections E.1–E.4 carries over without modification because we never imposed a specific shock distribution there. To obtain tractable analytical results in the new setting we extend the small-risk approximation of Section E.6 to also assume that π_d is small. Under this approximation, we can solve for δ_{si} as follows.

Lemma 13. *Suppose that Σ and π_d are small. Then*

$$\delta_{si} \approx \delta_{si}^\circ + H_{si}^{-1}(\mathcal{E}_{si}^\circ + \mathcal{F}_{si}^\circ), \quad (132)$$

where $\mathcal{E}_{si}^\circ = \Sigma D_{si}(\delta^\circ)$ and $\mathcal{F}_{si}^\circ = \pi_d D_{si}^d(\delta^\circ)$ with $D_{si}(\delta^\circ)$ and $D_{si}^d(\delta^\circ)$ defined in the proof, and where

$$H_{si} = (1 + \tau_{si}) \frac{1}{\bar{\chi}_{si} \bar{\omega}_s} \eta G_{si} + B_{si},$$

where $\bar{\chi}_{si}$ and $\bar{\omega}_s$ are evaluated at $\varepsilon = \varepsilon^d = 0$.

The structure mirrors Lemma 10: the firm's deviation from δ_{si}° is a marginal value of exposure scaled by an inverse curvature. The novelty is that the marginal value now has two additive pieces. The first, $\mathcal{E}_{si}^\circ$, is the same normal-risk premium as in Section E.6, scaled by Σ . The second, $\mathcal{F}_{si}^\circ$, is a new *disaster premium*, scaled by π_d . It captures, via the same SDF-weighted-revenue tilting structure as $\mathcal{E}_{si}^\circ$, how the firm's revenue shifts between the no-disaster and disaster regimes, multiplied by the disaster size. In the empirically relevant case where firms lose productivity in a disaster ($\delta_{si}^\circ + \bar{\delta} > 0$), the household's marginal utility is high in disaster states (consumption is low), so cash flows there are particularly valuable. Since $\mathcal{F}_{si}^\circ$ is negative, this reinforces the normal-risk premium: the firm reduces δ_{si} further than $\mathcal{E}_{si}^\circ$ alone would dictate. The curvature H_{si} is unchanged: as in Section E.6, it is the deterministic-equilibrium evaluation of the exact-CES Hessian.

GDP with disasters We can also express how GDP evolves in the economy with disasters.

Lemma 14. *Suppose that Σ and π_d are small. Suppose also that firm-specific shocks are small, such that $v_{si} \sim iid \mathcal{N}(0, \Sigma_{si}^v)$ with Σ_{si}^v small. Then, conditional on the disaster or no-disaster regime,*

⁷⁶We include this exogenous baseline to better fit the data. Without it, in a disaster the productivity of firms with large negative δ_{si} would rise sharply, potentially generating an economic boom. The exogenous baseline $\bar{\delta}$ limits that effect. We explain in Appendix B how we calibrate its value.

$r \in \{d, nd\}$, the log output is

$$y^r \approx \bar{y}^r + \Delta^r \varepsilon + \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si}^r v_{si}, \quad (133)$$

where $\bar{y}^r$ does not depend on ε and v_{si} , $\Delta^r = \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si}^r \delta_{si}$, and ς_{si}^r is defined in the proof.

Conditional on the regime, log GDP is linear in shocks just as in Lemma 12. The new feature is that all the regime-conditional objects (the deterministic level $\bar{y}^r$, the aggregate exposure Δ^r , and the firm weights ς_{si}^r) depend on whether we are in the disaster regime. Productivity drops in a disaster, so $\bar{y}^d < \bar{y}^{nd}$. Relative firm sizes shift, too: more disaster-exposed firms (those with larger $\delta_{si} + \bar{\delta}$) shrink in the disaster regime, and the aggregate weights ς_{si}^r tilt away from them. As a result, Δ^d generally differs from Δ^{nd} : the economy's aggregate exposure to normal risk is computed at the post-disaster firm composition, not the deterministic one.

Asset pricing We now turn to the asset-pricing implications of the CES model with disasters. The asset-pricing formulas below depend on two summary statistics of the regime-mixture distribution. The conditional variance of log GDP within regime $r \in \{d, nd\}$ is

$$\sigma_r^2 = (\Delta^r)^2 \Sigma + \sum_{s=1}^S \sum_{i=1}^{N_s} (\varsigma_{si}^r)^2 \Sigma_{si}^v,$$

combining the contribution of the normal aggregate factor and of all firm-specific shocks. The disaster's average impact on log GDP is the gap between the deterministic levels in the two regimes,

$$\vartheta = \bar{y}^{nd} - \bar{y}^d.$$

All the inputs needed to compute σ_r^2 and ϑ are derived in the proof of Lemma 14.

Lemma 15. *Suppose that the log output satisfies (133). Denote by $\gamma = \sum_{s=1}^S \tilde{\omega}_s \gamma_s$ the growth rate of GDP. Then the risk-free rate and the premium on the consumption claim are, respectively,*

$$\begin{aligned} \log E R_{f,t+1} &= r + \rho \gamma, \\ \log E R_{c,t+1} - \log E R_{f,t+1} &= \log M(-\rho) + \log \left[\varphi M(\rho) + (1 - \varphi) \frac{M(1)}{M(1-\rho)} \right], \end{aligned}$$

where $\varphi = \exp(-r - (\rho - 1)\gamma)$ and

$$M(x) = (1 - \pi_d) \exp\left(\frac{1}{2} x^2 \sigma_{nd}^2\right) + \pi_d \exp\left(-x\vartheta + \frac{1}{2} x^2 \sigma_d^2\right).$$

The risk-free rate formula $\log E R_{f,t+1} = r + \rho \gamma$ is the textbook CCAPM expression: the unconditional expected risk-free return is the time discount rate plus a growth premium. The equity

premium has more structure. The function M is the moment-generating function (MGF) of the de-trended log-GDP fluctuation $s_t := y_t - y_0 - \gamma t$:

$$M(x) = \mathbb{E}[\exp(xs_t)] = \underbrace{(1 - \pi_d) \exp\left(\frac{1}{2}x^2\sigma_{nd}^2\right)}_{\text{normal regime}} + \underbrace{\pi_d \exp\left(-x\vartheta + \frac{1}{2}x^2\sigma_d^2\right)}_{\text{disaster regime}}.$$

Each regime contributes a Gaussian MGF weighted by its probability, with the disaster regime shifted down by ϑ . In the no-disaster limit $\pi_d = 0$, $M(x)$ collapses to the standard log-normal MGF $\exp(\frac{1}{2}x^2\sigma_{nd}^2)$ and the equity premium reduces to the classical Lucas Jr (1978)/Mehra and Prescott (1985) formula, which is well known to be too small to match the data. The disaster term in $M(x)$ is what changes the picture. For $x = -\rho$, the disaster contribution to M scales as $\pi_d \exp(\rho\vartheta)$, exponential in the product of risk aversion and disaster size. With reasonable values ($\rho \approx 5$, $\vartheta \approx 30\%$), even a small disaster probability $\pi_d \approx 1\%$ adds an order of magnitude more to $M(-\rho)$ than the entire Gaussian variance contributes at typical GDP volatilities. This is the Rietz (1988) and Barro (2006) mechanism, and it is what allows the model to match the equity premium in the calibration.

E.9 Proofs

E.9.1 Lemma 10

Lemma 10. *Suppose that $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small. Then*

$$\delta_{si} \approx \delta_{si}^\circ + H_{si}^{-1} \mathcal{E}_{si}^\circ, \tag{129}$$

where $\mathcal{E}_{si}^\circ = \Sigma D_{si}(\delta^\circ)$ with $D_{si}(\delta^\circ)$ defined in the proof, and where

$$H_{si} = (1 + \tau_{si}) \frac{1}{\bar{\chi}_{si} \bar{\omega}_s} \eta G_{si} + B_{si},$$

where $\bar{\chi}_{si}$ and $\bar{\omega}_s$ are evaluated at $\varepsilon = 0$.

Proof. In (122), expectations are taken jointly over ε and v . The aggregate objects in the right-hand side (Λ , ω_s , $\bar{P}Y$, and the within-sector denominator of χ_{si}) depend on v only through within-sector θ -weighted sums of the form $\sum_{i=1}^{N_s} \theta_{si} f_{si}(v_{si})$. When firms within a sector are small, the law of large numbers makes these sums close to their expectations $\sum_i \theta_{si} \mathbb{E}_v[f_{si}(v_{si})]$. The numerator of χ_{si} is the only object that retains a genuine v_{si} -dependence. We can therefore first take expectations over v , replacing χ_{si} by $\mathbb{E}_v[\chi_{si}]$ and treating all aggregates as v -deterministic, and then take expectations over ε .

Using (108) and (107), we can write

$$\mathbb{E}_v \chi_{si} = \frac{\zeta_{si}}{\sum_{j=1}^{N_s} \zeta_{sj}},$$

where

$$\zeta_{si} = \theta_{si} \exp \left((\sigma_s - 1) \left(\delta_{si}^\top \varepsilon - b_i(\delta_{si}) - \log(1 + \tau_{si}) \right) + \frac{1}{2} (\sigma_s - 1)^2 \Sigma_{si}^v \right). \quad (134)$$

Next, from (122), we get

$$\delta_{si}^{(1)} = H_{si}^{-1} \Sigma \nabla_\varepsilon \log (\Lambda \mathbb{E}_v (\chi_{si}) \omega_s \bar{P}Y) \Big|_{\varepsilon=0},$$

where

$$H_{si} = \left[(1 + \tau_{si}) \eta \frac{\mathbb{E}(\Lambda \bar{P}Y)}{\mathbb{E}(\Lambda \mathbb{E}_v (\chi_{si}) \omega_s \bar{P}Y)} G_{si} + B_{si} \right]_{\varepsilon=0} = (1 + \tau_{si}) \eta \frac{1}{\bar{\chi}_{si} \bar{\omega}_s} G_{si} + B_{si}, \quad (135)$$

and $\bar{\chi}_{si}$ and $\bar{\omega}_s$ are evaluated at $\varepsilon = 0$, namely,

$$\bar{\chi}_{si} = \frac{\bar{\zeta}_{si}}{\sum_{j=1}^{N_s} \bar{\zeta}_{sj}} \text{ with } \bar{\zeta}_{si} = \theta_{si} \exp \left((\sigma_s - 1) (-\log(1 + \tau_{si})) + \frac{1}{2} (\sigma_s - 1)^2 \Sigma_{si}^v \right), \quad (136)$$

and

$$\bar{\omega}^\top = \beta^\top (I - \bar{\alpha})^{-1} \text{ with } \bar{\alpha}_{s's} = \hat{\alpha}_{s's} \sum_{i=1}^{N_{s'}} \frac{\bar{\chi}_{s'i}}{1 + \tau_{s'i}}.$$

We now need to compute $\nabla_\varepsilon \log (\Lambda \mathbb{E}_v (\chi_{si}) \omega_s \bar{P}Y) \Big|_{\varepsilon=0}$. Using (114) and (115), we get

$$\nabla_\varepsilon \log (\Lambda \mathbb{E}_v (\chi_{si}) \omega_s \bar{P}Y) \Big|_{\varepsilon=0} = -(\rho - 1) \nabla_\varepsilon \log C \Big|_{\varepsilon=0} + \nabla_\varepsilon \log \mathbb{E}_v (\chi_{si}) \Big|_{\varepsilon=0} + \nabla_\varepsilon \log \omega_s \Big|_{\varepsilon=0}.$$

From (134), we get

$$\nabla_\varepsilon \log \mathbb{E}_v (\chi_{si}) \Big|_{\varepsilon=0} = (\sigma_s - 1) \left(\delta_{si}^\circ - \sum_{j=1}^{N_s} \bar{\chi}_{sj} \delta_{sj}^\circ \right). \quad (137)$$

Next, from (117),

$$\nabla_\varepsilon \omega^\top \Big|_{\varepsilon=0} = \bar{\omega}^\top (\nabla_\varepsilon \tilde{\alpha} \Big|_{\varepsilon=0}) (I - \bar{\alpha})^{-1}, \quad (138)$$

where

$$\nabla_\varepsilon \tilde{\alpha}_{s's} \Big|_{\varepsilon=0} = \hat{\alpha}_{s's} (\sigma_{s'} - 1) \sum_{i=1}^{N_{s'}} \frac{\bar{\chi}_{s'i}}{1 + \tau_{s'i}} \left(\delta_{s'i}^\circ - \sum_{j=1}^{N_{s'}} \bar{\chi}_{s'j} \delta_{s'j}^\circ \right).$$

Finally, using (114), (119), and (109), we can write

$$\nabla_\varepsilon \log C|_{\varepsilon=0} = \frac{\sum_{s=1}^S \bar{\omega}_s \nabla_\varepsilon M_s|_{\varepsilon=0}}{1 - \sum_{s=1}^S \bar{M}_s \bar{\omega}_s} + \frac{\sum_{s=1}^S \bar{M}_s \nabla_\varepsilon \omega_s|_{\varepsilon=0}}{1 - \sum_{s=1}^S \bar{M}_s \bar{\omega}_s} + \sum_{s=1}^S \bar{\omega}_s \sum_{i=1}^{N_s} \bar{\chi}_{si} \delta_{si}^\circ, \quad (139)$$

where, as usual, $\tilde{\omega}^\top = \beta^\top (I - \hat{\alpha})^{-1}$, and where

$$\bar{M}_s = \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \bar{\chi}_{si}$$

and

$$\nabla_\varepsilon M_s|_{\varepsilon=0} = \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \bar{\chi}_{si} (\sigma_s - 1) \left(\delta_{si}^\circ - \sum_{j=1}^{N_s} \bar{\chi}_{sj} \delta_{sj}^\circ \right).$$

Therefore, we can write

$$\delta_{si}^{(1)} = H_{si}^{-1} \mathcal{E}_{si}^\circ,$$

where H_{si} is given by (135), and $\mathcal{E}_{si}^\circ = \Sigma D_{si}(\delta^\circ)$ with

$$D_{si}(\delta^\circ) = -(\rho - 1) \nabla_\varepsilon \log C|_{\varepsilon=0} + \nabla_\varepsilon \log E_v(\chi_{si})|_{\varepsilon=0} + \nabla_\varepsilon \log \omega_s|_{\varepsilon=0}, \quad (140)$$

where the elements of the sum are given by (137), (138), and (139). $\square$

E.9.2 Lemma 11

Lemma 11. *Suppose that $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small. Firms with higher steady-state Domar weights $\bar{\omega}_{si} = \bar{\chi}_{si} \bar{\omega}_s$ and lower wedges τ_{si} manage risk more aggressively, in the sense that*

$$\frac{\partial \left[(\mathcal{E}_{si}^\circ)^\top \delta_{si} \right]}{\partial \bar{\omega}_{si}} \geq 0 \quad \text{and} \quad \frac{\partial \left[(\mathcal{E}_{si}^\circ)^\top \delta_{si} \right]}{\partial \tau_{si}} \leq 0.$$

Proof. From (129), we can write

$$(\mathcal{E}_{si}^\circ)^\top \delta_{si} \approx (\mathcal{E}_{si}^\circ)^\top \delta_{si}^\circ + (\mathcal{E}_{si}^\circ)^\top H_{si}^{-1} \mathcal{E}_{si}^\circ,$$

where H_{si} is given by (135).

Note that $\mathcal{E}_{si}^\circ$ depends on $\bar{\omega}_{si}$ and τ_{si} only through sector- or economy-wide variables, and so in partial equilibrium, $\frac{\partial \mathcal{E}_{si}^\circ}{\partial \bar{\omega}_{si}} = \frac{\partial \mathcal{E}_{si}^\circ}{\partial \tau_{si}} = 0$. Differentiating with respect to $\bar{\omega}_{si}$, we get

$$\frac{\partial \left[(\mathcal{E}_{si}^\circ)^\top \delta_{si} \right]}{\partial \bar{\omega}_{si}} = -(\mathcal{E}_{si}^\circ)^\top H_{si}^{-1} \frac{\partial H_{si}}{\partial \bar{\omega}_{si}} H_{si}^{-1} \mathcal{E}_{si}^\circ = (\mathcal{E}_{si}^\circ)^\top H_{si}^{-1} (1 + \tau_{si}) \eta \frac{1}{\bar{\omega}_{si}^2} G_{si} H_{si}^{-1} \mathcal{E}_{si}^\circ \geq 0,$$

where the inequality follows since G_{si} is positive definite.

For the comparative static in τ_{si} , the markup enters H_{si} both directly through the $(1 + \tau_{si})$ factor and indirectly through $\bar{\chi}_{si}$, which itself depends on τ_{si} via (136). It is easy to see that $\partial \bar{\chi}_{si} / \partial \tau_{si} < 0$, and so $\partial H_{si} / \partial \tau_{si}$ is a positive definite matrix. The same identity used above then gives $\partial \left[(\mathcal{E}_{si}^\circ)^\top \delta_{si} \right] / \partial \tau_{si} = - (\mathcal{E}_{si}^\circ)^\top H_{si}^{-1} (\partial H_{si} / \partial \tau_{si}) H_{si}^{-1} \mathcal{E}_{si}^\circ \leq 0$. $\square$

E.9.3 Lemma 12

Lemma 12. *Suppose that $\varepsilon \sim \mathcal{N}(0, \Sigma)$ with Σ small. Suppose also that firm-specific shocks are small, such that $v_{si} \sim iid \mathcal{N}(0, \Sigma_{si}^v)$ with Σ_{si}^v small. Then*

$$y \approx \bar{y} + \Delta^\top \varepsilon + \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} v_{si}, \quad (130)$$

where $\bar{y}$ does not depend on shocks, $\Delta = \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} \delta_{si}$, and ς_{si} is defined in the proof.

Proof. Using (114), (119), and (109), we can write log real GDP as

$$y = -\log \left(1 - \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \chi_{si} \omega_s \right) + \beta^\top (I - \hat{\alpha})^{-1} \begin{pmatrix} \vdots \\ \frac{1}{\sigma_s - 1} \log \left(\sum_{i=1}^{N_s} \theta_{si} \left(\exp \left(\delta_{si}^\top \varepsilon - b_i(\delta_{si}) + \gamma_s t + v_{si} - \log(1 + \tau_{si}) \right) \right)^{\sigma_s - 1} \right) \\ \vdots \end{pmatrix}, \quad (141)$$

where, from (108),

$$\chi_{si} = \frac{\theta_{si} \exp \left((\sigma_s - 1) \left(\delta_{si}^\top \varepsilon - b_i(\delta_{si}) + v_{si} - \log(1 + \tau_{si}) \right) \right)}{\sum_{j=1}^{N_s} \theta_{sj} \exp \left((\sigma_s - 1) \left(\delta_{sj}^\top \varepsilon - b_j(\delta_{sj}) + v_{sj} - \log(1 + \tau_{sj}) \right) \right)}.$$

We linearize y with respect to ε and v for a given δ . Note that $\delta_{si}^\top \varepsilon + v_{si}$ always enter additively, and so

$$y \approx \bar{y} + \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si} \left(\delta_{si}^\top \varepsilon + v_{si} \right),$$

where $\bar{y}$ is (141) evaluated at $\varepsilon = 0$ and $v = 0$, and

$$\varsigma_{si} = \nabla_{v_{si}} y|_{\varepsilon=0, v=0}.$$

Differentiating (141) with respect to v_{si} at $\varepsilon = 0$ and $v = 0$, we get

$$\zeta_{s'i'} = \frac{\sum_{i=1}^{N_{s'}} \frac{\tau_{s'i}}{1+\tau_{s'i}} (\nabla_{v_{s'i'}} \chi_{s'i} |_{\varepsilon=0, v=0}) \bar{\omega}_{s'}}{1 - \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1+\tau_{si}} \bar{\chi}_{si} \bar{\omega}_s} + \frac{\sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1+\tau_{si}} \bar{\chi}_{si} (\nabla_{v_{s'i'}} \omega_s |_{\varepsilon=0, v=0})}{1 - \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1+\tau_{si}} \bar{\chi}_{si} \bar{\omega}_s} + \tilde{\omega}_{s'} \bar{\chi}_{s'i'}, \quad (142)$$

where $\tilde{\omega}^\top = \beta^\top (I - \hat{\alpha})^{-1}$, and

$$\bar{\chi}_{si} = \frac{\theta_{si} \exp(-(\sigma_s - 1)(b_i(\delta_{si}) + \log(1 + \tau_{si})))}{\sum_{j=1}^{N_s} \theta_{sj} \exp(-(\sigma_s - 1)(b_j(\delta_{sj}) + \log(1 + \tau_{sj})))}, \quad (143)$$

$$\bar{\omega}^\top = \beta^\top (I - \bar{\alpha})^{-1} \text{ with } \bar{\alpha}_{s's} = \hat{\alpha}_{s's} \sum_{i=1}^{N_{s'}} \frac{\bar{\chi}_{s'i}}{1 + \tau_{s'i}},$$

$$\nabla_{v_{s'i'}} \chi_{s'i} |_{\varepsilon=0, v=0} = \bar{\chi}_{s'i} (\sigma_{s'} - 1) (1\{i = i'\} - \bar{\chi}_{s'i'}),$$

$$\nabla_{v_{s'i'}} \tilde{\alpha}_{s's} |_{\varepsilon=0, v=0} = \hat{\alpha}_{s's} \sum_{i=1}^{N_{s'}} \frac{(\nabla_{v_{s'i'}} \chi_{s'i} |_{\varepsilon=0, v=0})}{1 + \tau_{s'i}},$$

$$\nabla_{v_{s'i'}} \omega_s |_{\varepsilon=0, v=0} = \bar{\omega}^\top (\nabla_{v_{s'i'}} \tilde{\alpha} |_{\varepsilon=0, v=0}) (I - \bar{\alpha})^{-1} \mathbf{1}_s.$$

The two terms in $\nabla_{v_{s'i'}} \chi_{s'i} |_{\varepsilon=0, v=0}$ reflect the two channels through which $v_{s'i'}$ enters $\chi_{s'i}$: directly via the numerator (when $i = i'$, contributing the $1\{i = i'\}$ term) and indirectly via the within-sector denominator (which contains $v_{s'i'}$ through the $j = i'$ term in the sum, contributing the $-\bar{\chi}_{s'i'}$ term). $\square$

E.9.4 Lemma 13

Lemma 13. *Suppose that Σ and π_d are small. Then*

$$\delta_{si} \approx \delta_{si}^\circ + H_{si}^{-1} (\mathcal{E}_{si}^\circ + \mathcal{F}_{si}^\circ), \quad (132)$$

where $\mathcal{E}_{si}^\circ = \Sigma D_{si}(\delta^\circ)$ and $\mathcal{F}_{si}^\circ = \pi_d D_{si}^d(\delta^\circ)$ with $D_{si}(\delta^\circ)$ and $D_{si}^d(\delta^\circ)$ defined in the proof, and where

$$H_{si} = (1 + \tau_{si}) \frac{1}{\bar{\chi}_{si} \bar{\omega}_s} \eta G_{si} + B_{si},$$

where $\bar{\chi}_{si}$ and $\bar{\omega}_s$ are evaluated at $\varepsilon = \varepsilon^d = 0$.

Proof. Using Taylor's approximation, we can write

$$\delta_{si} = \delta_{si}^\circ + \delta_{si}^{(n)} \Sigma + \delta_{si}^{(d)} \pi_d + o(\pi_d) + o(\Sigma).$$

Derivation of $\delta_{si}^{(n)}$ is analogous to that presented in the proof of Lemma 10. We therefore focus on deriving $\delta_{si}^{(d)}$ in this proof.

Denote $f_{si} = \Lambda \chi_{si} \omega_s \bar{P}Y$. Consider the right-most term of (122). To derive the first-order term

$\delta_{si}^{(d)}$, we need to evaluate this expression at $\varepsilon = 0$, which results in

$$\frac{\mathbb{E}(\Lambda_{\chi_{si}\omega_s}\bar{P}Y(\varepsilon + \varepsilon^d))}{\mathbb{E}(\Lambda_{\chi_{si}\omega_s}\bar{P}Y)} \stackrel{\varepsilon=0}{=} \frac{\mathbb{E}(f_{si}(\varepsilon = 0)\varepsilon^d)}{\mathbb{E}(f_{si}(\varepsilon = 0))} = -\frac{\pi_d d \bar{f}_{si}^d}{\pi_d \bar{f}_{si}^d + (1 - \pi_d) \bar{f}_{si}^{nd}} \approx -\pi_d d \frac{\bar{f}_{si}^d}{\bar{f}_{si}^{nd}}, \quad (144)$$

where the approximation holds under the small- π_d assumption, and where $\bar{f}_{si}^d$ and $\bar{f}_{si}^{nd}$ are the values of f_{si} at $\varepsilon = 0$ and $\varepsilon^d = -d$ and $\varepsilon^d = 0$, respectively.

Since (144) is linear in π_d , the first-order term $\delta_{si}^{(d)}$ is then given by

$$\delta_{si}^{(d)} = -H_{si}^{-1} d \frac{\bar{f}_{si}^d}{\bar{f}_{si}^{nd}},$$

where H_{si} is evaluated at $\varepsilon = \varepsilon^d = 0$ and given by (135). We, therefore, need to compute $\bar{f}_{si}^d / \bar{f}_{si}^{nd}$ to finish the derivation. Define an indicator function $1\{r = d\}$ which equals 1 in the disaster regime ($r = d$) and 0 in the no-disaster regime ($r = nd$). As in the proof of Lemma 10, we can write

$$\bar{\chi}_{si}^r = \frac{\bar{\zeta}_{si}^r}{\sum_{j=1}^{N_s} \bar{\zeta}_{sj}^r},$$

where

$$\bar{\zeta}_{si}^r = \theta_{si} \exp \left((\sigma_s - 1) \left(-(\delta_{si}^o + \bar{\delta}) 1\{r = d\} - \log(1 + \tau_{si}) \right) + \frac{1}{2} (\sigma_s - 1)^2 \Sigma_{si}^v \right).$$

Further, define

$$(\bar{\omega}^r)^\top = \beta^\top (I - \bar{\alpha}^r)^{-1} \text{ with } \bar{\alpha}_{s's}^r = \hat{\alpha}_{s's} \sum_i \frac{\bar{\chi}_{s'i}^r}{1 + \tau_{s'i}},$$

$$\bar{M}_s^r = \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \bar{\chi}_{si}^r.$$

Then, using (114), (119), and (109), we can write

$$\log \bar{C}^r = \sum_{s=1}^S \frac{1}{\sigma_s - 1} \bar{\omega}_s \log \left(\sum_{i=1}^{N_s} \bar{\zeta}_{si}^r \right) - \log \left(1 - \sum_{s=1}^S \bar{M}_s^r \bar{\omega}_s^r \right).$$

Using that, we can write

$$\frac{\bar{f}_{si}^d}{\bar{f}_{si}^{nd}} = \frac{\bar{\chi}_{si}^d \bar{\omega}_s^d (\bar{C}^d)^{1-\rho}}{\bar{\chi}_{si}^{nd} \bar{\omega}_s^{nd} (\bar{C}^{nd})^{1-\rho}},$$

where we can simplify $\bar{\chi}_{si}^d/\bar{\chi}_{si}^{nd}$ as

$$\frac{\bar{\chi}_{si}^d}{\bar{\chi}_{si}^{nd}} = \frac{\exp(-(\sigma_s - 1)\bar{\delta}_{si}^\circ d)}{\sum_{j=1}^{N_s} \bar{\chi}_{sj}^{nd} \exp(-(\sigma_s - 1)\bar{\delta}_{sj}^\circ d)}.$$

The exogenous baseline $\bar{\delta}$ does not appear in this ratio because the factor $\exp(-(\sigma_s - 1)\bar{\delta}d)$ is common to all firms in sector s and cancels between the numerator and denominator. $\square$

E.9.5 Lemma 14

Lemma 14. *Suppose that Σ and π_d are small. Suppose also that firm-specific shocks are small, such that $v_{si} \sim \text{iid } \mathcal{N}(0, \Sigma_{si}^v)$ with Σ_{si}^v small. Then, conditional on the disaster or no-disaster regime, $r \in \{d, nd\}$, the log output is*

$$y^r \approx \bar{y}^r + \Delta^r \varepsilon + \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si}^r v_{si}, \quad (133)$$

where $\bar{y}^r$ does not depend on ε and v_{si} , $\Delta^r = \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si}^r \delta_{si}$, and ς_{si}^r is defined in the proof.

Proof. In a given regime, $r \in \{d, nd\}$, we can approximate y following the same steps as in the proof of Lemma 12. Specifically, the analogue of (143) is

$$\bar{\chi}_{si}^r = \frac{\theta_{si} \exp(-(\sigma_s - 1)((\delta_{si} + \bar{\delta}) d \mathbf{1}\{r = d\} + b_i(\delta_{si}) + \log(1 + \tau_{si})))}{\sum_{j=1}^{N_s} \theta_{sj} \exp(-(\sigma_s - 1)((\delta_{sj} + \bar{\delta}) d \mathbf{1}\{r = d\} + b_j(\delta_{sj}) + \log(1 + \tau_{sj})))}.$$

With this $\bar{\chi}_{si}^r$, we can get ς_{si}^r using (142). Using (141), we can write $\bar{y}^r$ as

$$\begin{aligned} \bar{y}^r = & -\log \left(1 - \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \bar{\chi}_{si}^r \bar{\omega}_s^r \right) + \beta^\top (I - \hat{\alpha})^{-1} \times \\ & \times \begin{pmatrix} \vdots \\ \frac{1}{\sigma_s - 1} \log \left(\sum_{i=1}^{N_s} \theta_{si} (\exp(-(\delta_{si} + \bar{\delta}) d \mathbf{1}\{r = d\} - b_i(\delta_{si}) + \gamma_s t - \log(1 + \tau_{si})))^{\sigma_s - 1} \right) \\ \vdots \end{pmatrix}. \end{aligned} \quad (145)$$

Finally, note that $\vartheta = \bar{y}^{nd} - \bar{y}^d$ does not depend on t since the trend term cancels out. $\square$

E.9.6 Lemma 15

Lemma 15. *Suppose that the log output satisfies (133). Denote by $\gamma = \sum_{s=1}^S \tilde{\omega}_s \gamma_s$ the growth rate of GDP. Then the risk-free rate and the premium on the consumption claim are, respectively,*

$$\begin{aligned} \log E R_{f,t+1} &= r + \rho \gamma, \\ \log E R_{c,t+1} - \log E R_{f,t+1} &= \log M(-\rho) + \log \left[\varphi M(\rho) + (1 - \varphi) \frac{M(1)}{M(1-\rho)} \right], \end{aligned}$$

where $\varphi = \exp(-r - (\rho - 1)\gamma)$ and

$$M(x) = (1 - \pi_d) \exp\left(\frac{1}{2}x^2\sigma_{nd}^2\right) + \pi_d \exp\left(-x\vartheta + \frac{1}{2}x^2\sigma_d^2\right).$$

Proof. From (133), we have

$$y_t = y_0 + \gamma t + s_t,$$

where, from (145),

$$y_0 = -\log\left(1 - \sum_{s=1}^S \sum_{i=1}^{N_s} \frac{\tau_{si}}{1 + \tau_{si}} \bar{\chi}_{si}^{nd} \bar{\omega}_s^{nd}\right) + \beta^\top (I - \hat{\alpha})^{-1} \begin{pmatrix} \vdots \\ \frac{1}{\sigma_s - 1} \log\left(\sum_{i=1}^{N_s} \theta_{si} (\exp(-b_i(\delta_{si}) - \log(1 + \tau_{si})))^{\sigma_s - 1}\right) \\ \vdots \end{pmatrix},$$

and

$$s_t = \Delta^r \varepsilon + \sum_{s=1}^S \sum_{i=1}^{N_s} \varsigma_{si}^r v_{si} - \vartheta \mathbf{1}\{r = d\}.$$

Define

$$M(x) = \mathbb{E} \exp(x s_t) = (1 - \pi_d) \exp\left(\frac{1}{2}x^2\sigma_{nd}^2\right) + \pi_d \exp\left(-x\vartheta + \frac{1}{2}x^2\sigma_d^2\right).$$

Then

$$R_{f,t+1} = \frac{1}{\mathbb{E}_t(\exp(-r - \rho(y_{t+1} - y_t)))} = \frac{1}{\exp(-r - \rho\gamma + \rho s_t) M(-\rho)},$$

which further implies

$$\log \mathbb{E} R_{f,t+1} = r + \rho\gamma.$$

Next, conjecture that the price-consumption ratio of the consumption claim is $\phi_t = h \exp((\rho - 1)s_t)$. Then the Euler equation is

$$\phi_t = \mathbb{E}_t \left[e^{-r} \left(\frac{C_{t+1}}{C_t} \right)^{1-\rho} (\phi_{t+1} + 1) \right] \Leftrightarrow h = \frac{\varphi}{1 - \varphi} M(1 - \rho).$$

Then

$$R_{c,t+1} = \frac{C_{t+1}}{C_t} \frac{\phi_{t+1} + 1}{\phi_t} = \exp(\gamma + s_{t+1} - \rho s_t) \left(\exp((\rho - 1)s_{t+1}) + \frac{1}{h} \right),$$

which implies

$$\log \mathbb{E} R_{c,t+1} = \gamma + \log M(-\rho) + \log \left(M(\rho) + \frac{1 - \varphi}{\varphi} \frac{M(1)}{M(1 - \rho)} \right).$$

The equity premium is then

$$\log E R_{c,t+1} - \log E R_{f,t+1} = \log M(-\rho) + \log \left(\varphi M(\rho) + (1-\varphi) \frac{M(1)}{M(1-\rho)} \right).$$

□

F Additional analysis of the model

In this appendix, we provide additional analysis of the benchmark model presented in the main text.

F.1 Derivation of the stochastic discount factor

In this appendix, we derive equations (9) and (8). The household chooses how much to consume after uncertainty about ε is realized but how much risk-management resource to supply before uncertainty is realized. We first consider the problem of the household once the state ε has been realized. Its Lagrangian is

$$\mathcal{L} = \frac{\left(\left(\frac{C_1}{\beta_1} \right)^{\beta_1} \times \cdots \times \left(\frac{C_N}{\beta_N} \right)^{\beta_N} \right)^{1-\rho} (V(R))^{1-\rho}}{1-\rho} - \Lambda \left(\sum_{i=1}^N P_i C_i - W_L L - W_R R - \Pi \right),$$

where we assume that the household supplies $L = 1$ units of production labor inelastically. The first-order condition with respect to C_i is

$$\beta_i \mathcal{U}'(Y) Y (V(R))^{1-\rho} = \Lambda P_i C_i,$$

where $\mathcal{U}(Y) = \frac{Y^{1-\rho}}{1-\rho}$. Summing over i on both sides of this equation and using the binding budget constraint, we find

$$\mathcal{U}'(Y) Y (V(R))^{1-\rho} = \Lambda (W_L L + W_R R + \Pi). \quad (146)$$

Combining with the first-order condition implies

$$P_i C_i = \beta_i (W_L L + W_R R + \Pi). \quad (147)$$

Combining the first-order condition with $Y = \prod_{i=1}^N (\beta_i^{-1} C_i)^{\beta_i}$ yields

$$\begin{aligned} Y &= \prod_{i=1}^N (\beta_i^{-1} C_i)^{\beta_i} = \prod_{i=1}^N \left(\beta_i^{-1} \frac{\beta_i \mathcal{U}'(Y) Y (V(R))^{1-\rho}}{\Lambda P_i} \right)^{\beta_i} \Leftrightarrow \\ \Lambda &= \mathcal{U}'(Y) (V(R))^{1-\rho} \prod_{i=1}^N P_i^{-\beta_i}, \end{aligned} \quad (148)$$

which we can combine with (146) to find

$$Y = (W_L L + W_R R + \Pi) \prod_{i=1}^N P_i^{-\beta_i}. \quad (149)$$

This last equation implicitly defines a price index,

$$\bar{P} = \prod_{i=1}^N P_i^{\beta_i}, \quad (150)$$

such that $\bar{P}Y = W_L L + W_R R + \Pi$. Therefore, (148) is equivalent to (9).

Finally, we can also compute the household's first-order condition with respect to R :

$$\mathbb{E} [Y^{1-\rho}] (V(R))^{-\rho} V'(R) + \mathbb{E} [\Lambda] W_R = 0. \quad (151)$$

Using (148), we find

$$\mathbb{E} [Y^{1-\rho}] \frac{V'(R)}{V(R)} + \mathbb{E} \left[\frac{Y^{1-\rho}}{\bar{P}Y} \right] W_R = 0. \quad (152)$$

By Lemma 1, $\bar{P}Y = W_L L \Gamma_L^{-1}$. Given our choice of numeraire, $W_L L = 1$ and so $\bar{P}Y = \Gamma_L^{-1}$, which is non-stochastic by Lemma 1. Therefore, (152) simplifies to (8).

F.2 Choice of numeraire

In the baseline model, we set $W_L = 1$ in all states of the world. In this appendix, we show that this normalization is innocuous. Specifically, we treat W_L as a random variable and show that the firms' equilibrium risk-exposure decisions δ , the aggregate supply of risk management resources R and, hence, all macroeconomic aggregates are the same as in the baseline model.

The household's choice of R is governed by the first-order condition (151), which we can write as

$$W_R = -V'(R) (V(R))^{-\rho} \frac{\mathbb{E} [Y^{1-\rho}]}{\mathbb{E} [\Lambda]}, \quad (153)$$

where Λ is given by (148).

Next, given the Cobb-Douglas production function, the unit cost is given by

$$K_i(\delta_i, P, W_L) = \exp\left(-\delta_i^\top \varepsilon + b_i(\delta_i)\right) W_L^{1-\sum_{j=1}^N \alpha_{ij}} \prod_{j=1}^N P_j^{\alpha_{ij}}.$$

Since $P_i = (1 + \tau_i) K_i$, this implies that

$$p - w_L = -\tilde{\mathcal{L}}(\delta\varepsilon - b(\delta) - \log(1 + \tau)), \quad (154)$$

where $w_L = \log W_L$ and $\tilde{\mathcal{L}} = (I - \alpha)^{-1}$.

Firms' risk exposure solves (11). Taking the first-order condition with respect to δ_{im} , we get

$$\frac{dg_i(\delta_i)}{d\delta_{im}} = \frac{1}{W_R} \frac{\mathbb{E}\left[\Lambda Q_i K_i\left(\varepsilon_m - \frac{db_i(\delta_i)}{d\delta_{im}}\right)\right]}{\mathbb{E}[\Lambda]}.$$

Combining this equation with (153), we get

$$\frac{dg_i(\delta_i)}{d\delta_{im}} = -\frac{1}{V'(R)(V(R))^{-\rho}} \frac{\mathbb{E}\left[\Lambda K_i Q_i\left(\varepsilon_m - \frac{db_i(\delta_i)}{d\delta_{im}}\right)\right]}{\mathbb{E}[Y^{1-\rho}]}. \quad (155)$$

Next, from the market-clearing condition (13),

$$P_i Q_i = P_i C_i + \sum_j P_i X_{ji}.$$

Using (147) and the fact that $P_i = (1 + \tau_i) K_i$, we can derive that $\frac{K_i Q_i}{W_L + W_R R + \Pi} = \frac{\omega_i}{1 + \tau_i}$, where

$$\omega^\top = \beta^\top \left(I - [\text{diag}(1 + \tau)]^{-1} \alpha\right)^{-1}$$

is the non-stochastic vector of Domar weights. Hence, (155) can be rewritten as

$$\frac{dg_i(\delta_i)}{d\delta_{im}} = -\frac{1}{V'(R)(V(R))^{-\rho}} \frac{\omega_i}{1 + \tau_i} \frac{\mathbb{E}\left[\Lambda (W_L + W_R R + \Pi) \left(\varepsilon_m - \frac{db_i(\delta_i)}{d\delta_{im}}\right)\right]}{\mathbb{E}[Y^{1-\rho}]}$$

Combining this with (146), we get

$$\frac{dg_i(\delta_i)}{d\delta_{im}} = -\frac{V(R)}{V'(R)} \frac{\omega_i}{1 + \tau_i} \frac{\mathbb{E}\left[U'(Y) Y \left(\varepsilon_m - \frac{db_i(\delta_i)}{d\delta_{im}}\right)\right]}{\mathbb{E}[Y^{1-\rho}]} = -\frac{V(R)}{V'(R)} \frac{\omega_i}{1 + \tau_i} \frac{\mathbb{E}\left[Y^{1-\rho} \left(\varepsilon_m - \frac{db_i(\delta_i)}{d\delta_{im}}\right)\right]}{\mathbb{E}[Y^{1-\rho}]}. \quad (156)$$

From (149), we can write GDP Y as

$$Y = \left(\frac{W_L + W_R R + \Pi}{W_L} \right) \prod_{i=1}^N \left(\frac{P_i}{W_L} \right)^{-\beta_i}. \quad (157)$$

Furthermore, total profit in the economy is $\Pi = \sum_i \frac{\tau_i}{1+\tau_i} P_i Q_i - W_R R \Leftrightarrow \Pi + W_R R = (W_L + W_R R + \Pi) \sum_i \frac{\tau_i}{1+\tau_i} \omega_i$, which implies $W_R R + \Pi = W_L \frac{\sum_i \frac{\tau_i}{1+\tau_i} \omega_i}{1 - \sum_i \frac{\tau_i}{1+\tau_i} \omega_i}$. Therefore, (157) can be written as

$$Y = \frac{1}{1 - \sum_i \frac{\tau_i}{1+\tau_i} \omega_i} \prod_{i=1}^N \left(\frac{P_i}{W_L} \right)^{-\beta_i}.$$

Hence, using (154), (156) becomes

$$\frac{dg_i(\delta_i)}{d\delta_{im}} = - \frac{V(R)}{V'(R)} \frac{\omega_i}{1 + \tau_i} \frac{\mathbb{E} \left[\exp \left(-(\rho - 1) \Delta^\top \varepsilon \right) \left(\varepsilon_m - \frac{db_i(\delta_i)}{d\delta_{im}} \right) \right]}{\mathbb{E} [\exp \left(-(\rho - 1) \Delta^\top \varepsilon \right)]}.$$

Using the properties of the normal distribution to simplify the right-hand side of this equation, we can see that this equation coincides with (76). Therefore, setting $W_L = 1$ does not affect equilibrium risk exposures and hence moments of GDP.

F.3 Microfoundations for the risk management cost functions

In the main text, we treat the risk management cost functions b_i and g_i as exogenous. In reality, many firm decisions might affect their risk profiles (hiring, investment, research and development, location choices, etc.) and modeling all those individually would make the model intractable. Assuming general forms for b_i and g_i allows us to sidestep these issues, and lets us explore the impact of endogenous uncertainty on firm decisions and the macroeconomy in a simple way.

But for some applications, it might be helpful to consider specific risk-mitigation mechanisms, in which case one can microfound b_i and g_i based on the mechanisms of interest. Doing so could yield a greater understanding of the origin of the cost functions and of the features of the environment that affect them. In this appendix, we provide four possible microfoundations for b_i and g_i . These microfoundations are stylized and can certainly be generalized in many ways, at the cost of tractability. They are also only examples. It is possible to come up with many such microfoundations, especially if b_i and g_i are interpreted as approximations to more complicated functions.

F.3.1 Research and development

Consider a firm i that seeks to gain expertise in a new technology that will become operational soon (at the end of the period). The true quality ε of the technology is unknown, but it is expected to be distributed according to $\varepsilon \sim \mathcal{N}(\mu, \Sigma)$, where $\mu \gg 0$. Firms that gain expertise with the

technology will be able to use it to increase their productivity by an amount proportional to ε .

To gain expertise, firms must complete tasks (studies, experiments, etc.) that familiarize their workforce with the new technology. If it completes δ_i tasks, the firm will have productivity $\delta_i \varepsilon$ once the technology becomes available. Each task is associated with a difficulty level $t \geq 0$. Completing task t requires $\iota_g t$ units of resources and lowers the productivity of the firm by $\iota_b t$. Intuitively, resources might have to be purchased for experiments, and workers might need to be reallocated away from production to run those experiments, which lowers productivity. There is a continuum of tasks $t \geq 0$, and the firm is free to complete as many tasks as it wants. Suppose that the firm chooses to complete δ_i tasks. It obviously starts with the easiest ones. Its resource cost is therefore

$$g_i(\delta_i) = \int_0^{\delta_i} \iota_g t dt = \iota_g \frac{\delta_i^2}{2},$$

and similarly for its productivity cost. The total productivity of the firm is therefore

$$a_i(\varepsilon, \delta_i) = \delta_i \varepsilon - \iota_b \frac{\delta_i^2}{2},$$

where the first term reflects the productivity benefit of the new technology. As we can see, this simple task model of research and development fits into the general framework of the main text, and the cost functions also fit within the quadratic structure given by (4) and (5).

F.3.2 Transportation to market

Our second microfoundation involves a location decision by a firm. In the spirit of Hotelling (1929), suppose that there is a set of locations l on the unit interval $l \in [0, 1]$. A firm needs to locate its operation somewhere on $[0, 1]$. At $l = 1$ there is a city where customers live, but that city is located on the ocean shore and, as a result, is subject to potential natural catastrophes (floods, hurricanes, climate change, etc.). By being close to $l = 1$, the firm is closer to its customers, but also more exposed to catastrophes. We model this by assuming that if the firm is located at $l = \delta_i$, the stochastic part of its TFP is given by $\delta_i \varepsilon$, where the catastrophe shock ε is distributed according to $\varepsilon \sim \mathcal{N}(\mu, \Sigma)$ with $\mu \ll 0$.⁷⁷ If the firm is away from $l = 1$, it must transport its goods to the city. Transportation involves costs (drivers, gasoline, vehicles, repairs, etc.) and we model those as a combination of a variable cost and a fixed cost. The variable cost scales with the amount of goods sold, and can therefore be modeled as a productivity shifter $\kappa_b(1 - \delta_i)$. The firm's productivity is therefore

$$a_i(\varepsilon, \delta_i) = \delta_i \varepsilon - \kappa_b(1 - \delta_i).$$

⁷⁷Such a catastrophe risk can also be modeled as a disaster (rather than normal) shock, as in Section 9.5.

The fixed cost $\kappa_g(1 - \delta_i)$ does not vary with the size of the firm and is therefore akin to the risk management resource cost of the main text. It is natural to think that the functions κ_b and κ_g are convex: longer trips might require an overnight stay, fresh produce is more likely to go bad, mechanical problems away from the city incur larger costs, etc. In this case, approximating κ_b and κ_g using quadratic functions like b_i and g_i in the main text would be reasonable. In that case, this simple catastrophe and transportation model can provide a microfoundation for the general setup of the main text.

F.3.3 Workplace rules

Many organizations put rules into place to avoid bad outcomes. For instance, metal detectors are often installed at office entry points, large expenses need to be approved by managers, access to computer networks requires two-factor authentication, installing new software requires the approval of the IT department, etc. While these policies can help mitigate adverse outcomes (security incidents, thefts, hackers taking over computer networks, etc.), they might also impede normal work activity and lower productivity.

Suppose for instance that a firm has a unit mass of workers, and that each worker is endowed with an unobservable skill $s \sim U([0, 1])$. To produce, workers must complete tasks. There is a unit mass of tasks, each with a difficulty level $d \sim U([0, 1])$ that is observable. Tasks are assigned to workers at random. If a task d is assigned to a worker with skill $s \geq d$, the task is completed successfully and generates ι units of productivity for the firm. If a task is assigned to a worker with skill $s < d$, the task does not generate any productivity.

Any attempted task increases the importance of a potential problem $\varepsilon \sim \mathcal{N}(\mu, \Sigma)$, with $\mu \ll 0$, for the firm. To lower the amount of risk it is exposed to, the firm can impose a *rule* δ_i . With a rule δ_i , the workers do not attempt any tasks with a difficulty level $d \geq \delta_i$. Without rules ($\delta_i = 1$), the full unit mass of tasks is attempted, so that the random problem lowers productivity by $1 \times \varepsilon$, and half of those tasks are successfully completed, which generates productivity $1/2\iota$. In that case,

$$a_i(\varepsilon, \delta_i) = 1 \times \varepsilon + \frac{1}{2}\iota.$$

Under a rule δ_i , the firm faces some productivity losses from the tasks that are not attempted but that high-skill workers would have otherwise successfully completed. There are $(1 - \delta_i)^2/2$ such tasks.⁷⁸ On the positive side, the intensity of the random problem ε declines. Putting the pieces together, the productivity of the firm under rule δ is

$$a_i(\varepsilon, \delta_i) = \delta_i \varepsilon + \left(\frac{1}{2} - \frac{(1 - \delta_i)^2}{2} \right) \iota.$$

⁷⁸To get to that number, it helps to think of the square $[0, 1] \times [0, 1]$ of skills and difficulty. We need to compute the area of the upper triangle created by the horizontal line $d = \delta_i$ and the linear function $d = s$.

As we can see, rules imposed by organizations can serve as a microfoundation for the TFP process used in the main text (2) and the productivity risk management cost b_i ⁷⁹

$$b_i(\delta_i) = \frac{(1 - \delta_i)^2}{2} \iota,$$

which fits into our general quadratic formulation (4) with $\delta^\circ = 1$.

F.3.4 Skilled managers

To microfound the risk management cost functions, we can also assume that adjusting risk within a firm requires the use of managers. Managers differ in their skill level, and larger risk adjustments can only be done by managers with more advanced skills. In turn, high-skill managers are paid more, which leads to an increasing relationship between the size of a risk adjustment and its cost.

For simplicity, suppose that there is a single risk factor ε , and that firm i has a natural exposure δ_i° to that factor. To set $\delta_i \neq \delta_i^\circ$ requires a manager with a skill level $s = |\delta_i - \delta_i^\circ|$. Such a manager is paid an equilibrium wage of $W_R(s)$. The firm's problem (11) is therefore

$$\max_{\delta_i} \mathbb{E} [\Lambda(Q_i(P_i - K_i(\delta_i)) - W_R(s))],$$

where $s = |\delta_i - \delta_i^\circ|$.

It is straightforward to adapt the problem of the household to determine $W_R(s)$ in equilibrium. For instance, we can assume that the total cost R of supplying risk management resources is given by

$$R = \int_0^\infty R(s) \varphi(s) ds,$$

where $\varphi(s)$ is the cost of supplying a manager of skill s (from schooling, additional effort, training, etc.), and $R(s)$ is the mass of such managers. It is natural to think that φ is increasing and convex. Indeed, while almost anyone can learn basic management techniques, teaching new skills to high-level executives might incur larger and larger costs. In that case, we can approximate $\varphi(s) \approx \frac{h}{2} s^2$, for some $h > 0$.⁸⁰ For simplicity, we assume that this relationship holds with equality from now on. Household optimization then implies that $W_R(s) = \frac{h}{2} s^2 \bar{W}_R$, where $\bar{W}_R$ is the average manager wage in the economy.

Taking the equilibrium manager wage function into account, we can rewrite the profit maxi-

⁷⁹The constant 0.5ι in a_i can be thought of as a fixed exposure to a constant risk factor $\varepsilon_0 \sim \mathcal{N}(1, 0)$. It therefore fits into the general model of the main text as well.

⁸⁰We assume for simplicity here that the constant and the linear terms in the approximation are zero. These can be included without problems by defining a new $\tilde{\delta}_i^\circ$ as a shifter in the cost function g_i and by assuming that there is a constant risk factor $\varepsilon_0 \sim \mathcal{N}(1, 0)$ with fixed exposure.

mization problem of firm i as

$$\max_{\delta_i} \mathbb{E} \left[\Lambda \left(Q_i (P_i - K_i (\delta_i)) - \frac{h}{2} (\delta_i - \delta_i^\circ)^2 \bar{W}_R \right) \right].$$

Comparing this equation with (11), we see that we have microfounded the risk-management cost function as $g_i = \frac{h}{2} (\delta_i - \delta_i^\circ)^2$. Under this microfoundation, changes in the training cost of new managers would affect the risk profile of the firms and the aggregate economy.

F.4 Production network and aggregate risk exposure

The structure of the production network is a key determinant of the Domar weight vector. It therefore affects, among other things, how firms' individual exposure decisions contribute to Δ . The following proposition describes how a change in the network affects Δ when $\tau = 0$, in which case the forces at work are more transparent. The proof of the proposition also provides expressions for the general case.

Proposition 12. *Suppose that $\tau = 0$. Then the response of the equilibrium aggregate risk exposure Δ to a change in network connection α_{ij} is given by*

$$\frac{d\Delta}{d\alpha_{ij}} = \mathcal{H}^{-1} \bar{H} \sum_{k=1}^N \left(\frac{d\Delta^\circ}{d\tilde{\omega}_k} + \frac{d\bar{H}^{-1}}{d\tilde{\omega}_k} \mathcal{E} \right) \frac{d\tilde{\omega}_k}{d\alpha_{ij}}, \quad (158)$$

where the response of Domar weights to a change in α_{ij} is given by $\frac{d\tilde{\omega}_k}{d\alpha_{ij}} = \tilde{\omega}_i \tilde{\mathcal{L}}_{jk}$, the response of the natural exposure to a change in Domar weight is given by $\frac{d\Delta^\circ}{d\tilde{\omega}_k} = \delta_k^\circ$, and the response of the curvature of $\bar{h}$ to a change in Domar weight is given by $\frac{d\bar{H}^{-1}}{d\tilde{\omega}_k} = H_k^{-1} + \eta \frac{1}{\tilde{\omega}_k} H_k^{-1} G_k H_k^{-1}$.

Proof. Using (79), we can write

$$\Delta - \Delta^\circ - \bar{H}^{-1} \mathcal{E} = 0.$$

Using the implicit function theorem, we get

$$\frac{d\Delta}{d\alpha_{ij}} = \mathcal{H}^{-1} \bar{H} \left(\frac{d\Delta^\circ}{d\alpha_{ij}} + \frac{d\bar{H}^{-1}}{d\alpha_{ij}} \mathcal{E} \right). \quad (159)$$

Suppose that $\tau = 0$. We have $\frac{d\Delta^\circ}{d\alpha_{ij}} = \sum_{k=1}^N \frac{d\Delta^\circ}{d\tilde{\omega}_k} \frac{d\tilde{\omega}_k}{d\alpha_{ij}}$ and, from (31),

$$\frac{d\bar{H}^{-1}}{d\alpha_{ij}} = \sum_{k=1}^N \frac{d\tilde{\omega}_k}{d\alpha_{ij}} \left(H_k^{-1} + \eta \frac{1}{\tilde{\omega}_k} H_k^{-1} G_k H_k^{-1} \right).$$

By definition, $\Delta^\circ = \sum_{j=1}^N \delta_j^\circ \tilde{\omega}_j$ and $\tilde{\omega}_k = \beta^\top \tilde{\mathcal{L}} \mathbf{1}_k = \beta^\top (I - \alpha)^{-1} \mathbf{1}_k$. Then $\frac{d\Delta^\circ}{d\tilde{\omega}_k} = \delta_k^\circ$ and $\frac{d\tilde{\omega}_k}{d\alpha_{ij}} = \tilde{\omega}_i \tilde{\mathcal{L}}_{jk}$. Using these results, it is immediate to see that (158) follows from (159).

If $\tau \neq 0$, the only difference is that

$$\frac{d\bar{H}^{-1}}{d\alpha_{ij}} = \sum_{k=1}^N \frac{d\tilde{\omega}_k}{d\alpha_{ij}} \left(H_k^{-1} + \eta \frac{\kappa_k}{\tilde{\omega}_k} H_k^{-1} G_k H_k^{-1} \right) - \sum_{k=1}^N \frac{d\kappa_k}{d\alpha_{ij}} \eta H_k^{-1} G_k H_k^{-1},$$

where

$$\frac{d\kappa_k}{d\alpha_{ij}} = \frac{(1 + \tau_k)}{\omega_k} \frac{d\tilde{\omega}_k}{d\alpha_{ij}} - \frac{\tilde{\omega}_k (1 + \tau_k)}{\omega_k^2} \frac{d\omega_k}{d\alpha_{ij}},$$

and $\frac{d\omega_k}{d\alpha_{ij}} = \frac{\omega_i}{1 + \tau_i} \mathcal{L}_{jk}$. Therefore,

$$\frac{d\bar{H}^{-1}}{d\alpha_{ij}} = \sum_{k=1}^N \left\{ \frac{d\tilde{\omega}_k}{d\alpha_{ij}} H_k^{-1} + \eta \frac{\kappa_k}{\omega_k} \frac{d\omega_k}{d\alpha_{ij}} H_k^{-1} G_k H_k^{-1} \right\}.$$

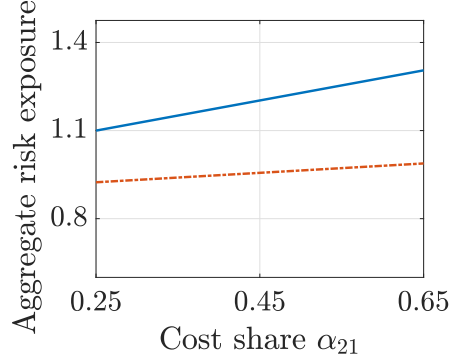
□

Equation (158) shows that when $\tau = 0$, the impact of α_{ij} on the aggregate risk exposure Δ operates exclusively through its effect on the Domar weights $\tilde{\omega}$ (last term in (158)). Since $\frac{d\tilde{\omega}_k}{d\alpha_{ij}} = \tilde{\omega}_i \tilde{\mathcal{L}}_{jk}$, an increase in i 's cost share of good j always leads to an increase in $\tilde{\omega}_k$. Indeed, a higher α_{ij} implies that firm j gains in importance as a supplier. It follows that the Domar weight of any firm k that supplies to j ($\tilde{\mathcal{L}}_{jk} > 0$) also benefits from the larger α_{ij} . The magnitude of that increase is larger when i is an important firm in the network (large $\tilde{\omega}_i$) or when k is an important supplier to j (large $\tilde{\mathcal{L}}_{jk}$).

This increase in Domar weights means that firms find it more cost-effective to manage risk. Consequently, the aggregate cost function becomes less curved, in the sense that $\frac{d\bar{H}^{-1}}{d\tilde{\omega}_k} = H_k^{-1} + \eta \frac{1}{\tilde{\omega}_k} H_k^{-1} G_k H_k^{-1}$ is a positive definite matrix. Through that channel, exposure to good risks tends to increase, while exposure to bad risks tends to decrease. Furthermore, the increase in Domar weights implies that the natural aggregate risk-exposure vector Δ° also adjusts. Specifically, an increase in Domar weight $\tilde{\omega}_k$ means that k 's natural exposure δ_k° matters more for Δ° . Both the adjustment in the curvature of $\bar{h}$ and in the natural aggregate risk exposure Δ° are propagated through the matrix $\mathcal{T} = \mathcal{H}^{-1} \bar{H}$ to shape the response of the aggregate risk exposure Δ to a change in α_{ij} .

Figure 21 shows the impact of increasing the importance of good 1 in the production of good 2 in the example economy of Figure 3. As α_{21} increases, firm 1 becomes a more important supplier and its Domar weight rises. Because of firm 1's rigid exposure to the good risk factor, Δ_1 increases. This triggers a response from firm 2, which reduces its own exposure to risk factor 1 to avoid creating too much correlated risk. Since G_2 is parametrized so that both risk factors are substitutes, firm 2's exposure to risk 2 increases, and Δ_2 rises as a result.

Figure 21: Changes in network in the example economy



Notes. The structure of the economy is given in panel (a) of Figure 3. Blue solid line shows $\Delta_1(\alpha_{21})$, and red dashed line shows $\Delta_2(\alpha_{21})$. Initial parametrization is as follows. Household: $\rho = 2$ and $\beta_2 = 0.8$, $\beta_1 = \beta_3 = 0.1$. Network: $\alpha_{21} = \alpha_{23} = 0.25$, all other entries of α are zero. Beliefs: $\mu = (0.75, 0)$, Σ is diagonal with $\text{diag}(\Sigma) = (0.5, 0.5)$. Risk exposures: $\delta_{11}^\circ = \delta_{32}^\circ = 1$, $\delta_{22}^\circ = 1.9$, $\delta_{12}^\circ = \delta_{21}^\circ = \delta_{31}^\circ = 0$, $G_1 = G_3$ are diagonal with very large entries on the main diagonals; $G_{2,11} = G_{2,22} = 1$, $G_{2,12} = G_{2,21} = 0.75$. We set $B_i = 0$ for all i . α_{21} changes from 0.25 to 0.65.

G Extensions

In this appendix, we extend the baseline model by allowing 1) multiple risk management resources, 2) general risk management cost functions, and 3) a general disutility from risk management.

G.1 Multiple risk management resources

In the baseline model, we assume that there is only one risk management resource. In this appendix, we show that this setup can be easily extended to handle multiple such resources. Specifically, suppose that there are K risk management resources, and that those can be supplied by the household at a cost in terms of utility. Denote by $R = (R_1, \dots, R_K)$ the vector of these resources. We assume that the representative household's utility function is $\mathcal{U}(Y)\mathcal{V}(R)$, where, as in the baseline model, $\mathcal{U}(Y) = \frac{Y^{1-\rho}}{1-\rho}$, and where $\mathcal{V}(R) = \exp\left((\rho-1)\sum_{k=1}^K \eta_k R_k\right)$.

The household's first-order conditions take the same form as before. Specifically, following the same steps as in Appendix F.1, we can derive

$$\Lambda = \left(\prod_{i=1}^N P_i^{\beta_i}\right)^{\rho-1} \Gamma_L^\rho \exp\left((\rho-1)\sum_{k=1}^K \eta_k R_k\right),$$

and

$$W_{R,k} = \eta_k \Gamma_L^{-1},$$

for all k , and where $W_{R,k}$ is the price of risk management resource k .

We assume that to achieve risk exposure δ_i , firm i must use $R_{ik} = g_{i,k}(\delta_i)$ units of resource k . As in the baseline model, we assume that $g_{i,k}(\delta_i)$ is quadratic with $g_{i,k}(\delta_i) = \frac{1}{2}(\delta_i - \delta_i^\circ)^\top G_{i,k}(\delta_i - \delta_i^\circ)$, where $G_{i,k}$ is a positive definite matrix. Consider now the problem of the firm. For a given risk

exposure δ_i , the cost minimization characterization is the same as in the baseline model (see Section 2.4). When choosing its risk exposure, firm i then solves

$$\delta_i^* \in \arg \max_{\delta_i \in \mathcal{A}_i} \mathbb{E} \left[\Lambda \left[Q_i(P_i - K_i(\delta_i, P)) - \sum_{k=1}^K g_{i,k}(\delta_i) W_{R,k} \right] \right].$$

Following the same steps as in the baseline model (see, in particular, the proof of Proposition 1), we can show that the equilibrium exposure decisions follow

$$\delta_i - \delta_i^\circ = \left(B_i + \frac{1 + \tau_i}{\omega_i} \sum_{k=1}^K \eta_k G_{i,k} \right)^{-1} \mathcal{E}. \quad (160)$$

The key difference between this extension and the model with a single risk management resource is that $\sum_{k=1}^K \eta_k G_{i,k}$ replaces ηG_i . Since both objects are constant, we can think of the Hessian matrix G_i from the main text as capturing the aggregated substitution patterns generated by the resource-specific Hessians $G_{i,k}$.

We can again characterize the equilibrium using a distorted planner's problem. Consider the maximization problem

$$\max_{\delta} \tilde{\omega}^\top \delta \mu - \tilde{\omega}^\top b(\delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L - \frac{1}{2} (\rho - 1) \tilde{\omega}^\top \delta \Sigma \delta^\top \tilde{\omega} - \sum_{i=1}^N \sum_{k=1}^K \eta_k \kappa_i g_{i,k}(\delta_i), \quad (161)$$

where $\kappa_i = \frac{\tilde{\omega}_i(1 + \tau_i)}{\omega_i}$. Taking first-order conditions with respect to δ_i , we get

$$\begin{aligned} 0 &= \tilde{\omega}_i \mu - \tilde{\omega}_i B_i(\delta_i - \delta_i^\circ) - (\rho - 1) \tilde{\omega}_i \Sigma \delta^\top \tilde{\omega} - \sum_{k=1}^K \eta_k \frac{\tilde{\omega}_i(1 + \tau_i)}{\omega_i} G_{i,k}(\delta_i - \delta_i^\circ) \Leftrightarrow \\ (\delta_i - \delta_i^\circ) &= \left(B_i + \frac{(1 + \tau_i)}{\omega_i} \sum_{k=1}^K \eta_k G_{i,k} \right)^{-1} \left(\mu - (\rho - 1) \Sigma \delta^\top \tilde{\omega} \right), \end{aligned}$$

which is equivalent to (160). It follows that any equilibrium must coincide with a solution to the optimization problem (161).

We can write (161) as

$$\max_{\Delta} \Delta^\top \mu - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L - \frac{1}{2} (\rho - 1) \Delta^\top \Sigma \Delta - \min_{\delta \text{ s.t. } \Delta = \delta^\top \tilde{\omega}} \left[\tilde{\omega}^\top b(\delta) + \sum_{i=1}^N \sum_{k=1}^K \eta_k \kappa_i g_{i,k}(\delta_i) \right],$$

or, equivalently, as

$$\mathcal{W}_{dist} := \max_{\Delta} \underbrace{\Delta^\top \mu - \bar{b}(\Delta) - \tilde{\omega}^\top \log(1 + \tau) - \log \Gamma_L}_{\mathbb{E}[y]} - \frac{1}{2} (\rho - 1) \underbrace{\Delta^\top \Sigma \Delta}_{\mathbb{V}[y]} - \bar{g}(\Delta), \quad (162)$$

where

$$\bar{b}(\Delta) := \tilde{\omega}^\top b(\delta(\Delta)) \text{ and } \bar{g}(\Delta) := \sum_{i=1}^N \sum_{k=1}^K \eta_k \kappa_i g_{i,k}(\delta_i(\Delta)), \quad (163)$$

where $\delta(\Delta)$ solves

$$\bar{h}(\Delta) = \min_{\delta \text{ s.t. } \Delta = \delta^\top \tilde{\omega}} \left[\tilde{\omega}^\top b(\delta) + \sum_{i=1}^N \sum_{k=1}^K \eta_k \kappa_i g_{i,k}(\delta_i) \right]. \quad (164)$$

As in the baseline model, we can solve for $\bar{h}(\Delta) = \bar{b}(\Delta) + \bar{g}(\Delta)$.

Lemma 16. *The aggregate cost function $\bar{h}$ is given by*

$$\bar{h}(\Delta) = \frac{1}{2} (\Delta - \Delta^\circ)^\top \nabla^2 \bar{h} (\Delta - \Delta^\circ), \quad (165)$$

where $\Delta^\circ = (\delta^\circ)^\top \tilde{\omega}$, and where the Hessian matrix of $\bar{h}$ is given by

$$\bar{H} = \left[\sum_{i=1}^N \tilde{\omega}_i H_i^{-1} \right]^{-1}, \text{ where } H_i = B_i + \frac{\kappa_i}{\tilde{\omega}_i} \sum_{k=1}^K \eta_k G_{i,k}. \quad (166)$$

Proof. The Lagrangian of problem (164) is

$$\mathcal{L} = \sum_{i=1}^N \tilde{\omega}_i b_i(\delta_i) + \sum_{i=1}^N \sum_{k=1}^K \eta_k \kappa_i g_{i,k}(\delta_i) - \sum_{m=1}^M \nu_m \left(\Delta_m - \mathbf{1}_m^\top \delta^\top \tilde{\omega} \right),$$

where ν_m is the Lagrange multiplier on the m th row of the constraint $\Delta = \delta^\top \tilde{\omega}$. The first-order condition with respect to δ_i is

$$\tilde{\omega}_i \nabla b_i(\delta_i) + \kappa_i \sum_{k=1}^K \eta_k \nabla g_{i,k}(\delta_i) = -\nu \tilde{\omega}_i,$$

which implies that

$$\nabla b_i(\delta_i) + \frac{\kappa_i}{\tilde{\omega}_i} \sum_{k=1}^K \eta_k \nabla g_{i,k}(\delta_i) = \nabla b_j(\delta_j) + \frac{\kappa_j}{\tilde{\omega}_j} \sum_{k=1}^K \eta_k \nabla g_{j,k}(\delta_j) = \nabla \bar{h},$$

for all i, j , where the last equality comes from the envelope theorem. Notice that

$$\begin{aligned} \left(B_i + \frac{\kappa_i}{\tilde{\omega}_i} \sum_{k=1}^K \eta_k G_{i,k} \right) (\delta_i - \delta_i^\circ) &= \left(B_j + \frac{\kappa_j}{\tilde{\omega}_j} \sum_{k=1}^K \eta_k G_{j,k} \right) (\delta_j - \delta_j^\circ) \Leftrightarrow \\ \delta_i &= \delta_i^\circ + \left(B_i + \frac{\kappa_i}{\tilde{\omega}_i} \sum_{k=1}^K \eta_k G_{i,k} \right)^{-1} \left(B_j + \frac{\kappa_j}{\tilde{\omega}_j} \sum_{k=1}^K \eta_k G_{j,k} \right) (\delta_j - \delta_j^\circ). \end{aligned}$$

Then, the constraint $\Delta = \delta^\top \tilde{\omega}$ can be rewritten as

$$\begin{aligned}\Delta &= \sum_{i=1}^N \tilde{\omega}_i \delta_i = \sum_{i=1}^N \tilde{\omega}_i \left(\delta_i^\circ + \left(B_i + \frac{\kappa_i}{\tilde{\omega}_i} \sum_{k=1}^K \eta_k G_{i,k} \right)^{-1} \left(B_j + \frac{\kappa_j}{\tilde{\omega}_j} \sum_{k=1}^K \eta_k G_{j,k} \right) (\delta_j - \delta_j^\circ) \right) \\ &= \Delta^\circ + \sum_{i=1}^N \tilde{\omega}_i \left(B_i + \frac{\kappa_i}{\tilde{\omega}_i} \sum_{k=1}^K \eta_k G_{i,k} \right)^{-1} \left(B_j + \frac{\kappa_j}{\tilde{\omega}_j} \sum_{k=1}^K \eta_k G_{j,k} \right) (\delta_j - \delta_j^\circ) \Leftrightarrow \\ \delta_j - \delta_j^\circ &= \left(B_j + \frac{\kappa_j}{\tilde{\omega}_j} \sum_{k=1}^K \eta_k G_{j,k} \right)^{-1} \bar{H} (\Delta - \Delta^\circ),\end{aligned}$$

where $\bar{H}$ is given by (166). Combining this last expression with (164) yields the result. $\square$

Since (162) and (163) fully characterize the equilibrium, it follows that the only difference between the baseline model and the multiple resources model comes from the cost function $\bar{h}$. In turn, the cost function in the baseline model is identical to (165) if we impose that

$$\eta G_i = \sum_{k=1}^K \eta_k G_{i,k}$$

for all i . Under that restriction, both models behave identically. It follows that we can simply think of the unique resource in the baseline model as an aggregate of various risk management resources, each with its own supply elasticity η_k .

G.2 General cost function

In this appendix, we relax our assumption on the functional form of the cost functions $b_i(\delta_i)$ and $g_i(\delta_i)$. Specifically, we assume that $b_i(\delta_i)$ and $g_i(\delta_i)$ are strictly convex—but not necessarily quadratic—functions. The Hessians of $b_i(\delta_i)$ and $g_i(\delta_i)$ are denoted by B_i and G_i , respectively, and both of them are positive definite matrices. Unlike in the baseline model, B_i and G_i are not necessarily constant, in the sense that they can depend on δ_i .

All the results of Sections 2, 3, 4, and 5 remain unchanged. However, (30) does not hold when b_i and g_i are non-quadratic. Nevertheless, the Hessian of the aggregate cost function $\bar{h}$ is still given by (31), as the following result shows.

Lemma 17. *The Hessian matrix of $\bar{h}$ is given by (31).*

Proof. Equations (70), (71), and (72) do not use the assumption that the cost functions are quadratic and, hence, hold in this more general case. Differentiate (72) with respect to Δ_j to get

$$\frac{d^2 \bar{h}}{d\Delta_m d\Delta_j} = \mathbf{1}_m^\top \left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i \right) \frac{d\delta_i}{d\Delta_j} \quad (167)$$

and

$$\left(B_k + \eta \frac{\kappa_k}{\tilde{\omega}_k} G_k\right) \frac{d\delta_k}{d\Delta_j} = \left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i\right) \frac{d\delta_i}{d\Delta_j}$$

for all i, k .

Next, we can differentiate the constraint $\Delta = \delta^\top \tilde{\omega}$ with respect to Δ_j to find $\sum_{k=1}^N \tilde{\omega}_k \frac{d\delta_k}{d\Delta_j} = \mathbf{1}_j$. Combining this with the last equation, we find

$$\left(B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i\right) \frac{d\delta_i}{d\Delta_j} = \left(\sum_{k=1}^N \tilde{\omega}_k \left(B_k + \eta \frac{\kappa_k}{\tilde{\omega}_k} G_k\right)^{-1}\right)^{-1} \mathbf{1}_j. \quad (168)$$

Plugging this into (167) yields (31). $\square$

G.2.1 Risk exposures

Turning to the results of Section 6, we are no longer able to solve for Δ and δ in closed form. Nevertheless, we can still derive similar comparative statics results. We start by characterizing how δ depends on aggregate risk exposure Δ . We then characterize how Δ depends on various primitives of the model.

Proposition 13. *The response of firm i 's risk exposure δ_i to a change in the aggregate risk exposure Δ_j is given by*

$$\frac{d\delta_i}{d\Delta_j} = H_i^{-1} \bar{H} \mathbf{1}_j, \quad (169)$$

where $H_i = B_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} G_i$.

Proof. This result follows immediately from (168) and (31). $\square$

In the model with general cost functions, (33) still implicitly defines the equilibrium Δ . Proposition 3 immediately follows by applying the implicit function theorem to (33). In what follows, we prove analogues of Propositions 4 and 12.

Proposition 14. *The response of the equilibrium aggregate risk exposure Δ to a change in wedge τ_i is given by*

$$\frac{d\Delta}{d\tau_i} = -\mathcal{H}^{-1} \bar{H} \sum_{j=1}^N \eta \frac{d\kappa_j}{d\tau_i} H_j^{-1} (\nabla g_j), \quad (170)$$

where $\mathcal{H}$ is given by (37), $\bar{H}$ is given by (31), $d\kappa_j/d\tau_i$ is given by (80), and ∇g_j is evaluated at the equilibrium δ_j .

Proof. By the implicit function theorem applied to (33),

$$\frac{d\Delta}{d\tau_i} = -\mathcal{H}^{-1} \frac{\partial \nabla \bar{h}}{\partial \tau_i} = -\mathcal{H}^{-1} \sum_{j=1}^N \frac{\partial \nabla \bar{h}}{\partial \kappa_j} \frac{d\kappa_j}{d\tau_i}, \quad (171)$$

where $\mathcal{H}$ is given by (37) and $\frac{d\kappa_j}{d\tau_i}$ is given by (80).

Next, from the envelope theorem applied to (70), we have $\frac{\partial \bar{h}}{\partial \kappa_j} = \eta g_j(\delta_j)$, and so

$$\frac{d}{d\Delta_k} \frac{\partial \bar{h}}{\partial \kappa_j} = \eta (\nabla g_j)^\top \frac{d\delta_j}{d\Delta_k}.$$

Using (169), we get

$$\frac{\partial \nabla \bar{h}}{\partial \kappa_j} = \eta \bar{H} H_j^{-1} \nabla g_j. \quad (172)$$

Plugging this into (171), we get

$$\frac{d\Delta}{d\tau_i} = -\mathcal{H}^{-1} \bar{H} \sum_{j=1}^N \eta H_j^{-1} \nabla g_j \frac{d\kappa_j}{d\tau_i},$$

which is the result. $\square$

Next, we derive an analogue of Proposition 12.

Proposition 15. *The response of the equilibrium aggregate risk exposure Δ to a change in network connection α_{ij} is given by*

$$\frac{d\Delta}{d\alpha_{ij}} = -\mathcal{H}^{-1} \bar{H} \sum_{k=1}^N \left[\eta H_k^{-1} (\nabla g_k) \frac{d\kappa_k}{d\alpha_{ij}} - \left(\delta_k + \eta \frac{\kappa_k}{\bar{\omega}_k} H_k^{-1} \nabla g_k \right) \frac{d\bar{\omega}_k}{d\alpha_{ij}} \right], \quad (173)$$

where $\frac{d\kappa_k}{d\alpha_{ij}} = \frac{(1+\tau_k)}{\omega_k} \frac{d\bar{\omega}_k}{d\alpha_{ij}} - \frac{\bar{\omega}_k(1+\tau_k)}{\omega_k^2} \frac{d\omega_k}{d\alpha_{ij}}$, $\frac{d\bar{\omega}_k}{d\alpha_{ij}} = \tilde{\omega}_i \tilde{\mathcal{L}}_{jk}$, $\frac{d\omega_k}{d\alpha_{ij}} = \frac{\omega_i}{1+\tau_i} \mathcal{L}_{jk}$, $\mathcal{H}$ is given by (37), $\bar{H}$ is given by (31), and ∇g_k is evaluated at the equilibrium δ_k .

Proof. By the implicit function theorem applied to (33),

$$\frac{d\Delta}{d\alpha_{ij}} = -\mathcal{H}^{-1} \sum_{k=1}^N \left(\frac{\partial \nabla \bar{h}}{\partial \kappa_k} \frac{d\kappa_k}{d\alpha_{ij}} + \frac{\partial \nabla \bar{h}}{\partial \bar{\omega}_k} \frac{d\bar{\omega}_k}{d\alpha_{ij}} \right), \quad (174)$$

where $\mathcal{H}$ is given by (37), $\frac{d\bar{\omega}_k}{d\alpha_{ij}} = \tilde{\omega}_i \tilde{\mathcal{L}}_{jk}$,

$$\frac{d\kappa_k}{d\alpha_{ij}} = \frac{(1+\tau_k)}{\omega_k} \frac{d\bar{\omega}_k}{d\alpha_{ij}} - \frac{\bar{\omega}_k(1+\tau_k)}{\omega_k^2} \frac{d\omega_k}{d\alpha_{ij}},$$

and $\frac{d\omega_k}{d\alpha_{ij}} = \frac{\omega_i}{1+\tau_i} \mathcal{L}_{jk}$.

Next, from (70), we obtain $\frac{\partial \bar{h}}{\partial \bar{\omega}_k} = b_k + \nu^\top \delta_k$, where $\nu = -\nabla \bar{h}$ from (72). Therefore,

$$\frac{d}{d\Delta_m} \frac{\partial \bar{h}}{\partial \bar{\omega}_k} = (\nabla b_k)^\top \frac{d\delta_k}{d\Delta_m} - \mathbf{1}_m^\top \bar{H} \delta_k - (\nabla \bar{h})^\top \frac{d\delta_k}{d\Delta_m}.$$

Using (169), this equation can be rewritten as

$$\frac{\partial \nabla \bar{h}}{\partial \tilde{\omega}_k} = -\bar{H} \delta_k - \bar{H} H_k^{-1} (\nabla \bar{h} - \nabla b_k). \quad (175)$$

Plugging (172), (72), and (175) into (174), we get (173). $\square$

G.2.2 GDP and welfare

As can be seen from the analysis of the baseline model, we do not use the fact that b_i and g_i are quadratic in Section 7, with the exception of deriving the expression for $\nabla \bar{h} - \nabla \bar{h}_V$, which is given by (86). Below, we derive an analogue of this equation under general cost functions. Using the definition of $\bar{h}_V$, we can derive

$$\frac{d\bar{h}_V}{d\Delta_m} = \sum_{i=1}^N \left(\tilde{\omega}_i (\nabla b_i)^\top + \eta (\nabla g_i)^\top \right) \frac{d\delta_i}{d\Delta_m}.$$

From (72), we have $\nabla b_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} \nabla g_i = \nabla \bar{h}$. Furthermore, $\frac{d\delta_i}{d\Delta_m}$ is given by (169). Then we have

$$\nabla \bar{h} - \nabla \bar{h}_V = (\nabla^2 \bar{h}) \sum_{i=1}^N \eta (\kappa_i - 1) H_i^{-1} \nabla g_i.$$

In our baseline model with quadratic cost functions, we have $\nabla g_i = G_i (\delta_i - \delta_i^\circ) = G_i H_i^{-1} \mathcal{E}$ (from (24)), such that in a diagonal economy, $\text{sign}(\nabla g_i) = \text{sign}(\mathcal{E}) \forall i$. In the model with general cost functions, it is not always true that the signs of ∇g_i and $\mathcal{E}$ coincide. Thus, Lemmas 7, 8, and 9 will generally look different. One sufficient condition under which the signs of ∇g_i and $\mathcal{E}$ coincide is the following. Suppose that $b_i(\delta_i) = v_i g_i(\delta_i)$ where v_i is a non-negative constant. Then, from (72) and (33),

$$\left(v_i + \eta \frac{\kappa_i}{\tilde{\omega}_i} \right) \nabla g_i = \nabla \bar{h} = \mathcal{E},$$

such that the signs of ∇g_i and $\mathcal{E}$ coincide. Then the results of Section 7 hold even if b_i and g_i are non-quadratic.

G.3 A general specification of the disutility from risk management

In the main model, we assume that the household's disutility is given by (6), where $\mathcal{V}(R) = \exp(-\eta(1-\rho)R)$. In this appendix, we consider a general disutility from supplying risk management resources. Specifically, the household's utility function is given by

$$\mathcal{U} \left(\frac{Y^{1-\rho}}{1-\rho}, R \right),$$

where $Y = \prod_{i=1}^N (\beta_i^{-1} C_i)^{\beta_i}$. Denote by $\mathcal{U}_1$ and $\mathcal{U}_2$ the derivatives of $\mathcal{U}(\cdot, \cdot)$ with respect to its first and second arguments, respectively. Following the same steps as in the baseline model (see Appendix F.1), we can derive

$$\mathbb{E} [\mathcal{U}_2] = -W_R \mathbb{E} [\Lambda], \quad (176)$$

where

$$\Lambda = \Gamma_L^\rho \mathcal{U}_1 \left(\prod_{i=1}^N P_i^{\beta_i} \right)^{\rho-1}. \quad (177)$$

As in the baseline model, the labor share of income Γ_L is given by $\Gamma_L := \frac{W_L L}{P Y} = 1 - \tau^\top (\text{diag}(1 + \tau))^{-1} \omega$, and real GDP is given by $Y = \Gamma_L^{-1} \left(\prod_{i=1}^N P_i^{-\beta_i} \right)$ (see Lemma 1).

For a given risk-exposure decision, a firm's cost minimization does not depend on the household's utility function. Therefore, the unit cost for firm i is given by (10), and the vector of prices is given by (15). The choice of risk exposure for firm i is described by (11), which can be rewritten as

$$\delta_i^* \in \arg \min_{\delta_i} \mathbb{E} [\Lambda [Q_i K_i (\delta_i, P) + g_i (\delta_i) W_R]].$$

Taking the first-order condition with respect to δ_{im} , we get

$$\mathbb{E} \left[\Lambda Q_i \frac{dK_i}{d\delta_{im}} \right] + \mathbb{E} [\Lambda] W_R \frac{dg_i}{d\delta_{im}} = 0.$$

Using the expression for the unit cost (10) together with equations (176) and (177), this expression can be simplified as

$$\frac{dg_i}{d\delta_{im}} = -K_i Q_i \frac{\mathbb{E} \left[\Gamma_L^\rho \mathcal{U}_1 \left(\prod_{i=1}^N P_i^{\beta_i} \right)^{\rho-1} \left(\varepsilon_m - \frac{db_i}{d\delta_{im}} \right) \right]}{\mathbb{E} [\mathcal{U}_2]}.$$

Recall that in equilibrium, $K_i Q_i = \frac{\omega_i \Gamma_L^{-1}}{1 + \tau_i}$. Using the expression for prices (15), we can rewrite the equation above as

$$\begin{aligned} \frac{dg_i}{d\delta_{im}} = & - \frac{\omega_i \Gamma_L^{\rho-1}}{1 + \tau_i} \exp \left((\rho - 1) \tilde{\omega}^\top (b(\delta) + \log(1 + \tau)) \right) \frac{\mathbb{E} \left[\mathcal{U}_1 \exp \left(-(\rho - 1) \tilde{\omega}^\top \delta \varepsilon \right) \left(\varepsilon_m - \frac{db_i}{d\delta_{im}} \right) \right]}{\mathbb{E} [\mathcal{U}_2]}. \end{aligned} \quad (178)$$

The system of equations (178) for all i, m implicitly defines the equilibrium risk-exposure matrix δ . In our baseline model, we show that the equilibrium allocation is a solution to a distorted planner's problem (32). This distorted planner chooses the aggregate risk exposure Δ , and then chooses δ to minimize the aggregate cost function (29) subject to the $\Delta = \delta^\top \tilde{\omega}$ constraint. For a general utility function, we are not able to write the equilibrium allocation as a solution to a distorted planner's

problem. Therefore, to characterize the equilibrium, we need to work with equation (178) directly. In what follows, we use the implicit function theorem to characterize how equilibrium δ changes in response to a change in the environment. It is then straightforward to characterize how the aggregate risk exposure $\Delta = \delta^\top \tilde{\omega}$ responds.

In general, one cannot compute the expectations on the right-hand side of (178) analytically. Therefore, it is generally infeasible to compute the derivatives of δ with respect to changes in the environment in closed form. We consider two special cases for which this is possible.

G.3.1 Multiplicative disutility

Suppose that the household's utility function is $\mathcal{U}\left(\frac{Y^{1-\rho}}{1-\rho}, R\right) = \frac{Y^{1-\rho}}{1-\rho} V(R)$, where $V(R)$ takes positive values and is increasing and convex. One special case, considered in the baseline model, is $V(R) = \exp((\rho - 1)\eta R)$. Under this specification, (178) becomes

$$\frac{dg_i}{d\delta_{im}} = (\rho - 1) \frac{\omega_i}{1 + \tau_i} \left(\mu_m - (\rho - 1) \mathbf{1}_m^\top \Sigma \delta^\top \tilde{\omega} - \frac{db_i}{d\delta_{im}} \right) \frac{V(R)}{V'(R)}.$$

Market clearing implies that $R = \sum_{i=1}^N g_i(\delta_i)$. While we cannot solve for δ in closed form for a general $V(\cdot)$ function, we can characterize how δ changes when one of the parameters changes using the implicit function theorem. Specifically, write the first-order condition for δ_{im} as

$$F_{im} = \frac{dg_i}{d\delta_{im}} - (\rho - 1) \frac{\omega_i}{1 + \tau_i} \left(\mu_m - (\rho - 1) \mathbf{1}_m^\top \Sigma \delta^\top \tilde{\omega} - \frac{db_i}{d\delta_{im}} \right) \frac{V\left(\sum_{i=1}^N g_i(\delta_i)\right)}{V'\left(\sum_{i=1}^N g_i(\delta_i)\right)} = 0.$$

Then, by the implicit function theorem,

$$\begin{pmatrix} \frac{d\delta_{11}}{d\chi} \\ \vdots \\ \frac{d\delta_{1M}}{d\chi} \\ \vdots \\ \frac{d\delta_{N1}}{d\chi} \\ \vdots \\ \frac{d\delta_{NM}}{d\chi} \end{pmatrix} = - \begin{pmatrix} \frac{\partial F_{11}}{\partial \delta_{11}} & \cdots & \frac{\partial F_{11}}{\partial \delta_{1M}} & \cdots & \frac{\partial F_{11}}{\partial \delta_{N1}} & \cdots & \frac{\partial F_{11}}{\partial \delta_{NM}} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial F_{1M}}{\partial \delta_{11}} & \cdots & \frac{\partial F_{1M}}{\partial \delta_{1M}} & \cdots & \frac{\partial F_{1M}}{\partial \delta_{N1}} & \cdots & \frac{\partial F_{1M}}{\partial \delta_{NM}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial F_{N1}}{\partial \delta_{11}} & \cdots & \frac{\partial F_{N1}}{\partial \delta_{1M}} & \cdots & \frac{\partial F_{N1}}{\partial \delta_{N1}} & \cdots & \frac{\partial F_{N1}}{\partial \delta_{NM}} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial F_{NM}}{\partial \delta_{11}} & \cdots & \frac{\partial F_{NM}}{\partial \delta_{1M}} & \cdots & \frac{\partial F_{NM}}{\partial \delta_{N1}} & \cdots & \frac{\partial F_{NM}}{\partial \delta_{NM}} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial F_{11}}{\partial \chi} \\ \vdots \\ \frac{\partial F_{1M}}{\partial \chi} \\ \vdots \\ \frac{\partial F_{N1}}{\partial \chi} \\ \vdots \\ \frac{\partial F_{NM}}{\partial \chi} \end{pmatrix}, \quad (179)$$

where

$$\begin{aligned} \frac{\partial F_{im}}{\partial \delta_{jl}} &= \frac{d^2 g_i}{d\delta_{im} d\delta_{il}} \mathbf{1}_{j=i} + \\ &(\rho - 1) \frac{\omega_i}{1 + \tau_i} \left((\rho - 1) \Sigma_{ml} \tilde{\omega}_j + \frac{d^2 b_i}{d\delta_{im} d\delta_{jl}} \mathbf{1}_{j=i} \right) \frac{V(R)}{V'(R)} - \frac{dg_i}{d\delta_{im}} \frac{dg_j}{d\delta_{jl}} \left(\frac{V'(R)}{V(R)} - \frac{V''(R)}{V'(R)} \right), \end{aligned}$$

and $\frac{\partial F_{im}}{\partial \chi}$ is the derivative of F_{im} with respect to parameter χ . For example, if $\chi = \mu_j$, we have

$$\frac{\partial F_{im}}{\partial \mu_j} = -\mathbf{1}_{j=m} (\rho - 1) \frac{\omega_i}{1 + \tau_i} \frac{V(R)}{V'(R)}.$$

Similarly, one can compute

$$\begin{aligned} \frac{\partial F_{im}}{\partial \Sigma_{jl}} &= (\rho - 1)^2 \frac{\omega_i}{1 + \tau_i} \frac{1}{2} (\mathbf{1}_{j=m} \Delta_l + \mathbf{1}_{l=m} \Delta_j) \frac{V(R)}{V'(R)}, \\ \frac{\partial F_{im}}{\partial \tau_j} &= (\rho - 1) \frac{1}{1 + \tau_i} \frac{\omega_j}{1 + \tau_j} \mathcal{L}_{ji} \left(\mu_m - (\rho - 1) \mathbf{1}_m^\top \Sigma \Delta - \frac{db_i}{d\delta_{im}} \right) \frac{V(R)}{V'(R)}, \\ \frac{\partial F_{im}}{\partial \alpha_{jl}} &= -(\rho - 1) \frac{\mathcal{L}_{li}}{1 + \tau_i} \frac{\omega_j}{1 + \tau_j} \left(\mu_m - (\rho - 1) \mathbf{1}_m^\top \Sigma \Delta - \frac{db_i}{d\delta_{im}} \right) \frac{V(R)}{V'(R)} + \\ &(\rho - 1)^2 \frac{\omega_i}{1 + \tau_i} \mathbf{1}_m^\top \Sigma \left(\sum_{k=1}^N \delta_k \tilde{\mathcal{L}}_{lk} \right) \tilde{\omega}_j \frac{V(R)}{V'(R)}, \end{aligned}$$

where we have used that $\frac{d\tilde{\omega}_k}{d\alpha_{ij}} = \tilde{\omega}_i \tilde{\mathcal{L}}_{jk}$ and $\frac{d\omega_k}{d\alpha_{ij}} = \frac{\omega_i}{1 + \tau_i} \mathcal{L}_{jk}$ (see the proof of Proposition 12), and $\frac{d\left(\frac{\omega_j}{1 + \tau_j}\right)}{d\tau_i} = -\frac{\mathcal{L}_{ij}}{1 + \tau_j} \frac{\omega_i}{1 + \tau_i}$.

G.3.2 Additive disutility

Suppose that the household's utility function is $\mathcal{U}\left(\frac{Y^{1-\rho}}{1-\rho}, R\right) = \frac{Y^{1-\rho}}{1-\rho} - V(R)$, where $V(R)$ is an increasing convex function. Under this specification, (178) becomes

$$\begin{aligned} \frac{dg_i}{d\delta_{im}} &= \frac{\omega_i \Gamma_L^{\rho-1}}{1 + \tau_i} \exp\left((\rho - 1) \tilde{\omega}^\top (b(\delta) + \log(1 + \tau))\right) \mathbb{E} \left[\exp\left(-(\rho - 1) \tilde{\omega}^\top \delta \varepsilon\right) \left(\varepsilon_m - \frac{db_i}{d\delta_{im}}\right) \right] \frac{1}{V'(R)} \\ &= \frac{\omega_i \Gamma_L^{\rho-1}}{1 + \tau_i} \left(\mu_m - \frac{db_i}{d\delta_{im}} - (\rho - 1) \mathbf{1}_m^\top \Sigma \delta^\top \tilde{\omega} \right) \times \\ &\quad \exp\left(-(\rho - 1) \tilde{\omega}^\top (\delta \mu - b(\delta) - \log(1 + \tau)) + \frac{1}{2} (\rho - 1)^2 \tilde{\omega}^\top \delta \Sigma \delta^\top \tilde{\omega}\right) \frac{1}{V'(R)}. \end{aligned}$$

Market clearing implies that $R = \sum_{i=1}^N g_i(\delta_i)$. As in the previous section, we cannot solve for δ in closed form. However, we can similarly characterize how δ changes when one of the parameters

changes using the implicit function theorem. Specifically, write the first-order condition for δ_{im} as

$$F_{im} = \frac{dg_i}{d\delta_{im}} - \frac{\omega_i \Gamma_L^{\rho-1}}{1 + \tau_i} \left(\mu_m - \frac{db_i}{d\delta_{im}} - (\rho - 1) \mathbf{1}_m^\top \Sigma \delta^\top \tilde{\omega} \right) \times \\ \exp \left(-(\rho - 1) \tilde{\omega}^\top (\delta \mu - b(\delta) - \log(1 + \tau)) + \frac{1}{2} (\rho - 1)^2 \tilde{\omega}^\top \delta \Sigma \delta^\top \tilde{\omega} \right) \frac{1}{V' \left(\sum_{i=1}^N g_i(\delta_i) \right)} = 0.$$

We can again use the implicit function theorem to get (179), where

$$\frac{\partial F_{im}}{\partial \delta_{jl}} = \frac{d^2 g_i}{d\delta_{im} d\delta_{il}} \mathbf{1}_{j=i} + \frac{dg_i}{d\delta_{im}} \times \\ \left\{ \frac{\frac{d^2 b_i}{d\delta_{im} d\delta_{il}} \mathbf{1}_{j=i} + (\rho - 1) \Sigma_{ml} \tilde{\omega}_j}{\mu_m - \frac{db_i}{d\delta_{im}} - (\rho - 1) \mathbf{1}_m^\top \Sigma \delta^\top \tilde{\omega}} + \frac{V''(R)}{V'(R)} \frac{dg_j}{d\delta_{jl}} + (\rho - 1) \tilde{\omega}_j \left(\mu_l - \frac{db_j}{d\delta_{jl}} - (\rho - 1) \mathbf{1}_l^\top \Sigma \delta^\top \tilde{\omega} \right) \right\},$$

and $\frac{\partial F_{im}}{\partial \chi}$ is the derivative of F_{im} with respect to parameter χ . For example, if $\chi = \mu_j$, we have

$$\frac{\partial F_{im}}{\partial \mu_j} = \left(-\frac{\mathbf{1}_{j=m}}{\mu_m - \frac{db_i}{d\delta_{im}} - (\rho - 1) \mathbf{1}_m^\top \Sigma \delta^\top \tilde{\omega}} + (\rho - 1) \tilde{\omega}^\top \delta \mathbf{1}_j \right) \frac{dg_i}{d\delta_{im}}.$$

Similarly, one can compute derivatives of F_{im} with respect to Σ_{jl} , τ_j and α_{jl} in a straightforward way.

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